

SIMPLIFIED CHEMISTRY NOTES

O'LEVEL

(S.1 – S.4)

2026 Edition

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CHEMISTRY AND SOCIETY

1.0.0 Introduction to Chemistry

Welcome to Chemistry, the Science concerned with the study of matter.

In simplest terms, chemistry is the science of matter.

Anything that can be touched, tasted, smelled, seen or felt is made of chemicals.



Tasting, feeling and smelling

Different things feel, taste, smell and appear differently- like the fresh vegetables. If you are blindfolded and asked to identify potatoes, onions, carrots or cabbages from a pack, chances are you could, based on interactions with them in your home kitchen.



Fresh vegetables

Chemistry will supply you with the knowledge and understanding to engage as informed citizens with science based issues.

You will use contemporary and relevant contexts of interest such as environmental issues to gain greater scientific knowledge.

You will study the patterns and properties of the elements and how these combine to produce useful materials, such as air and water among others.

Throughout the course you will develop practical skills, powers of analysis and the ability to apply chemical concepts to unfamiliar situations.

Being a Good Chemistry Student

You are beginning the study of one of the most interesting subjects that you will ever come across.

As is the case with all subjects, what you will get out of Chemistry will depend upon what you put into it.

Chemistry can intrigue and enlighten you.

It all depends upon the effort you are willing to put into your studies.

If you keep an open mind, and listen to your teacher, a course in Chemistry will change the way you view the world! Chemistry is a skill based course, and many of the most important lessons will come in the first few weeks.

You must start off with good study habits from the very beginning.

Below are some tips to help ensure that you get the most out of the course.

Do the reading faithfully.

Don't try to get through the course by just listening to your teacher. Your teacher will assume that you are coming to class having read the text, and he or she will expect you to have that background knowledge required to follow the lessons.

Most importantly, you will not be able to ask clarifying questions, if you have never read the book.

Ask as many questions as you want to

Like any teacher, your Chemistry teacher uses the feedback from the class to determine whether or not his or her lessons are clear enough. The types of questions that you and your classmates ask will tell your teacher more than the results of any exam.

Some students are embarrassed to admit that something is unclear, but this subject is new to you.

There is no shame in admitting that you are not sure about a subject is new to you!

Get help early and often.

As soon as you notice that you are not following along with the class discussion you should address the problem.

There are so many sources for extra help, and you should take advantage of them.

Make an appointment to see your teacher after school, or ask one of your classmates for help.

Search the Internet for sites with additional information.

Don't stop getting extra help until you feel confident in your grasp of the subject.

Take advantage of the learning opportunities in Laboratory classes.

Many of the lab activities that you will do will seem exciting. Violent chemical reactions will cause every heart to beat a little faster.

However, if you don't understand the concepts behind the activities than the activities become nothing more than a magic show! Strive to understand every aspect of these activities and you will find that they are a fun way to learn Chemistry.

Do your written homework correctly.

It may seem that you are saving time when you copy homework from a friend or from the back of a textbook, but you are really causing yourself some trouble.

Each assignment is designed to reinforce a specific concept. If you don't do the homework correctly, you may not master the concepts.

Remember that your task is never to complete a specific set of problems, but rather to master a topic or skill.

Study in groups.

Chemistry is much harder when you try to learn it in isolation. Form study groups from the first week of school.

Having people to talk to about Chemistry will help you avoid the frustration that comes from feeling you are the only one having difficulty with a particular concept or type of Math problem.

You will also be more likely to ask questions when you see that an idea may not be clear to others as well.

Some Careers in Chemistry

Chemists are the people who transform the everyday materials around us into amazing things.

Some chemists work on cures for cancer while others monitor the ozone protecting us from the sun.

Still others discover new materials to make our homes warmer in the winter, or new textiles to be used in the latest fashions.

The knowledge gained through the study of chemistry opens many career pathways. Here are just a few of the careers chosen by chemists.

- Agricultural Chemistry
- Biochemistry
- Chemical Education
- Chemical Engineering
- Consumer Product Chemistry
- Environmental Chemistry
- Food and Flavor Chemistry
- Forensic Chemistry
- Medicinal Chemistry

Some Career Descriptions:

1. Biochemistry

Biochemistry is the study of the structure, composition, and chemical reactions of substances in living systems.



Biochemists at work

Biochemistry emerged as a separate discipline when scientists combined biology with organic, inorganic, or physical chemistry and began to study such topics as how living

things obtain energy from food, the chemical basis of heredity, and what fundamental changes occur in disease.

Biochemistry is applied to medicine, dentistry, veterinary medicine and food science.

2. Chemical Engineering

Chemical engineers apply the principles of chemistry, math, and physics to the design and operation of large-scale chemical manufacturing processes.



Chemical synthesis

They translate processes developed in the lab into practical applications for the production of products such as plastics, medicines, detergents, and fuels; design plants to maximize productivity and minimize costs; and evaluate plant operations for performance and product quality.

Chemical engineers are employed by almost all companies in the chemical process industry.

3. Forensic Chemists

A forensic chemist is a professional chemist who analyzes evidence that is brought in from crime scenes and reaches a conclusion based on tests run on that piece of evidence.



A forensic chemist at work

A forensic chemist's job is to identify and characterize the evidence as part of the larger process of solving a crime. Forensic chemists rarely conduct any investigative work; they handle the evidence collected from the crime scene.

4. Medicinal Chemistry

Medicinal chemistry is the application of chemical research techniques to the synthesis of pharmaceuticals.



Some common medicines

During the early stages of medicinal chemistry development, scientists were primarily concerned with the isolation of medicinal agents found in plants.

Today, scientists in this field are also equally concerned with the creation of new synthetic drug compounds. Medicinal chemistry is almost always geared toward drug discovery and development.

5. Food and Flavor Chemists

Food chemistry focuses on the chemistry of foods, their deterioration, and the principles underlying the improvement of foods for consumers.

It applies chemistry to developing, processing, packaging, preserving, storing, and distributing foods and beverages to obtain safe, economical, and aesthetically pleasing food supplies.



Processed and unprocessed foods

Few people recognize the science behind the food they consume. While food science involves chemistry, biology, physics, biochemistry, microbiology, nutrition, and engineering, the major portion of a food science curriculum is chemistry. Food chemistry encompasses everything from agricultural raw materials to consumer end-use products.

General Chemistry Laboratory

safety rules and regulations

Your school science laboratory is set up so that you can perform science experiments in safety provided that you follow the proper procedures and safety precautions listed below.

Your teacher will give you specific information about the safety routines used in your school.



It is essential for all concerned that certain rules be followed while in the lab. Read the following carefully and ask questions necessary for clarity.

1. Goggles will be worn at all times. No exceptions. Failure to wear goggles will result in expulsion from laboratory.
2. Full shoes are required. No sandal, flip-flops, etc. are allowed.
3. Lab apron is required when wearing shorts, tank tops, etc.
4. Keep locker drawers closed when not in use.
5. Do not leave flames unattended. Turn burners off when not in use.
6. Remember that most chemicals are flammable, toxic, carcinogenic or all three. Treat them accordingly. Do not ingest chemicals.
7. Acquaint yourself with the eyewash station, safety shower and fire-fighting equipment. You are responsible for knowing their location and use.
8. No smoking, chewing, eating or drinking allowed in the laboratory.

If you are taking a prescription or other drug that will affect your alertness, notify your instructor before going into lab.

9. No students are allowed in the stockroom. No lab visitors without permission of the lab instructor.
10. Report all accidents or injuries to the instructor immediately!
11. If you do not understand a procedure or you cannot read a label, contact the instructor.

Do not gamble with your (and others) safety when there is a question. What you don't know can hurt you. Ditch the foolish notion that asking questions will make you look stupid.

Some Further Explanations

1. Do not pipette by mouth

You say, "But it's only water." Even if it is, how clean do you think that glassware really is? Using disposable pipettes? I know lots of people who rinse them and put them back! Learn to use the pipette bulb or automated pipetter.

A Material Safety Data Sheet (MSDS) should be available for every chemical you use in lab.

Read these and follow the recommendations for safe use and disposal of the material.

2. Dress appropriately (for chemistry lab, not fashion or the weather)

No sandals, no clothes you love more than life, no contact lenses, and long pants are preferable to shorts or short skirts.

Tie long hair back. Wear safety goggles and a lab coat.

Even if you aren't clumsy, someone else in the lab probably is.

If you take even a few chemistry courses you will probably see people set themselves on fire, spill acid on themselves, others, or notes, splash themselves in the eye, etc.

Don't be the bad example to others, remembered for all time for something stupid!

3. Identify the Safety Equipment

And know how to use it! Given that some people (possibly you) will need them, know the locations of the fire blanket, extinguishers, eyewash, and shower.

Ask for demonstrations! If the eyewash hasn't been used in a while the discoloration of the water is usually sufficient to inspire use of safety glasses.

4. Don't Taste or Sniff Chemicals

For many chemicals, if you can smell them then you are exposing yourself to a dose that can harm you! If the safety information says that a chemical should only be used inside a fume hood, then don't use it anywhere else. This isn't cooking class - don't taste your experiments!

5. Don't casually dispose of chemicals down the drain

Some chemicals can be washed down the drain, while others require a different method of disposal.

If a chemical can go in the sink, be sure to wash it away rather than risk an unexpected reaction between chemical 'leftovers' later.

6. Don't eat or drink in lab

It's tempting, but dangerous... just don't do it.

7. Don't play mad scientist



Don't haphazardly mix chemicals! Pay attention to the order in which chemicals are to be added to each other and do not deviate from the instructions. Even chemicals that mix to produce seemingly safe products should be handled carefully.

For example, hydrochloric acid and sodium hydroxide will give you salt water, but the reaction could break your glassware or splash the reactants onto you if you aren't careful.

8. Take data during lab

Put data directly in your lab book rather than transcribing from another source (e.g., notebook or lab partner).

Not after lab, on the assumption that it will be neater.

There are lots of reasons for this, but the practical one is that it is much harder for the data to get lost in your lab book. For some experiments, it may be helpful to take data before lab.

Student Activity

Crash, bang , wallop! Here is a disaster zone! But a real school chemistry laboratory is one of the safest places in which to work. Good chemists always work carefully and safely.

The rules of the laboratory have been forgotten by these students.



1. Look carefully at the disaster zone above.
 - a. Make a list of as many of the dangers as you can
 - b. From this list, make a set of Rules of the Laboratory
2. Draw a rough sketch of the laboratory you normally use. Mark on it where the following items are usually kept.
 - a) Bunsen burners
 - b) Clamps and retort stands
 - c) Beakers
 - d) Test tube racks

3. Very often, safety rules in the laboratory are written negatively and start with don't do this, Don't do that, you should NOT

Rewrite your Rules of the laboratory positively by starting with Do this, Always, You should

The scientific method

The scientific method is a set of ideas or a procedure that scientists use to investigate things they want to understand. By using the method, you can be sure you're carrying out your project correctly.

The scientific method allows you to investigate an experiment in a step-by-step method.

Problem: What are you going to solve in the lab? The problem or purpose explains exactly what you hope to accomplish in the investigation.

Hypothesis: How do you think it is going to turn out? Use the facts you already know to come up with a guess that might really make sense.

Materials and Apparatus: List what equipment you will need to complete the experiment. (Include diagrams of set up apparatus if required to do so in this section.

Procedure: What you must do to complete the experiment. Write down the steps you need to follow.

Data and Work: Include the tables, observations and work you did during the experiment.

This section is where you keep very careful notes on everything you do and everything you find out.

Be sure you write down or draw what really happened, even if it's not what you thought would happen.

At the end, you look over all your data and think about it very hard.

You think of the results of your procedure, or how everything turned out. Analysis Questions: You do not need to write out questions, but you must answer in sentences that include the question.

Conclusion: You must say what you found out during the lab. You figure out whether your results agreed with your hypothesis or not. Put everything you observed together and try to make some sense out of it.

****Hint**** The conclusion should answer the problem.

Remember!

1. Be neat. Your lab should be organized and easily readable.
2. Hand in a good copy. You can take a rough copy of data tables into the lab and then recopy for your hand-in report.
3. Hand in lab reports on time!

Learning activity

1. Name two basic storage and chemical handling apparatus.
2. Name one accurate apparatus for measuring liquid volume.
3. How do you determine the mass of an object?
4. Name two safety devices to prevent chemical splash.
5. Very often, safety rules in the laboratory are written negatively and start with ‘don’t do this, don’t do that, you should not.....

Rewrite your Rules of the Laboratory positively by starting with ‘Do this, Always, You should.....

Names of Common Laboratory Apparatus

Which apparatus?

You wouldn’t dream of trying a chicken using a spoon, or of drying your hair in over a stove! In everyday life, we need to use the right tools for the job we are doing.

The same is true in the science lab.

There are lots of types of different chemical apparatus, all designed to do different jobs.

Using the right apparatus makes your practical work safer.

The pictures below show some of the apparatus you will be expected to know about and use in a chemistry laboratory.

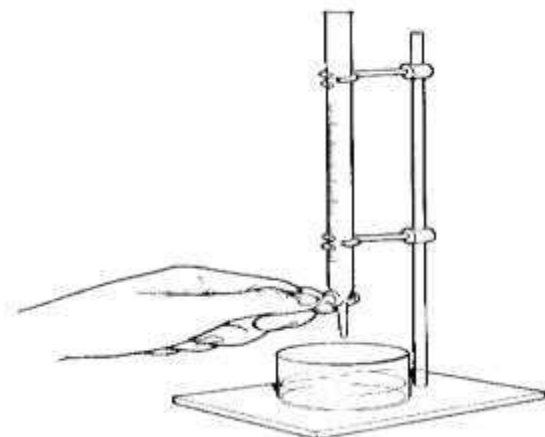
**Burette**

Made from accurate uniform wall tubing to insure the stipulated capacity tolerances. Features durable, permanent markings; fine, sharp lines and large, easy-to-read numbers.

The stopcock is carefully ground and finished to assure a leak-free operation.



A burette is used to deliver solution in precisely measured, variable volumes.



To fill a burette, close the stopcock at the bottom and use a funnel. You may need to lift up on the funnel slightly, to allow the solution to flow in freely.

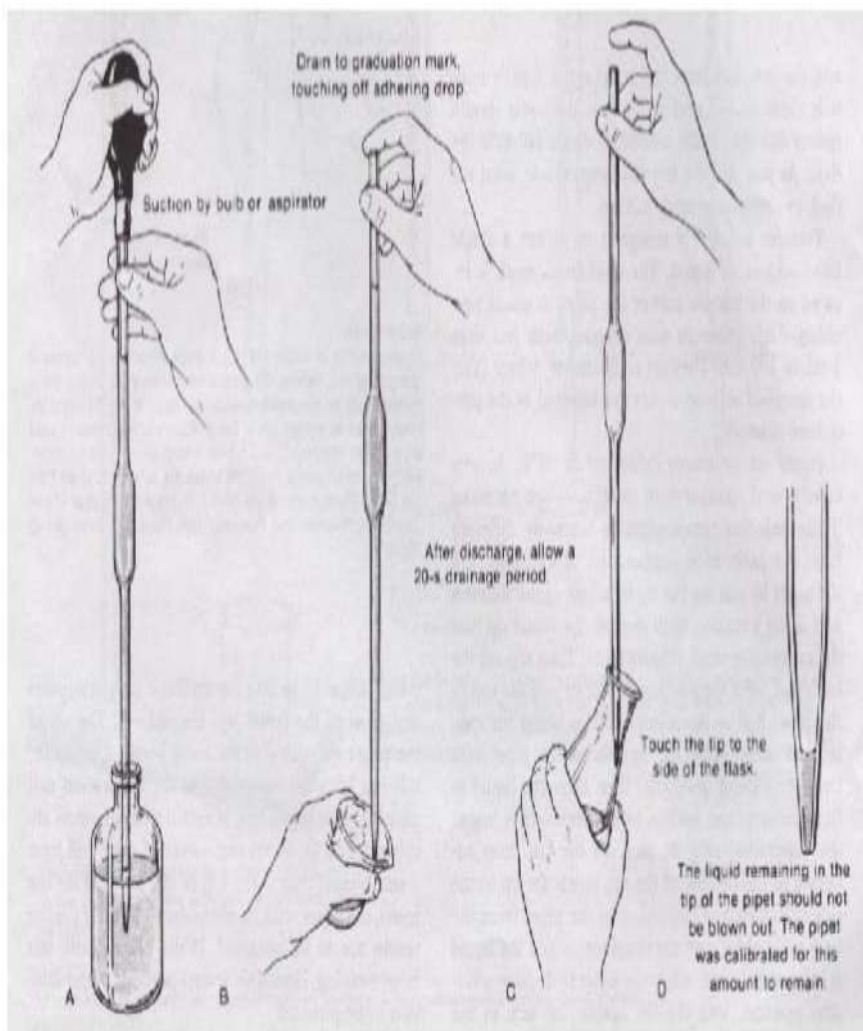
Pipette

A pipette is used to measure small fixed amounts of solution very accurately. A pipette bulb is used to draw solution into the pipette.



A pipette has a mark that shows how much volume it can draw. It cannot be used for transferring any other volume unless the one specified on it.

Using a Pipette



3. Beakers

Made of heat resistant borosilicate glass, withstanding temperatures up to 680°C, the beaker features a large marking spot, graduations and pouring spout.

The primary function of a beaker is to hold and work with liquids.

If graduated, it can serve to make approximate measurements of liquid volume.

The beaker is made of specialized glass so it can be heated and cooled without breaking. This type of glass makes the beaker brittle and it must be handled carefully.



Evaporating Dish



resembles a shallow bowl with a spout. Evaporating dishes are used to evaporate solvent to concentrate a solution; however they can also

This porcelain item resembles a shallow bowl with a spout. Evaporating dishes are traditionally used to evaporate solvent to concentrate a solution; however they can also be used to hold sand for a sand bath, as a small water bath, or as a drying dish (like a watch glass).

Brushes

**Glass rods**

These rods are used in all science laboratories to stir and mix substances. 5mm in diameter and 200 mm (approx. 8 inches) in length.

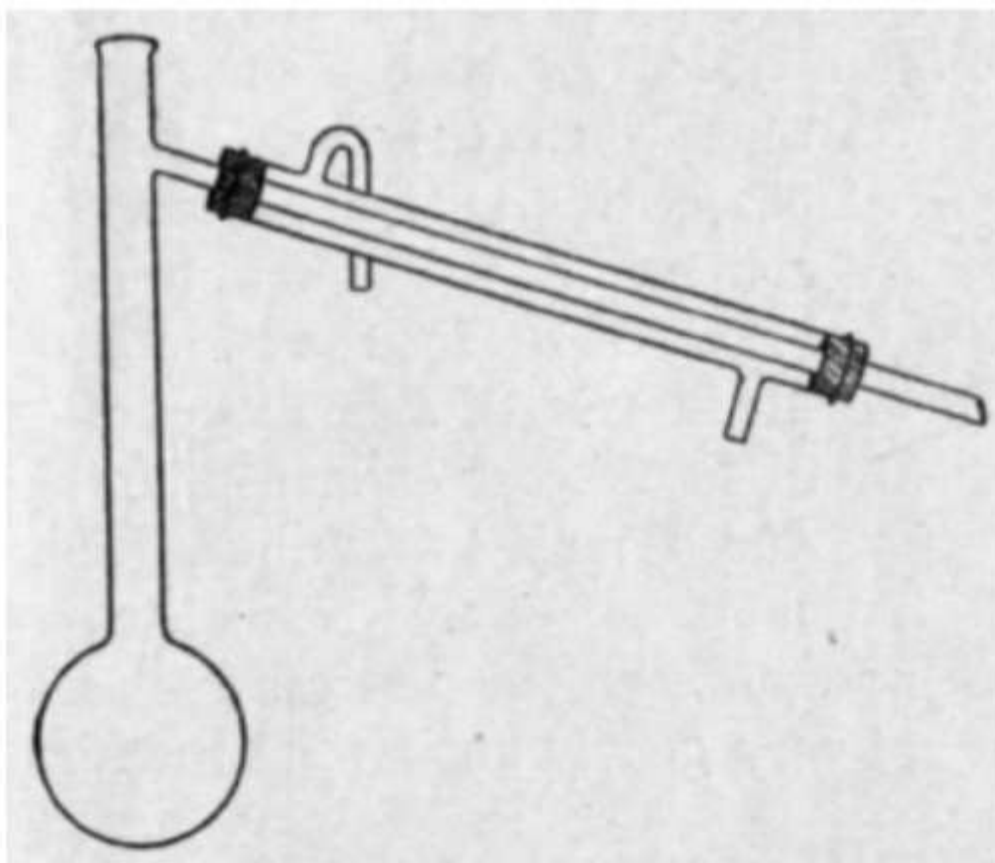
Condenser

This apparatus is made of high quality borosilicate glass. It can withstand up to 600°C temperature.

This instrument is suitable for pure water distillation and other kinds of distillation in the laboratory.

The Graham condenser has both inner and outer joints. For use in vacuum distillation, the jacket is sealed to the inner tube.

Drip tip is located at the end. Hose connections are 10mm overall diameter. Top opening is 29mm. Cold water enters from the lower tap and exits from the top.



4. Goggles.

Eye protection is a priority in any science laboratory setting. When fumes are involved (and you will be working with A LOT of those in chemistry) and well as possible splashing, you generally need something that does not allow air to vent through.

Protective goggles that are meant specifically for chemistry have plugs to keep those fumes/liquids out.

It's also why you have eyewash stations in your lab in case someone does not wear their goggles and need to wash the chemicals out of their eyes.



5. Stirrer

The function of a stirrer is to agitate liquids for speeding up reactions or improving mixtures.



6. Measuring Cylinder

Graduated or measuring cylinders are specifically designed to make accurate liquid volume measurements.

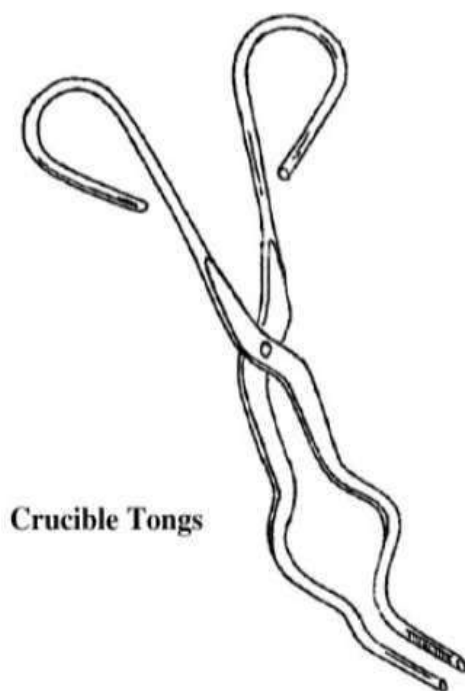


The volume is read from the lowest portion of the meniscus of the liquid; that is, the lowest portion of the convex dip of the liquid as it sits in the graduated cylinder. Graduated cylinders are available in a number of sizes.

7. Crucible Tongs

Used for handling or manipulating hot crucibles and similar apparatus.

They have corrugated tips for easy handling; riveted joints and the finger openings are large enough for a firm grip. They are made of plated steel.



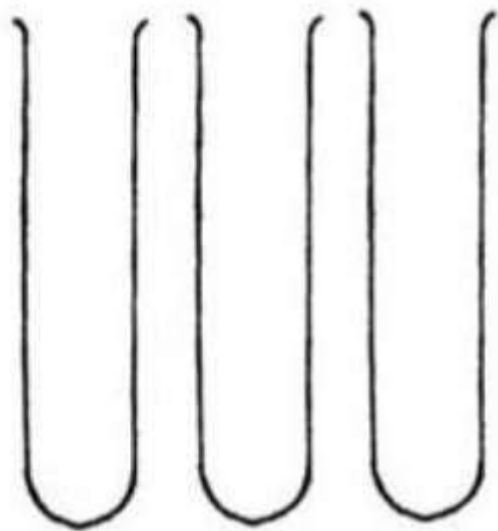
7. Mortar and Pestle



Mortar and pestles are used to grind solids into powders.

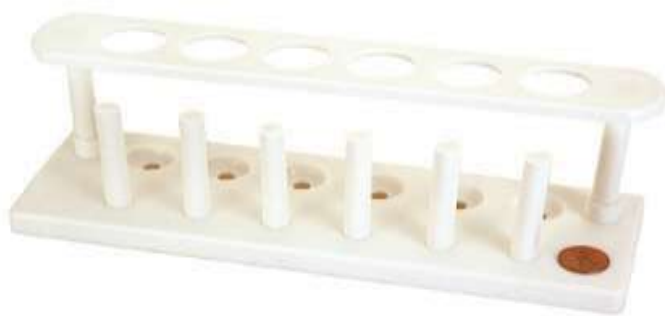
8. Test Tube

The function of a test tube is to hold a small experiment, which would be used to conduct an investigation.



The test tube is made of specialized glass so it can be heated and cooled without breaking. This same type of glass makes the test tube brittle and it must be handled carefully.

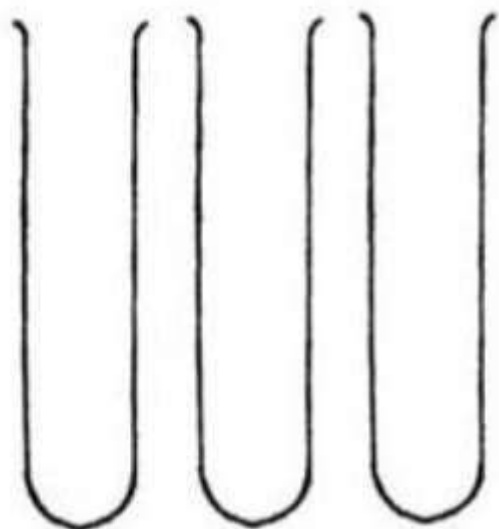
Test-tube Rack



9. Thermometer

The thermometer is an instrument for determining temperature.

Most laboratory thermometers are calibrated in the SI scale (degrees Celsius).



10. Thistle Tube

Thistle tubes are usually used in experiments involving semi-permeable membranes. Osmosis and diffusion can be illustrated well.



Thistle tubes are also used in separation experiments as well as other chemical applications.

11. Balance

The function of a balance is to mass objects.

[**NOTE:** If the device is measuring an object using springs against gravity, the devise is correctly referred to as a scale and its measurements are described as weight not mass.]

A balance uses a comparison of a known substance or calibration with the unknown object to determine the unknown object's mass.

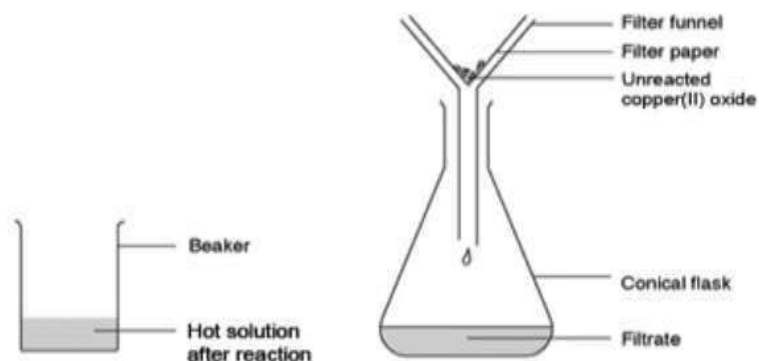


12. Fume Hood

Fume hoods protect laboratory workers from fumes and potentially dangerous chemical reactions by continuously vacuuming air out of the lab and by providing a glass shield.

Experiments can be clearly seen by the user, yet the user is protected from splatter and harmful fumes.

13. Funnel



A funnel is used to aid in the transfer liquid from one vessel to another.

14. Dropper

A dropper is used to transfer a small volume of liquid (less than one mL).



15. Volumetric Flask

A volumetric flask is used to measure very precisely one specific volume of liquid (100 ml, 250 ml, etc., depending on which flask you use).

This flask is used to prepare a solution of known concentration.

A volumetric flask should not be used to heat substances or store solutions, and you should avoid pipetting directly from the volumetric flask.

Our volumetric flasks provide precise volume measurement.

The necks are tooled and have glass stoppers.

The graduation line is sharp and permanent and there is an easy to read marking spot.

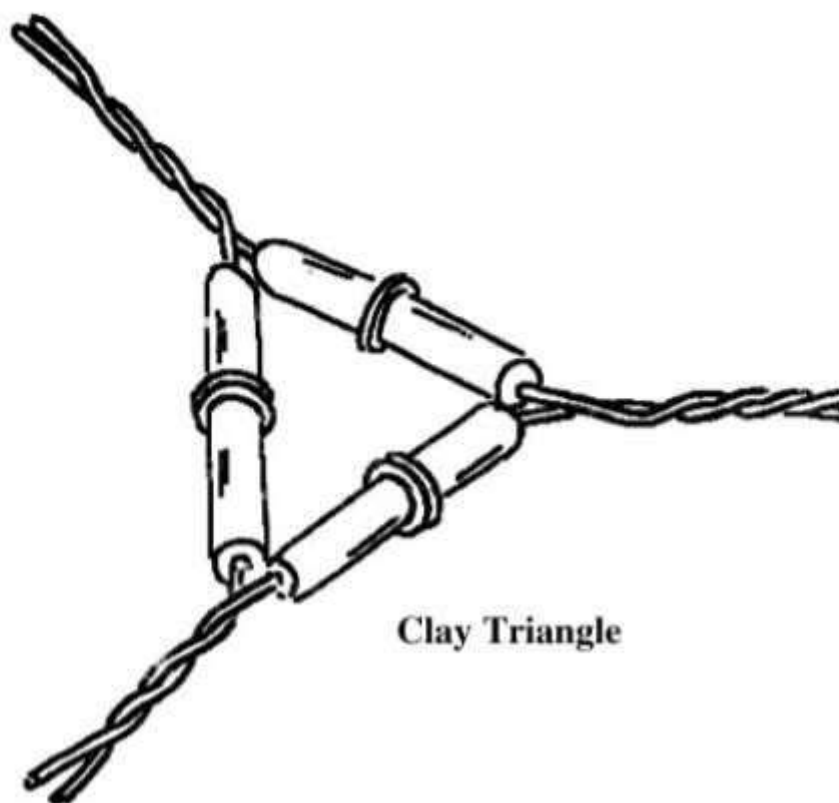


For solutions of solid in a liquid solvent, first dissolve the solid in a vessel other than the volumetric flask, in case the solid must be heated or crushed in order to be dissolved.

After the solid dissolves, pour the liquid into the volumetric flask and use the solvent to rinse all of the solution from the vessel into the flask. However, if the solid is known to be very soluble, it may be added directly to the flask.

Clay Triangles

Clay Pipe Triangle

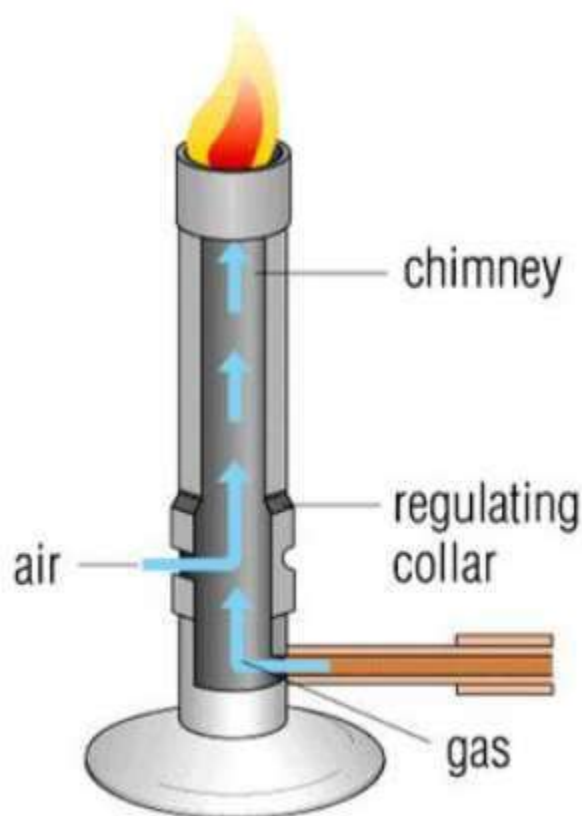


Used for supporting crucibles and dishes when heating on gas or alcohol burners. It has three porcelain pipe stem attached to galvanized iron wire.

Bunsen burner

A Bunsen burner is a laboratory device designed to heat substances for various experiments.

In essence, a Bunsen burner is a small gas burner with an adjustable flame, manipulated at the base by controlling the amount of gas and air admitted into the burner.



The design of a Bunsen burner includes a vertical metal tube which is connected to a weighted base.

The base includes a nozzle to connect with a fuel source, as well as a gas valve and a flue adjuster to control how much air is admitted through small air holes at the base of the tube.

The gas mixes with air at the bottom of the tube and then rises to the top of the Bunsen burner, where it can be lit with a match or lighter.

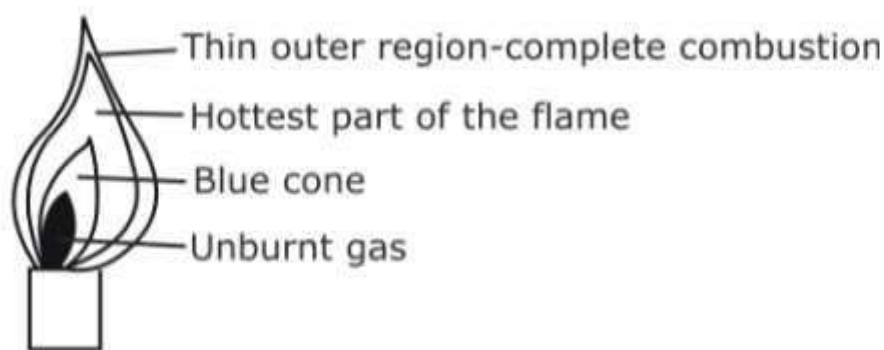
In general, the scientist should make sure that hair and clothing are secure, and unlikely to fall into the flame.

In addition, flammable chemicals should be kept away from the Bunsen burner, and someone should remain by the burner at all times to supervise it.

The flexible rubber hose connecting the Bunsen burner to the gas nozzle on the lab bench should also be secure, with no evidence of leaking, and people should be cautious about touching things which have been exposed to the often considerable heat of a Bunsen burner, especially glass objects.

Flame

A typical flame consists of three zones that are easily distinguished.



-luminous cone, is composed of a gas-air mixture at a comparatively low temperature.

second, or luminous, cone, hydrogen and carbon monoxide are produced by decomposition and begin to react with oxygen to form water and carbon dioxide, respectively.

In this cone the temperature of the flame—about 590° to 680° C is great enough to dissociate the gases in the flame and produce free particles of carbon, which are heated to incandescence and then consumed.

The incandescent carbon produces the characteristic yellow light of this portion of the flame.

isible cone in which the remaining carbon monoxide and hydrogen are finally consumed.

If a cold object is introduced into the outer portions of a flame, the temperature of that part of the flame will be lowered below the point of combustion, and unburned carbon and carbon monoxide will be given off.

Thus, if a porcelain dish is passed through a candle flame, it will receive a deposit of carbon in the form of soot.

Operation of any kind of flame-producing stove in a room that is unventilated is dangerous because of the production of carbon monoxide, which is poisonous.

All combustible substances require a definite proportion of oxygen for complete burning. (A flame can be sustained in an atmosphere of pure chlorine, although combustion is not complete.)

In the burning of a candle, or of solids such as wood or coal, this oxygen is supplied by the surrounding atmosphere.

In blowpipes and various types of gas burners, air or pure oxygen is mixed with the gas at the base of the burner so that the carbon is consumed almost instantaneously at the mouth of the burner.

For this reason such flames are non-luminous.

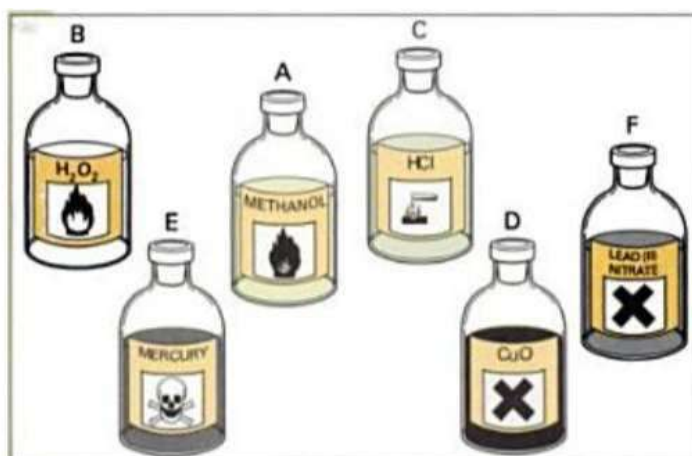
They also occupy a smaller volume and are proportionately hotter than a simple candle flame.

The hottest portion of the flame of a Bunsen burner has a temperature of about 1600°C . Such flames have a bluish-green cone in place of the luminous cone.

If the oxygen supply is reduced, such flames have four cones: nonluminous, bluish-green, luminous, and invisible.

Hazard Warning Symbols

Hazardous Material Safety Reference Sheets and Material Safety Data Sheets are produced to remind laboratory users of the potential hazards of the materials at hand. The table below shows some common hazard symbols and some of the chemicals on which these may be found.



Hazard symbol	Meaning of symbol	Examples of what might be labelled/classified with this hazard warning sign
	Irritant	Most acidic and alkaline solutions unless very dilute; acidic gases like chlorine, sulphur dioxide, nitrogen dioxide; bleaches
	Harmful	Some acids e.g. nitric acid; acidic gases like chlorine, sulphur dioxide, nitrogen dioxide; bleaches; heavy metal ions e.g. of lead, barium; some salts e.g. silver nitrate
	Corrosive	All concentrated acidic and alkaline solutions
	Highly flammable	Most organic solvents, petrol and other hydrocarbon fuels
	Toxic	Chlorine, sulphur dioxide, hydrogen cyanide

		
	Oxidising	Chlorine and oxygen gases, potassium manganate (VII), potassium chlorate (in some weed killers)
	Radioactive	Radioisotopes
	Explosive	TNT, hydrogen, fireworks
	Biohazard	organisms and viruses infectious to humans, animals or plants (e.g. parasites, viruses, bacteria, fungi); and biologically active agents (i.e. toxins, allergens, venoms)

SAMPLE SCENARIOS FOR CHAPTERS ONE AND TWO**ITEM 1**

Chemistry is one of the most prestigious science subjects studied at the secondary level of Education. When you study chemistry, you do not only acquire knowledge about many chemical reactions and other substances, but you can also take up various careers after school.



Biochemical work



Some common medicines



Processed and unprocessed foods

(a) Identify any three careers that you can take up through studying chemistry.

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b) Briefly describe how the knowledge of chemistry has been applied in the following field in the Ugandan economy;

(i) Bio chemistry.

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(ii) Medicine

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(iii) Forensic industry

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(iv) . Chemical Engineering

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(iv) Food and flavors chemistry

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c) Engineers need to know that substances expand when they are heated. Mention three areas where engineers use this knowledge when designing things.

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ITEM 2

Crash, bang , wallop! Here is a disaster zone! But a real school chemistry laboratory is one of the safest places in which to work. Good chemists always work carefully and safely.

The rules of the laboratory have been forgotten by these students.

A. Look carefully at the disaster zone above.

a. Make a list of as many of the dangers as you can

b. From this list, make a set of Rules of the Laboratory



B. Draw a rough sketch of the laboratory you normally use. Mark on it where the following items are usually kept.

- a) Bunsen burners
- b) Clamps and retort stands
- c) Beakers
- d) Test tube racks

C. Very often, safety rules in the laboratory are written negatively and start with don't do this, Don't do that, you should NOT

Rewrite your Rules of the laboratory positively by starting with Do this, Always, You should

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[illegible]

S. 1 CHEMISTRY GUIDE EOT 1, 2026**ITEM 1****(a) Careers from studying Chemistry (03 marks)****(Any 3, 1 mark each)**

Medical doctor

Pharmacist

Chemical engineer

Laboratory technician

Teacher (science/chemistry)

Forensic scientist

Food scientist

Environmental scientist

(b) Applications of Chemistry in fields**(i) Biochemistry**

Study of chemical processes in living organisms

Used in understanding enzymes, digestion, metabolism

Helps in disease diagnosis and nutrition

(ii) Medicine

Manufacture of drugs and vaccines

Understanding body chemistry and diseases

Used in treatment and prevention of illnesses

(iii) Forensic industry

Identification of chemicals in crime scenes

Analysis of blood, drugs, poisons

Helps in criminal investigations

(iv) Chemical Engineering

Design of industrial processes

Production of chemicals, fuels, plastics

Large-scale manufacturing and processing

(v) Food and Flavour Chemistry

Production and preservation of food

Development of flavours and additives

Ensures food safety and quality

(c) Areas where expansion is used (03 marks)**(Any 3)**

Railway lines (gaps left for expansion)

Bridges (expansion joints)

Thermometers (liquid expansion)

Electric cables (allow slack)

Road construction

ITEM 2**A (a) Dangers in laboratory (Any relevant points – 6–8 marks)****Possible dangers:**

Running in the lab

Spilled chemicals

Broken glassware

Heating test tubes facing people

Not wearing protective gear

Eating/drinking in lab

Mixing chemicals carelessly

Open flames near flammable substances

Poor storage of chemicals

A (b) Rules of laboratory (6–8 marks)

Do not run in the laboratory

Wear protective clothing (lab coat, goggles)

Handle chemicals carefully

Do not taste chemicals

Keep work area clean

Follow teacher's instructions

Use apparatus correctly

Report accidents immediately

B (Diagram – 4–6 marks)**Award marks for:**

Clear sketch of lab

Correct labeling of:

Bunsen burners

Retort stands

Beakers

Test tube racks

Logical arrangement

(Teacher awards based on attempt and labeling)

C Positive rules (6–8 marks)**Convert to positive form:**

Always walk carefully in the laboratory

Always wear protective equipment

Always follow instructions carefully

Always handle chemicals with care

Always keep your work area clean

Always report accidents immediately

Always use apparatus properly

Always store chemicals safely

S1 New Curriculum chemistry**Theme: Particle Nature of Matter****S1 New Curriculum Chemistry Chapter 3 – States and change of states of matter****Matter**

Matter is anything that occupies space and has weight. Matter is made up of particles that are too small. These tiny particles cannot be seen by the naked eye. The tiny particles are called molecules. The molecules consist of more tiny particles called atoms. Atoms are made of small particles called protons, electrons and neutrons

Examples are stone, clouds, water etc.



There are majorly four states of matter

- (i) **Solids** e.g. chairs, tables, stone wall etc. Here the molecules are closely packed together and are vibrating within fixed positions, but are not free to move due to the strong forces of attraction between the molecules. Solid state has a definite shape and volume.

- (ii) **Liquids** e.g. water, paraffin, cooking oil. In liquids, molecules are further apart compared to solids. The particles are free to move and are in constant random motion but within the surface of the liquid. The forces of attraction between the particles are weaker compared to solid state. Liquids take up the shape of the container in which they are put. So liquids do not have a definite shape but have definite volume.
- (iii) **Gas** e.g. air. The particles are much further apart compared to liquids and solids. These particles can move independently and at a high speed compared to solids and liquids. The intermolecular forces of attraction are almost negligible. Gases have no particular shape and volume.
- (iv) **Plasma**: it is made of gaseous positive ions and electrons only at very high temperatures. That is, plasma is an ionized gas, a good conductor of electricity with indefinite shape and indefinite volume.

Summary table

Properties	Solid	Liquid	Gas	Plasma
Distance of separation	Least	Closer	Larg	Largest
Intermolecular forces	Strongest	Strong	Wea	weakest
Potential energy	Highest	Low	Lower	Lowest
Kinetic energy	Lowest	High	Higher	Highest

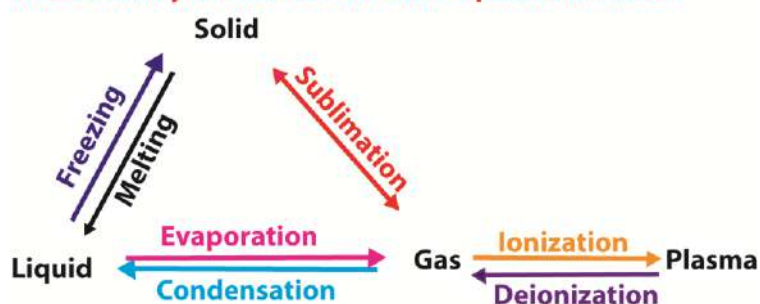
Effect of heat on matter

- (a) Increases the speed of movement of particle or kinetic energy of particles
- (b) Lead to change from solid to liquid to gas and finally to plasma.

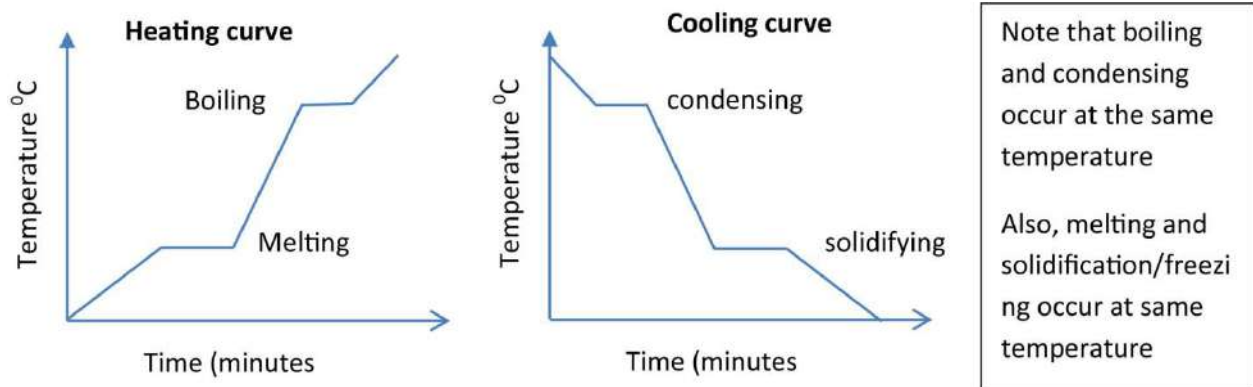
When a solid is heated, the particles gain kinetic energy and turns into a liquid by a process called **melting** or into a gas by the process called **sublimation**.

Heating a liquid turns it into a gas by the process called **evaporation**.

Cooling a gas turns into a solid by the process of **sublimation** or into a liquid by **condensation** and a liquid turns into a solid by **freezing** or **solidification**.

The summary for interconversion of phase of matter

Note that melting/solidification and boiling/condensing occur at constant temperatures. The characteristic heating and cooling curves are shown below;



The cooling effect of evaporation

Evaporation causes cooling because it removes heat from a surface or substance because

- **Energy Absorption:** When a liquid evaporates, its molecules gain energy from their surroundings to escape into the air as vapor.
- **Loss of High-Energy Molecules:** The fastest-moving (hottest) molecules leave first, reducing the average energy of the remaining liquid.
- **Lower Temperature:** With fewer high-energy molecules, the temperature of the liquid and the surface it touches drops.

Real-Life Examples:

- **Sweating:** Sweat evaporates from the skin, cooling the body.
- **Evaporation of alcohol from skin:** When alcohol/ethanol is poured on the skin it feels cold.
- **Wet Clothes Drying:** As water evaporates, the fabric feels cooler.
- **Cooling Effect Near Water Bodies:** Water evaporation from lakes or rivers helps lower surrounding temperatures.

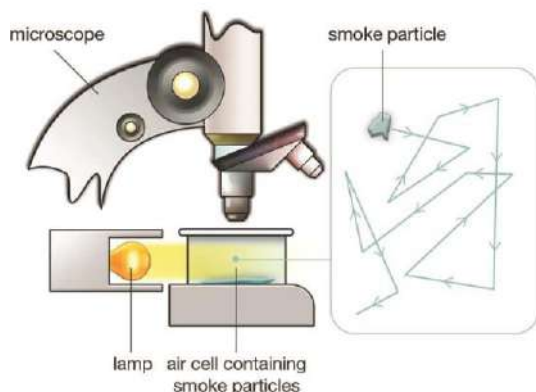
Evidences for the motion of particles in matter

1. Diffusion of gases.

This is the movement of particles from areas of high concentration to a region of low concentration. For instance, when a person wearing a perfume enters a room, the whole room is filled by the scent.

2. Brownian motion

This is the random movement of smoke particles in a glass cell as seen through a microscope.



Explanation

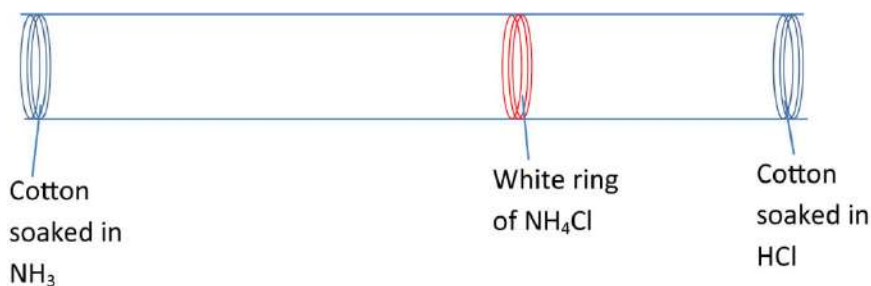
Air particles are in random movement; these collide with smoke particles and cause them to move randomly.

When temperature is increased, the kinetic energy of the air particles increase which increases the speed of smoke particles.

The Brownian motion can be observed by the dust particles through light rays i

3. Formation of white fumes of ammonium chloride (NH_4Cl) from ammonia (NH_3) and hydrogen chloride gas (HCl) from hydrochloric acid (HCl)

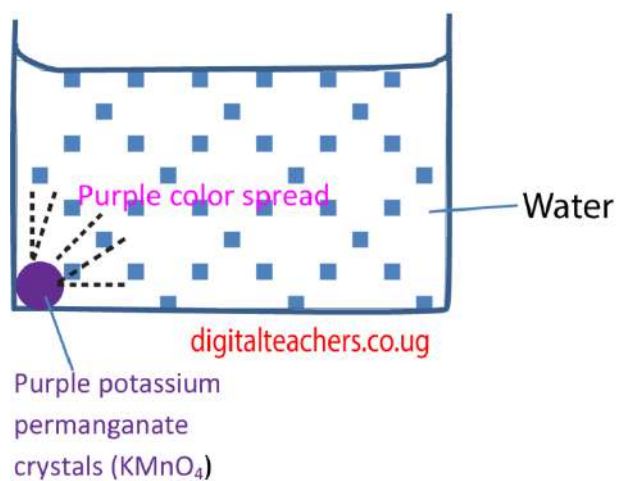
When a cotton wool soaked in ammonia and another soaked in concentrated hydrochloric acid are placed in opposite end of a long glass tube, a white ring form in a tube near the side that contains cotton wool soaked in concentrated hydrochloric acid.



Conclusion

- (i) Ammonia and hydrochloric acid contains particle that move to meet each other and react to form white fumes of ammonium chloride
- (ii) Ammonia and hydrochloric acid are volatile i.e. can easily turn into vapour.
- (iii) The white ring forms near the cotton wool soaked in HCl because ammonia molecules are lighter and therefore diffuse/move faster. Or ammonia molecule move a bigger distance than hydrogen chloride molecules in the same time.

4. Presence and movement of particles in liquids is demonstrated by diffusion of purple permanganate color in water



Revision question

1. What is matter
2. List states of matter
3. How can we show that matter is made of particles?
4. Arranges the states of matter starting from the state with least energy.
5. In terms of kinetic theory, explain condensation and melting.

CHAPTER: USING MATERIALS.

Introduction

In everyday life, we use a wide variety of materials like cotton in your clothes, cellulose in your books, plastic in utensils, steel in buildings and tools etc. Although these materials look different and serve different purposes based on their different properties. In science and engineering, we don't just look at what a material is, but how it behaves. This is known as Material Science.

What is a material?

A material is any substance or mixture of substances that makes up an object. Like wool in blankets, polyethylene in polythene bags.

The four main types of materials.

Type	Key characteristics	Common examples
Metals	Strong, shiny and great at conducting heat and electricity	Iron, Aluminium, Gold, Copper
Polymers	Light weight, flexible, and usually chemically unreactive	Plastics, Rubber, DNA, Wool
Ceramics	Hard, brittle, and resistant to high temperatures	Glass, Porcelain, Graphite
Composites	Combination of two or more materials to get the best of both	Fiberglass, Carbon Fibre, Concrete, Reinforced Concrete

CLASSIFICATION OF MATERIALS

At the most basic level, we classify materials by their source. Are they found ready to use in nature, or do they require heavy chemical processing in a lab.

Understanding how materials are classified helps us:

- i) Choose the right material for a specific use
- ii) Predict how a material will behave
- iii) Manage resources responsibly (e.g., recycling)

Materials are broadly classified into two main groups:



A. Natural Materials

These are materials obtained directly from nature with little or no chemical processing. They are found in nature (from plants, animals or the ground) and under go little modification before use

Sources of Natural Materials

Source	Example	Use
Plants	Wood, Cotton, Paper, Starch	Wood used to make furniture Cotton used to make clothings Paper used to make books
Animals	Wool, Silk, Leather	Wool used to make beddings and clothings Silk used to make clothings Leather used to make shoes.
Earth (Ground)	Clay, Sand, Metals, Stones	Used in building construction Used in utensils

Characteristics of natural materials

They are usually biodegradable (can decay naturally)

They often environmentally friendly

They may have limited durability compared to synthetic materials

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B. Synthetic Materials

These are man-made materials produced through chemical processes, often in industries. Most of them are plastic in nature

Examples include polythene, polystyrene, polyester, synthetic rubber, nylon

General Characteristics of synthetic materials

They are usually strong and durable

They are Often resistant to water and chemicals

Many are non-biodegradable, causing environmental concerns

Comparing Natural and Synthetic Materials

Feature	Natural materials	Synthetic materials
Source	Plants, Animals or Earth	Man made in factories
Biodegradability	Usually biodegradable	Often Non biodegradable
Durability	Less durable in most cases	More durable
Examples	Cotton, Wool, Wood	Nylon, Polyester, Polyethylene

Classification Based on Physical Properties

Apart from origin, materials can also be grouped according to their physical properties.

Common Physical Properties

Property	Description	Example
Strength	Ability to withstand force without breaking	Steel is stronger than wood
Flexibility	Ability to bend without breaking	Rubber is flexible
Elasticity	Ability to return to original shape after stretching	Elastic bands
Water absorption	Ability to absorb water	Cotton absorbs water, plastic does not
Conductivity	Ability to allow heat and electricity to pass through	Metals are good conductors of heat and electricity

Why Classification is Important (Real-Life Application)

Classification helps us make practical decisions in daily life.

Examples:

Cotton is chosen to make clothings because it absorbs sweat

Nylon is used for raincoats because it does not absorb water

Metals are used for bridges due to high strength

Wood is used for furniture because it is easy to shape

Plastics are used because they are lightweight and waterproof

Activity

Learners can:

Collect materials from home (paper, plastic, cloth, wood, metals etc.)

Group them into:

Natural vs synthetic

Based on properties (strong, flexible, absorbent and conductivity)

Record results in a table

This helps them see classification in practice, not just theory.

Check Your Understanding

1. What is the main difference between natural and synthetic materials?
2. Give two examples of materials obtained from animals.
3. Why is plastic widely used for packaging?
4. Name one material that absorbs water and one that does not.

PHYSICAL PROPERTIES OF MATERIALS AND THEIR USES

Materials are not chosen randomly for use. Each material has specific physical properties that make it suitable for a particular purpose.

For example, you would not use paper to make a cooking pot or glass to make a rope. The reason is simple their physical properties are different.

Understanding these properties helps us;

- i) select appropriate materials for different uses
- ii) Explain why certain materials are preferred in real-life situations
- iii) Predict how materials will behave under different conditions

Key Physical Properties of Materials

Below are five important physical properties, clearly explained with practical meaning:

Property	Description	Examples
Strength	This is the ability of a material to withstand force without breaking. There are two types; Compressive strength which is the ability of a material to resist to being crushed. Tensile strength which is the ability of a material to withstand pulling or stretching force without breaking	Steel rods are used in buildings because they can carry heavy loads without breaking. Steel cables on suspension bridges have high tensile strength Concrete pillars or bricks have high compressive strength
Flexibility	This is the ability of a material to bend easily without breaking	Rubber pipes can bend around without breaking. Copper wires can easily be coiled
Elasticity	This is the ability of a material to return to its original shape after being stretched or compressed	A rubber band and steel spring stretch and return to their original length
Water absorption	This is the ability of a material to absorb or resist water	Cotton absorbs water and plastics do not.
Hardness	This is the ability of a material to resist scratching, cutting or wear	Glass is harder than plastics and resists scratching better

Common Materials and Their Properties (Natural & Synthetic)

The table below shows common materials and how their physical properties relate to their uses:

Material	Properties	Use
Wood	It is strong Fairly hard Slightly flexible Less light	Used to make furniture Roofing trusses Construction artistic designs <i>Reason: it can support weight and easy to shape</i>
Cotton	It is soft Flexible High water absorption	Used to make clothes and beddings. Reason: it absorbs sweat and is comfortable on skin
Wool	It is elastic Soft Good heat insulator	Used to make sweaters and blankets Reason: it traps heat and keeps the body warm
Silk	It is smooth Flexible Strong (high tensile strength)	Used in making fine garments Reason: strong yet light weight and comfortable
Clay	It soft when wet Hard when dry Mouldable	It is used to make pots, bricks Reason: it can be shaped easily and becomes strong after heating
Glass	It is hard Brittle Transparent Non absorbent	It is used to make windows, doors and bottles Reason: it allows light through and does not absorb liquids
Polythene	It is flexible Water proof Light weight	It is used to make packaging materials and carrier bags Reason: it prevents water penetration, does not add extra weight and easy to carry
Nylon	It is strong Elastic Light	It is used to make ropes fishing nets and fabrics Reason: it can withstand pulling forces without breaking
Polyster	It is strong Wrinkle resistant Low water absorption	It is used to make clothes and curtains Reason: it is durable and dries quickly
Rubber	Highly elastic Flexible Water proof	Used to make tyres seals elastic bands Reason: it can stretch and return to shape

Polystyrene	It is light Poor conductor of heat Brittle	It is used in packaging and insulation Reason: protects items and reduces heat transfer
Metal (eg iron/steel)	It is very strong Hard High tensile strength	It is used in building, bridges and tools Reason: it can carry heavy loads without breaking

Linking Properties to Uses

The most important concept we must understand is that the use of a material depends on its physical properties.

Examples:

Roofing materials → need strength and low water absorption

Clothing materials → need flexibility and comfort

Packaging materials → need to be lightweight and waterproof

Simple Investigation Activity

Testing properties of materials using simple methods:

Different materials

What to do

- Try to pull or press the material
- Now bend the material
- Stretch the material
- Drop each material in water and observe
- Scratch with another object
- Record all your observation in a table and the property being investigated.
- Make Conclusions about best uses of each material

Check Your Understanding

- Why is cotton suitable for making towels?
- Which property makes rubber useful for tyres?
- Why is glass not suitable for making ropes?
- Name a material that is strong but not flexible.
- Explain why polythene is used for packaging.

POLYMERS AND THEIR USES

Many of the materials we use every day plastics, fabrics, rubber belong to a special group called polymers.

A polymer is a substance made of very long chains of repeating small units joined together. These long chains give polymers their unique properties such as strength, flexibility, and durability.

Think of a polymer like a chain made of many identical links. Each link is small, but when joined together, they form a strong and useful structure.

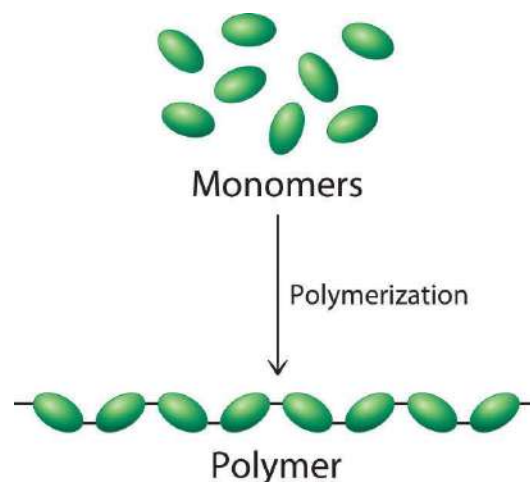
What is a Polymer?

A polymer is a large molecule made of many repeating units

The small repeating units are called monomers

Simple Illustration

Monomer + Monomer + Monomer → Polymer (long chain)



Types of Polymers

Polymers are classified into two main types:

Natural Polymers

These occur naturally in plants and animals.

Examples

Polymer	Monomer
Cotton	Glucose
Wool	Amino acids
Starch	Glucose
Natural rubber	Isoprene
Silk	Amino acid

Synthetic Polymers

These are man-made through chemical processes in factories. The table below has some examples

Polymer	Monomer
Polythene/ polyethylene	Ethene
Polyester	Ester
Nylon	Diamine and dicarboxylic acid
Polyvinyle Chloride (PVC)	Vinyl chloride
Teflon	Tetrafluoroethene
Polystyrene	Styrene

Common Polymer Materials and Their Uses

Here are at least 10 common polymers, linked to their properties and uses:

Polymer	Properties	Uses
Polythene	It is flexible, Waterproof Light weight	Its is used to make carrier bags and packaging materials
Nylon	It is strong, Elastic, Durable	It is used to make ropes, fishing nets and fabrics
Polyster	It is strong, Wrinkle resistant, Low water absorption	It is used to make curtains and clothes
Polyvinyl chloride	It is hard, Durable Water resistant	It is used to make water pipes, Cable insulations
polystyrene	It is lightweight Poor heat conductor brittle	It is used to make packaging materials and disposable cups
Teflon	It is heat resistant	Used to make non stick cooking pans
Synthetic ruber	It is elastic, Flexible, Water proof	It is used to make tyres and foot wear
Cotton	It absorbs water, Soft Breathable	Used to make clothings and beddings
Wool	It is elasstic Insulating Soft	It is used to make sweaters and blankets
Silk	It is strong, smooth Flexible	It is used to make garments

Advantages and Disadvantages of synthetic Polymers

Advantages

They are Durable

They are Lightweight

They are Cheap to produce

They are Versatile (many uses)

Disadvantages

Many are non-biodegradable

They cause environmental pollution

They can release harmful gases when burnt

Check Your Understanding

1. What is a polymer?
2. Give two examples of natural polymers and two synthetic polymers.
3. Why is nylon suitable for making ropes?
4. What property makes Teflon useful in cooking pans?
5. Why are plastics considered an environmental problem?

MOLECULAR STRUCTURE AND MATERIAL PROPERTIES

So far, you have learned that materials have different physical properties and that polymers are made of long chains. The next important step is to understand why materials behave the way they do.

The answer lies in their molecular structure.

Key Idea:

The way particles (atoms and molecules) are arranged and held together determines the properties of a material and therefore its uses.

What is Molecular Structure?

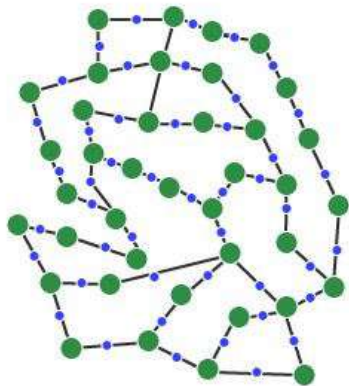
Molecular structure refers to how atoms are arranged in a material, how they are bonded together, and how closely they are packed

There are two main things to focus on: Arrangement of particles (orderly or random) **and** Forces holding particles together (strong or weak)

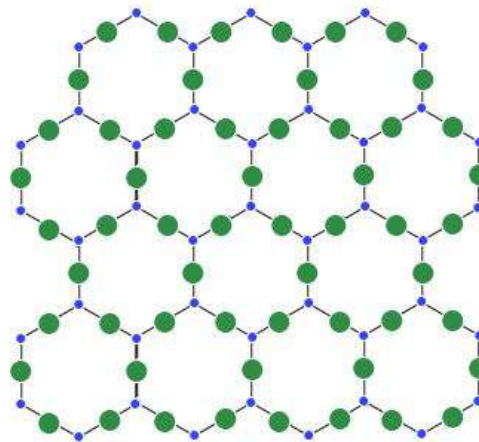
Types of Structures in Materials

Materials can have different internal structures. The most relevant at this level are:

AMORPHOUS STRUCTURE



CRYSTALLINE STRUCTURE



A. Crystalline Structure

Particles are arranged in a regular, repeating pattern. Very orderly and closely packed. For Example; Metals (iron, copper), Salt crystals, Some forms of solid plastics etc

Properties

They are usually strong and hard

They have a definite shape

They often have high melting points

Explanation (reason for properties)

This is because particles are tightly packed and strongly bonded, it is difficult to break or deform the material.

B. Amorphous Structure

Particles are arranged in a random, irregular manner. No fixed pattern. For Example; Glass, Rubber and Most plastics

Properties

They are usually softer

They can be easily shaped or bent

They do not have a sharp melting point (they soften gradually)

Explanation

Since the arrangement is irregular, particles can move past each other more easily.

Structure of Polymers

Polymers have a chain-like molecular structure, and this greatly affects their properties.

Types of Polymer Structures

1. Linear Polymers

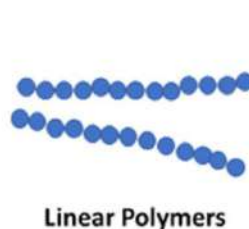
These have long, straight chains that lie close together *for example* high density polythene (HDPE)

Properties

They are strong

They can be tightly packed

They are sometimes rigid



2. Branched Polymers

These have chains which have side branches

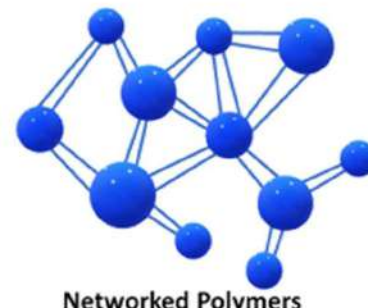
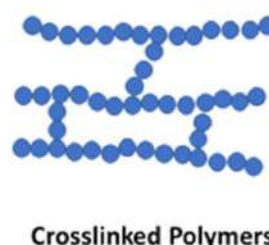
for example low density polythene (LDPE)

Properties

They are less tightly packed

They are more flexible

They have a lower strength compared to linear polymers



3. Cross-linked Polymers

These have chains which are connected by strong links between them *for example* vulcanised rubber, Bakelite (used in electrical fittings)

Properties

They are very strong

They are rigid

They are heat resistant

How Molecular Structure Affects Physical Properties

Now let's connect structure directly to properties learners already know.

Strength

Materials with strong bonds and closely packed particles are strong

Example: Metals are strong due to tightly packed structure. Cross-linked polymers are very strong due to bonds between chains

Flexibility

Materials with loosely packed or sliding particles are flexible

Example: Polythene chains slide over each other making it flexible

Elasticity

Materials that can stretch and return to shape have coiled or flexible chains

Example: Rubber chains uncoil when stretched and recoil when released thus can stretch and return to shape

Hardness

Materials with rigid and tightly packed structures are hard

Example: Glass and metals resist scratching due to tightly packed particles

Melting Behavior

Strongly bonded structures have high melting point and weakly bonded structures have low melting point

Example: Metals melt at high temperatures **while** Plastics soften easily when heated

Real-Life Applications (Linking Structure to Use)

Understanding structure helps explain real-world choices:

Category of use	Use related to molecular structure
Construction Materials	Steel has a strong crystalline structure therefore used in buildings Cement forms rigid structure hence supports heavy loads
Clothing Materials	Cotton has loosely arranged fibres making it soft and absorbent

	Polyester is tightly bonded hence durable and wrinkle-resistant
Household Items	Glass is amorphous and transparent but brittle Plastic containers have flexible polymer chains hence do not break easily
Electrical Uses	PVC has cross-linked structure making it a good insulator Bakelite has a rigid structure hence used in switches

Simple Analogy for Understanding

Think of molecular structure like arrangement of people in a room:

Tightly packed people and holding their hands (strong bonds) makes moving hard → **strong material**

Loosely standing people can easily move → **flexible material**

People connected with elastic ropes can stretch and return → **elastic material**

Check Your Understanding

1. What is meant by molecular structure?
2. Why are crystalline materials usually strong?
3. How does the structure of rubber make it elastic?
4. Why do plastics soften when heated?
5. Give one example where structure determines the use of a material.

SCIENTIFIC INVESTIGATION OF MATERIAL PROPERTIES

Science is not only about knowing facts, it is about finding out how and why things happen through investigation.

When studying materials, learners must go beyond theory and Test properties, Observe results and Draw conclusions based on evidence

Scientific investigations help us make reliable decisions about which materials are suitable for different uses.

The Scientific Method (Step-by-Step)

Every investigation should follow a clear structure:

1. Aim (Purpose)

What are you trying to find out? For example: To determine which material is strongest.

2. Hypothesis and variables

A prediction based on prior knowledge. For example: I think metal will be stronger than wood.

3. Apparatus/Materials

List all items needed. For example: Wood pieces, plastic strips, metal rods, weights.

4. Procedure (Method)

Step-by-step instructions of what to do. Steps you follow when carrying out the investigation.

5. Results (Observations)

What you see or measure and Recorded in tables

6. Conclusion

What you have discovered. Whether the hypothesis was correct

Key Variables in Experiments

To ensure fair testing, learners must understand Independent Variable as the factor you change and Dependent Variable as the factor you measure. Controlled Variables are factors you keep the same

Example (Testing Strength)

Independent variable: Type of material

Dependent variable: Maximum weight held

Controlled variables: Size of material, method of testing

Common Experiments on Material Properties

Below are practical investigations learners can carry out.

Experiment 1: Testing Strength of Materials

Aim

To compare the strength of different materials.

Hypothesis

The metal strip will hold the most weight compared to wood and plastic, making it the strongest.

Variables

Independent: type of material

Dependent: weight held

Controlled: size of weights, size of the material strips.

Materials

Wood, plastic, metal strips

Weights (stones or masses)

String

Procedure

- a) Tie one end of the wood securely using a string on a retort stand and ensure its firm.
- b) Gradually add weights on the wood until it breaks or bend
- c) Record the maximum weight the wood held before breaking
- d) Repeat steps a to c using the plastic and metal strips.
- e) Record all your results in a table

Results Table

Material	Wood	Plastic	Metal
Maximum Weight Held			

Conclusion

The material that holds the most weight is the strongest

Discussion question

- h) What are some of the safety precautions you took while carrying out this experiment and why?
- i) With a reasons, identify the strongest and weakest material.

Experiment 2: Testing Flexibility

Aim

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To determine which material is most flexible.

Hypothesis

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.....

Variables

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Materials

Rubber, plastic, wood.

Risks and mitigations

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Procedure

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Results table

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Conclusion

The easiest to bend is the most flexible

Experiment 3: Testing Water Absorption

Aim

To find out which material absorbs water.

Hypothesis

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Variables

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Materials

Cotton, paper, plastic

Water

Risks and mitigations

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Procedure

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Results Table

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Conclusion

Materials like cotton absorb water, plastics do not

Experiment 4: Testing Insulation Properties

Aim

To compare heat insulation of materials.

Hypothesis

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Variables

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Materials

Metal sheet, wood, cloth

Hot water containers

Risks and mitigations

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Procedure

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Conclusion

Material that retains heat best is the best insulator

IMPORTANT TO NOTE

Recording Results Properly

Learners should:

Use tables for clarity

Include units where necessary

Write clear observations, not guesses

Drawing Conclusions

A good conclusion: Refers back to the aim, uses evidence from results and avoids guessing

Example: “Metal was the strongest because it held the highest weight.”

Importance of Scientific Investigation

Scientific investigations help learners:

Develop critical thinking skills

Make evidence-based decisions

Understand real-life applications

Safety Precautions

During experiments, learners must:

Handle materials carefully

Avoid sharp or hot objects without supervision

Wear protective gear if necessary

Check Your Understanding

1. What is the purpose of a hypothesis?
2. Differentiate between independent and dependent variables.
3. Why must some variables be controlled?
4. In a strength test, what is being measured?
5. Why is it important to record results in tables?

EFFECTS OF HEATING ON MATERIALS

Heating is one of the most common processes we apply to materials in everyday life eg cooking food, firing bricks, melting plastics, or shaping metals.

When materials are heated, they undergo changes in their structure and properties. These changes can either be: Temporary (reversible), or Permanent (irreversible)

Heating provides energy to particles, causing them to move, rearrange, or even form new substances.

What Happens When Materials Are Heated?

When heat is applied, Particles gain energy. They begin to move faster and the arrangement of particles may change This leads to observable effects such as: Expansion, Change of state (solid → liquid → gas), Change in colour and Change in strength or texture

Types of Changes Caused by Heating

Physical Changes (Reversible Changes)

A physical change does not form a new substance and can usually be reversed **for example** ice melting into water, wax melting, heating metal causing expansion etc.

Chemical Changes (Irreversible Changes)

A chemical change produces a new substance and cannot be easily reversed **for example** Burning paper, Cooking food, Firing clay into bricks etc.

Observable Effects of Heating different materials

When materials are heat, they may show the following signs:

Change in Colour: Indicates a possible chemical change **for example** Heating iron forms reddish-brown rust (after reaction with oxygen)

Change in Shape or Size: Materials may expand or deform **for example** Metal expands when heated

Release of Gas or Smoke: Indicates a chemical reaction **for example** Burning wood produces smoke and gases

Change in State: Solid to Liquid to Gas **for example** Ice to water to steam

Change in Strength or Hardness: Materials may become stronger or weaker **for example** Clay becomes hard after heating (used in bricks and pots)

Effects of Heating on Common Materials.

Material	What happens	Importance
Metals	Expand when heated May soften at high temperatures	Blacksmiths heat metals to shape them
Plastics (Polymers)	Soften when heated Can be moulded into different shapes	Making bottles, containers
Rubber	Becomes softer when heated May lose elasticity if overheated	
Glass	Softens when heated Can be reshaped	Glass blowing to make bottles
Clay	Soft when wet Hardens permanently when heated	Making bricks, pottery
Wood	Burns when heated strongly Produces ash, smoke, and heat	Used as a fuel

Experiment: Heating Different Materials

Aim

To observe the effects of heating on various materials

Materials

Candle

Metal spoon

Plastic piece

Paper

Procedure

- a) Carefully heat each material (under supervision)

b) Observe changes

Observations

Material	Observation
Metal	Heats up, no major observable change
Plastic	Softens
Paper	Burns to ash
Candle	Melts

Conclusion

Different materials respond differently to heat

Safety Precautions

Always supervise heating activities

Use tongs or holders for hot objects

Avoid inhaling fumes from burning materials

Keep flammable materials away

Check Your Understanding

1. What happens to particles when a material is heated?
2. Give two examples of physical changes caused by heating.
3. Why is burning considered an irreversible change?
4. What happens to plastics when heated?
5. Why is heating clay important in making bricks?

ENVIRONMENTAL IMPACT OF MATERIALS

Materials are essential for modern life used in packaging, construction, clothing, and technology. However, the way materials are produced, used, and disposed of can have serious effects on the environment.

Some materials are environmentally friendly, while others cause long-term pollution. Understanding this helps us make responsible choices.

What is Environmental Impact?

Environmental impact refers to the effect that materials have on the environment, including Land (soil), Water bodies (rivers, lakes, oceans), and Air

These effects may be Positive (e.g., biodegradable materials) or Negative (e.g., plastic pollution)

Types of Materials Based on Environmental Impact

Biodegradable Materials

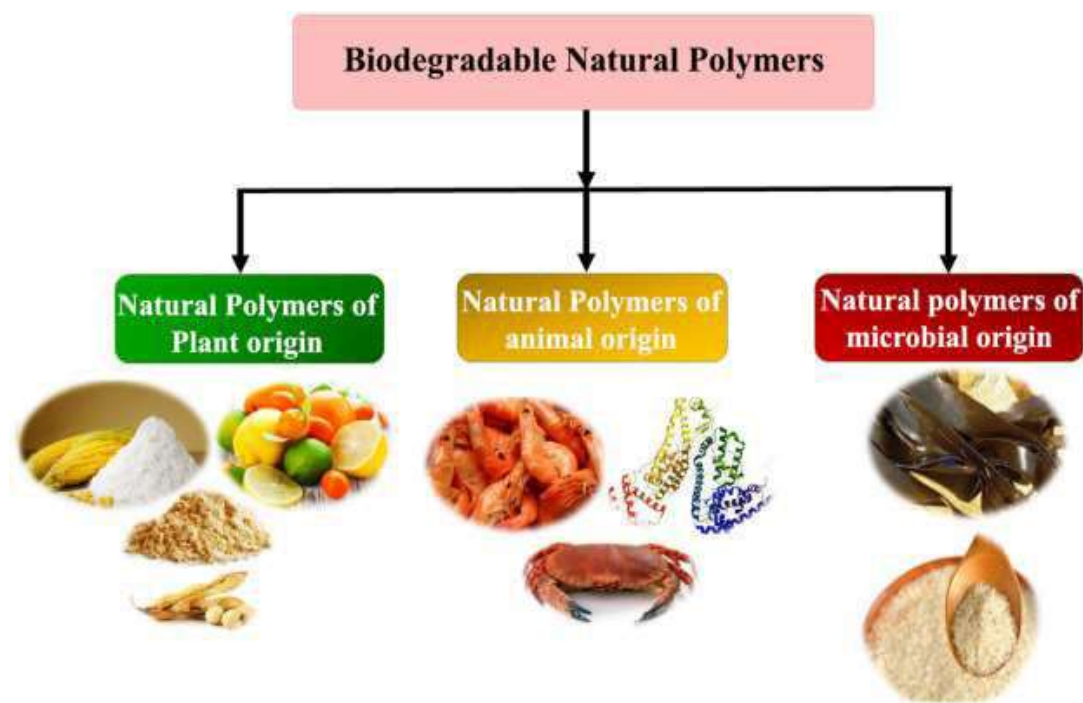
These are materials that can be broken down naturally by microorganisms such as bacteria and fungi. For examples Paper, Cotton, Food waste, Wood etc.

Characteristics

They decompose naturally over time

They do not accumulate in the environment

They are environmentally friendly



Real-Life Meaning

If you throw away a cotton cloth, it will eventually rot and return to the soil.

Non-Biodegradable Materials

These are materials that do not break down easily or take a very long time to decompose. For example Plastics (polythene, polystyrene), Glass, Metals

Characteristics

They remain in the environment for many years

They cause accumulation of waste

They can be harmful to living organisms

Real-Life Meaning

Plastic bags can remain in the soil or water for decades without decomposing.

Major Environmental Problems Caused by Materials

Land Pollution caused by dumping waste materials on land like plastic bags scattered in communities, old electronics dumped in open areas making soil unhealthy harming plants and animals

Water Pollution occurring when materials enter water bodies such as plastics in rivers and lakes, and oil spills harming aquatic life and also make water unsafe for drinking.

Air Pollution caused by burning materials for example burning plastics releases toxic gases into the air, Industrial emissions. These result into respiratory problems in people and global warming

Harm to Living Organisms whereby animals may eat plastic materials for example fish and livestock may die after ingesting plastic materials.

Why are Plastics a Major Concern in our communities?

Plastics are widely used because they are cheap, durable, lightweight. However, they are mostly non-biodegradable and accumulate in large quantities polluting land and water.

Important Note:

Improper disposal of plastics is one of the biggest environmental challenges today.

Responsible Use of Materials

Learners should understand practical ways to reduce environmental harm:

Reduce: Use fewer materials to avoid unnecessary packaging

Reuse: Use items multiple times for example reusing plastic containers

Recycle: Convert waste into new materials for example recycling paper into new books

Real-Life Applications for responsible use of materials

Area	Practice
Community practice	Sorting waste into biodegradable and non biodegradable
School practices	Using dustbins correctly Avoiding littering
House hold practices	Compositing organic waste Reusing containers

Activity: Waste Classification

Task

Learners collect waste items and classify them into:

Biodegradable

Non-biodegradable

Check Your Understanding

1. What is meant by environmental impact?
2. Give two examples of biodegradable materials.
3. Why are plastics harmful to the environment?
4. How does burning materials affect air quality?
5. Explain one way learners can reduce environmental pollution.

RECYCLING AND WASTE MANAGEMENT

As you have learned, many materials especially plastics and metals do not decompose easily and can harm the environment if not properly handled. This makes waste management an essential part of modern life.

Proper waste management helps to reduce pollution, conserve resources, and protect living organisms.

What is Waste?

Waste refers to any material that is no longer useful and is thrown away.



Common Sources of Waste

Homes (food remains, packaging)

Schools (paper, plastic bottles)

Industries (chemicals, scrap materials)

Types of Waste

Waste can be classified into two main types i.e

Biodegradable Waste that can be broken down naturally by microorganisms. Examples Food waste, Paper, Leaves, Cotton

Characteristics

It decomposes quickly

It can be used to make compost (organic fertilizer)

Non-Biodegradable Waste that does not break down easily, Examples Plastics, Glass, Metals, Electronics (e-waste)

Characteristics

It remains in the environment for long periods

It requires proper disposal or recycling

What is Recycling?

Recycling is the process of collecting and converting waste materials into new useful products.

Instead of throwing materials away, they are processed and reused.

Common Recyclable Materials

Material	Recycled product
Paper	New books, cardboards, art pieces
Plastic (polythene, polystyrene, polyester etc)	New Bottles, containers
Glass	New bottles,jars
Metals	New Cans, tools
Fabrics	Cleaning materils

None recyclable materials

These are mostly natural materials. While they are often environmentally friendly, they are typically not recycled in a factory. Instead, they are biodegradable (decompose naturally) or are reused.

Examples include, wood, cotton, silk, starch, proteins, rubber etc

The Recycling Process (Step-by-Step)

1. Collection: Waste materials are gathered from homes, schools, and industries
2. Sorting: Materials are separated into categories, Plastics, Metals, Paper and glass
3. Cleaning: Removing dirt or impurities
4. Processing: Materials are broken down, Plastics are melted, Paper is pulped, Metals are melted
5. Manufacturing: New products are made from recycled materials

Benefits of Recycling

It reduces Pollution as less waste is dumped in the environment

It conserves Natural Resources since fewer raw materials are needed for example recycling paper reduces cutting down of trees

It saves Energy since recycling often uses less energy than producing new materials

It reduces landfill Waste preventing overcrowding of dumping sites.

Waste Management Methods

Landfilling, here waste is buried in the ground though it has a side effect of causing soil and water pollution

Incineration (Burning) waste is burned in incinerators reducing volume of waste

Composting where biodegradable waste is allowed to decay producing compost (used as fertilizer)

Recycling which converts waste into useful products and this is the best method for non-biodegradable materials

The 3Rs Principle

A very important concept in waste management is the 3Rs:

1. Reduce

Minimize the amount of waste produced

Example:

Avoid unnecessary packaging

2. Reuse

Use items more than once

Example:

Reusing containers

3. Recycle

Convert waste into new products

Order matters:

Reduce → Reuse → Recycle (best to least preferred)

Activity: Waste Sorting Exercise

Task

Provide learners with mixed waste items and ask them to:

Sort into:

Biodegradable

Non-biodegradable

Recyclable

Check Your Understanding

1. What is waste?
2. Give two examples of recyclable materials.
3. Why is recycling important?
4. What is the difference between biodegradable and non-biodegradable waste?
5. Explain the meaning of the 3Rs.



S1 New Curriculum chemistry**Theme: Temporary and permanent changes to Materials****S1 New Curriculum Chemistry Chapter 5 – Temporary and permanent changes**

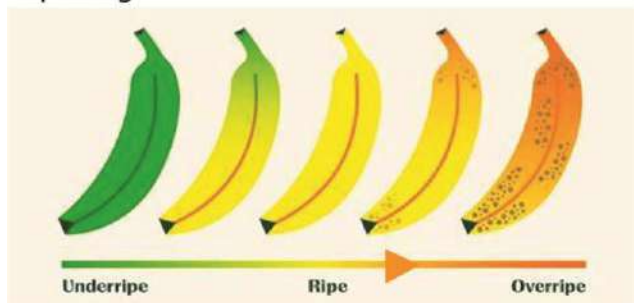
Change is the process of becoming different—whether it's gradual or sudden, small or monumental. It can happen in countless ways: in nature, in society, in our personal lives, even within us. Some changes are external, like growing of plant while others are internal, like development in a stem.

Examples of changes

Growth of a plant



Ripening



Types of changes

1. **Temporary changes:** These are changes that occur for a short period of time and are reversible. Examples of temporary changes include freezing, melting or ice, butter or cheese, evaporation, and sublimation

Melting of butter



2. **Permanent changes** are changes that remain for a long time and are not reversible for example growth, burning, rotting

Burning



3. **Chemical changes** are changes that occur in the chemical properties of a substance like reactivity, flammability, stability. Most chemical changes are irreversible.
4. **Physical changes** are changes that occur in the physical properties of a substance such as expansion, contraction, melting, and evaporation. Most physical changes are reversible.

Revision questions

1. Tick in the table below which changes are reversible

Melting of ice		Boiling of water into steam	
Egg to chicken		Cutting tree into logs	
Ripening of fruits		Dissolving salt into water	
Burning of candle		Burning sugar	
Wetting cloth		cooling	

S1 New Curriculum chemistry

Theme: Air and environment

S1 New Curriculum Chemistry Chapter 7 – Air

Air is a **mixture of gases** that surrounds the Earth and makes life possible. It is mostly composed of:

- **Nitrogen (78%)** – Helps plants grow and is essential for life.
- **Oxygen (21%)** – Vital for breathing and combustion.
- **Argon (0.93%)** – An inert gas used in industry.
- **Carbon dioxide (0.04%)** – Used by plants for photosynthesis and plays a role in climate change.
- **Other gases** – Includes small amounts of neon, helium, methane, and water vapor.

Oxygen

Uses of oxygen

Oxygen in air plays a crucial role in many natural and industrial processes. Here are some key uses: It is obtained by fractional distillation of air from which water and carbon dioxide have been removed by freezing.

- **Breathing and Respiration** – Oxygen is essential for humans and animals to survive. It helps in cellular respiration, where food is converted into energy.

Glucose + Oxygen

Energy + Carbon dioxide + water

- **Combustion** – Fuels like wood, gas, and coal burn in the presence of oxygen to produce heat and energy.
- **Medical Applications** – Oxygen is used in hospitals to support patients with breathing difficulties.
- **Water Purification** – Oxygen helps maintain water quality in lakes and rivers by supporting aquatic life.
- **Industrial Use** – It is used in welding, metal cutting, and the production of steel.
- **Rocket Fuel** – Liquid oxygen is used in rocket engines as an oxidizer to help burn fuel.
- **Ozone Formation** – Oxygen molecules form ozone (O₃), which protects the Earth from harmful ultraviolet rays.

Experiment to measure the percentage of oxygen in air

(a) Using a candle and water

We can measure the percentage of **oxygen in air** using a simple experiment with a burning candle and a gas jar. This experiment relies on the fact that oxygen supports combustion, and when consumed, the air volume decreases.

Materials Needed

- A small candle
- A gas jar or transparent container
- A shallow dish filled with water
- Matches

Procedure

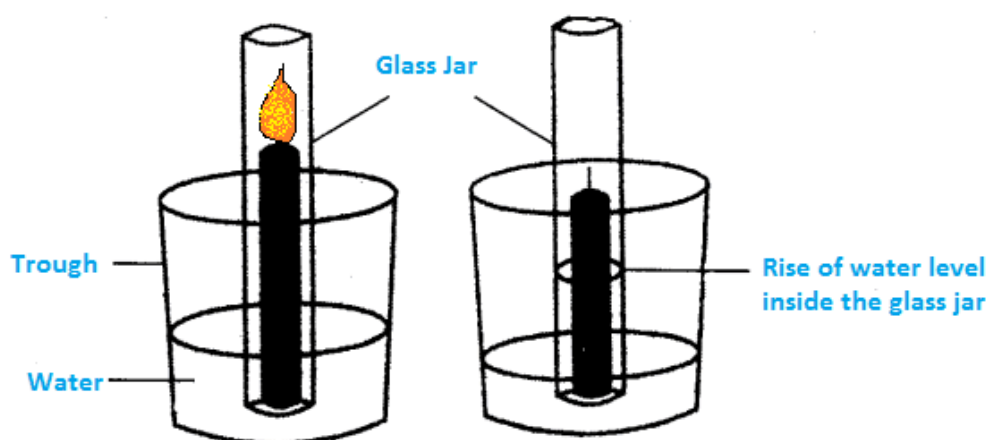
- Fill the Dish** – Pour water into the shallow dish.
- Place the Candle** – Stand the candle upright in the center of the dish.
- Light the Candle** – Use matches to ignite the candle flame.
- Cover with the Gas Jar** – Quickly place 100cm³ jar over the burning candle, sealing it in.
- Observe Water Movement** – As the oxygen inside the jar is used up, the flame goes out, and water rises into the jar to replace the lost oxygen.

Observation

The water level in the jar rises to 21cm³ in the jar

Hence volume of oxygen used up = 21cm³

Percentage of oxygen in air = —



Conclusion

Air is 21% oxygen

(b) Using copper fillings

Materials

Hard glass

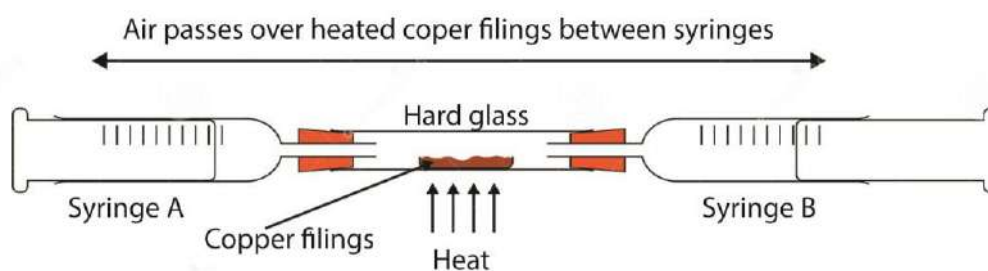
Copper filings

2 gas syringes

Source of heat

Setup

A tube of hard glass is connected to two gas syringes A and B. the tube is packed with copper filings. At the start syringe A contains 100cm^3 of air

**Procedure**

- (i) Heat copper filings while forcing air continuously over the filings from syringe A to syringe B and from syringe B to syringe A until there is not further change in the volume of air.
- (ii) Leave the apparatus to cool
- (iii) Push all air to syringe A and measure the remaining air.

Observations

Starting volume of air = 100cm^3

Final volume of air = 79cm^3

Volume of oxygen used = $100 - 79 = 21\text{cm}^3$

Percentage of oxygen in air = $\frac{21}{100} \times 100 = 21\%$. Hence percentage of oxygen in air = 21%

Conclusion: Air is 21% oxygen

Carbon dioxide

Uses of carbon dioxide from air

Carbon dioxide (CO₂) in the air has several important uses across nature and industry.

- **Photosynthesis** – Plants absorb CO₂ from the air to produce oxygen and food through photosynthesis, supporting all life on Earth.
- **Carbonated Drinks** – Used to add fizz to soda, sparkling water, and beer.
- **Fire Extinguishers** – CO₂ helps put out fires by displacing oxygen and cooling flames.
- **Refrigeration** – Used as a cooling agent in food preservation and transport.
- **Industrial Manufacturing** – Used in producing chemicals, dry ice, and welding processes.
- **Medicine** – Helps regulate breathing in patients and is used in medical treatments.
- **Enhanced Oil Recovery** – CO₂ is injected into oil wells to push out remaining oil for extraction.

Experiment to show that exhaled air contains carbon dioxide

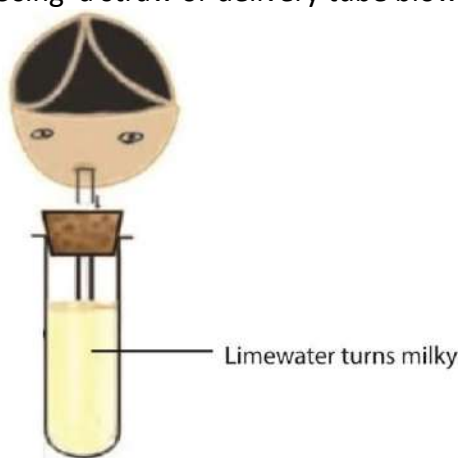
You can perform a simple experiment to show that **exhaled air contains carbon dioxide (CO₂)** using **limewater**. Limewater turns **cloudy** when it reacts with CO₂, proving its presence in exhaled air.

Materials Needed

- **Limewater** (a dilute solution of calcium hydroxide)
- **Two test tubes**
- **Drinking straw or delivery tube**
- **Source of fresh air** (optional for comparison)

Procedure

- Using a straw or delivery tube blow air through lime water as shown below



Observation

The limewater in the test tube turns cloudy due to the reaction between **carbon dioxide** and **limewater**, forming **calcium carbonate**.

Conclusion

The exhaled air contains carbon dioxide

Water**Experiment to show that exhaled air contains water****Materials Needed**

- A **clean mirror or glass**
- A **cool surface** (can be refrigerated for a few minutes)
- Your **breath**

Procedure

1. Cool a mirror in a refrigerator for a few minutes.
2. Breathe out onto the cold mirror

Observation

A **thin layer of mist** forms on the surface.



Mirror with condensed water droplet

Explanation of the Observation

The mist is caused by **water vapor** in breath condensing on the cool surface.

Conclusion

Exhaled air contains water vapour.

Nitrogen

Nitrogen, which makes up about **78% of the air**, has many important uses in nature and industry. It is obtained by fractional distillation of air from which water and carbon dioxide have been removed by freezing. Nitrogen is inert because a lot of energy is needed to break the strong bond between the two nitrogen atoms

Uses of nitrogen

- **Plant Growth** – Plants need nitrogen to produce proteins and chlorophyll. It's absorbed from the soil in the form of nitrates.
- **Food Preservation** – Used in food packaging to keep snacks and other products fresh by preventing oxidation.
- **Medical Applications** – Liquid nitrogen is used to freeze and remove unwanted tissue, such as warts.
- **Industrial Use** – Helps in welding, metal cutting, and making ammonia for fertilizers.
- **Fire Prevention** – Used in environments where reducing oxygen levels prevents fire hazards.
- **Oil & Gas Industry** – Used for pressurizing pipelines and removing impurities.
- **Cryogenics** – Liquid nitrogen cools and preserves biological samples, including blood and tissues.
- **Electronics Manufacturing** – Helps in semiconductor production for devices like computers and smartphones.

Air pollution

Air pollution refers to the presence of harmful substances in the air, negatively affecting health, the environment, and climate. It can come from both natural and human activities.

Causes of Air Pollution

1. **Burning Fossil Fuels** – Emissions from vehicles, factories, and power plants release carbon dioxide, sulfur dioxide, and nitrogen oxides.
2. **Industrial Waste** – Factories produce chemicals and particulates that pollute the air.
3. **Deforestation** – Fewer trees mean less absorption of carbon dioxide, increasing pollution.
4. **Agricultural Activities** – Pesticides, fertilizers, and methane from livestock contribute to pollution.
5. **Natural Causes** – Volcanic eruptions, forest fires, and dust storms release pollutants into the air.

Effects of Air Pollution

- **Health Problems** – Causes respiratory issues, heart disease, and lung cancer.
- **Climate Change** – Increases global warming and extreme weather events.
- **Acid Rain** – Damages buildings, soil, and aquatic ecosystems.
- **Harm to Wildlife** – Polluted air affects animals and plant life.

Ways to Reduce Air Pollution

- Use clean energy sources like solar and wind power.
- Reduce vehicle emissions by using public transport, cycling, or electric cars.
- Plant trees to absorb carbon dioxide.
- Adopt sustainable farming practices.
- Implement stricter environmental regulations.

Combustion

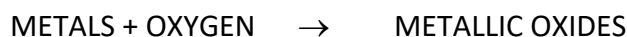
Combustion is a chemical reaction of a substance with oxygen.

Oxides

These are products of reaction of elements with oxygen. These reactions usually require heat.

Reactions of metals with oxygen

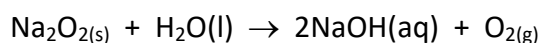
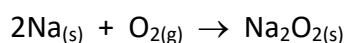
Oxygen combine vigorously with many metals to form basic oxides i.e. the solution of metal oxide in water turns red litmus paper blue



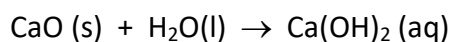
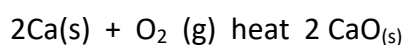
Examples

(a) Sodium (Na)

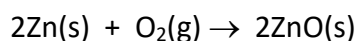
Burns with a bright yellow flame and form a yellowish solid



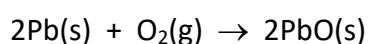
(b) Calcium (Ca): Burns with a bright red flame and form a white solid



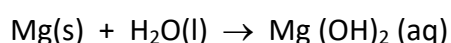
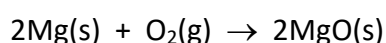
(c) Zinc: On burning it decomposes to a yellow solid when hot and turns white on cooling.



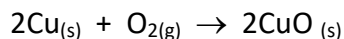
(d) Lead: Melts and then forms yellow oxides when hot and yellow when cold



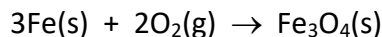
(e) Magnesium: burns in oxygen with a very bright light and forms a light white ash.



- (f) **Copper:** burns with a green flame



- (g) **Iron filings** (or wire): burn with a shower of bright sparks and forms a blue-black solid insoluble in water

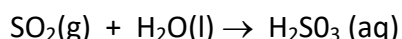
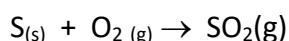


Non-metals

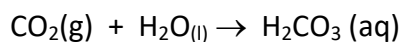
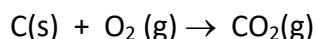
Oxygen combine vigorously with many metals to non-metals to form acidic oxides, i.e. the solution of oxides of non-metals in water turn blue litmus paper red.



- (a) **Sulphur:** Burns with a bright blue flame and forms cloudy fumes with a choking smell



- (b) **Carbon:** burns with an orange flame and makes bright sparks



Types of oxides

Basic oxide: is a metallic oxide, which react with acid to produce a salt and water only. When dissolved in waters, it forms an alkaline solution or solutions that turn red litmus paper blue. Example of basic oxides are metal oxides such calcium oxide (CaO) and sodium oxide (Na₂O)

An acidic oxide: is a non-metallic oxide when combined with element of water produces an acid or solutions that turn blue litmus paper red. Examples of acidic oxides are sulphur dioxide (SO₂).

Amphoteric oxide: is metallic oxide, which can show both basic and acidic properties. (it reacts with both acids and bases to produce water only and a salt e.g. ZnO, Al₂O₃)

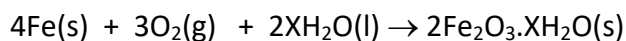
Neutral oxide: - is an oxide which shows neither basic nor acidic character e.g. H₂O, CO

Rusting



Rust: is a brown layer/coating that form on iron/steel when left in damp air

Rust is hydrated Iron III oxide.



Disadvantages of rusting

Rusting is a major problem for metal objects and structures, leading to several disadvantages:

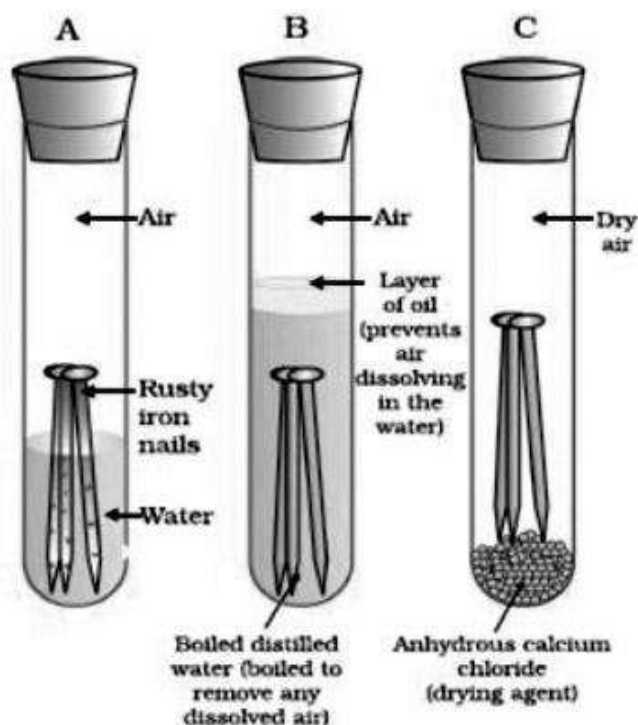
- **Weakens Structures** – Rust makes iron and steel brittle, causing bridges, buildings, and vehicles to weaken over time.
- **Economic Loss** – Industries spend billions on maintenance, repairs, and replacements due to rust damage.
- **Aesthetic Issues** – Rust makes metal surfaces look unattractive, reducing their value.
- **Malfunction of Machinery** – Rust can cause machines and tools to stop working properly, leading to breakdowns.
- **Environmental Pollution** – Rusted metal can flake off and contaminate soil and water sources.
- **Safety Hazards** - Rusted structures and vehicles can collapse or fail, posing dangers to people.

To investigate the conditions necessary for Iron to rust

Materials Needed

- Three test tubes
- Iron nails
- Water
- Boiled water + oil
- Dry calcium chloride (a drying agent)
- Labels

Procedure



1. **Test tube A: Air and Water Present**
 - Place an **iron nail in tap water**, leave space for air and stopper.
 - This exposes iron to both **oxygen** and **water**, allowing rusting.
2. **Test Tube B: Water Present, No Oxygen**
 - Place an **iron nail in boiled water** and add a **layer of oil** to prevent oxygen from entering.
 - Without oxygen, rust cannot form.
3. **Test Tube C: Oxygen Present, No Water**
 - Place an **iron nail in a test tube with dry calcium chloride** and **seal the tube**.
 - The calcium chloride **absorbs moisture**, keeping the nail dry.

Observations

- **Test Tube A** – The nail rusts because both **oxygen** and **water** are present.
- **Test Tube B** – No rust forms because **oxygen is absent**.
- **Test Tube c** – No rust forms because **moisture is absent**.

Conclusion

Iron rusts **only when both oxygen and water are present**.

Factors that influence rusting

- Nature of the metal
- pH
- temperature

- humidity
- surface state of metal
- nature of protective film

Method used to prevent rusting:

- (i) **Oil or Grease:** A layer of oil or grease is put on the metal.
- (ii) **Paint or tar:** paint is put on the iron metal e.g door, windows etc. and tar is put on iron that we do not see e.g. at the bottom of the bridges ship.
- (iii) **Tinplate:** made by dipping a sheet of iron into molten tin.
- (iv) **Galvanized iron:** Made by dipping a sheet of iron in molten zinc.
- (v) **Chromium- plate:** Iron coated with chromium (and nickel)
- (vi) **Alloying:** Stainless steel: Is iron that contains carbon, chromium, nickel and other elements.

Revision questions

1. Name two major components of air.
2. Give the uses of the following gases
 - (i) Oxygen
 - (ii) Nitrogen
 - (iii) Carbon dioxide
 - (iv) Argon
3. Describe an experiment to show that air contains carbon dioxide and water vapour.
4. Describe an experiment to determine the percentage of oxygen in air
5. Describe an experiment to show the conditions necessary for rusting.
6. Mentions ways of minimizing rusting

S1 New Curriculum Chemistry Chapter 8 – Water

Water is the essence of life!

Water is an important natural resource required by all living things to survive. For instance, man is made up between 70% to 90% water. Water covers 71% of the Earth's surface.

Chemically, water is made by reacting hydrogen to oxygen in ratio 2:1. Hence, the chemical formula of water is H_2O .

States of water

Water exists in **solid** as **ice**, in **liquid** as **water** and in **gas** as **steam**.

Physical properties of water

- (i) It is colorless
- (ii) It is odorless
- (iii) And tasteless
- (iv) Density of water is $1g/cm^3$. Ice is less dense than water.

Experiment to determine the density of water

Apparatus

- Clean water in a beaker (about $20cm^3$)
- Measuring cylinder ($25cm^3$ or $50cm^3$)
- Weighing scale

Procedure

- Weigh a dry, empty measuring cylinder = x g
- Pour and weigh $20cm^3$ of water in measuring cylinder = y g

Calculation

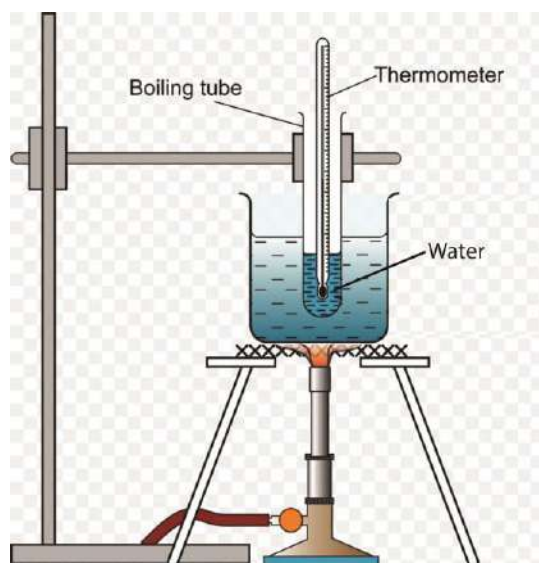
Mass of 20cm^3 of water = $(x - y)$ g

Density of water = $\frac{(y-x)}{20}$ g/cm³

- (v) Pure water freezes at **0°C** and boils at **100°C**.

Determining the boiling point of water

Setup



Procedure

Heat the beaker gently, till water boils and the temperature becomes constant. The constant temperature is the boiling point. The boiling point of water at Kampala is between **96°C – 97°C**.

Chemical test for water

Water **turns white anhydrous copper Sulphate powder blue**.

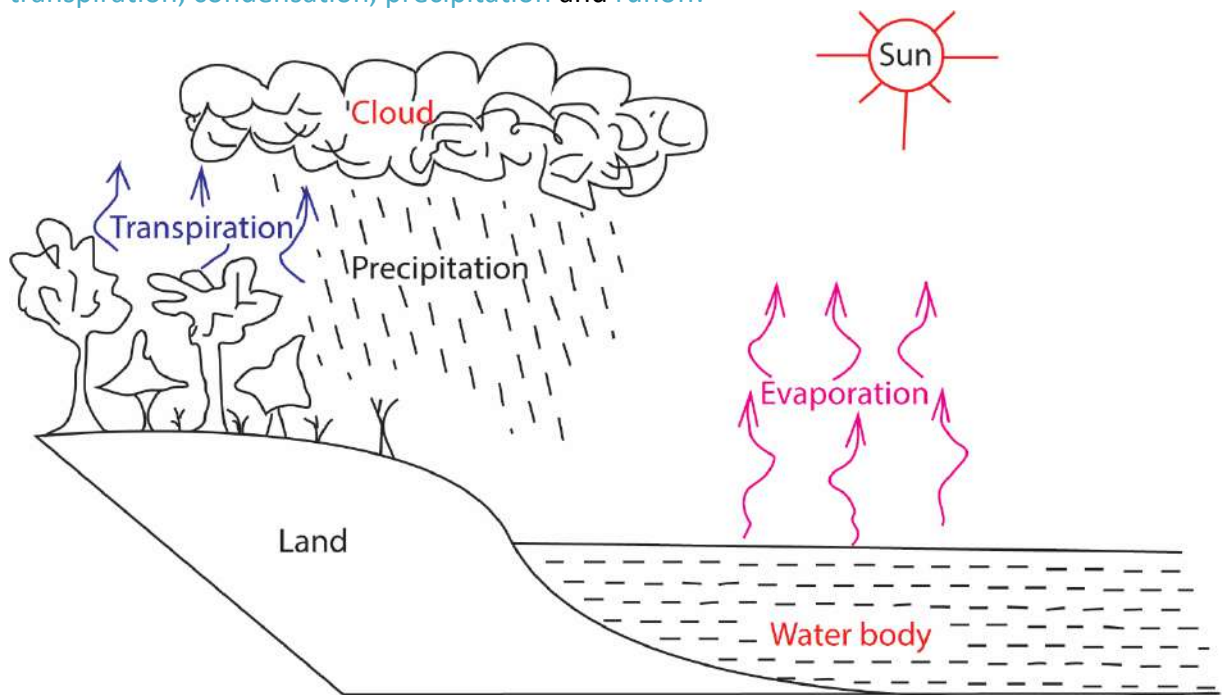
Some of the uses of water in daily life

- **Drinking & Hydration** – Keeps the body functioning, regulates temperature, and supports overall health.
- **Cooking & Food Preparation** – From boiling to washing ingredients, it's essential in the kitchen.
- **Cleaning & Hygiene** – Used for bathing, washing hands, doing laundry, and maintaining sanitation.

- **Agriculture & Gardening** – Vital for growing crops, watering plants, and sustaining livestock.
- **Transportation** – Rivers, lakes, and oceans enable travel and shipping.
- **Energy Production** – Hydropower generates electricity for homes and industries.
- **Manufacturing & Industry** – Used in production processes, cooling systems, and creating materials like paper and textiles.

Water cycle

- The **water cycle** is the path that all water follows as it moves around our planet.
- Of the many processes involved in the water cycle, the most important are **evaporation**, **transpiration**, **condensation**, **precipitation** and **runoff**.



- **Evaporation** is the change of a liquid into vapour below its boiling point.
- **Transpiration** is the loss of water vapour from the plant through the stomata.
- **Condensation** is process by which water vapor turns into a liquid.
- **Precipitation** is any liquid or frozen water that forms in the atmosphere and falls back to the Earth in form of rain.
- **Runoff**, is the waters that travel over the land surface and through channels to reach a stream.

Common source of water for daily life

Water comes from various sources, each serving different needs. Here are some of the most common ones:

- **Rainwater** – Collected in tanks or reservoirs, especially in areas with limited access to other sources.
- **Wells** – Groundwater extracted through wells, a vital source for many communities.

- **Rivers and Streams** – Often used for drinking, irrigation, and industrial purposes.
- **Lakes and Ponds** – Natural reservoirs that supply water for consumption, fishing, and recreation.
- **Boreholes** – Deep underground water sources tapped for clean and sustainable supply.
- **Municipal Supply** – Treated water from reservoirs, dams, or rivers delivered through pipelines.
- **Springs** – Naturally occurring sources of fresh water from underground.

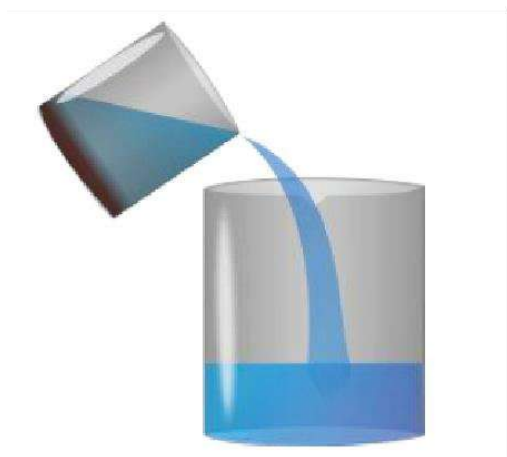
Each source requires different levels of treatment to ensure safety for human use.

Methods used to make water safe

- decanting,
- filtering,
- boiling,
- distilling
- adding chemicals.

(a) Decanting

Here dirty water is left in a container to settle and then clean water is poured into another container leaving settle particles behind. The particles that settle at the bottom of the container are called **sediments**. **Decanted water contain germs and is not fit for consumption until it is boiled**



(b) Filtration

Unclean water can be made clean by passing the water through a piece of cloth or a filter paper.

The cloth or filter paper acts as a sieve.

Water produced by this method is not fit for drinking because it may contain germs.

Homemade filtration apparatus



(c) Boiling:

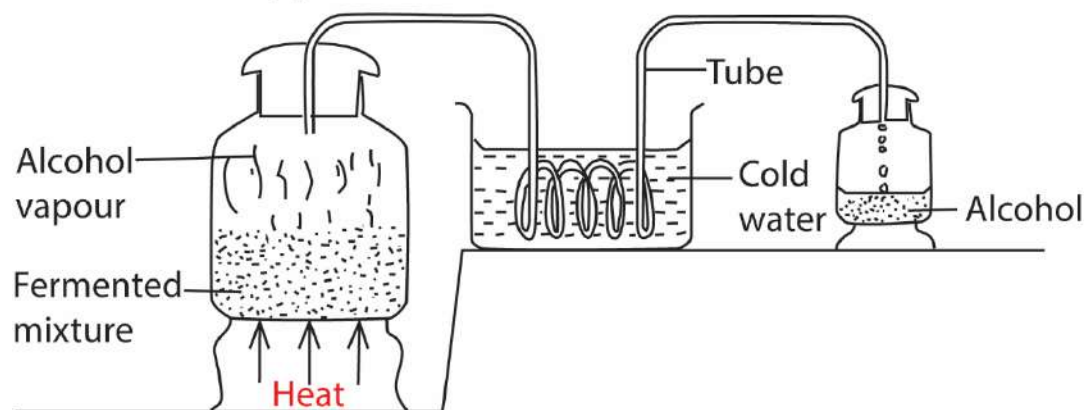
Boiling kills germs

(d) Distillation

Here water is boiled to form vapor/steam that is cooled to form water or distillate. This method produces the purest water. This water is no good for drinking because it lacks minerals.

Distillation is also used to prepare alcohol from its fermented mixture as shown in the diagram below

Distillation of apparatus for alcohol



The mixture is boiled, alcohol with low boiling point forms vapour first; the vapour is condensed with cold water to form liquid alcohol.

Stages in water treatment plant

A commercial water treatment plant follows several key stages to ensure water is safe for consumption and industrial use. Here's a breakdown of the process:

1. **Collection & Screening** – Water is sourced from lakes, rivers, or reservoirs and screened to remove large debris like leaves, plastic, and fish.
2. **Coagulation & Flocculation** – Chemicals like aluminum sulfate or ferric chloride are added to bind small particles together into larger clumps (floc).
3. **Sedimentation** – The floc settles at the bottom of a tank, separating from the cleaner water.
4. **Filtration** – Water passes through layers of sand, gravel, and charcoal to remove remaining impurities, including bacteria and algae.
5. **Disinfection** – Chlorine or other disinfectants are added to kill harmful microorganisms and ensure water remains safe during distribution.
6. **Storage & Distribution** – Treated water is stored in tanks before being delivered through pipelines to homes, businesses, and industries.

Sewage

Sewage refers to wastewater that comes from households, industries, and businesses, containing human waste, chemicals, and other pollutants. In Kampala, Uganda, sewage management is primarily handled by the **National Water and Sewerage Corporation (NWSC)**, which provides water and sanitation services.

Stages in sewage management

Sewage management involves several key stages to ensure wastewater is treated effectively before being released back into the environment or reused. Here's a breakdown of the process:

1. **Preliminary Treatment** – Large debris like plastics, leaves, and grit are removed using screens and grit chambers to protect equipment from damage.
2. **Primary Treatment** – Wastewater is held in sedimentation tanks where heavier solids settle at the bottom as sludge, while oils and grease float to the surface for removal.
3. **Secondary Treatment** – Biological processes break down organic matter using bacteria and microorganisms, often through activated sludge systems or biofilters.
4. **Tertiary Treatment** – Advanced filtration and chemical treatments remove remaining contaminants, including nutrients like nitrogen and phosphorus, to improve water quality.

5. **Sludge Treatment** – The sludge collected from earlier stages is processed through digestion, drying, or incineration to reduce volume and recover useful byproducts.

Reasons for sewage treatment

Sewage treatment is essential for protecting public health, the environment, and water resources.

- **Prevents Water Contamination** – Untreated sewage contains harmful bacteria, viruses, and chemicals that can pollute rivers, lakes, and groundwater, making water unsafe for drinking and agriculture.
- **Reduces Disease Spread** – Proper sewage management prevents outbreaks of waterborne diseases like cholera, dysentery, and typhoid.
- **Protects Ecosystems** – Without treatment, sewage can damage aquatic life by depleting oxygen in water bodies, leading to dead zones where fish and plants cannot survive.
- **Supports Sustainable Water Use** – Treated wastewater can be safely reused for irrigation, industrial processes, and even drinking after advanced purification.
- **Minimizes Foul Odors & Pollution** – Sewage buildup in communities can cause bad smells and unsanitary living conditions, which effective treatment helps prevent.
- **Compliance with Regulations** – Governments enforce sewage treatment standards to maintain a clean and healthy environment for everyone.

Revision Questions

1. (a) Name five sources of natural water.

(b) How is this water from natural sources made safe for drinking.
2. Describe an experiment to show that a give liquid is water.
3. Describe the process of recycling water naturally.
4. List the physical properties of water.
5. (a) What is sewage?

(b) List stages of sewage treatment

(c) Why is sewage treatment necessary?

S1 New Curriculum chemistry**Theme: Earth and space****S1 New Curriculum Chemistry Chapter 9 – Rocks and minerals****Rocks**

Rocks are solid, natural substances composed of minerals. They make up Earth's crust and come in different types based on how they form:

1. **Igneous Rocks** – Formed from cooled magma or lava. Examples include granite and basalt.
2. **Sedimentary Rocks** – Created by the accumulation and compression of sediments over time. Examples include limestone and sandstone.
3. **Metamorphic Rocks** – Formed when existing rocks undergo extreme heat and pressure, changing their structure. Examples include marble and slate.

Examples of Igneous Rocks



Granite



Basalt

Examples of Sedimentary Rocks



Limestone



Sand stone

Examples of Metamorphic Rocks



Marble



Slate

Uses of rocks

Construction – Used in buildings, roads, bridges, and dams.

Jewelry & Decoration – Precious stones like diamonds, rubies, and emeralds are cut and polished for jewelry.

Manufacturing – Limestone is used in cement production, while granite and marble are used for countertops and flooring.

Energy Production – Some rocks, like coal, are burned for fuel, while others contain minerals used in nuclear energy.

Agriculture – Crushed rocks improve soil quality and drainage.

Tools & Equipment – Hard rocks like flint were historically used for making tools.

Water Filtration – Certain porous rocks help purify water.

Landscaping – Used for pathways, garden features, and erosion control.

Some Examples of uses of rocks



Decoration made out of rocks



Dining set of rocks

Minerals

Minerals are naturally occurring substances with a **specific chemical composition** and **crystalline structure**. They are the building blocks of rocks and are essential for various industries and biological functions.

Characteristics of minerals

- **Inorganic** – Minerals are not formed by biological processes.
- **Solid** – They exist in solid form under normal conditions.
- **Naturally Occurring** – Found in nature, not artificially created.
- **Definite Composition** – Each mineral has a unique chemical formula.
- **Crystalline Structure** – Their atoms are arranged in an orderly pattern.

Examples include **quartz, feldspar, calcite, gold, diamond, and hematite**.

Minerals are used in construction, technology, jewelry, and even nutrition (like calcium and iron in the human body).

Differences between Rocks and Minerals

Minerals and rocks are closely related but have distinct differences. Here's how they compare:

- **Composition** – Minerals are pure substances with a definite chemical formula, while rocks are made up of multiple minerals and other materials.
- **Structure** – Minerals have a crystalline structure, meaning their atoms are arranged in a specific, orderly pattern. Rocks, on the other hand, can have varied textures and structures depending on how they form.
- **Formation** – Minerals form through natural processes like crystallization, evaporation, or cooling. Rocks are classified into igneous, sedimentary, and metamorphic types based on how they are formed.
- **Identification** – Minerals have specific properties such as hardness, color, and luster, while rocks are identified based on their mineral content and texture.
- **Uses** – Minerals are used in jewelry, industry, and nutrition, whereas rocks are primarily used in construction, sculptures, and historical artifacts.

Weathering

Weathering is the process by which rocks and minerals break down due to natural forces like wind, water, temperature changes, and biological activity. It plays a crucial role in shaping landscapes and forming soil.

Types of weathering:

- (i) **Physical Weathering** – Breakdown of rocks without chemical changes, caused by temperature fluctuations, freeze-thaw cycles, and abrasion from wind or water.
- (ii) **Chemical Weathering** – Changes in the chemical composition of rocks due to reactions with water, oxygen, acids, or other substances. This can lead to the formation of new minerals.
- (iii) **Biological Weathering** – Caused by living organisms like plants, fungi, and bacteria. Roots can grow into cracks, breaking rocks apart, while microbes produce acids that dissolve minerals.

Revision questions

1. (a) What is a rock?

(b) Name three major types of rocks and explain how they are formed
2. Distinguish between rocks and minerals.
3. Give the various uses of rocks and minerals

S2 New Curriculum chemistry**Theme: Acid and Alkalis****Chapter 1 – Acid and Alkalis****Acids**

An acid is a substance which, when dissolved in water produces hydrogen ions, H^+ as the only positively charged ions. That is

Examples of nature foods that contain acidic juices**Lemons****Tomatoes****Pineapples****Oranges****Examples of acids include**

Names	Formula	Ions
Sulphuric acid	H_2SO_4	$2H^+$, SO_4
Hydrochloric acid	HCl	H^+ , Cl^-
Nitric acid	HNO_3	H^+ , NO_3
Carbonic acid	H_2CO_3	$2H^+$, CO_3
Ethanoic acid/acetic acid	CH_3COOH	H^+ , CH_3COO^-

Properties of acids**Properties of acidic solutions**

- (i) Acids have a **sour taste**, which is why substances like lemon juice and vinegar taste sharp. (But you should never taste solution in a lab unless told to do so)
- (ii) Acid solutions turn blue litmus paper red



Blue litmus paper turns **red**

- (iii) They liberate carbon dioxide from carbonates e.g. potassium and sodium carbonate
- (iv) They liberate carbon dioxide from bicarbonates e.g. potassium and sodium bicarbonates
- (v) They ionize in water to produce H^+ as the only positive ions examples:

$$H_2SO_4 + (aq) \longrightarrow 2H^+(aq) + SO_4^{2-}(aq)$$

$$HCl + (aq) \longrightarrow H^+(aq) + Cl^-(aq)$$

$$HNO_3 + (aq) \longrightarrow H^+(aq) + NO_3^-(aq)$$

(vi) **Corrosive action**

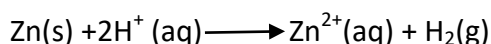
Some acids are corrosive e.g. H_2SO_4 . A **corrosive substance** is a material that can cause irreversible damage or destruction to other substances and skin upon contact, often through chemical reactions.



Acid burns - **be careful**

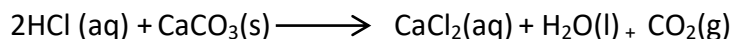
(vii) **Action with metals**

Some metals react with dilute hydrochloric acid or dilute sulphuric acid to liberate hydrogen gas. Examples of these metals are Zn, Mg, Al.



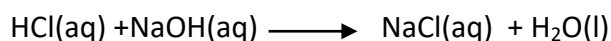
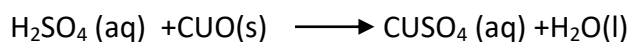
(viii) **Action with carbonates and hydrogen carbonates**

Almost all acids liberate carbon dioxide from carbonates and hydrogen carbonates.



(ix) **Reaction with Oxides and hydroxides.**

Acids react with oxides and hydroxides of metals to form salts and water only.



Types of acids

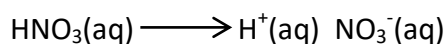
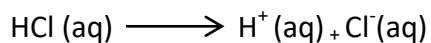
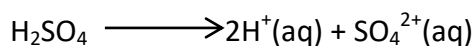
Strong and weak acids

A **strong acid** is one which is completely ionised in water or dilute solution.

Examples of strong acid

Sulphuric acid (H_2SO_4), hydrochloric acid (HCl), nitric acid (HNO_3)

Ionisation is as follows:



Weak acid

A **weak acid** is one which is slightly ionised in water/ dilute solution

That is one which is not fully ionised in dilute solution. Its ionisation is irreversible.

Example of weak acids

Carbonic acid H_2CO_3

Phosphoric acid H_3PO_4

Ethanoic acid CH_3COOH

Methanoic acid HCOOH ,

Sulphurous acid H_2SO_3 etc.

Its ionisation is as follows

Basicity of an acid

Basicity is the number of hydrogen ions in a molecule of an acid.

Monobasic acids have only one ionisable hydrogen ion and are called monobasic acid eg.

Hydrochloric acid (HCl), Nitric acid (HNO₃) and ethanoic acid (CH₃COOH)

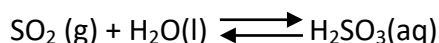
Dibasic acid have two ionisable hydrogen ions e.g. Sulphuric acid (H₂SO₄).

Tribasic acids have three ionisable hydrogen ion e.g. phosphoric acid (H₃PO₄).

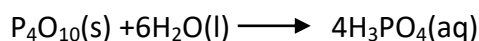
Preparations of acids

By the reaction between an acid anhydride (the acidic oxide of a non-metal) and water, e.g.

Sulphurous acid

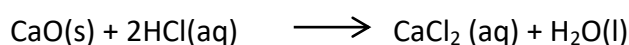


Phosphoric acid

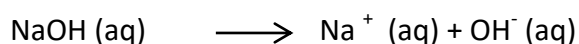
**Bases and alkalis**

A **base** is a substance that reacts with acids to form water and salt only. A base can be an oxide or hydroxide

Calcium oxide + Hydrochloric acid $\longrightarrow$ Calcium chloride + water



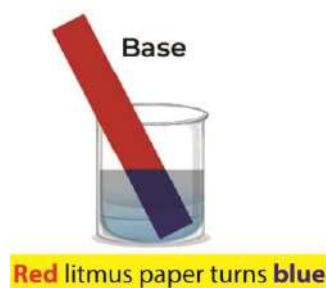
An **alkali** is a compound which, when dissolved in water forms hydroxide ions as the only negatively charged ions for example sodiumhydroxide (NaOH)



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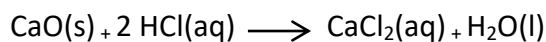
Properties of bases and alkalis

- (a) Bases tend to have a bitter flavor.
- (b) Solutions of bases are like soap, feel slippery when touched.
- (c) They dissolve in water to form solutions that turn red litmus paper blue.

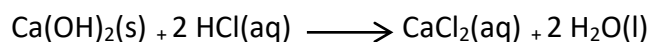


- (d) They react with acid to produce salts and water only in a reaction called
NEUTRALISATION reaction

Base

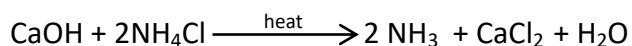


Alkalis



Neutralisation is the combination of the oxide or hydroxide ions of a base with the hydrogen ions of an acid to form water, and a salt.

- (e) They evolve ammonia when warmed with ammonium salts.



Examples of foods with basic juices i.e. Juices with pH greater than 7

- **Bananas** – Naturally alkaline and gentle on digestion.
- **Coconut Water** – Has a slightly alkaline effect and is hydrating.
- **Melons** – Watermelon, cantaloupe, and honeydew are alkaline-forming.

- **Avocados** – Rich in healthy fats and naturally alkaline.
- **Spinach & Kale** – Leafy greens that promote alkalinity.
- **Cucumber** – Refreshing and alkaline-forming.
- **Sweet Potatoes** – A starchy vegetable with alkaline properties.
- **Baking powder**

Uses of neutralizing reaction

- Lime is used neutralize acidic soils in agriculture
- Antacids are used to treat acidic stomach ulcers
- Bee stings are acidic and can be soothed by basic household substances such as baking powder

Acid – base indicators are substances that show different colours in acids and alkalis.

Examples of Indicator

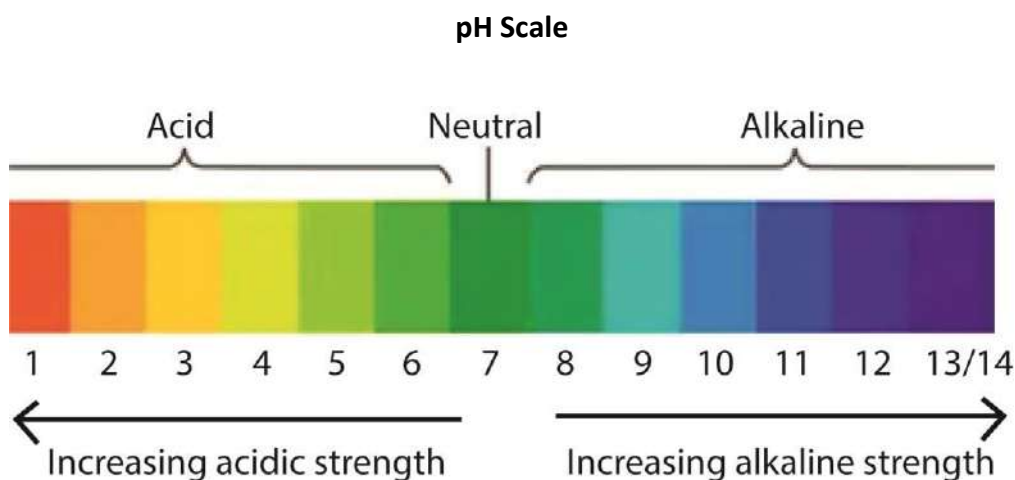
Indicator	Acids	Alkalis
Litmus	Red	blue
Methyl orange	Red	Yellow
Phenolphthalein	Colourless	Red

pH

pH is the measure of the degree of acidity of a solution.

The pH scale

The pH scale is one way of measuring acidity or alkalinity in dilute solution. The PH scale is made of values that run from 1 -14. The acid range is from 1-6 while alkalis (bases) range from 8-14. At PH 7, the solution is neutral that is neither acid nor alkaline.



Revision questions

1. (a) Give two properties of acids
(b) Define the term base and give one example.
(c) Define the term alkali and give an example.
2. (a) What is pH
(b) State the pH range
(c) Name one property of substance with pH less than 7.
(b) Give an example of substance with pH = 7
3. (a) What is neutralization?
(b) Write any word equation for a neutralization reaction.
4. Sulphuric acid is a strong acid while ethanoic acid is a weak acid
(a) What is a strong and weak acids.
(b) Two other examples of strong acids.

5. Litmus is a dye made from plant called lichen.

It is red in acid

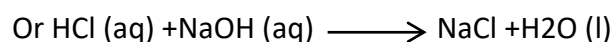
It s blue in alkalis

It is purple in neutral solution

- (a) What is the scientific term for a substance that changes colour in this way?
- (b) What colour is litmus in a solution of pH 10?
- (c) What colour is litmus in a solution of pH 4?
- (d) What colour is litmus in pure water.

S2 New Curriculum chemistry Theme: Acid and Alkalis Chapter 2 – Salts**SALTS**

A **salt** is a compound formed when an acid reacts with a base, a carbonate or a metal. For example common salt (NaCl) is formed when hydrochloric acid reacts with sodium hydroxide, i.e. Hydrochloric acid + sodium hydroxide $\longrightarrow$ Sodium chloride + water

**Types of salts**

There are two different types of salts

Normal salt

Normal salt is the one in which all the replaceable hydrogen from the acid has been replaced by the metal. Examples of normal salts are:

Sodium chloride NaCl

Sodium sulphate Na₂SO₄

Calcium carbonate CaCO₃

Calcium sulphate CaSO₄

Calcium chloride CaCl_2

Ammonium chloride NH_4Cl

Acidic salt

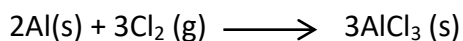
Acidic salt is one in which part of replaceable hydrogen of the acid has been replaced by a metal. Examples acidic salts include

- sodium hydrogen carbonate (NaHCO_3)
- Sodium hydrogen sulphate NaHSO_4
- Calcium hydrogen sulphate $\text{Ca}(\text{HSO}_4)_2$
- Calcium hydrogen carbonate $\text{Ca}(\text{HCO}_3)_2$ etc.

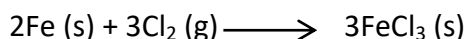
Preparation of salts

1. By direct combination of metals and non-metal. For example:

Aluminum + Chlorine $\longrightarrow$ Aluminium chloride

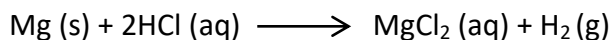


Iron + Chlorine $\longrightarrow$ Iron chloride



2. By reaction of excess metal with an acid, for example:

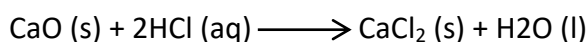
Magnesium + Hydrochloric acid $\longrightarrow$ Magnesium chloride



The excess metal is filtered off and the solution of the salt is evaporated to dryness to recover the salt.

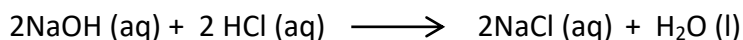
3. By reacting excess metal oxide with an acid, for example

Calcium oxide + Hydrochloric acid $\longrightarrow$ Calcium chloride + water



4. By neutralisation of an alkali by an acid, for example:

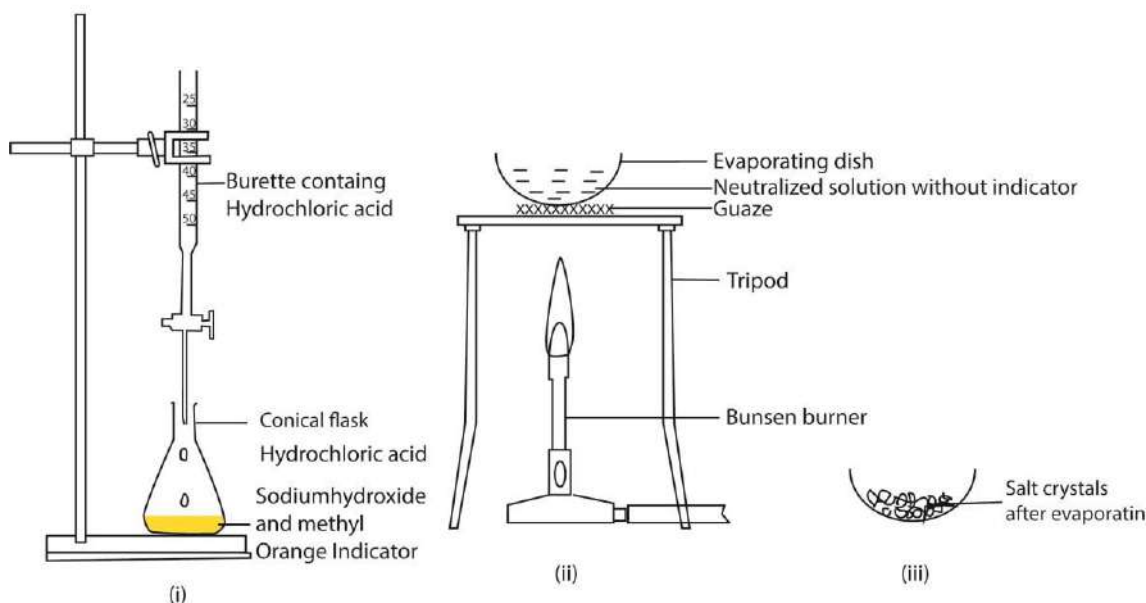
Sodium hydroxide + Hydrochloric acid $\longrightarrow$ Sodium chloride + water



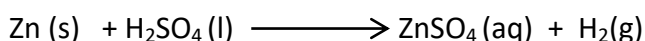
Procedure

- (i) 50 cm^3 of sodium hydroxide solution titrated with dilute hydrochloric acid solution using methyl orange indicator to find the volume $x\text{ cm}^3$ of the acid required to neutralize the alkali. The using fresh solutions mix 50 cm^3 of sodium hydroxide and $x\text{ cm}^3$ of hydrochloric acid to form a neutralized solution without indicator
- (ii) Evaporate the neutralized solution to near dryness in an evaporating dish.
- (iii) Leave the concentrated solution cool and form crystals

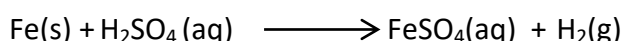
Preparation of salt by neutralization

**Preparation of zinc sulphate from zinc**

- Add dilute $\text{H}_2\text{SO}_4(\text{s})$ mixed with little copper sulphate and zinc in a beaker
- Add more zinc to saturate the solution
- Filter to remove the undissolved zinc
- Warm the filtrate to concentrate it
- Cool to room temperature
- Filter the crystals of zinc sulphate-7-water
- Wash the crystals with little cold distilled water
- Dry the crystals between filter papers

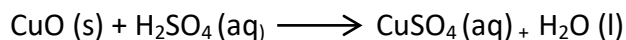
**Preparation of iron (II) sulphate-7-water($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$)**

- Add warm dilute sulphuric acid to iron filling in a beaker
- Add more iron fillings to saturate the solution
- Filter to remove the un dissolved iron filings
- Warm the filtrate slightly to concentrate it
- Cool to room temperature
- Filter the crystals of iron(II) sulphate-7-water
- Wash the crystals with cold distilled water
- Dry the crystals between filter papers

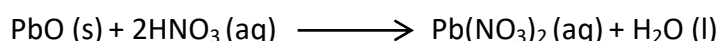
**Preparation of copper (II) sulphate-5-water from copper oxide**

- Pour dilute sulphuric acid in a beaker, warm gently. (to increase on the solubility)
- Add copper oxide to the warm acid.

- Add more copper oxide to saturate the solution.
- Filter to remove the un dissolved copper oxide
- Evaporate the filtrate concentrate it cool to room temperature
- Filter the crystals of copper(II) sulphate-5-water
- Wash the crystals with little cold distilled water
- Dry the crystals between the filter paper

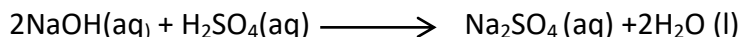


N.B: Magnesium sulphate, zinc sulphate, and lead (II) nitrate can be prepared in the same way.

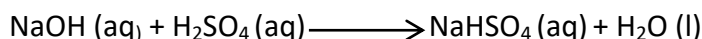


Preparation of sodium hydrogen sulphate.

- Add 20cm³ of 2M NaOH in a clean conical flask.
- Titrate with 2M H₂SO₄ from a burette using methyl orange indicator.
- The solution formed contains sodium sulphate.



- Note the volume of H₂SO₄ used, say x cm³.
- Measure another 20 cm³ of 2M NaOH in a clean beaker.
- Add 2x cm³ of 2M H₂SO₄
- The solution formed contains sodium hydrogen sulphate.



Solubility of salts

Solubility is a measurement of how much of a substance will dissolve in a given volume of a liquid at a given temperature.

- All sulphates are soluble, except barium sulphate, calcium sulphate and lead (II) sulphate
- All chlorides are soluble except silver chloride. Lead (ii) chloride dissolves in hot water but reforms on cooling
- All carbonates are insoluble except potassium, sodium and ammonium carbonates
- All nitrates are soluble

Some examples where salts are useful

Salts are incredibly versatile and play a crucial role in various fields. Here are some examples of their usefulness:

- **Food Industry** – Sodium chloride (table salt) enhances flavor and preserves food.

- **Agriculture** – Fertilizers contain salts like ammonium nitrate and potassium chloride to support plant growth.
- **Medicine** – Salts like magnesium sulfate (Epsom salt) are used for muscle relaxation, while sodium bicarbonate helps with indigestion.
- **Water Treatment** – Salts help soften hard water, making it more suitable for household use.
- **Chemical Manufacturing** – Sodium chloride is a key ingredient in producing chlorine and caustic soda.
- **Deicing Roads** – Rock salt effectively melts ice, ensuring safer travel during winter months.
- **Textile Industry** – Sodium sulfate is used in dyeing fabrics.
- **Photography** – Silver nitrate is used in film development.
- **Construction** – Calcium carbonate is a component in cement and plaster.
- **Electronics** – Some salts are used in batteries and circuit boards.

Revision questions

1. (a) Define the following terms. Giving two examples in each case;
 - (i) Normal salt
 - (ii) Acid salt(b) Write an equation for the reaction between NaOH(aq) and dilute H_2SO_4
2. Normal salts can be prepared by neutralization
 - (a) What is neutralization?
 - (b) What are the products of neutralization?
 - (c) What are application of neutralization reaction in every day life
3. (a) What is solubility?
 - (b) In the table below tick the appropriate box.

Type of salt	Soluble	insoluble
Barium sulphate		
Copper sulphate		
Barium nitrate		
Potassium carbonate		
Sodium chloride		
Lead carbonate		

4. Give one use of each of salt s in the table below

Ammonium chloride	
Ammonium nitrate	
Copper (II) sulphate	
Lead nitrate	
Magnesium sulphate	
Sodium chloride	

ACIDS, BASES AND SALTS

Definition of an acid	What causes the acidic properties of acids?
An acid is a compound which when dissolved in water produces hydrogen ions as the only positively charged ions	The hydrogen ions (H ⁺) cause the acidic properties and these are formed in the presence of water.

Another term to refer to an acid	Why an acid is also called a proton donor?
An acid is called a <u>proton donor</u>	It's because an acid provides protons or hydrogen ions (H ⁺) to other substances during the reaction.

Substance to which an acid provides protons	A base is then called	Why the base is called a proton acceptor?	Equation for the reaction between acid and base
BASE	PROTON ACCEPTOR	Because the base accepts hydrogen ions from acids	$\text{H}^+_{(\text{aq})} + \text{OH}_{(\text{aq})} \longrightarrow \text{H}_2\text{O}_{(\text{l})}$

Common laboratory acids		These three common laboratory acids are also called	Why these acids are also called mineral acids?
Hydrochloric acid	HCl	MINERAL ACIDS	They are derived from mineral salts ie <i>chlorides</i> for HCl, <i>sulphates</i> for H ₂ SO ₄ and <i>nitrates</i> for HNO ₃
Sulphuric acid	H ₂ SO ₄		
Nitric acid	HNO ₃		

Other mineral acids known		Mineral salts from which the acid is derived	Organic acids known	Naturally occurring acids
Sulphurous acid	H ₂ SO ₃	Derived from sulphites	Ethanoic acid (CH ₃ COOH) Methanoic acid (HCOOH)	CITRIC ACID from lemons
Carbonic acid	H ₂ CO ₃	Derived from carbonates		TARTARIC ACID from grapes
Phosphoric acid	H ₃ PO ₄	Derived from phosphates		ACETIC ACID from vinegar
Nitrous acid	HNO ₃	Derived from nitrites		LACTIC ACID from sour milk
				Hydrochloric acid from digestive juices

Whenever an acid is dissolved in water, it produces	Term given to the number of hydrogen ions produced by one molecule of an acid	Definition of basicity of an acid
HYDROGEN IONS	BASICITY OF AN ACID	BASICITY of an acid is the <u>number of hydrogen ions</u> produced by <u>one molecule</u> of an acid <u>in aqueous solution</u> .

<u>Basicity can also be defined as</u>	<u>Categorization of acids depending on basicity</u>
BASICITY of an acid is the <u>number of hydrogen ions</u> produced by <u>one molecule</u> of an acid <u>when dissolved in water</u> .	Monobasic acids
	Dibasic acids
	Tribasic acids

<u>Definition of</u>	<u>Its basicity</u>	<u>Examples of acids</u>	<u>Ionization equations of acids</u>
<u>Monobasic acid</u> is an acid whose one molecule produces one hydrogen ion when dissolved in water. OR <u>Monobasic acid</u> is an acid whose one molecule produces one hydrogen ion when in aqueous solution.	Basicity of monobasic acids is ONE	Nitric acid	$\text{HNO}_{3(\text{aq})} \longrightarrow \text{H}^+_{(\text{aq})} + \text{NO}_{3(\text{aq})}^-$
		Hydrochloric acid	$\text{HCl}_{(\text{aq})} \longrightarrow \text{H}^+_{(\text{aq})} + \text{Cl}_{(\text{aq})}^-$
		Nitrous acid	$\text{HNO}_{2(\text{aq})} \longrightarrow \text{H}^+_{(\text{aq})} + \text{NO}_{2(\text{aq})}^-$
		Ethanoic acid	$\text{CH}_3\text{COOH}_{(\text{aq})} \longrightarrow \text{H}^+_{(\text{aq})} + \text{CH}_3\text{COO}_{(\text{aq})}^-$
		Hypochlorous acid	$\text{HOCl}_{(\text{aq})} \longrightarrow \text{H}^+_{(\text{aq})} + \text{OCl}_{(\text{aq})}^-$
		Methanoic acid	$\text{HCOOH}_{(\text{aq})} \longrightarrow \text{H}^+_{(\text{aq})} + \text{HCOO}_{(\text{aq})}^-$
		Have general formula of HX	
<u>Dibasic acid</u> is an acid whose one molecule produces two hydrogen ions when dissolved in water. OR <u>Dibasic acid</u> is an acid whose one molecule produces two hydrogen ions when in aqueous solution.	Basicity of Dibasic acids is TWO	Sulphuric acid	$\text{H}_2\text{SO}_{4(\text{aq})} \longrightarrow 2\text{H}^+_{(\text{aq})} + \text{SO}_{4(\text{aq})}^{2-}$
		Carbonic acid	$\text{H}_2\text{CO}_{3(\text{aq})} \rightleftharpoons 2\text{H}^+_{(\text{aq})} + \text{CO}_{3(\text{aq})}^{2-}$
		Sulphurous acid	$\text{H}_2\text{SO}_{3(\text{aq})} \rightleftharpoons 2\text{H}^+_{(\text{aq})} + \text{SO}_{3(\text{aq})}^{2-}$
		Have general formula of H₂X	The double half arrows ($\rightleftharpoons$) imply that the ionization of such an acid is reversible. Double half arrows ($\rightleftharpoons$) mainly apply to weak acids
<u>Tribasic acid</u> is an acid whose one molecule produces three hydrogen ions when dissolved in water. OR <u>Tribasic acid</u> is an acid whose one molecule produces three hydrogen ions when in aqueous solution.	Basicity of Tribasic acids is THREE	Phosphoric acid	$\text{H}_3\text{PO}_{4(\text{aq})} \rightleftharpoons 3\text{H}^+_{(\text{aq})} + \text{PO}_{4(\text{aq})}^{3-}$
		Have general formula of H₃X	<i>Important note on basicity of an acid</i> Basicity of an acid is not necessarily the number of hydrogen atoms contained in one molecule of the acid. Basicity refers to the number of hydrogen atoms capable of ionization in an acid for example; In CH_3COOH , its only one hydrogen atom that can ionize. The other three hydrogen atoms are incapable of ionization, THUS CH_3COOH has basicity of 1

TYPE OF ACIDS	DEFINITION 1	DEFINITION 2
Strong acids	A strong acid is an acid which when dissolved in water produces ALL the hydrogen ions it contains.	A strong acid is an acid which completely ionizes in dilute solution.
Weak acids	A weak acid is an acid which when dissolved in water produces PART of the hydrogen ions it contains	A weak acid is an acid which only slightly ionizes in dilute solution

Examples of strong acids	Ionization equation for the acid when dissolved in water
Hydrochloric acid	$\text{HCl}_{(\text{aq})} \longrightarrow \text{H}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})}$
Sulphuric acid	$\text{H}_2\text{SO}_{4(\text{aq})} \longrightarrow 2\text{H}^+_{(\text{aq})} + \text{SO}_4^{2-}_{(\text{aq})}$
Nitric acid	$\text{HNO}_{3(\text{aq})} \longrightarrow \text{H}^+_{(\text{aq})} + \text{NO}_3^{-}_{(\text{aq})}$

Examples of weak acids	Ionization equation for the acid when dissolved in water
carbonic acid	$\text{H}_2\text{CO}_{3(\text{aq})} \rightleftharpoons 2\text{H}^+_{(\text{aq})} + \text{CO}_3^{2-}_{(\text{aq})}$
phosphoric acid	$\text{H}_3\text{PO}_{4(\text{aq})} \rightleftharpoons 3\text{H}^+_{(\text{aq})} + \text{PO}_4^{3-}_{(\text{aq})}$
Ethanoic acid	$\text{CH}_3\text{COOH}_{(\text{aq})} \rightleftharpoons \text{H}^+_{(\text{aq})} + \text{CH}_3\text{COO}^-_{(\text{aq})}$
Methanoic acid	$\text{HCOOH}_{(\text{aq})} \rightleftharpoons \text{H}^+_{(\text{aq})} + \text{HCOO}^-_{(\text{aq})}$

PROPERTIES OF ACIDS																							
PHYSICAL PROPERTIES			CHEMICAL PROPERTIES																				
Physical properties of acids are those properties of acids that can be seen, felt and smelt.			Chemical properties of acids are those properties of acids when they are involved in chemical reactions.																				
1.	Have a sour and sharp taste		1.	Reaction with carbonates and hydrogencarbonates. <table><tr><td>Observation made</td><td colspan="3">Products of reaction</td></tr><tr><td>Effervescence of a colourless gas that turns lime- water milky.</td><td>SALT</td><td>WATER</td><td>Carbon dioxide</td></tr></table> <u>Examples illustrated by equations</u> $\text{Na}_2\text{CO}_{3(\text{s})} + 2\text{HCl}_{(\text{aq})} \longrightarrow 2\text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} + \text{CO}_{2(\text{g})}$ $\text{K}_2\text{CO}_{3(\text{s})} + 2\text{HCl}_{(\text{aq})} \longrightarrow 2\text{KCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} + \text{CO}_{2(\text{g})}$	Observation made	Products of reaction			Effervescence of a colourless gas that turns lime- water milky.	SALT	WATER	Carbon dioxide											
Observation made	Products of reaction																						
Effervescence of a colourless gas that turns lime- water milky.	SALT	WATER	Carbon dioxide																				
2.	Acids change colour of indicators. <u>Definition of an indicator</u> An indicator is a substance that has different colours in acidic and alkaline solutions. <table><tr><th>Indicator</th><th>Colour in acidic solution</th><th>Colour in alkaline solution</th></tr><tr><td>Phenolphthalein</td><td>Colourless</td><td>Pink</td></tr><tr><td>Methyl orange</td><td>Pink</td><td>Yellow</td></tr><tr><td>Red litmus</td><td>Red</td><td>Blue</td></tr><tr><td>Blue litmus</td><td>Red</td><td>Blue</td></tr></table>		Indicator	Colour in acidic solution	Colour in alkaline solution	Phenolphthalein	Colourless	Pink	Methyl orange	Pink	Yellow	Red litmus	Red	Blue	Blue litmus	Red	Blue	2.	Reaction with oxides and hydroxides <table><tr><td colspan="2">Products of the reaction</td></tr><tr><td>SALT</td><td>WATER</td></tr></table> <u>Examples illustrated by equations</u> $\text{CuO}_{(\text{s})} + \text{H}_2\text{SO}_{4(\text{aq})} \longrightarrow \text{CuSO}_{4(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$ $\text{NaOH}_{(\text{aq})} + \text{HCl}_{(\text{aq})} \longrightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$	Products of the reaction		SALT	WATER
Indicator	Colour in acidic solution	Colour in alkaline solution																					
Phenolphthalein	Colourless	Pink																					
Methyl orange	Pink	Yellow																					
Red litmus	Red	Blue																					
Blue litmus	Red	Blue																					
Products of the reaction																							
SALT	WATER																						
3.	Dilute acids are colourless solutions.		3.	Reaction with metals <table><tr><td>Observation made</td><td colspan="2">Products of reaction</td></tr><tr><td>Effervescence of a colourless gas that burns with a pop sound.</td><td>SALT</td><td>Hydrogen</td></tr></table> <u>Examples illustrated by equations</u> $\text{Mg}_{(\text{s})} + \text{H}_2\text{SO}_{4(\text{aq})} \longrightarrow \text{MgSO}_{4(\text{aq})} + \text{H}_{2(\text{g})}$	Observation made	Products of reaction		Effervescence of a colourless gas that burns with a pop sound.	SALT	Hydrogen													
Observation made	Products of reaction																						
Effervescence of a colourless gas that burns with a pop sound.	SALT	Hydrogen																					
4.	Concentrated acids are oily liquids eg concentrated sulphuric acid.																						

STATEMENT(S)		REASON(S)
When a piece of aluminium foil is placed in a test tube containing cold dilute hydrochloric acid, no reaction occurs.	BECAUSE	A thin protective layer forms on aluminium as soon as the metal is exposed to moist air, which prevents any reaction.
When a piece of aluminium foil is placed in a test tube containing cold dilute sulphuric acid, no reaction occurs.	BECAUSE	A thin protective layer forms on aluminium as soon as the metal is exposed to moist air, which prevents any reaction.
If a piece of aluminium foil is placed in a test tube containing warm acid, a reaction occurs after a short while.	BECAUSE	The oxide layer on aluminium dissolves in the warm acid exposing the metal which reacts with the acid.
Copper does not liberate hydrogen with dilute acids.	BECAUSE	Copper is below hydrogen in the electrochemical series, thus cannot displace it from dilute acids.
Nitric acid does not liberate hydrogen with nitric acid except magnesium.	BECAUSE	Nitric acid is a strong oxidizing agent. It oxidizes the hydrogen formed immediately into water.

NOTE;

Magnesium only reacts with nitric acid when it is very dilute.



BASES AND ALKALIS

Definition of a base	In general terms, bases are;	A reaction where an acid reacts with a base is called
A base is a substance which reacts with an acid to form a salt and water only.	Oxides of metals	NEUTRALIZATION REACTION
	Hydroxides of metals	
	Ammonium hydroxide	

Definition of neutralization;

Neutralization is a reaction in which an acid reacts with a base to form a salt and water only.

Many bases exist but only a few are soluble in water

Examples of bases that are soluble in water		These soluble bases are called	Alkalis are also called
Sodium hydroxide	NaOH	ALKALIS	SOLUBLE BASES
Potassium hydroxide	KOH		
Calcium hydroxide	Ca(OH) ₂		
Aqueous ammonia	NH ₄ OH	Alkalis are categorized into STRONG and WEAK alkalis	

Definition of alkalis;

Alkalis are substances which when dissolved in water produce hydroxide ions as the only negatively charged ions.

TYPES OF ALKALIS

STRONG ALKALIS	WEAK ALKALIS
These are electrovalent compounds that completely ionize in both aqueous solution and in solid state.	These are covalent compounds that partly ionize in aqueous solution and their ionization is reversible.
Examples of strong alkalis 1. Sodium hydroxide solution $\text{NaOH}_{(\text{aq})} \longrightarrow \text{Na}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})}$ 2. Potassium hydroxide solution $\text{KOH}_{(\text{aq})} \longrightarrow \text{K}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})}$ 3. Calcium hydroxide solution $\text{Ca(OH)}_{2(\text{aq})} \longrightarrow \text{Ca}^{2+}_{(\text{aq})} + 2\text{OH}^-_{(\text{aq})}$	Example of weak alkalis 1. Aqueous ammonia It is also called <u>ammonia solution</u> Aqueous ammonia is also called <u>ammonium hydroxide solution</u> $\text{NH}_{3(\text{g})} + \text{H}_2\text{O}_{(\text{l})} \rightleftharpoons \text{NH}_4^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})}$

PROPERTIES OF ALKALIS

Physical properties	Chemical properties
Have a bitter taste	React with acids to form a salt and water only $\text{NaOH}_{(\text{aq})} + \text{HCl}_{(\text{aq})} \longrightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$
Have a soapy feeling to touch	Alkalis precipitate insoluble metallic hydroxides from solutions of their salts. $2\text{NaOH}_{(\text{aq})} + \text{Pb(NO}_3)_2_{(\text{aq})} \longrightarrow \text{Pb(OH)}_{2(\text{s})} + 2\text{NaNO}_3_{(\text{aq})}$
Change colours of indicators	
Form colourless solutions	

COLOUR OF METAL HYDROXIDES

Metal hydroxide	Colour
Potassium hydroxide	White
Sodium hydroxide	White
Calcium hydroxide	White
Magnesium hydroxide	White
Zinc hydroxide	White
Aluminium hydroxide	White
Lead (ii) hydroxide	White
Copper (ii) hydroxide	Blue
Iron (ii) hydroxide	Green
Iron (iii) hydroxide	Brown

pH SCALE OF ACIDITY AND ALKALINITY

pH scale	pH is related to	pH number
Is a scale of numbers from 1 to 14, to express acidity and alkalinity	HYDROGEN ION concentration in solution	Is a measure of the hydrogen ion concentration

APPROXIMATE pH VALUES OF COMMON SOLUTIONS

pH 1	pH 2 – pH 6	pH 7	pH 8 – pH 13	pH 14
<i>Strong acid</i>	<i>Weak acid</i>	<i>Neutral</i>	<i>Weak alkali</i>	<i>Strong alkali</i>
Dilute sulphuric acid	Lemon juice (pH 2)	Sodium chloride	Baking powder (pH 9)	Sodium hydroxide
Dilute nitric acid	Sour milk (pH 5)	Pure water	Wood ash (pH 10)	Potassium hydroxide

NOTE 1	Acidic solutions have pH values less than seven. The smaller the pH value, the more acidic the solution is ie the larger the concentration of hydrogen ions.
NOTE 2	When distilled water is added to an acid, the pH value of the acid increases towards seven. The solution becomes less acidic.
NOTE 3	Water and other solutions have a pH of seven.
NOTE 4	Any solution of pH greater than seven is alkaline. The higher the pH value, the more alkaline the solution is ie the larger the concentration of hydroxyl or hydroxide ions.
NOTE 5	When distilled water is added to an alkaline solution, the pH value of the alkali decreases towards seven. The solution becomes less alkaline.

UNIVERSAL INDICATOR

<i>Definition</i>	<i>Forms in which universal indicator occurs</i>	<i>Uses of universal indicator</i>
Universal indicator is a mixture of indicators.	✓ In solution form ✓ In paper form	Determines whether the solution is acidic or alkaline. Used to determine the degree of acidity and alkalinity.

pH scale	1 -2	3	4	5	6 - 8	9 - 10	11 - 12	13 - 14
Colour	Red	Pink	Brown	Yellow	Green	Blue	Indigo	Violet

SIGNIFICANCE OF pH MEASUREMENTS

1. It helps to know that the final product in soap industry is neutral.
2. Too acidic soils are harmful in agriculture, and this can be determined by measuring the pH of the soil.
3. Various drugs are prepared at pHs which must be determined

SALTS

DEFINITION OF SALT	TYPES OF SALTS	
A salt is a substance formed when <u>all</u> or <u>part</u> of the replaceable hydrogen of an acid is replaced by a metal or metallic radical.	Acid salts	Normal salts

Definition of normal salt	Definition of acid salt
A normal salt is a salt formed when <u>all</u> the replaceable hydrogen of an acid is replaced by a metal or metallic radical.	An acid salt is a salt formed when <u>part</u> of the replaceable hydrogen of an acid is replaced by a metal or metallic radical.
Examples of normal salts	Examples of acid salts
✓ Sodium sulphate	✓ Calcium hydrogencarbonate
✓ Sodium carbonate	✓ sodium hydrogencarbonate
✓ Potassium nitrate	✓ Calcium hydrogensulphate
✓ Potassium sulphate	✓ Potassium hydrogencarbonate
✓ Calcium nitrate	✓ Calcium hydrogenphosphate
✓ Aluminium sulphate	✓ Magnesium hydrogencarbonate
Formation of a normal salt	Formation of an acid salt
Zinc granules reacting with dilute sulphuric acid. $\text{Zn}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{ZnSO}_{4(aq)} + \text{H}_{2(g)}$	Sodium chloride reacting with concentrated sulphuric acid. $\text{H}_2\text{SO}_{4(l)} + \text{NaCl}_{(s)} \longrightarrow \text{NaHSO}_{4(aq)} + \text{HCl}_{(g)}$

STATEMENT(S)		REASON(S)
Monobasic acids do not form acid salts	BECAUSE	Monobasic acids contain only one atom of replaceable hydrogen per acid molecule.
Sodium ethanoate, CH_3COONa is a normal salt.	BECAUSE	The hydrogen it contains does not form ions and cannot be replaced by a metal

TYPES OF SALTS AND ACIDS FROM WHICH THEY ARE FORMED		
ACID	TYPE OF SALT	EXAMPLE OF THE SALT
Sulphuric acid	Sulphates	Iron (ii) sulphate
Hydrochloric acid	Chlorides	Sodium chloride
Carbonic acid	Carbonates	Potassium carbonate
Nitric acid	Nitrates	Calcium nitrate
Sulphuric acid	hydrogencarbonates	Sodium hydrogencarbonate

CATEGORIZING SALTS DEPENDING ON THEIR SOLUBILITY IN WATER

SOLUBLE SALTS		INSOLUBLE SALTS	
Soluble salts are salts which completely dissolve in water to form a solution.		Insoluble salts are salts which do not dissolve completely in water and they form a residue when filtered.	
Soluble salts are usually prepared by methods which involve crystallization.		Insoluble salts are usually prepared by methods which involve precipitation (double decomposition)	
Crystallization is a process by which a solution is saturated by evaporating some of the solvent,		Precipitation is a process where an insoluble solid is formed when two aqueous solutions are mixed together.	
On cooling, the excess salt dissolved in the hot solution is deposited as crystals.		The insoluble salt is formed as a precipitate in an aqueous solution which is filtered off, washed and dried.	
A crystal is a solid that has solidified in a definite regular shape		A precipitate is an insoluble solid that separates from the solution.	
EXAMPLES OF SOLUBLE SALTS	COLOUR	EXAMPLES OF INSOLUBLE SALTS	COLOUR
Lead (ii) nitrate	White	Lead (ii) carbonate	White
Copper (ii) sulphate	Blue	Copper (ii) carbonate	Green
Copper (ii) nitrate	Blue	Lead (ii) sulphate	White
Iron (ii) sulphate	Green	Iron (ii) carbonate	Green
Iron (ii) chloride	Green	Barium sulphate	White
Iron (ii) nitrate	Green	Calcium sulphate (slightly soluble in water)	White
Iron (iii) sulphate	Brown/ yellow	Silver chloride	White
Iron (iii) chloride	Brown/ yellow	Magnesium carbonate	White
Iron (iii) nitrate	Brown/ yellow	Zinc carbonate	White
<u>Formation of zinc sulphate</u> $\text{Zn}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{ZnSO}_{4(aq)} + \text{H}_{2(g)}$		<u>Formation of zinc carbonate by precipitation</u> $\text{Na}_2\text{CO}_{3(aq)} + \text{ZnCl}_{2(aq)} \longrightarrow \text{NaCl}_{(aq)} + \text{ZnCO}_{3(s)}$	
<u>Formation of zinc sulphate</u> $\text{CuO}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{CuSO}_{4(aq)} + \text{H}_2\text{O}_{(l)}$		<u>Formation of copper (ii) carbonate by precipitation</u> $\text{Na}_2\text{CO}_{3(aq)} + \text{CuSO}_{4(aq)} \longrightarrow \text{CuCO}_{3(s)} + \text{Na}_2\text{SO}_{4(aq)}$	

SOLUBLE SALTS	INSOLUBLE SALTS
ALL Sodium salts	NO Nitrate salt is insoluble
ALL potassium salts	
ALL ammonium salts	
ALL NITRATES form soluble salts	
ALL sulphates	Lead (ii) sulphate
Except	Barium sulphate
	Calcium sulphate
ALL chlorides	Lead (ii) chloride (soluble in hot water)
Except	Silver chloride
	Mercury (i) chloride
Sodium carbonate	Except ALL CARBONATES are insoluble
Potassium carbonate	
Ammonium carbonate	

PREPARATION OF SOLUBLE SALTS BY

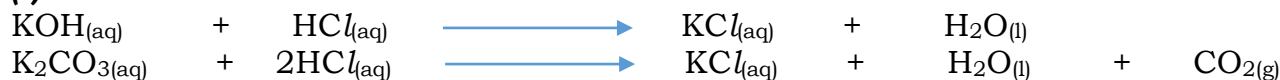
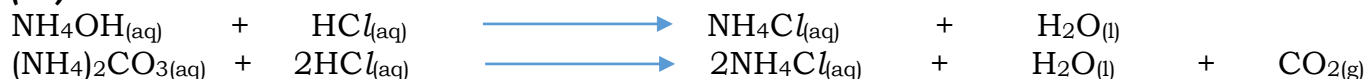
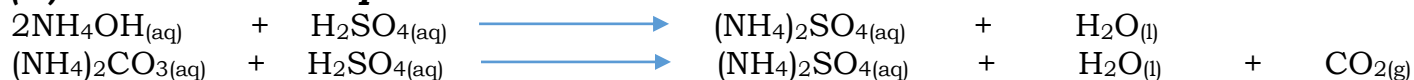
1. ACTION OF AN ACID ON A METAL		
General equation	Metal + Acid	Salt + Hydrogen
This method is not suitable for preparation of salts of highly reactive metals for example; • Potassium • Sodium • Calcium	BECAUSE	The metals of potassium, sodium and calcium react explosively with dilute acids
This method is only used to prepare salts of less reactive metals such as • Aluminium • Zinc • Magnesium • Iron		
Magnesium sulphate can be prepared by using magnesium and dilute sulphuric acid $\text{Mg}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{MgSO}_{4(aq)} + \text{H}_{2(g)}$		
Iron (ii) sulphate can be prepared by using iron filings and dilute sulphuric acid $\text{Fe}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{FeSO}_{4(aq)} + \text{H}_{2(g)}$		
EXPERIMENT: <u>Preparation of zinc sulphate crystals</u> ✓ Dilute sulphuric acid is poured in a beaker and granulated zinc is added. ✓ Effervescence occurs ✓ If the reaction is slow, a little copper (ii) sulphate solution is added as a catalyst and the reactants are warmed gently. $\text{Zn}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{ZnSO}_{4(aq)} + \text{H}_{2(g)}$ ✓ When the reaction stops, more zinc is added to make sure that the acid is not left in considerable amounts. ✓ Excess zinc granules are filtered off. ✓ The filtrate is gently heated in an evaporating dish to boil off some water until crystals begin to form, when the filtrate cools, on a glass rod, which is dipped into the filtrate at regular intervals. ✓ The crystals are filtered off and then pressed gently between filter papers to dry.		

PREPARATION OF SOLUBLE SALTS BY

2. ACTION OF AN ACID ON SOLUBLE HYDROXIDE OR CARBONATE

This method is used to prepare salts of potassium, sodium and ammonium

PREPARATION OF;

(i) Potassium chloride**(ii) Sodium nitrate****(iii) Ammonium chloride****(iv) Ammonium sulphate****EXPERIMENT: Preparation of sodium sulphate crystals**

- ✓ A known volume of sodium hydroxide solution is pipetted into a conical flask and 2 drops of phenolphthalein added.
- ✓ Dilute sulphuric acid is added from the burette to conical flask at intervals until the colour of the indicator changes to pink.



- ✓ Having noted the volume of the acid used, the solution is poured away as the indicator would colour the salt obtained from it.
- ✓ The whole process is repeated using the same volume of the solution of sulphuric acid and sodium hydroxide solution without adding the indicator.
- ✓ The solution is evaporated until it forms crystals when it cools, on a clean glass rod, which is dipped into the solution at regular intervals.
- ✓ The crystals are filtered off and then pressed gently between filter papers to dry.

PREPARATION OF SOLUBLE SALTS BY

3. ACTION OF AN ACID ON INSOLUBLE OXIDES OR HYDROXIDES

This method is used to prepare Magnesium sulphate, zinc sulphate and lead (ii) nitrate

PREPARATION OF;

(i) Magnesium sulphate**(ii) Zinc sulphate****(iii) Lead (ii) nitrate****EXPERIMENT: Preparation of copper (ii) sulphate crystals**

- ✓ Copper (ii) oxide is added to a beaker of warm dilute sulphuric acid and the mixture stirred gently.
 - ✓ More of the oxide is added, little at a time until no more reacts, showing that all the acid has been neutralized.
- $$\text{CuO}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{CuSO}_{4(aq)} + \text{H}_2\text{O}_{(l)}$$
- ✓ Excess copper (ii) oxide is filtered off and the filtrate evaporated until crystals begin to form when it cools, on a clean glass rod, which is dipped into the filtrate at regular intervals
 - ✓ The crystals are filtered off and then pressed gently between filter papers to dry.

PREPARATION OF SOLUBLE SALTS BY

4. ACTION OF AN ACID ON SOLUBLE INSOLUBLE CARBONATES

The salts of copper (ii) sulphate, copper (ii) nitrate, magnesium sulphate, zinc sulphate, calcium chloride and calcium nitrate are prepared by this method.

Calcium chloride and calcium nitrate are deliquescent and do not form crystals. Their solutions must be evaporated to dryness

PREPARATION OF;	(i) Copper (ii) sulphate $\text{CuCO}_{3(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{CuSO}_{4(aq)} + \text{H}_2\text{O}_{(l)} + \text{CO}_{2(g)}$
	(ii) Copper (ii) nitrate $\text{CuCO}_{3(s)} + 2\text{HNO}_{3(aq)} \longrightarrow \text{Cu}(\text{NO}_3)_{2(aq)} + \text{H}_2\text{O}_{(l)} + \text{CO}_{2(g)}$
	(iii) Magnesium sulphate $\text{MgCO}_{3(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{MgSO}_{4(aq)} + \text{H}_2\text{O}_{(l)} + \text{CO}_{2(g)}$
	(iv) Zinc sulphate $\text{ZnCO}_{3(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{ZnSO}_{4(aq)} + \text{H}_2\text{O}_{(l)} + \text{CO}_{2(g)}$
	(v) Calcium (ii) nitrate $\text{CaCO}_{3(s)} + 2\text{HNO}_{3(aq)} \longrightarrow \text{Ca}(\text{NO}_3)_{2(aq)} + \text{H}_2\text{O}_{(l)} + \text{CO}_{2(g)}$
	(vi) Calcium chloride $\text{CaCO}_{3(s)} + 2\text{HCl}_{(aq)} \longrightarrow \text{CaCl}_{2(aq)} + \text{H}_2\text{O}_{(l)} + \text{CO}_{2(g)}$

EXPERIMENT: Preparation of lead (ii) nitrate crystals

- ✓ Lead (ii) carbonate is added little at a time to dilute nitric acid in a beaker.
- ✓ Effervescence occurs as carbon dioxide is evolved.
- ✓ More carbonate is added until no more reacts, showing that all the acid has reacted.



- ✓ The excess carbonate is filtered off and the filtrate evaporated until crystals begin to form when it cools, on a clean glass rod, which is dipped into the filtrate at regular intervals.
- ✓ The crystals are filtered off and then pressed gently between filter papers to dry.

5. PREPARATION OF SALTS BY DIRECT SYNTHESIS

Salts to which this method applies	APPLIES to both <i>soluble</i> and <i>insoluble</i> salts
Another name for this method	Direct synthesis
Salts prepared by direct synthesis	Used to prepare binary salts, for example; ✓ Chlorides eg anhydrous iron (iii) chloride ✓ Bromides eg aluminium bromide ✓ Sulphides eg iron (ii) sulphide
Definition of direct synthesis	<i>Direct synthesis is the method of preparing soluble and insoluble salts directly from their elements.</i>

PREPARATION OF ANHYDROUS IRON (III) CHLORIDE

Conditions for the reaction	▪ Dry chlorine gas ▪ Heating is required
Equation for the reaction	$2\text{Fe}_{(s)} + 3\text{Cl}_{2(g)} \longrightarrow 2\text{FeCl}_{3(s)}$
Colour of iron (iii) chloride	BROWN

PREPARATION OF ALUMINIUM CHLORIDE

Conditions for the reaction	▪ Dry chlorine gas ▪ Heating is required
Equation for the reaction	$2\text{Al}_{(s)} + 3\text{Cl}_{2(g)} \longrightarrow 2\text{AlCl}_{3(s)}$
Colour of aluminium chloride	WHITE

PREPARATION OF IRON (II) SULPHIDE

Conditions for the reaction	▪ Heating is required
Observation made	The mixture glows when heated forming a black solid.
Equation for the reaction	$\text{Fe}_{(s)} + \text{S}_{(g)} \longrightarrow \text{FeS}_{(s)}$
Colour of iron (ii) sulphide	BLACK

6. PREPARATION OF INSOLUBLE SALTS BY PRECIPITATION

Method also called	Double decomposition reaction
What is involved in this method?	In this method, two soluble salts are mixed together to give a mixture of a soluble salt and an insoluble salt (precipitate)
PREPARATION OF	<i>(i) Barium sulphate</i> $\text{BaCl}_{2(\text{aq})} + \text{Na}_2\text{SO}_{4(\text{aq})} \longrightarrow \text{BaSO}_{4(\text{s})} + 2\text{NaCl}_{(\text{aq})}$
	<i>(ii) Lead (ii) chloride</i> (It is soluble in hot water) $\text{Pb}(\text{NO}_3)_{2(\text{aq})} + \text{NaCl}_{(\text{aq})} \longrightarrow \text{PbCl}_{2(\text{s})} + 2\text{NaNO}_{3(\text{aq})}$
	<i>(iii) Calcium carbonate</i> $\text{CaCl}_{2(\text{aq})} + \text{Na}_2\text{CO}_{3(\text{aq})} \longrightarrow \text{CaCO}_{3(\text{s})} + 2\text{NaCl}_{(\text{aq})}$
	<i>(iv) Lead (ii) bromide</i> $\text{Pb}(\text{NO}_3)_{2(\text{aq})} + 2\text{NaBr}_{(\text{aq})} \longrightarrow \text{PbBr}_{2(\text{s})} + 2\text{NaNO}_{3(\text{aq})}$
	<i>(v) Calcium sulphate</i> $\text{Ca}(\text{NO}_3)_{2(\text{aq})} + \text{Na}_2\text{SO}_{4(\text{aq})} \longrightarrow \text{CaSO}_{4(\text{s})} + 2\text{NaNO}_{3(\text{aq})}$
EXPERIMENT: Preparation of lead (ii) sulphate crystals ✓ Dilute sulphuric acid is added to warm lead (ii) nitrate solution in a beaker and the mixture is stirred. ✓ The white precipitate formed is heated to enable rapid filtration. ✓ After filtration, the precipitate is washed several times with hot distilled water to remove soluble impurities. ✓ The precipitate is allowed to dry on a filter paper. $\text{Pb}(\text{NO}_3)_{2(\text{aq})} + \text{H}_2\text{SO}_{4(\text{aq})} \longrightarrow \text{PbSO}_{4(\text{s})} + 2\text{HNO}_{3(\text{aq})}$	
NOTE; SODIUM SULPHATE solution may be used instead of SULPHURIC ACID $\text{Pb}(\text{NO}_3)_{2(\text{aq})} + \text{Na}_2\text{SO}_{4(\text{aq})} \longrightarrow \text{PbSO}_{4(\text{s})} + 2\text{NaNO}_{3(\text{aq})}$	

HYDROLYSIS OF SALTS

<i>Why does a solution of potassium carbonate show basic characteristics?</i>	This is because when in aqueous solution, potassium carbonate hydrolyses in water to form a mixture of a strong alkali (KOH) and a weak acid (H ₂ CO ₃). The resultant solution is alkaline because the concentration of hydroxyl ions from the strong alkali is greater than the concentration of hydrogen ions from the weak acid. The strong alkali completely ionizes in solution and the weak acid under goes incomplete ionization.
<i>Equation for the reaction</i>	$\text{K}_2\text{CO}_{3(s)} + 2\text{H}_2\text{O}_{(l)} \longrightarrow 2\text{KOH}_{(aq)} + \text{H}_2\text{CO}_{3(aq)}$

<i>Why does a solution of ammonium chloride show acidic characteristics?</i>	This is because when in aqueous solution, ammonium chloride hydrolyses in water to form a mixture of a strong acid (HCl) and a weak alkali (NH ₄ OH). The resultant solution is acidic because the concentration of hydrogen ions from the strong acid is greater than the concentration of hydroxyl ions from the weak alkali. The strong acid completely ionizes in solution and the weak alkali under goes incomplete ionization.
<i>Equation for the reaction</i>	$\text{NH}_4\text{Cl}_{(s)} + 2\text{H}_2\text{O}_{(l)} \longrightarrow \text{NH}_4\text{OH}_{(aq)} + \text{HCl}_{(aq)}$

REACTIONS FOR IONIC SALTS

When an ionic salt dissolves in water, its ions separate into free ions. For example; when zinc sulphate is dissolved in water, zinc ions (Zn²⁺) and sulphate ions (SO₄²⁺)

IONIC EQUATIONS

Ionic equations describe chemical changes by showing only the reacting ions.

Three steps are followed when writing ionic equations;

STEP 1	Write the formal equation.
STEP 2	Write down all the ions in the equation.
STEP 3	The ionic equation is written by omitting the identical ions which appear on both sides of the equation.

Important notes to take;

NOTE 1	Gases do not ionize.
NOTE 2	Solids do not ionize. (Precipitates do not ionize)
NOTE 3	Water does not ionize.
NOTE 4	Only aqueous solutions ionize.

EXERCISE

Write ionic equations for the following reactions

- i) *Copper (ii) sulphate solution is added to sodium carbonate solution*
- ii) *Sodium hydroxide solution is added to lead (ii) nitrate solution*
- iii) *Zinc powder is added to copper (ii) sulphate solution*
- iv) *Chlorine gas is passed through a solution of iron (ii) chloride*
- v) *Dilute hydrochloric acid is added to solid calcium carbonate*

ATOMIC STRUCTURE

An atom is the smallest particle of an element that can take part in chemical reactions.

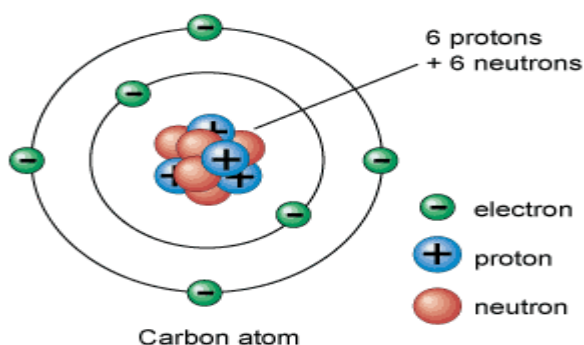
A typical atom is composed of protons, neutrons and electrons.

Protons and neutrons together occupy the core of the atom called the nucleus and they are collectively called nucleons.

Electrons on the other hand, occupy proximal space but in orbits called shells or energy levels.

Only a hydrogen atom lacks neutrons.

THE STRUCTURE OF CARBON-12 ISOTOPE



A proton has a relative mass of 1 and a charge of +1

An electron has a relative mass of approximately zero and a charge of -1.

A neutron has a relative mass of nearly 1 (normally considered to be 1) and a charge of zero making it neutral. That's why it is called a neutron.

SUB ATOMIC PARTICLE	RELATIVE MASS	CHARGE
Proton(P)	1	+1
Neutron(n)	1	0
Electron (e)	0	-1

This fact makes the mass of an atom be determined entirely by the nucleus since only the nucleons (protons and neutrons have mass).

In any atom, the number of protons is equal to the number of electrons, making the atom electrically neutral.

ATOMIC NUMBER (Z)

This refers to the number of protons in the nucleus of an atom. It is denoted by Z.

MASS NUMBER (A)

This is the number of protons and neutrons in the nucleus of an atom. It is denoted by A. Mass number is sometimes called **Atomic Mass**.

An atom represented by a symbol X can be fully represented as A_ZX . eg ${}^{40}_{20}\text{Ca}$, ${}^{12}_6\text{C}$, ${}^1_1\text{H}$.

The number of neutrons can be calculated if atomic number and mass number of an element are known i.e. Mass number $A = \text{atomic number}(\text{no. of protons}) + \text{number of neutrons}$

$$\text{i.e. } A = Z + n$$

$$\text{Hence } n = A - Z$$

EXAMPLES

ATOM	ATOMIC NUMBER	NUMBER OF PROTONS	NUMBER OF ELECTRONS	NUMBER OF NEUTRONS	MASS NO
Oxygen, $^{16}_8\text{O}$,	8	8	8	$16 - 8 = 8$	16
Magnesium, $^{24}_{12}\text{Mg}$	12	12	12	$24 - 12 = 12$	24
Sulphur, $^{32}_{16}\text{S}$					
Chlorine, $^{35}_{17}\text{Cl}$	17	17	17	$35 - 17 = 18$	35
Sodium, $^{23}_{11}\text{Na}$					

Complete the rest of the table above.

ELECTRONIC CONFIGURATION

This is the arrangement of electrons in energy levels round the nucleus. Electronic configuration is also called electron structure of an atom.

GUIDELINES TO WRITING ELECTRONIC CONFIGURATION

- The innermost energy level accommodates a maximum of two (2) electrons, while the rest of the energy levels accommodate a maximum of eight (8) electrons.
- Before electrons occupy outer energy levels, inner energy levels must first get filled to their capacity.
- Energy levels are separated using a comma(,) or the symbol)

EXAMPLES.

Write the electronic configuration of the following:

- a) Hydrogen (atomic number 1)

Ans. 1, since the hydrogen atom has 1 electron.

- b) Sodium (Atomic number 11)

Ans: 2,8,1 or 2)8)1. I encourage you to use the former.

- c) Calcium (Atomic number 20)

Ans: 2,8,8,2

d) Chlorine (Atomic number 17)

Ans: 2,8,7

Activity: Draw structures to show electron arrangement for these atoms above.

ION FORMATION

An ion is any particle that carries a charge.

The charge of an ion may be negative in which case this ion has more electrons than protons, and it is called an **ANION**.

The charge may be positive implying that the ion has more protons than electrons, and it is called a **CATION**.

All metals form cations by losing all the electrons in the outermost energy level thus attaining a stable electron structure of a noble/inert gas.

All nonmetal atoms form anions by gaining electrons to fill the outermost energy level to its capacity, attaining an electron structure of a noble/inert gas. Noble gases are inert (unreactive) because they have fully filled energy levels.

Atoms of metals lose electrons which in number are equal to their valency while atoms of non-metals gain electrons which in number are equal to their valencies.

ATOM	No of protons in the atom	No of electrons in the atom	Electronic configuration of the atom	No. of Protons in the ion	No of electrons in the ion	Electronic configuration of the ion	Formula of the ion
$^{16}_8\text{O}$	8	8	2,6	8	10(since 2 are gained)	2, 8	O^{2-}
$^{23}_{11}\text{Na}$	11	11	2,8,1	11	10(since 1 is lost)	2,8	Na^+
$^{14}_7\text{N}$	7	7	2,5	7	10(since 3 are gained)	2,8	N^{3-}
$^{27}_{13}\text{Al}$	13	13	2,8,3	13	10(since 3 are lost)	2,8	Al^{3+}
$^{39}_{19}\text{K}$	19	19	2,8,8,1	19	18(since 1 is lost)	2,8,8	K^+

ISOTOPY

Isotopy is the occurrence/existence of two or more atoms of the same element having the same atomic number but with different mass numbers resulting from having different numbers of neutrons.

Isotopes therefore, are different atoms of the same element which have the same atomic number but with different mass numbers.

EXAMPLES OF ISOTOPES

ELEMENT	ISOTOPES
Hydrogen	${}^1_1\text{H}$ (Hydrogen), ${}^2_1\text{H}$ or ${}^2_1\text{D}$ (<i>Deuterium</i>), and ${}^3_1\text{H}$ or ${}^3_1\text{T}$ (<i>Tritium</i>). Hydrogen-1 is the most abundant.
Oxygen	${}^{16}_8\text{O}$, ${}^{17}_8\text{O}$ and ${}^{18}_8\text{O}$, Oxygen-16 is the most abundant
Carbon	${}^{12}_6\text{C}$, ${}^{13}_6\text{C}$, ${}^{14}_6\text{C}$. <i>${}^{14}_6\text{C}$ is the radioactive isotope of carbon while ${}^{12}_6\text{C}$ is the most abundant isotope</i>
Chlorine	${}^{35}_{17}\text{Cl}$, ${}^{36}_{17}\text{Cl}$, ${}^{37}_{17}\text{Cl}$. Chlorine-35 isotope is the most abundant

Activity: In the remaining space in the table, list other isotopic elements and their isotopes.

THE PERIODIC TABLE

The Periodic Table is the arrangement of elements in order of increasing atomic numbers. The Periodic Table is made up of rows and columns. The horizontal are called **periods** and are assigned numbers according to the number of electron shells/energy levels that are occupied by electrons. All elements in the same period have the same number of electron shells /energy levels. The vertical columns are referred to as **groups** and the group number is written in **Roman numeral** on top of the group (column).

The transition metals (a strip between groups II and III) belong to no group.

Kindly correct the error in the Periodic Table below as follows:

- Remove the numbers 3,4,5,...,12 on top of the columns of transition metals ie from the

- column containing Scandium, Sc to the column containing zinc, Zn.
- ii. Change the numbers in Arabic numerals i.e 1,2,13,14,15,16,17,18 on top of the columns to Roman numerals I, II, III, IV,V,VI,VII and VIII respectively. These Roman numerals represent the correct groups of the elements in these columns.
 - iii. **Note:** Elements in a particular group resemble each other chemically because they have the **same number of electrons** in the outer most shell/energy level.

Periodic table of the elements

EXAMPLES

Group I elements	Atomic Number	Electronic configuration
Lithium	3	2,1
Sodium	11	2,8,1
Potassium	19	2,8,8,1

These metals are **very reactive** and are kept under oil in the laboratory where they have no contact with water and air. They include **lithium (Li)**, **sodium (Na)** and **potassium (K)**. Atoms of all these elements each have a single electron on the outer most shell, which is easily lost during chemical reactions leaving a single positively charged ion.

All elements in the **same period** have the same number of energy levels/electron shells. The periods are written in **roman numbers** as 1, 2, 3, etc.

Also from one element to the next, across the period, the number of electrons in the outer most energy level increases by one.

Period 3 elements	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Electronic Configuration	2,8,1	2,8,2	2,8,3	2,8,4	2,8,5	2,8,6	2,8,7	2,8,8

The elements in the periodic table are classified as metals, metalloid, non-metals and noble/inert/rare gases.

- i) **Metals:** these are on the left of the periodic table in group I, II and III. Metals react by loss of electrons.
- ii) **Semi metals (metalloids):** These are found in the middle of the periodic table in group IV.
- iii) **Non-metals:** they are on the right of the periodic table in groups V, VI and VII. Non-metals react by gain of electrons.
- iv) **Noble/rare or inert gases:** They are at the extreme right of the periodic table in group VIII (O). Non-metals are not reactive because they are very stable.

Transition elements: these are found between group II and III and towards the bottom of the periodic table. Transition elements form colored compounds and have variable valencies. Examples of transition elements include copper, iron, chromium, manganese, lead, cobalt etc.

General Trends in the periodic table

1. Elements in the same group react in similar ways because they have the same number of electrons in their outer most energy level/ shell.
2. Reactivity increases as you go **down the group** of metals because of **increase** in **atomic size (atomic radius)**. As the size of atoms increase, the outer most electrons become loosely held because they are no longer strongly attracted by the nucleus hence such electrons are easily lost. Because metals react by loss of electrons, the more easily a metal loses electrons, the more reactive that metal is.
3. As you go down the group of non-metals, reactivity decreases due to increase in **atomic size (atomic radius)**. As the atomic sizes increase, the forces of attraction of the nucleus for the electrons decrease hence electrons cannot be gained readily. So, the more easily a non-metal gains electrons, the more reactive that non-metal is.
4. **Across the period** of elements from **left to right**, reactivity **decreases** from group I to group IV and **then increases** up to group VII and falls sharply in group VIII. This is because, as you go across the period, the effective force of the nucleus increases hence the outer most electrons become more strongly attracted.

TRENDS

ATOMIC RADIUS

The atomic radius of an element is the measure of the size of an atom. It is defined as the distance between the center of the nucleus of an atom and its outermost shell.

Variation in atomic radius for elements in the same period

Consider **period 2** elements; **Lithium** has the **least nuclear charge**. Thus its outermost electrons experience the **least nuclear attraction** which results in the **greatest** distance between its nucleus and outermost electrons.

Oxygen has the **highest nuclear charge**, so its outermost electrons experience the **greatest** nuclear attraction. This results in the smallest distance between its nucleus and outermost electrons.

Conclusion

Therefore generally, atomic radius decreases across the period from left to right.

Variation in atomic radius down the group

Consider group I elements, Lithium has the smallest atomic radius because its outer most shell are the least shielded from nuclear attraction.

Potassium has the largest atomic radius because its outermost electrons are the most shielded from the nucleus.

In addition, down the group, an extra shell of electrons is added and these extra electrons increase the repulsive force on outer electrons causing them to be pushed outwards. This causes an increase in atomic size.

Variation in metallic character

Metallic character depends on how readily atoms of elements lose their electrons to form cations **(positively charged ions)**.

Increase in atomic size increases the tendency of metals to lose electrons, hence increasing the metallic character. This is because when the atomic radius or size is bigger, outermost electrons are farther away from the nucleus and therefore the nuclear attraction on these outermost electrons is lower, making it easier to remove these outermost electrons, vice versa.

Decrease in atomic size decreases the tendency to lose electrons, hence decreasing the metallic character. Metallic character is the tendency of metals (or elements) to lose outermost electrons.

Activity: Complete the table below by stating whether the properties indicated increase or decrease.

Property	Trend down the group	Trend across the period from left to right
Atomic size		
Tendency to lose electrons		
Metallic character		

Variation in melting and boiling points of period 3 elements

Some substances require more heat than others to change their physical state (melting). **Why is this so?**

Melting points and boiling points depend on the strengths of forces which exist between the particles which make up a substance.

Trend

Melting and boiling points increase among metals across the period and generally decrease among

non-metals from left to right.

Consider the table below;

Period 3 elements	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Melting point ($^{\circ}\text{C}$)	97.8	650	660.3	1410	44.1	112.8	-101.5	-189.4
Boiling point ($^{\circ}\text{C}$)	882.8	1091	2470	2355	280.5	444.6	-34.04	-185.8

Question: On the same axes, draw a graph of melting and boiling point for the elements from Na to Ar against atomic number

Factors which contribute to the above trend in the melting and boiling point

- Number of electrons in the outermost shell (for metals)
- Atomic radius
- Molecular size for non-metals

Explanation

The melting points and boiling points **increase across the three metals** (Sodium, Magnesium, and Aluminium) because of **increasing strength** of the **metallic bonds**.

Silicon adopts a 3-dimensional structure (**giant covalent structure**) with strong interlinking covalent bonds that give it extra stability. These require high energy to break hence the exceptionally high melting and boiling points.

Phosphorus, sulphur and chlorine exist as simple molecules held by **Vander Waal's forces of attraction**, whose strength increases with increase in molecular size. Therefore their melting points are low because their bonds require little energy for melting and boiling to take place.

Density

All group I elements have very low densities. Density **increases** as we go down the group because the **increase in atomic mass** is **more than** the **increase in atomic size**. Potassium is an exception to this trend as it has a lower density than sodium.

The density of period 3 elements increases among metals and generally decreases among non-metals.

Increase in densities from sodium to Aluminium is due to decrease in atomic radius, which favours formation of stronger metallic bonds, hence closer packing of the atoms.

Summary of the physical properties of Group I Elements

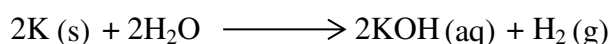
1. The metals are soft and can be cut with a knife. The softness decreases down the group.
(Hardness)
2. Their **melting and boiling points** are low. The melting point decreases down the group.
3. They have **low densities** and can float on water.
4. They have shiny surfaces when freshly cut. The shiny surfaces soon tarnish due to reaction with oxygen to form the oxides.
5. They are good conductors of electricity and heat.

Chemical properties of Group I Metals

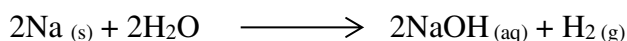
All group I elements have one electron in the outer most shell therefore undergo similar chemical reactions.

1. Reaction with water

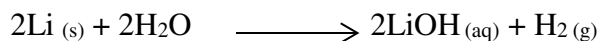
Potassium reacts very vigorously, darts on water and burns with a bright purple (lilac) flame producing fumes of hydrogen gas and an alkaline solution of potassium hydroxide.



Sodium melts into a silvery ball, darts on water reacting vigorously with it producing fumes of hydrogen gas with a hissing sound and an alkaline solution of sodium hydroxide. The reducing in size as the reaction proceeds until when it finally disappears.

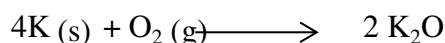


Lithium reacts slowly with cold water liberating hydrogen gas.



2. Reaction with air (oxygen)

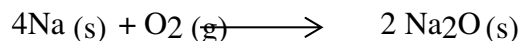
Potassium burns with a bright purple flame producing white solid of potassium oxide.



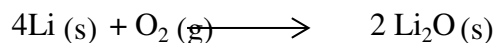
Sodium burns in air with a bright yellow flame forming yellow solids of sodiumperoxide in plenty of air.



In limited in air, sodium forms white solids of sodium oxide.

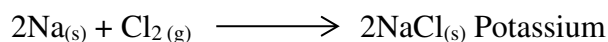


Lithium burns with a red flame forming white residue of lithium oxide.

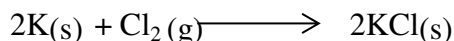


3. Reaction with chlorine

The alkaline metals react with chlorine to form metal chloride salts e.g. Sodium continues to burn in chlorine to form white fumes of sodium chloride.



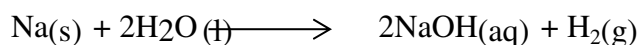
burns in chlorine to form fumes of potassium chloride.



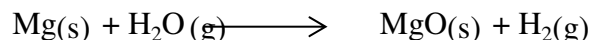
Chemical properties of period 3 elements

1. Reaction with water

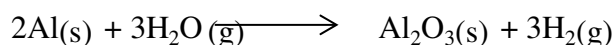
Sodium vigorously reacts with cold water to produce sodium hydroxide solution and hydrogen gas.



Magnesium has a very slight reaction with cold water but burns in steam with a white flame and produces magnesium oxide and hydrogen gas



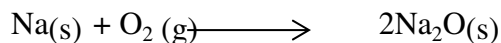
Aluminium also reacts with steam to produce Aluminium oxide and hydrogen gas.



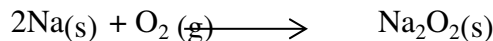
1. Reaction with oxygen

Note: All elements in Period 3 except chlorine and argon combine directly with oxygen to form oxides.

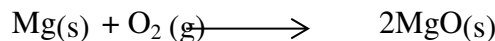
Sodium burns in oxygen with an orange flame to produce sodium oxide.



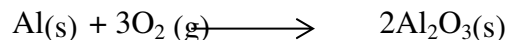
In excess oxygen, sodium burns to produce sodium peroxide.



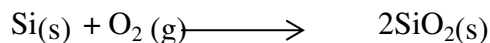
Magnesium burns in oxygen with an intense white flame to form a white solid of magnesium oxide.



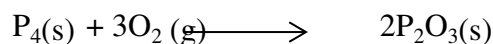
Powdered **aluminium** burns when sprinkled in a bunsen flame to produce white solid of aluminium oxide.



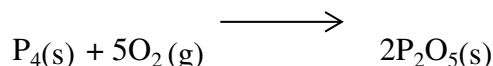
Silicon burns in oxygen when heated strongly to form silicon dioxide or silicon(IV) oxide



Phosphorus burns in oxygen with a white flame to produce phosphorus trioxide (phosphorus (III) oxide)



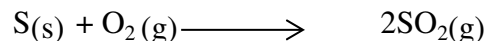
In excess oxygen, **phosphorus** burns to produce phosphorus pentoxide (phosphorus (V) oxide)



NOTE:

- Phosphorus occurs as a tetratomic molecule, P_4
- Oxides of phosphorus normally occur as dimers ie P_4O_6 for P_2O_3 and P_4O_{10} for P_2O_5 . These dimers are still named as phosphorus trioxide and phosphorus pentoxide respectively.

Sulphur burns in oxygen on gentle heating with a blue flame to form sulphur dioxide gas



Chlorine does not react directly with oxygen. **Argon** does not either.

Note: Sodium and magnesium form **basic oxides**, aluminium forms an **amphoteric oxide** whereas silicon, phosphorus and sulphur form acidic oxides.

THEME: CARBON IN THE ENVIRONMENT**TOPIC 7: CARBON IN THE ENVIRONMENT****SENIOR: 2 TERM: 2 PERIODS: 36****PAGE: 28-29**

COMPETENCY: The learner investigates the diversity of carbon compounds in the environment.

LEARNING OUTCOMES AND**SUGGESTED LEARNING ACTIVITIES**

From learning outcome A-J: The learner should be able to:

A Understand how and why carbon compounds are used as fuels (k,u)

• Draw on your experience and research to discuss the main uses of common carbon-based fuels in Uganda: charcoal, paraffin, petrol, firewood and diesel.

• Use prior learning to write a word equation to show the reaction when carbon-based fuels burn in oxygen

HOW CARBON-BASED FUELS ARE USED:**1. Firewood:**

- Burned directly in fireplaces, stoves, or furnaces to produce heat for warmth, cooking, or water heating.

- Combustion process: Firewood + Oxygen → Heat + Carbon Dioxide + Water Vapor

2. Charcoal:

- Burned in stoves, grills, or furnaces to produce heat for cooking, warmth, or industrial processes.

- Combustion process: Charcoal + Oxygen → Heat + Carbon Dioxide

3. Paraffin (Kerosene):

- Burned in lamps, heaters, or stoves to produce light and heat.

- Combustion process: Paraffin + Oxygen → Heat + Light + Carbon Dioxide + Water Vapor

- Also used as a fuel for jet engines and rockets.

4. Petrol (Gasoline):

- Burned in internal combustion engines to power vehicles, generators, and machinery.

- Combustion process: Petrol + Oxygen → Energy (mechanical or electrical) + Carbon Dioxide + Water Vapor

5. Diesel:

- Burned in internal combustion engines to power vehicles, generators, and machinery.

- Combustion process: Diesel + Oxygen → Energy (mechanical or electrical) + Carbon Dioxide + Water Vapor

- Diesel engines operate at higher compression ratios than petrol engines, making them more efficient for heavy-duty applications.

In each case, the fuel is combined with oxygen and ignited, releasing energy in the form of heat, light, or mechanical work, and producing carbon dioxide and water vapor as byproducts.

REASONS WHY CARBON-BASED FUELS ARE USED:

1. Firewood:

- Availability and accessibility
- Low cost
- Renewable resource (if sustainably harvested)
 - Can be used for cooking, heating, and power generation
 - Does not require complex processing or refining
 - Can be used in traditional and rural settings

2. Charcoal:

- High energy density
- Portable and easy to store
- Can be used for cooking, heating, and industrial processes
 - Does not produce smoke or toxic fumes when burned
 - Can be made from waste wood or biomass
 - Has a long shelf life

3. Paraffin (Kerosene):

- High energy density
- Clean-burning and smokeless
- Can be used for lighting, cooking, and heating
 - Has a long shelf life
 - Does not require complex infrastructure for transportation and storage
 - Can be used in aviation and rocket fuel

4. Petrol (Gasoline):

- Has high energy density

- Easy to refine and process
- Can be used in internal combustion engines for transportation and power generation
 - Has a high octane rating, making it suitable for high-compression engines
 - Can be blended with other fuels to improve performance
 - Has a wide range of applications, from small engines to industrial power generation

5. Diesel:

- High energy density
 - High compression ratio, making it efficient for heavy-duty applications
 - Can be used in internal combustion engines for transportation, power generation, and industrial processes
 - Has a high cetane rating, making it suitable for compression ignition engines
 - Can be used in marine and aviation applications
 - Has a wide range of applications, from small generators to large industrial equipment.
- These reasons highlight the advantages and benefits of each fuel, making them suitable for various applications and uses.

B Know and appreciate the difference between renewable and non-renewable fuels and understand that non-renewable fuels are not sustainable (k, u)

- Discuss and report on the meaning of the terms 'renewable' and 'non-renewable' as applied to fuels, and then use data to estimate how long the world's coal, oil and natural gas reserves are likely to last at the current rates of usage.
- Research on the idea of 'sustainability' and write a report on how the use of fuel in your locality can be made more sustainable.

RENEWABLE FUEL

A renewable fuel is a fuel that can be replenished naturally over time, typically from natural resources like plants, trees, or waste materials.

Illustration: A tree farm where trees are harvested for firewood or wood pellets. As the trees are harvested, new trees are planted to replace them, making the fuel source renewable.

Example: Biomass fuels like firewood, charcoal (made from sustainable wood sources), biofuels, and biogas.

NON-RENEWABLE FUEL

A non-renewable fuel is a fuel that cannot be replenished naturally over time, typically from finite resources like fossil fuels.

Illustration: An oil well extracting crude oil from the ground. Once the oil is extracted, it cannot be replaced or replenished.

Example: Fossil fuels like petrol, diesel, paraffin, and coal.

FUEL SUSTAINABILITY

Fuel sustainability refers to the ability of a fuel source to meet the energy demands of the present without compromising the ability of future generations to meet their own energy demands.

Illustration: A sustainable fuel cycle where fuel is produced, used, and then replaced or replenished, ensuring a continuous and sustainable supply.

Example: A forest where trees are harvested for firewood, and new trees are planted to replace them, ensuring a sustainable supply of fuel.

In summary, renewable fuels are sustainable and can be replenished naturally, while non-renewable fuels are finite and cannot be replaced once depleted. Fuel sustainability ensures that fuel sources meet present demands without compromising future energy needs.

RENEWABLE AND NON-RENEWABLE CARBON-BASED FUELS

1. **Firewood:** Renewable fuel

- Reason: Firewood is a biomass fuel that can be sustainably harvested from forests or plantations, and it can be replenished naturally.

2. **Charcoal:** Renewable fuel

- Reason: Charcoal is made from wood, and if the wood is sourced sustainably,

charcoal can be considered a renewable fuel.

3. **Petrol:** Non-renewable fuel

- Reason: Petrol is a fossil fuel derived from crude oil, which takes millions of years to form and is finite.

4. **Diesel:** Non-renewable fuel

- Reason: Diesel is also a fossil fuel derived from crude oil, making it finite and non-renewable.

5. **Paraffin:** Non-renewable fuel

- Reason: Paraffin, also known as kerosene, is a refined product derived from crude oil, making it a non-renewable fuel source.

WAYS TO MAKE FUEL USE MORE SUSTAINABLE

Major ways can significantly contribute to more sustainable fuel use, reducing environmental impacts and promoting a cleaner energy future as explained below:

Fuel Efficiency and Conservation:

Improve vehicle fuel efficiency, use fuel-efficient technologies, and implement fuel-saving measures like turning off engines, using public transport, or carpooling.

Fuel Switching and Blending: Switch from fossil fuels to renewable fuels like biofuels, biogas, or electricity, and blend fossil fuels with renewable fuels to reduce emissions and dependence on fossil fuels.

Sustainable Production and Land Use:

Ensure sustainable production and harvesting practices for renewable fuel sources like biomass or biofuels, and

promote sustainable land use practices for biofuel crops or biomass production.

Alternative Energy Sources and Technologies:

Promote alternative energy sources like solar, wind, or hydro power to reduce dependence on fuels, and continuously research and develop new sustainable fuel technologies and alternatives.

Carbon Capture and Storage: Capture and store carbon emissions from fuel use to reduce greenhouse gas emissions, and utilize waste materials as fuel sources through waste-to-energy technologies.

C Know and appreciate the impact of burning carbon-based fuels on the environment (k, u)

HOW THE BURNING OF CARBON-BASED FUELS IMPACT THE ENVIRONMENT

Air Pollution: Burning carbon-based fuels releases harmful pollutants like particulate matter, nitrogen oxides, sulfur dioxide, and volatile organic compounds, contributing to poor air quality and negative health effects.

Mitigation: Implement emission controls like scrubbers, electrostatic precipitators, and fabric filters; promote cleaner fuels like natural gas or renewable energy.

Climate Change: Burning carbon-based fuels releases carbon dioxide, a greenhouse

gas that contributes to global warming and climate change.

Mitigation: Transition to renewable energy sources like solar, wind, or hydro power; implement carbon capture and storage technologies; increase energy efficiency.

Acid Rain: Burning carbon-based fuels releases sulfur dioxide and nitrogen oxides, which combine with water and oxygen to form acid rain, damaging ecosystems and infrastructure.

Mitigation: Implement emission controls like scrubbers; switch to cleaner fuels like natural gas or renewable energy; add alkaline substances to neutralize acidity.

Water Pollution: Burning carbon-based fuels can contaminate water sources through wastewater disposal or cooling system discharges.

Mitigation: Implement proper wastewater treatment and disposal; use dry cooling systems or hybrid cooling towers; promote water conservation.

Land Degradation: Extracting, processing, and transporting carbon-based fuels can lead to habitat destruction, soil erosion, and land subsidence.

Mitigation: Implement sustainable extraction practices; restore degraded lands through reclamation and reforestation; promote renewable energy sources.

Health Impacts: Burning carbon-based fuels can cause respiratory problems, cardiovascular disease, and other health issues due to air pollution.

Mitigation: Promote cleaner fuels and energy sources; implement emission controls; increase public awareness and education on air quality and health impacts.

By understanding these environmental impacts and implementing mitigation strategies, we can reduce the harmful effects of burning carbon-based fuels and transition towards a more sustainable energy future.

Understand the processes of making charcoal, but recognise that the use of charcoal as a fuel is cheap, efficient, and sustainable only if it is made from wood that can be regrown easily (u, s)

- Research on the process of making charcoal from waste organic material, and energy- saving charcoal stoves, and then explain the chemistry of what happens when organic material burns in a limited supply of oxygen.

MAKING CHARCOAL BRIQUETTES FROM WASTE ORGANIC MATERIALS:

Materials Needed:

- Waste organic materials (e.g., sawdust, rice husks, coconut shells)
- Charcoal dust or powder
- A binding agent (e.g., starch, clay)
- Water
- A mixing device (e.g., hammer mill, blender)
- A briquette machine or press
- A drying rack or oven

Procedure:

1. Charcoal Production: Produce charcoal from wood through carbonization.
2. Crushing and Milling: Crush and mill the charcoal into a fine powder.
3. Mixing: Mix the charcoal powder with a binding agent and water to create a uniform blend.
4. Briquetting: Use a briquette machine or press to shape the blend into briquettes.
5. Drying: Dry the briquettes to remove excess moisture.
6. Curing: Allow the briquettes to cure for several hours or days to harden.

Purification Process:

- Sieving: Sieve the briquettes to remove any impurities or irregularly shaped briquettes.

Applications:

- Fuel: Use the charcoal briquettes as a fuel for cooking, heating, or industrial processes.
- Barbecues and Grills: Use the charcoal briquettes as a fuel for barbecues and grills.

Note: The specific steps and purification processes may vary depending on the desired charcoal briquette quality and application. Additionally, proper safety measures should be taken during the process to avoid accidents and ensure a safe working environment.

MAKING CHARCOAL FROM WOOD:**Materials Needed:**

- Trees (preferably hardwoods like oak, maple, or beech)
- Axe or chainsaw for felling and cutting

- Kiln or pit for carbonization
- Heat source (e.g., firewood, diesel)
- Ventilation system
- Water for cooling
- Sieves or screens for purification

Procedure:

1. Harvesting: Select and fell trees, then cut them into manageable logs.
2. Splitting: Split logs into smaller pieces to increase surface area.
3. Drying: Dry wood pieces to reduce moisture content.
4. Loading: Load wood pieces into kiln or pit.
5. Carbonization: Heat wood in kiln or pit to high temperatures (200-300°C) in absence of oxygen.
6. Cooling: Allow charcoal to cool before removing from kiln or pit.

Purification Process:

- Sieving: Sieve charcoal through screens or sieves to remove impurities and achieve uniform size.

Applications:

- Fuel: Use charcoal as fuel for cooking, heating, or industrial processes.
- Water Filtration: Use charcoal as filter medium to remove impurities from water.
- Agriculture: Use charcoal as soil amendment to improve soil fertility and structure.

Note: The specific steps and purification processes may vary depending on desired charcoal quality and application. Additionally, proper safety measures should be taken during the process to avoid accidents and ensure a safe working environment.

THE CHEMISTRY BEHIND BURNING ORGANIC MATERIAL BURNS IN A LIMITED SUPPLY OF OXYGEN

When organic material burns in a limited supply of oxygen, it undergoes a process called pyrolysis or carbonization.

Decomposition: The organic material (e.g., wood) breaks down into simpler molecules, releasing volatile gases and liquids.

Dehydration: Water molecules (H_2O) are released, leaving behind a carbon-rich material.

Decarboxylation: Carbon dioxide (CO_2) is released, leaving behind a carbon-rich material.

Pyrolysis: The carbon-rich material breaks down into:

- Volatile compounds (e.g., methane, CH_4 ; hydrogen, H_2 ; and other hydrocarbons)
- Char (carbon, C): the solid residue, which is the charcoal
- Tar: a liquid mixture of hydrocarbons

Carbonization: The char (carbon) is the final product, which is the charcoal.

Chemical equations:

$C_6H_{12}O_6$ (wood) $\rightarrow C$ (char) + H_2O (water) + CO_2 (carbon dioxide) + CH_4 (methane) + other hydrocarbons

Note: The specific chemical equations and reactions may vary depending on the type of organic material and the conditions of the pyrolysis process.

In summary, burning organic material in a limited supply of oxygen leads to pyrolysis, which breaks down the material into

simpler molecules, releasing volatile gases and liquids, and leaving behind a carbon-rich solid residue, which is the charcoal.

E Know and appreciate the physical properties and uses of carbon dioxide (k, u)

- From your prior learning of making carbon dioxide by reacting limestone with dilute hydrochloric acid; use apparatus to collect the gas, and then explain the chemical reaction and the property when tested with a lighted splint.

CARBON DIOXIDE GAS

Molecular Formula: CO_2

Elemental Composition: Carbon & Oxygen

Chemical Structure: $O=C=O$

Molecular Weight: 44.01 g/mol

Category: Covalent compound (Formed by sharing of electrons)

Properties:

Color: Colorless

Odor: Odorless

Taste: Slightly acidic

State: Gas at room temperature and pressure

Density: 1.98 kg/m^3

Test with a lit splint: Extinguishes a lit splint with a "hissing" sound, indicating the presence of CO_2 .

Occurrence:

- Found in Earth's atmosphere (0.04%)
- Released through human activities (fossil fuel combustion, deforestation)
- Present in volcanic gases and ocean water

Word Equation of Carbon Dioxide**Production:**

Limestone (Calcium Carbonate) + Dilute Hydrochloric Acid → Calcium Chloride + Water + Carbon Dioxide

In chemical symbols:

CaCO_3 (limestone) + HCl (dilute hydrochloric acid) → CaCl_2 (calcium chloride) + H_2O (water) + CO_2 (carbon dioxide)

USES OF CARBON DIOXIDE

Food and Beverages: CO_2 is used to create the fizz in soft drinks, beer, and sparkling water, making them refreshing and carbonated.

Industrial Applications: CO_2 is used in oil recovery, chemical synthesis, and power generation, helping to extract resources and produce energy.

Cooling and Agriculture: CO_2 is used as a refrigerant in commercial and industrial cooling systems, and in greenhouses to enhance plant growth and photosynthesis.

Safety and Cleaning: CO_2 is used in fire extinguishers to put out electrical fires, and as a solvent for cleaning surfaces and equipment.

Environmental and Energy Applications: CO_2 is used in carbon capture and storage, enhanced oil recovery, and other technologies to reduce greenhouse gas emissions and promote sustainable energy solutions.

F Understand how the increase in carbon dioxide in the air can cause the atmosphere and the oceans to get warmer (u)

• Research and report on how an increase in carbon dioxide in the air can cause the atmosphere and the oceans to get warmer; what does this tell us about the nature of carbon dioxide? Then, explain the effect of carbon dioxide on climate.

ATMOSPHERIC AND OCEAN WARMING

The increase in carbon dioxide in the air leads to atmospheric and oceanic warming through several mechanisms:

Trapping heat: CO_2 absorbs infrared radiation, trapping heat in the atmosphere and preventing it from being released into space.

Greenhouse effect: CO_2 is a greenhouse gas, enhancing the natural greenhouse effect and leading to an increase in global temperatures.

Ocean absorption: The oceans absorb some of the excess CO_2 , which increases their acidity and temperature.

Feedback loops: Warmer oceans release more CO_2 , amplifying the warming effect.

Therefore this tells us that the nature of carbon dioxide is that of a potent greenhouse gas, capable of altering the Earth's energy balance and leading to significant climate changes.

IMPACTS OF CARBON DIOXIDE ON CLIMATE

Global Warming: CO₂ traps heat, increasing global temperatures and altering climate patterns. (Example: 2020 was the hottest year on record, with temperatures 1.2°C above pre-industrial levels.)

Extreme Weather Events: CO₂-influenced climate change leads to more frequent and intense heatwaves, droughts, and storms. (Example: Hurricane Harvey in 2017 was intensified by CO₂-driven climate change.) These impacts demonstrate the significant effects of carbon dioxide on the Earth's climate, leading to rising temperatures and extreme weather events.

Understand what greenhouse gases are, where they come from, and how they are affecting climate (u)

• Discuss and produce posters to explain why air pollution is a global problem and why it can only be properly controlled if all the countries of the world agree to collaborate.

GREENHOUSE GASES:

Greenhouse gases (GHGs) are gases that trap heat in the Earth's atmosphere, leading to global warming. Examples:

- Carbon dioxide (CO₂)
- Methane (CH₄)
- Nitrous oxide (N₂O)
- Ozone (O₃)
- Fluorinated gases (F-gases)

SOURCES OF GREENHOUSE GASES

- Human activities: burning fossil fuels, deforestation, agriculture, industrial processes
- Natural sources: volcanic eruptions, wildfires, ocean releases

WAYS GREENHOUSE GASES ARE AFFECTING THE CLIMATE

Major ways Greenhouse Gases are affecting the climate include:

Rising Global Temperatures: The average global temperature has risen by about 1°C since the late 1800s, leading to:

- Increased heatwaves, droughts, and water scarcity
- Melting of polar ice caps, glaciers, and sea-level rise
- Changes in seasonal patterns and weather extremes.

Extreme Weather Events: Greenhouse gases intensify weather events, leading to:

- More frequent and intense storms, hurricanes, and typhoons
- Severe droughts, wildfires, and heatwaves
- Unpredictable and dangerous weather patterns

Sea-Level Rise: Melting ice caps and glaciers cause sea levels to rise, resulting in:

- Coastal erosion, flooding, and saltwater intrusion
- Loss of coastal ecosystems, habitats, and biodiversity

- Displacement of coastal communities and cities

Changes in Precipitation Patterns:

Greenhouse gases alter global precipitation patterns, leading to:

- Droughts, water scarcity, and impacts on agriculture
- Floods, landslides, and infrastructure damage
- Changes in ecosystems, biodiversity, and food chains.

Loss of Biodiversity: Climate change affects ecosystems, leading to:

- Extinctions, habitat loss, and species migrations
- Disruptions to food chains, fisheries, and agriculture
- Decreased ecosystem resilience and adaptability

CONTROLLING THE NEGATIVE EFFECTS OF GREENHOUSE GASES

- Transition to renewable energy sources
- Increase energy efficiency and conservation
- Electrify transportation and industry
- Carbon capture and storage technologies
- International cooperation and climate policies (e.g., Paris Agreement)

All countries can control GHG emissions by:

- Implementing policies and regulations
- Investing in clean energy and technology
- Promoting sustainable land use and forestry practices
- Encouraging individual actions and behavior change

- Collaborating globally to address this shared challenge

H Understand the origin of hard water in limestone areas and investigate how it can be softened (u, s)

- Research on how rainwater becomes hard as it soaks through limestone.
- Describe how hard water affects soap.
- Discuss, research and explain how adding washing soda (sodium carbonate) results in reducing the water 'hardness'.

WATER HARDNESS

Hard water is water that contains high levels of dissolved minerals, specifically calcium (Ca) and magnesium (Mg) ions. These minerals make water "hard" because they react with soap and other cleaning agents, making it difficult to lather or foam. Hard water can also lead to scaling, which is the deposition of mineral deposits in pipes and appliances.

HOW RAINWATER BECOMES 'HARD'

In limestone areas, hard water originates from the dissolution of calcium carbonate (CaCO_3) and magnesium carbonate (MgCO_3) from limestone rocks in the ways below;

1. Limestone formation: Limestone is formed from the accumulation of calcium carbonate (CaCO_3) from ancient marine organisms, such as coral, shells, and skeletons.

2. Rainwater absorption: Rainwater absorbs CO_2 from the atmosphere, forming a weak carbonic acid (H_2CO_3).

3. Reaction with limestone: The acidic rainwater reacts with limestone, dissolving

The calcium carbonate (CaCO_3) and releasing calcium (Ca) and bicarbonate (HCO_3^-) ions.

4. Groundwater flow: The water, now rich in calcium and bicarbonate ions, flows through the ground and into aquifers.

5. Magnesium contribution: Magnesium, present in smaller amounts in limestone, also dissolves and contributes to the hardness of the water.

6. Hard water emergence: The water, now containing high levels of calcium and magnesium ions, emerges as hard water in wells, springs, or surface water sources.

In summary, hard water in limestone areas originates from the dissolution of calcium and magnesium carbonates in limestone rocks, which are then carried into the water supply by rainwater and groundwater flow.

TYPES OF WATER HARDNESS

Two types exist i.e:

1. Temporary Hardness (Carbonate Hardness)

- Caused by calcium (Ca) and bicarbonate (HCO_3^-) ions

- Sources:

- Surface water (rivers, lakes, reservoirs)
- Groundwater (wells, springs) in limestone or dolomite aquifers
- Municipal water supplies that use surface or groundwater sources
- Rainwater that has passed through carbonate-rich soils or rocks

2. Permanent Hardness (Non-Carbonate Hardness)

- Caused by calcium (Ca) and magnesium (Mg) sulfates, chlorides, and nitrates

- Sources:

- Groundwater (wells, springs) in aquifers with high sulfate or chloride concentrations
- Seawater or brackish water
- Municipal water supplies that use desalination or wastewater treatment
- Industrial processes that use water with high levels of dissolved minerals.

Note that some water sources may exhibit both temporary and permanent hardness, depending on the specific geology and water chemistry.

SOFTENING WATER HARDNESS

Temporary Hardness:

Boiling:

- Simply boil the water to precipitate calcium carbonate (CaCO_3) out of solution.
- Effective for small quantities of water.

Lime Softening:

- Add lime (calcium oxide, CaO) to the water to remove calcium and bicarbonate ions.
- Forms calcium carbonate precipitate, which can be removed.

Precipitation with Washing Soda:

- Add washing soda (sodium carbonate, Na_2CO_3) to the water to precipitate calcium carbonate.

Permanent Hardness**Distillation:**

- Vaporize the water and then condense it, leaving minerals behind.
- Effective, but energy-intensive.

Chemical Softening:

- Add chemicals like sodium carbonate (Na_2CO_3) or sodium hydroxide (NaOH) to remove calcium and magnesium ions.
- Forms precipitates or complexes with the minerals, making the water softer.

Ion Exchange (IX):

- Use a resin to exchange calcium and magnesium ions for sodium or potassium ions.
- Regenerate the resin with salt (sodium chloride) periodically.

Reverse Osmosis (RO):

- Use a semipermeable membrane to remove calcium and magnesium ions from the water.
- Produce a permeate stream (soft water) and a concentrate stream (hard water).

Electrodialysis (ED):

- Use an electric field to remove calcium and magnesium ions from the water.
- Produce a dilute stream (soft water) and a concentrate stream (hard water).

HOW WATER HARDNESS AFFECTS SOAP**Reduced Lather Formation:**

- Hard water ions (Ca^{2+} and Mg^{2+}) react with soap to form a sticky scum, reducing lather.

- Illustration: Soap + Hard Water = Scum + Less Lather

Soap Scum Formation

- Hard water ions combine with soap to form a white, sticky scum.

- Example: Soap scum buildup on skin, hair, and surfaces.

Increased Soap Consumption

- Hard water requires more soap to produce a lather.

- Illustration: Hard Water + Soap = More Soap Needed

J Understand how the properties and uses of the allotropes of carbon relate to their structures (u)

- Research and make models to explain the structures of diamonds and graphite, and explain how their properties determine the uses of carbon structures such as carbon fibres and graphene.

ALLOTROPY

Allotropy refers to the existence of two or more different physical forms of the same element with the same atomic number but differ in their atomic arrangement, crystal structure, or molecular configuration. These different forms are called allotropes. Allotropes are different physical forms of the same element, which can exhibit varying physical and chemical properties.

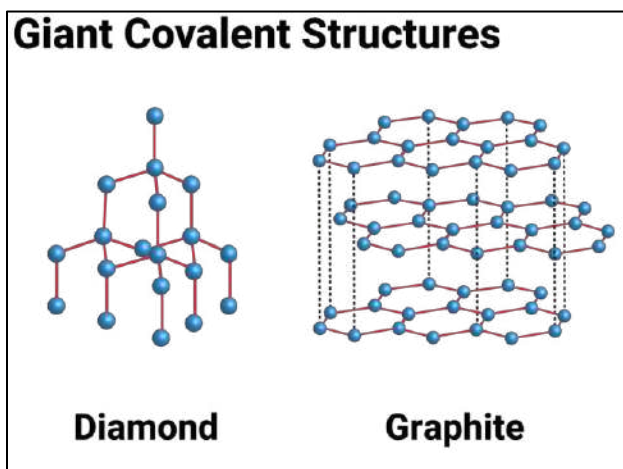
Examples:

- **Carbon** allotropes includes:
 - Diamonds
 - Graphite (used in pencils)
 - Fullerenes (hollow spheres)
 - Nanotubes (tiny tubes)

-Phosphorus allotropes includes:

- White phosphorus (toxic and waxy)
- Red phosphorus (less reactive and more stable)
- Black phosphorus (graphite-like structure)

THE STRUCTURE OF DIAMONDS AND GRAPHITE AS ALLOTROPES OF CARBON:



Diamonds:

- Crystal structure: Face-centered cubic (FCC)
- Each carbon atom is bonded to four neighboring atoms in a strong, three-dimensional lattice
- Extremely hard and rigid, with high thermal conductivity

Graphite:

- Crystal structure: Hexagonal or layer lattice
- Each carbon atom is bonded to three neighboring atoms in a planar, hexagonal arrangement
- Weak van der Waals forces hold layers together, making graphite soft and slippery.

These different structures result in distinct properties, making diamonds ideal for industrial cutting tools and graphite perfect for pencil lead and lubricants.

THE PROPERTIES OF DIAMONDS AND GRAPHITE THAT DETERMINE THEIR USES:

The following properties make diamonds and graphite versatile materials with various industrial, technological, and artistic applications.

Diamonds:

Hardness: Diamond is the hardest substance known, making it ideal for:

- Cutting and drilling tools (e.g., diamond-tipped saw blades)

Thermal Conductivity: Diamond is an excellent heat conductor, making it suitable for:

- Electronic heat sinks and thermal management

Optical Transparency: Diamond is highly transparent, making it perfect for:

- High-pressure windows and optical components

Chemical Inertness: Diamond is resistant to chemicals, making it ideal for:

- Chemical reaction vessels and containers

High Dispersion: Diamond has a high dispersion coefficient, making it suitable for brilliant gemstones and jewelry

Graphite:

Softness: Graphite is soft and slippery, making it perfect for:

- Pencil lead and lubricants (e.g., graphite powder in locks)

Conductivity: Graphite is a good electrical conductor, making it suitable for:

- Electrodes in batteries and electrolysis

Thermal Resistance: Graphite has low thermal conductivity, making it ideal for:

- Heat shields and thermal insulation

Chemical Inertness: Graphite is resistant to chemicals, making it suitable for:

- Crucibles and reaction vessels

Lubricity: Graphite is highly lubricating, making it perfect for:

- Bearings and gears in machinery

APPLICABLE SCENARIOS**Scenario 1:**

Kampala International University wants to reduce its carbon footprint by 20% within the next year. The university's current carbon footprint is 10,000 metric tons of CO₂ equivalent (MTCO₂e).

Task: As a chemistry learner, make a written advice to achieve this goal.

Scenario 2:

A coastal community dependent on shellfish farming is experiencing declines in production due to ocean acidification caused by excess carbon dioxide absorption. The dependants are angry of the consequences. Their leader has consulted you for advice he can use in the workshop already organized.

Task: As a chemistry learner, develop a plan to be used.

Scenario 3:

Part of Kampala town wants to increase its urban forest cover to sequester more carbon dioxide and mitigate the urban heat island effect. Currently, the town has 10% tree cover, and the goal is to increase it to 20% within the next two years. The town mayor responsible for the task has approached you for more details from what he knows.

Task: As a chemistry learner, make a detailed write up to satisfy the town mayor

END

Reactivity Series

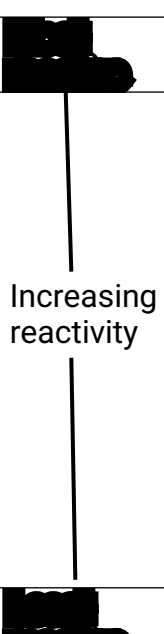
Reactivity Series refers to an arrangement of elements beginning with the most reactive and ending with the least reactive element. The series is based on the reactions with air, water and dilute acids.

Importance of the Reactivity Series

Reactivity series is used to check which element is more or less reactive. A metal higher in the reactivity series displaces that which is lower from its compound.

The most reactive elements are placed at the top of the series and the least reactive elements are placed at the bottom.

Commonly Used Reactivity Series

Element	Symbol	
Potassium	K	
Sodium	Na	
Calcium	Ca	
Magnesium	Mg	
Aluminium	Al	
Carbon (metalloid)	C	
Zinc	Zn	
Iron	Fe	
Tin	Sn	
Lead	Pb	
Hydrogen (nonmetal)	H	
Copper	Cu	
Mercury	Hg	
Silver	Ag	
Gold	Au	

1) Reactions of Metals

Potassium	}	Reacts with cold water
Sodium		
Calcium		
Magnesium	}	reacts with steam
Aluminium		
Zinc		

Iron	
Lead	} do not react with water in any form
Hydrogen	
Copper	
Mercury	
Silver	
Gold	

Investigating the reaction of potassium and sodium with cold water.

What you need

Water, trough, a pair of tongs, a knife, filter papers litmus papers, sodium metal and potassium metal.

Safety precaution

1. The size of the metal to be put in cold water should be very small, as large pieces would result in an explosion.
2. Use gloves while handling sodium and potassium metals
3. Safety goggles should be worn before and during the experiment
4. Observations should be made at a reasonable distance.

What to do

1. Pour water in the trough.
2. Use a dry knife to cut a tiny piece of sodium metal and use a filter paper to remove the oil from the metal.
3. Use a pair of tongs to drop it in to a trough of water.
4. Carefully observe what happens to the metal.
5. Test the liquid in the trough using blue and red litmus papers and describe what happens to each paper.
6. Record your observations in the table below.
7. Repeat procedures 1-6 using potassium metal.

Reaction of potassium and sodium with cold water.

Metal	Observation	Effect on litmus paper
Sodium	Sodium floats, melts in to silvery ball	Blue litmus paper remains

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	which darts on the surface of water producing a hissing sound as it dissolves to form a colourless solution.	blue and red litmus paper turns blue due to formation of sodium hydroxide solution which is an alkaline.
Potassium	Potassium floats, melts in to silvery ball which darts on the surface of water producing a hissing sound and busts in to a purple flame as it dissolves to form a colourless solution.	Blue litmus paper remains blue and red litmus paper turns blue due to formation of potassium hydroxide solution which is an alkaline.

Conclusion

Potassium reacts violently with cold water to form potassium hydroxide and hydrogen gas.

Potassium + water $\longrightarrow$ potassium hydroxide + hydrogen gas



Sodium also reacts vigorously with cold water to form sodium hydroxide and hydrogen gas.

Sodium + water $\longrightarrow$ Sodium hydroxide + hydrogen gas



NOTE:

- (i) The reaction of potassium metal with cold water produces a lot of heat, which causes both potassium and hydrogen gas to burn producing a purple flame or a lilac flame.
- (ii) Sodium and cold water also produce a lot of heat but not enough to burn sodium and hydrogen gas, showing that potassium is more reactive than sodium.
- (iii) Potassium and sodium are highly reactive therefore are kept under oil to prevent them from reacting with moist air.

Reaction of other metals with cold water.

Investigating the reaction of other metals in the series with cold water.

What you need

Calcium metal, magnesium ribbon. Aluminium, zinc, iron, copper, a beaker, a filter funnel, boiling tube, a spatula, water.

What to do

1. Put a spatula endful of calcium at the bottom of the beaker containing some water.

Prepared by Sir Salim

2. Invert a funnel over calcium.
3. Add enough water to cover the whole funnel.
4. Quickly invert the boiling tube full of water and collect the gas as shown in figure below.
5. Carefully observe what happens to the metal.
6. When the boiling tube is full of hydrogen, close it with cork while still under water.
7. Test the gas burning placing the burning splints at the mouth of the test tube.
8. Test the liquid using blue and read litmus paper.
9. Record and discuss your observations.
10. Repeat procedures 1-9 using magnesium, aluminium, zinc, iron and copper.

Set up

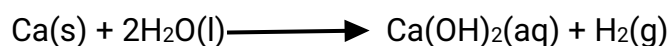
Reaction of other metals with cold water.

Metal	Observation	Effect on litmus paper
Calcium	Grey white powder dissolves moderately with bubbles of a colourless gas to form a cloudy solution of calcium hydroxide.	Blue litmus paper remains blue and red litmus paper turns blue due to formation of calcium hydroxide solution which is an alkaline.
Magnesium	Magnesium reacts very slowly with cold water forming cloudy solution of magnesium hydroxide and hydrogen gas.	Blue litmus paper remains blue and red litmus paper turns blue due to formation of Magnesium hydroxide solution which is an alkaline.
Aluminium, zinc, iron, and copper	Do not react with cold water.	-

Conclusion

Calcium reacts moderately/ less violently with cold water. The heat produced is not sufficient for hydrogen gas to catch fire. Calcium sinks at the bottom of the beaker and reacts moderately with cold water producing bubbles of hydrogen gas and a cloudy solution of calcium hydroxide.

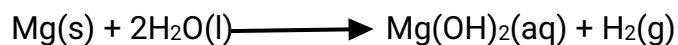
Calcium + water $\longrightarrow$ Calcium hydroxide + hydrogen gas.



Magnesium reacts very slowly with cold water producing bubbles of hydrogen and magnesium hydroxide solution.

Prepared by Sir Salim

Magnesium + water $\longrightarrow$ Magnesium hydroxide + hydrogen gas.



Aluminium, zinc, iron and copper do not react with cold water.

Metals which are very reactive, react with cold water to form corresponding hydroxide and hydrogen gas.

From the above reactions of metals with cold water, Potassium is the most reactive whereas Magnesium is the least reactive.

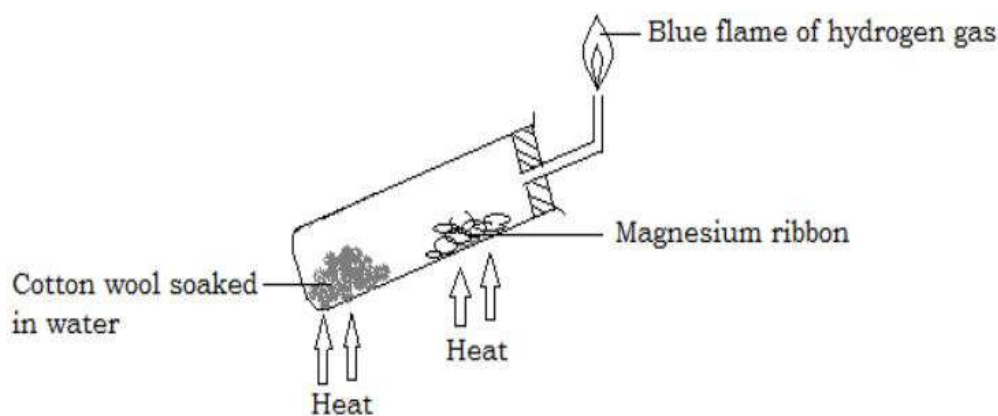
Investigating the reaction of Magnesium, Aluminium, Zinc, Iron and Copper with steam.

What you need

Magnesium ribbon, Aluminium Zinc, Iron, Copper, wet glass wool, a heat source, sandpaper, a cork with a glass tube, boiling tube.

What to do

1. Soak glass/cotton wool in water and insert it at the bottom of the boiling tube.
2. Clean a piece of magnesium ribbon using sandpaper and insert it in the middle of the boiling tube.
3. Stopper the boiling tube with cork containing glass tubing.
4. Clamp the boiling tube as shown in the figure below.



5. Heat the magnesium ribbon strongly and then heat the wet cotton wool gently.
6. Estimate the time for the mixture to glow red hot.

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7. Carefully record your observations.

8. Repeat procedure 1-7 using Aluminium Zinc, Iron and Copper.

Metal	Observation
Magnesium	Grey solid glows red hot and forms a white solid. Magnesium + steam $\longrightarrow$ magnesium oxide + hydrogen gas $\text{Mg(s)} + \text{H}_2\text{O(g)} \longrightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$
Zinc	Grey solid glows red hot and forms yellow solid when hot and white when cold. Zinc + steam $\longrightarrow$ Zinc oxide + hydrogen gas $\text{Zn(s)} + \text{H}_2\text{O(g)} \longrightarrow \text{ZnO(s)} + \text{H}_2\text{(g)}$
Iron	Grey solid glows red hot and forms a blue-black solid. Iron + steam $\longrightarrow$ Tri iron tetra oxide + hydrogen gas $3\text{Fe(s)} + 4\text{H}_2\text{O(g)} \longrightarrow \text{Fe}_3\text{O}_4\text{(s)} + 4\text{H}_2\text{(g)}$.
Copper	Does not react steam.

Conclusion

Magnesium takes the shortest time to react with steam, while iron takes the longest time.

Metals which are moderately reactive do not react with cold water but react with steam to form corresponding metallic oxides and hydrogen gas. Those which are least reactive such as copper do not react with cold water or steam.

NOTE:

- (i) Magnesium ribbon is cleaned first in order to remove the coating of magnesium oxide.
- (ii) Cotton wool is heated to generate steam that will react with the heated metal.
- (iii) Aluminium does not react with water or steam due to the formation of an insoluble oxide layer which prevents contact between water and the metal inhibiting any further reaction.

Summary of the reactions of the metals with cold water and steam.

Metal	Reaction with cold water or steam	Products formed
Potassium	Very violent with cold water and catches fire	Potassium hydroxide and hydrogen gas
Sodium	Violent with cold water	Sodium hydroxide and hydrogen gas

Prepared by Sir Salim

Calcium	Less violent with cold water	Calcium hydroxide and hydrogen gas
Magnesium	Very slow with cold water but vigorously with steam	Solid magnesium oxide and hydrogen with steam Magnesium hydroxide and hydrogen with cold water.
Aluminium	No reaction	No product
Zinc	No reaction with cold water but slow with steam	Zinc oxide and hydrogen gas
Iron	No reaction with cold water but very slow with steam	Tri Iron tetra oxide and hydrogen gas
Lead	No reaction	No product
Copper	No reaction	No product
Silver	No reaction	No product
Gold	No reaction	No product

2. Reaction of the metals with acids

Metals reacts with acids under different conditions. Some metals react with acids forming metal salts and liberating hydrogen gas.

General equation

Metal + acid $\longrightarrow$ metal salt + hydrogen

The speed at which the reaction occurs depends on the metal and its position in the reactivity series.

Investigating the reaction of the metals with dilute hydrochloric acid

What you need

Zinc, Calcium, Magnesium, Aluminium, Iron, Copper, Lead, trough, filter funnel, a gas jar, a round bottomed flask, water, thistle funnel, and delivery tubes.

What to do

1. Put zinc pieces in a round bottomed flask.
2. Add dilute hydrochloric acid into the flask using a thistle funnel.

Prepared by Sir Salim

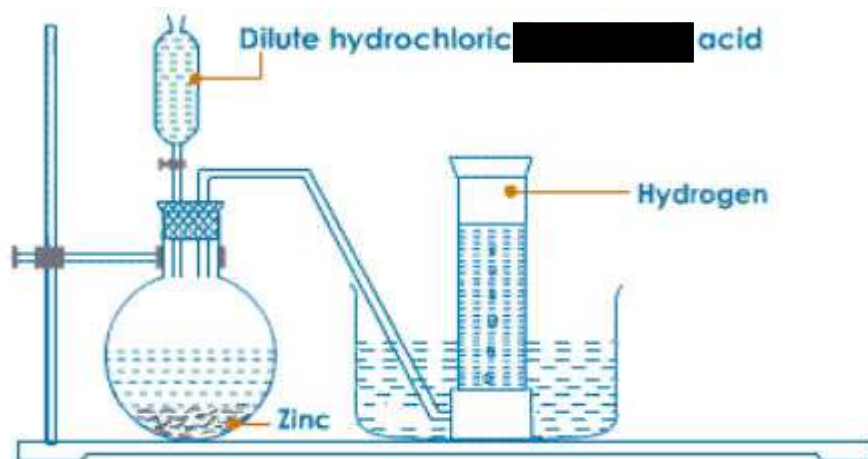


Fig. 9.7 Preparation of hydrogen from zinc with dilute acids

3. Observe what takes place in the round bottomed flask and the rate at which the gas jar is being filled with hydrogen,
4. Repeat procedure 1-3 using Calcium, Magnesium, Aluminium, Iron, Copper, Lead.
5. Record your observations in the table below.

Observations

Metal	Rate of reaction	products
Calcium	Vigorous	Calcium chloride and hydrogen gas Calcium + hydrochloric acid $\rightarrow$ calcium chloride + hydrogen $\text{Ca(s)} + 2\text{HCl(aq)} \rightarrow \text{CaCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
Magnesium	Vigorous	Magnesium chloride and hydrogen gas Magnesium + hydrochloric acid $\rightarrow$ Magnesium chloride + hydrogen $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
Aluminium	No reaction	No product
Zinc	Moderate	Zinc chloride and hydrogen gas

Prepared by Sir Salim

		Zinc + hydrochloric acid $\rightarrow$ Zinc chloride + hydrogen $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
Iron	Moderate	Iron (II) chloride and hydrogen gas Iron + hydrochloric acid $\rightarrow$ Iron (II) chloride + hydrogen $\text{Fe(s)} + 2\text{HCl(aq)} \rightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
Lead	Slow if acid is concentrated	Lead (II) chloride and hydrogen gas
Copper	No reaction even if acid is concentrated	No product

Conclusion

Calcium and magnesium react rapidly with dilute acids. Metals like zinc and iron react moderately and lead reacts slowly with dilute acids, whereas copper does not react at all. The reaction is fastest in calcium and none in copper.

Water and acids contain hydrogen atoms that can be replaced and therefore when they react with metals, hydrogen atoms are displaced and combine to form hydrogen gas.

NOTE:

- (i) The most reactive metal takes shorter time and the least reactive takes longer time to produce hydrogen gas/ or to dissolve in the acid.
- (ii) Aluminium does not react with dilute acids due to the formation of an insoluble oxide layer which prevents contact between acid and the metal inhibiting any further reaction.
- (iii) Potassium and sodium should never be reacted with acids because the reactions are fast, explosive and produce a lot heat energy which may break the apparatus.
- (iv) Copper does not react with water or acids under all conditions because it is less reactive than hydrogen and so cannot displace hydrogen from its compounds.

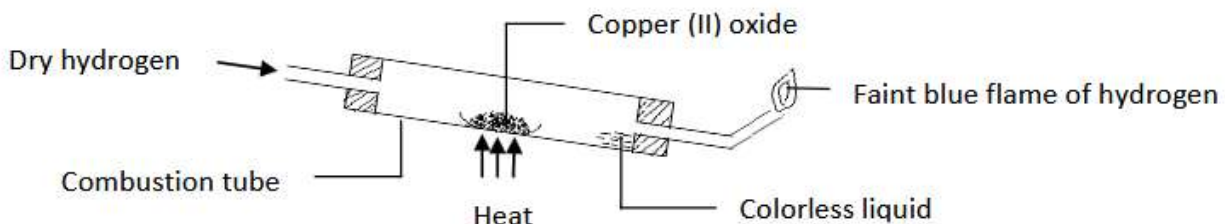
Investigating the reaction between copper (II) oxide and hydrogen gas

Prepared by Sir Salim

What you need

Combustion tube, dry hydrogen, copper (II) oxide, heat source.

Set up



What to do

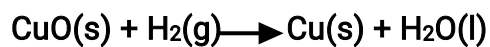
1. Set up apparatus as shown in the figure above.
2. Put copper (II) oxide in a combustion tube and heat it strongly.
3. Pass dry hydrogen gas through the apparatus until the pop sound stops.
4. Record your observations in the combustion tube.

Observation

The black powder glows red and changes to a brown solid. Colourless condensate(liquid) is formed on the cooler parts of the combustion tube.

Equation

Copper (II) oxide + hydrogen $\longrightarrow$ copper + water



Conclusion

Hydrogen displaces copper atoms from their oxide which is black to form copper atoms which are brown. Hydrogen combines with oxygen to form water which is colourless. This shows that hydrogen is more reactive than copper.

3. Reaction of metals with oxygen

The vigour with which each element reacts with oxygen depends on its position in the reactivity series. The more vigorous a metal burns in oxygen, the higher its position in the reactivity series.

Metals react with oxygen to form metal oxides.

Prepared by Sir Salim

General equation

Metal + oxygen → metal oxide

Investigating the reaction of metals with oxygen**What you need**

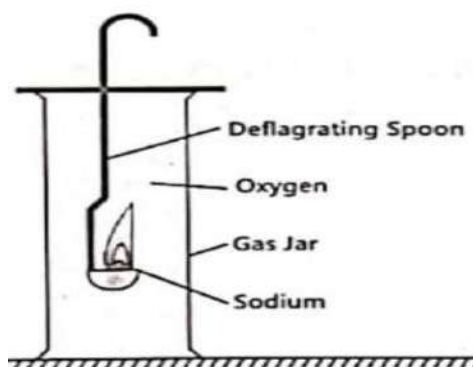
The metals, deflagrating spoon, gas jar, heat source

Safety precautions

➤ The size of the potassium and sodium should be very small as burning large pieces would result in an explosion.

Steps taken/what to do

1. Hold a piece of sodium using a clean dry pair of tongs and place it on a petri dish.
2. Cut a small piece of the sodium.
3. Place it on a deflagrating spoon and warm it gently until it starts to burn.
4. Lower it into a gas jar containing oxygen as shown in figure below



5. Add a little water to the product and shake.
6. Carefully observe and record your observations
7. Test the resultant solution with blue and red and red litmus papers and state the effect of solution on each litmus paper.
8. Repeat procedure 1-7 using potassium, calcium and magnesium ribbon, aluminium, zinc, iron, lead, copper and silver and gold. But for magnesium ribbon, aluminium, zinc, iron, lead, copper and silver and gold strong heating is required.

Reactions of the metals with oxygen

Metal	Observation	Effect on litmus paper
-------	-------------	------------------------

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Potassium	Potassium burns in oxygen with a lilac flame (purple flame) forming white ash/solid of potassium oxide. Potassium + Oxygen $\longrightarrow$ potassium oxide $4K(s) + O_2(g) \longrightarrow 2K_2O(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Sodium	Sodium burns with a bright yellow flame forming a yellow solid. Sodium + Oxygen $\longrightarrow$ Sodium peroxide $2Na(s) + O_2(g) \longrightarrow Na_2O_2(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Calcium	Calcium burns with a bright red flame forming white solids of calcium oxide. Calcium + oxygen $\longrightarrow$ calcium oxide $2Ca(s) + O_2(g) \longrightarrow 2CaO(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Magnesium	Magnesium burns with a brilliant white flame forming white ash (powder) of magnesium oxide. Magnesium + oxygen $\longrightarrow$ Magnesium oxide $2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Aluminium,	Aluminium glows red and forms a white solid of aluminium oxide. Aluminium + oxygen $\longrightarrow$ Aluminium oxide $4Al(s) + 3O_2(g) \longrightarrow 2Al_2O_3(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Zinc	Zinc glows red and forms a yellow solid when hot and white when cold (zinc oxide) $2Zn(s) + O_2(g) \longrightarrow 2ZnO(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Iron	Iron burns in oxygen with bright sparks to form blue-black beads of tri-iron tetra oxide. $3Fe(s) + 2O_2(g) \longrightarrow Fe_3O_4(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Lead	Lead glows red and forms a reddish-brown solid when hot and yellow when cold (lead (II) oxide) $2Pb(s) + O_2(g) \longrightarrow 2PbO(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Copper	The brown solid glows red and forms a black solid of copper (II) oxide. $2Cu(s) + O_2(g) \longrightarrow 2CuO(s)$	Blue litmus paper remains blue and red litmus paper turns blue due to formation of an alkaline solution.
Silver and gold	No reaction	-

Conclusion

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Aluminium, zinc and Lead are covered with a thin protective layer of oxide which prevents the metal from further oxidation. Therefore, they must be cleaned to remove the protective layer of the oxide in order to allow them combine with oxygen on strong heating.

Competition for oxygen and ions between metals

This is also called displacement; where a more reactive metal displaces a less reactive metal from its compound. For example.

Competition for oxygen between magnesium and copper.

What you need

Copper (II) oxide, magnesium metal powder, magnesium oxide, copper metal powder, crucible (saucepan), heat source.

What to do

1. Transfer one small spatula endful of magnesium powder in a crucible.
2. Add one spatula endful of copper (II) oxide powder to the same crucible.
3. Mix the two powders together using the spatula
4. Heat the mixture strongly.
5. Note and record your observations

Observation

The mixture glows red and the black powder changes to a brown solid

6. Repeat procedure 1-5 using a mixture of magnesium oxide and copper powder

Observation

No observable change

Identify with a reason the more reactive between the two elements

Answer.

Magnesium is more reactive than copper; this is because magnesium displaces copper from copper (II) oxide while copper does not displace magnesium from magnesium oxide.

NOTE: The most reactive metal will always displace the less reactive metal from its compound.

Investigating the reaction between iron and copper (II) sulphate.

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What you need

Iron metal/wool, copper (II) sulphate solution, a beaker, a stop clock/watch and a measuring cylinder.

What to do:

1. Measure 100cm³ of copper (II) sulphate solution using a measuring cylinder and pour it in a clean beaker.
2. Add iron wool to the copper (II) sulphate solution.
3. Leave it to stand for one minute.
4. Note and record your observations

Observation

The blue solution turns green and reddish-brown solid is formed on the surface of the iron wool.

Explanation

Iron is more reactive than copper and therefore displaces copper (II) ions which are blue to form copper metal which is reddish-brown and iron (II) sulphate solution/ Iron (II) ions which are green.

Equation for the reaction.

Copper (II) sulphate + Iron $\longrightarrow$ copper + iron (II) sulphate



OR

Copper (ii) ions + iron $\longrightarrow$ copper + iron (II) ions

**Applications of reactivity series in daily life**

1. Extraction of some metals
2. Making simple cells
3. Galvanizing iron
4. The sacrificial protection of iron

ALLOYS

What is an alloy?

An alloy; is a uniform mixture of two or more metals. Alloys are formed by thoroughly mixing molten metals or carbon. It has been found that alloying produces metallic substance with more useful properties than the original pure metal it is made from.

Qn. Explain why alloys are preferred to pure metals.

Alloys are; harder, strong, have a high resistance and are more flexible for manipulating in to different forms.

Examples of alloys

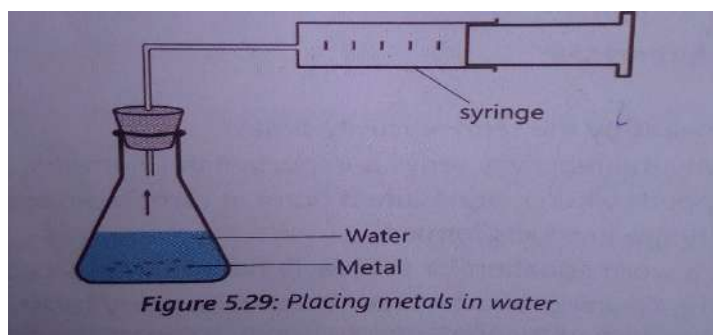
Alloy	Composition	Properties	uses
Brass	copper and Zinc	Harder and stronger	In making of musical instruments and kitchenware
Bronze	copper and Tin	-hard and strong -does not corrode easily -has shiny surface	-to build statues and monuments and ornaments like necklace, sculptures. -in the making of medals, swords and artistic materials. -In Making screws.
Steel	Iron and Carbon	Hard and strong	-in construction of buildings and bridges. -in the building of body of cars and railway tracks. -making cutting tools.
Stainless steel	Iron, chromium, nickel and carbon	-shiny -strong -doesn't rust	-to make cutlery, razor blades and surgical instruments
Solder	Lead and Tin	Strong	-Joining /connecting metals -Connecting electrical wiring
Duralumin	Aluminium, copper, magnesium and manganese	-light -strong -doesn't rust	-to make the body of aeroplanes and bullet trains - making bicycle parts and small boats.
Tungsten steel	Iron and Tungsten	strong	-making edges of high-speed cutting tools.
Pewter	Tin, Copper	-Lustrous	-making plates, ornaments and

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	and Antimony	-strong	drinking mugs.
Magnalium	Magnesium and Aluminium	-Light -strong -doesn't rust	Aircraft construction and small boats
Alnico (ferromagnetic)	Aluminium, Nickel and Cobalt.	Strong and hard	-making generators, electric motors and mass spectrometer.
Nichrome	Nickel and Chromium	Strong and hard	-making electrical heating elements.

Revision items

1. While investigating how fast metals can react with water, learners of Ombokolo high school placed 6 metals (M, T, N, U, B, C) in each conical flask containing 200cm^3 of water and measured the volume of hydrogen gas produced after 2 minutes for every metal as shown in the figure below.



Results

Metal	Volume of hydrogen produced after 2minutes (cm^3)
M	0
T	5
N	10
U	8
B	20
C	29

Task

- Arrange the elements in order of decreasing reactivity
- Given that the elements were selected from group I and group II of the periodic

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table, identify any two elements that are likely to be members of Group I.

c) For the two elements identified in (b) above, state the element that is likely to be from

(i) Period 3

(ii) Period 4 of the periodic table

(d) Give a reason for your choice in c(ii).

2. (a) what happens when:

(i) dilute sulphuric acid is poured on a copper plate

(ii) iron nails are placed in copper (II) sulphate solution

(b) write equations for the reactions involved, if any.

3. Normally, copper is used to make hot water tanks and not steel (an alloy of Iron). Explain why.

4. Discuss why gold is preferred for making jewellery.

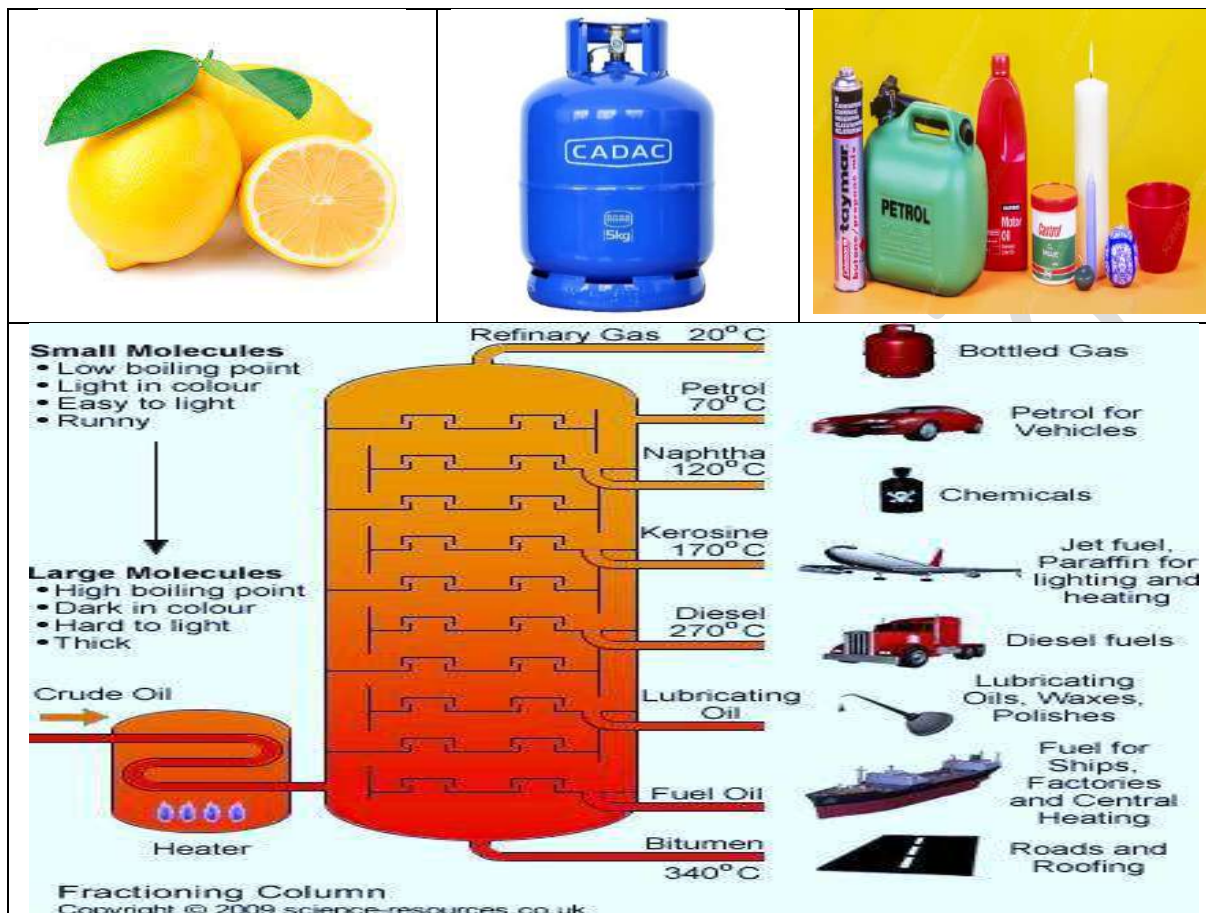
5. Can you store lemon slices in an aluminium foil. Explain.

6. would a ring of gold react if you put it into a beaker of dilute hydrochloric acid. Explain your answer.

7. Identify in life the objects made using alloys and their corresponding uses in daily life.

8. Identify the uses of alloys in our everyday life.

CHAPTER 15: CARBON IN LIFE



KEY WORDS

- Petroleum oils
- Hydrocarbons
- Homologous series
- Cracking
- Functional groups.
- Alkanes.
- Alkenes.
- Alkanols. (Alcohols)
- Carboxylic acids.
- Fermentation
- Organic compounds.
- Structural formulae.
- Fractional distillation.

LEARNING OUTCOMES (L.O)

- By the end of this chapter, you should be able to:
- Recognise that there is a diversity of carbon – based compounds in living things and materials derived from living things, and that these maybe classified into groups,
 - Understand that crude oil is a mixture of different fractions of alkanes which can be separated by fractional distillation, that these products are used in fuels and chemicals for feedstocks.
 - Know and appreciate that natural gas deposits are found all over the world, that their main constituent is methane gas which is a useful source of fuel and chemical feedstocks.
 - Appreciate that biogas is a carbon – based fuel useful for cooking and lighting.
 - Know some common synthetic and natural carbon – based polymers and how their properties relate to their use.
 - Understand how ethanol is made naturally by fermentation of sugars and other organic chemicals, and awareness of the dangers of abuse of ethanol as drugs.

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○ Polymers. ○ Polymerization. ○ Soapless detergents ○ Soaps.	g) Know the process of making soaps and Soapless detergents from natural fats and oils and appreciate that soaps are effective in removing organic stains. h) Know that Soapless detergents are made from crude oil and that Soapless detergents are better cleaning agents in hard water than soapy detergents but have a more deleterious effect on the environments i) Understand how organic compounds can be grouped into homologous series, each of which has similarities in structures and properties.
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COMPETENCY: Learner will appreciate the diversity of carbon organic compounds including the alkanes, alkenes, alkanols, and carboxylic acids.

15.1. INTRODUCTION.

In this chapter, you will be learning about the significance of carbon –containing compounds called **organic compounds**. They may be natural, synthetic polymers, hydrocarbons, organic acids, alcohols, foodstuffs, drinks, soaps and soap less detergents

15.2.PETROLEUM OILS. What is petroleum?

Petroleum oil is a smelly mixture of hundreds of different alkanes. They are organic compounds, which means they contain carbon along with hydrogen. In fact, most petroleum oils are called **hydrocarbons**, since they contain carbon and hydrogen only. Petroleum is formed from the remains of **dead organisms (plants and animals)** that fell to the ocean floor, and were buried under thick sedimentary rocks. High pressures slowly converted them to petroleum, over millions of years.

ACTIVITY: 1.2(a)

What to do:

1. In groups or individually discuss more about petroleum oils.
2. Share and compare your findings with the rest of members in your class.
3. Write down your work into your chemistry notebook.

Discussion questions:

- | | |
|---|--|
| a) What is meant by the term “petroleum oil?” | d) Why are crude oils sometimes called fossil fuels? |
| b) What is another name for petroleum oils? | e) Name the process by which fossil oils are formed. |
| c) Explain how petroleum oils are formed? | |

15.2.1. Refining crude oils. What does refining mean?

15.2.2. Refining crude oils.

In a refinery, the fractional distillation is carried out in a tower that is kept very hot at the base, and cooler towards the top. Petroleum is pumped in at the base. The compounds start to boil off. Those with the smallest molecules boil off first, and rise to the top of the tower, depending on their boiling points, and then condense. As the molecules get larger, the fractions get less runny, or more viscous: from gas at the top of the tower to solid at the

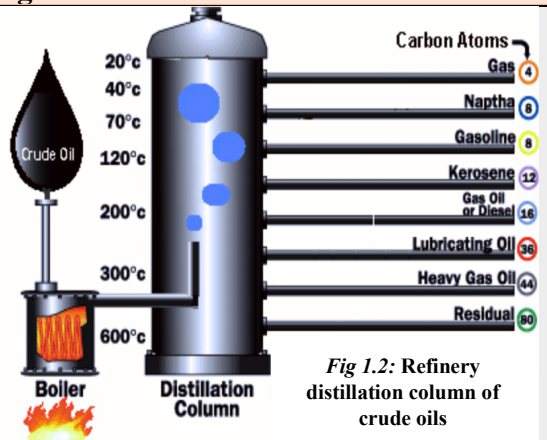


Fig 1.2: Refinery distillation column of crude oils

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bottom. They also get less flammable. So, the last two fractions in the distillation column are not used as fuels. Why?

ACTIVITY:15.2(b)**What to do:**

In groups or individually, visit a computer library or using your smartphone for internet, conduct the following research to find out more about refinery of crude oils. Share and compare your findings with the rest of the class. Write down your work into your notebook.

Discussion questions:

- Draw and label the refinery distillation process of crude oils in figure 1.2.
- Explain how the different fractions of crude oils are refined.
- Copy and complete the table below correctly. Name the fractions of the crude oils which possess the following number of carbon atoms using the following boiling points. In each case, state two uses of the fractions in our day – to – day life activities.

Fraction	Name of fraction	Number of carbon atoms	Uses of each fraction
1 st		C ₁ to C ₄ (Below 25°C)	
2 nd		C ₅ to C ₁₀ (Below 70°C)	
3 rd		C ₁₀ to C ₁₅ (Below 180°C)	
4 th		C ₁₅ to C ₂₀ (Below 200°C)	
5 th		C ₂₀ to C ₃₀ (Below 250°C)	
6 th		C ₃₀ to C ₅₀ (Below 300°C)	
7 th		C ₅₀ upwards (Above 350°C)	

- Copy and complete the following. Write your answers in the spaces provided. Use the following words. *Easily flow or slowly flow; volatile or less volatile; easily burn or burns with difficulty. No smoke and ash or smokey and ashy.*

Fraction	Boiling point range	How easily does it flow?	How volatile is it?	How easily does it burn?	Level of smoke or ash on combustion
1 st Natural gas	Below 25°C				
2 nd Gasoline	Below 70°C				
3 rd Naphtha	Below 180°C				
4 th Kerosene	Below 200°C				
5 th Diesel	Below 250°C				

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6 th Fuel oil	Below 300°C				
7 th Lubricating oil	Below 350°C				

15.2.3. Cracking Hydrocarbons: What does cracking hydrocarbons mean?**After fractional distillation...**

Petroleum is separated into fractions by fractional distillation. The fractions all need further treatment before they can be used.

1. They contain impurities – mainly sulfur and lead compounds. If left in the fuels, these will burn to form harmful sulfur dioxide gas and leaded fuels respectively.
2. Some fractions are separated further into single compounds, or smaller groups of compounds. For example, the gas fraction is separated into methane, ethane, propane, and butane.
3. Part of a fraction may be cracked. Cracking breaks molecules down into smaller ones.

Cracking is a **petrochemical method** that involves breaking long chain hydrocarbons to short chained hydrocarbon molecules and other useful products.

Why is steam or thermal cracking used in petrochemical process?

a) Steam cracking: is a petrochemical process in which long chain saturated hydrocarbons are broken down into smaller, often unsaturated, hydrocarbons. It is the principal industrial method used in the production of lighter alkenes (or commonly known as olefins), including ethene (or ethylene) and propene (or propylene).

b) Thermal cracking: Is a method of breaking large hydrocarbon molecules into smaller useful molecules by heating them to high temperatures in the absence of oxygen. It is the most recommended method that help increase the production of valuable products like gasoline, diesel from crude oil.

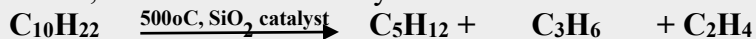
c) Catalytic cracking: Is method of breaking large hydrocarbons into small useful molecules in the presence of catalysts. Such catalysts include Aluminosilicate (Zeolites).

Why is cracking Hydrocarbons important?

- Cracking makes some petroleum products be put into the best use. Suppose you have too much of the naphtha fraction, and too little of the gasoline fraction. You can crack some naphtha to get much quantities of gasoline fraction in the right size for petrol.
- Cracking always produces short-chain compounds with a carbon–carbon double or triple bond. This bond makes the compounds reactive. So, they can be used as feedstocks for making plastics and other substances.

Examples of cracking Hydrocarbons:

1. **Cracking naphtha fraction:** Compounds in the naphtha fraction are often cracked in the presence of aluminium oxide and silicon dioxide catalyst and high temperatures of 500°C, to form short chain hydrocarbons. This is the kind of reaction that occurs.



2. **Cracking kerosene fraction:** Kerosene fraction has long chain hydrocarbons that can still be cracked at high temperatures in the presence of a catalyst to produce useful short chain hydrocarbons and other products.



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ACTIVITY:1.2(c) Laboratory cracking of Hydrocarbons.**What you need**

Kerosene (liquid paraffin), aluminium oxide, sand (mineral wool), test tube, heat source, delivery tube, water, tough and rubber bungs.

What to do:

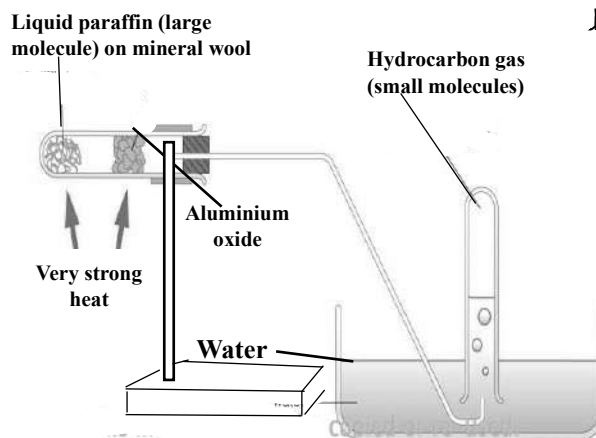
In groups or individually carryout the following procedure to investigate the products of cracking hydrocarbons.

Procedure:

1. Arrange the set up of the apparatus as shown above.
2. Apply heat on the soaked liquid paraffin in mineral wool until the long chain hydrocarbon is cracked into short chain gaseous products.
3. Pass the short gaseous alkane and alkene over heated aluminium oxide catalyst. NB aluminium oxide can obtained from broken pieces of clay pot.
4. Collect the gaseous products over water.
5. Add a drop or few drops of potassium manganate (VII) solution into the gas collected in the inverted test tube. Make observation.

Discussion Questions:

- a) Identify the:
 - (i) Gas that evolved in the product
 - (ii) Colour of flame on igniting the gaseous product.
- b) State with a reason the role of aluminium oxide in the above experiement.
- c) Suggest the name of the process that aided the formation of gaseous product in (b) above.

**ACTIVITY: 1.2(d)****Discussion Questions:**

- a) Crude oil is a mixture of different fractions of hydrocarbons that maybe catalytically cracked at varying temperatures. What is meant by the following terms:
 - (i) Hydrocarbon.
 - (ii) Cracking.
- b) Name the impurities that should be removed from the fuels after refinery. Give a reason for your answer.
- c) After fractional distillation, the fractions obtained may contain complex hydrocarbons. This means there is need for them to be broken down into useful products.
 - (i) Identify the process that aids the breakdown of long chain hydrocarbons.
 - (ii) Why is this process important?
- d) Explain the process of petroleum oil refinery. Diagram not essential.

15.3.NATURAL GAS. What is the major component of natural gas?

Natural gas is mainly methane. It is often found with petroleum. It is formed in the same way. But high temperatures and high pressures caused the compounds to break down to gas.

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Natural gas is a fossil fuel formed over millions of years from dead remains of ancient marine organisms and other organic matter such as zoo or Phyto-plankton, algae that settled on the ocean floor buried by layers of sediments.

Overtime, high heat and pressure transformed this dead organic matter into hydrocarbons (crude oils), which eventually produce natural gas.

Composition of Natural gas

Natural gas constitutes of mainly 90% methane and 5% ethane, and propane.

Common impurities found along with natural gas may include carbon dioxide gas, nitrogen gas, and hydrogen sulphide.



Fig 1.2 shows the processing plant for natural gas

15.4.BIOGAS.

Biogas is a gaseous renewable energy source produced from raw materials such as agricultural waste, manure, municipal waste, plant material, sewage, green waste, wastewater, and food waste. Biogas is produced by anaerobic digestion with anaerobic organisms or methanogens inside an anaerobic digester, biodigester or a bioreactor. . The gas composition is primarily methane (CH_4) and carbon dioxide (CO_2) and may have small amounts of hydrogen sulfide (H_2S), moisture and siloxanes.

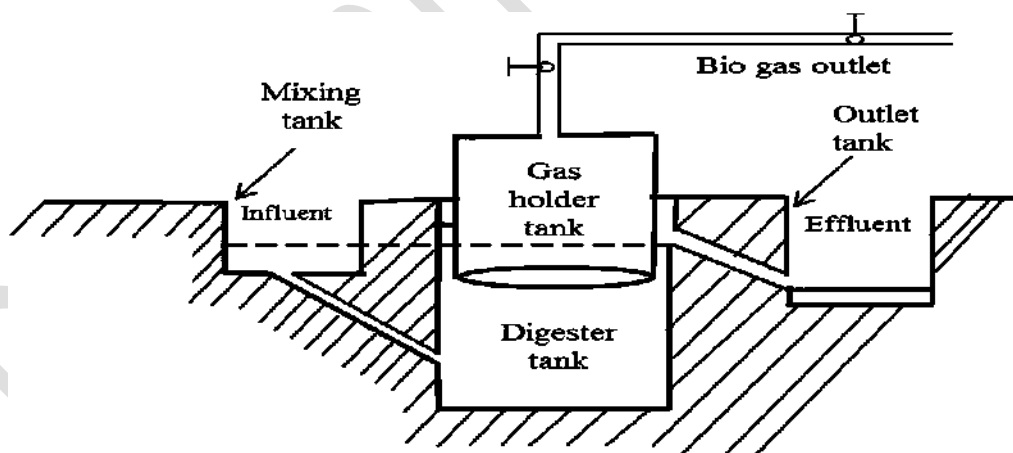


Fig 1.4. A simple schematic of a biogas production plant

ACTIVITY: 1.4(a).

Discussion Questions:

- Biogas is always obtained from biomass in the presence of anaerobic digesters. Define the term "**Biogas**" as an energy source.
- Explain why biogas is termed as a **renewable energy** source whereas natural gas is called **non – renewable energy** source.
- Name the different sources of biomass from which biogas fuel is obtained.
- Identify the:

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- (i) main gaseous component(s) of biogas.
- (ii) other gaseous components produced as impurities that must be removed when formed.
- e) Explain how biogas is formed from biomass.
- f) Differentiate between **influent** and **effluent** of biomass in a biogas digester.
- g) Describe the different uses of Biogas in our day – to – day life activities.

ACTIVITY: 1.4(b) Project work on constructing Biogas digester.**What you need:**

Use locally available materials around you.

What to do:

In groups or individually construct your own biogas digester.

1. Suggest a title (theme) of your project to meet your study findings.
2. Do research and gather information (data) about your project title.
3. List down the requirements you need to achieve the intended project.
4. Plan and design your own procedural steps for the project.
5. Follow up your instructions to start building your intended project.
6. Make test and troubleshoot on your project, to help you evaluate where the problem or failure might be coming from.
7. Compile and document your project using “word” or Pdf for future reference.

NOTE: *In every single step of your project, always present them first before your teacher for some advice or a go ahead with that activity or to make some adjustment.*

15.5. CLASSIFICATION OF CARBON –COMPOUNDS.

Carbon is one of the most successful elements on the planet Earth. Its existence on Earth has supported many lives and without its life would not have come into existence. Carbon has the ability to form many structural and useful organic compounds which are found in the hydrocarbons, food stuffs, bodies of living organisms but excluding inorganic carbon-compounds like carbonates, oxides.

Carbon has the ability to join each other and other elements to form rings or long chains in their structures called **Organic compounds**. The study of organic compounds and its applications as a branch of Chemistry is called **Organic Chemistry**.

The carbon – organic containing compounds are hydrocarbons, alcohols, carboxylic acids etc.

15.6. HOMOLOGOUS SERIES:

Homologous series: Refers to a group of organic compounds with the same general formula, similar structural formulae, similar chemical properties in which the successive members differ from the next by methylene group. (– CH₂ – group).

Examples include **alkanes**, **alkenes**, **alcohols**, and **carboxylic acids** etc.

Each homologous series differs in their chemical properties from the next series by a “**functional group**”.

Functional group: refers to an atom or a compound which determine the chemical properties of any given homologous series.

General characteristics of Homologous series.

Property	Alkanes	Alkenes	Alcohols	Carboxylic acids
Physical state at room temperature	Gases (first four lower molecular mass) liquids (medium molecular	Gases (first four lower molecular mass) liquids (medium molecular	Most of them exist as liquids and some as solids	Most of them exist as liquids and some as solids

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	weight) and solids (higher molecular weight)	weight) and solids (higher molecular weight)		
Solubility in water	Immiscible with (insoluble in) water but soluble in organic solvents	Immiscible with (insoluble in) water but soluble in organic solvents	Miscible with (soluble in) water and organic solvents to form solution	Miscible with (soluble in) water and organic solvents to form solution but the higher members become immiscible with water
Boiling point	Higher boiling points than alkenes	Lower boiling points than alkanes	Higher boiling points than alkanes due to the presence of hydrogen bonding	Higher boiling points than alkanes due to the presence of hydrogen bonding
Electrical conductivity	Does not conduct electricity	Does not conduct electricity	Does not conduct electricity	Conducts electricity
Odour (smell)	Odorless (has no distinct smell)	Typically, pungent or sweet	Typically, pungent or sweet	Typically, pungent
Density	Less dense than water	Less dense than water	Almost equal to density of water	Dense than water

15.7. HYDROCARBONS

The groups (families) of homologous series such as alkanes and alkenes belong to a group of organic compounds called **Hydrocarbons**.

Hydrocarbons are organic compounds made up of carbon and hydrogen only.

There are two major types of hydrocarbons, namely:

1. Saturated hydrocarbons
2. Unsaturated Hydrocarbons

Properties	a) Saturated Hydrocarbons	b) Unsaturated hydrocarbons
Type of bond	single C – C bond	double C = C bond
Family	Alkanes	Alkenes
Suffix	– ane	– ene
General formula	C_nH_{2n+2}	C_nH_{2n}

ACTIVITY: 1.6(a)

What to do:

In groups or individually, carryout the following research about homologous series of organic compounds. share and compare your findings with each other group in your class. Write them down in your notebook.

Discussion Questions:

- Different organic compounds are placed into group called the homologous series because of the difference in their functional groups. Define the following terms:
 - Homologous series.
 - Functional group.
- How does:
 - each member of the same homologous series differs from the other successive members?

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- (ii) each member of the same homologous series similar to the other successive members?

Table showing the general formulae, functional groups of the first homologous series

Homologous series	Alkanes	Akenes	Alcohols	Carboxylic acids
General formula	C_nH_{2n+2}	C_nH_{2n}	$C_nH_{2n+1}OH$	$C_nH_{2n} + O_2$
Functional group	C – C <i>single bonds</i>	$\diagup C = C \diagdown$ <i>double bond</i>	– OH <i>Hydroxyl group</i>	– COOH <i>Carboxyl group</i>
n = C ₁	CH ₄	None	CH ₃ OH	HCOOH
n = C ₂	C ₂ H ₆	C ₂ H ₄	C ₂ H ₅ OH	CH ₃ COOH
n = C ₃	C ₃ H ₈	C ₃ H ₆	C ₃ H ₇ OH	C ₂ H ₅ COOH
n = C ₄	C ₄ H ₁₀	C ₄ H ₈	C ₄ H ₉ OH	C ₃ H ₇ COOH
n = C ₅	C ₅ H ₁₂	C ₅ H ₁₀	C ₅ H ₁₁ OH	C ₄ H ₉ COOH
n = C ₆	C ₆ H ₁₄	C ₆ H ₁₂	C ₆ H ₁₃ OH	C ₅ H ₁₁ COOH

15.8. NOMENCLATURE (NAMES) OF HOMOLOGOUS SERIES

In Uganda, every born baby is usually given a sir name, in line to the tribe from which the biological parents belong. You may have learnt that a tribe is often placed under “**ethnic group**” like-wise any names of any given homologous series must be in line to the group from which the parent structure originated, we normally use *IUPAC* naming system (International Union of Pure and Applied Chemistry), to provide universal names that avoid confusion amongst chemists.

To name members of any alkanes, alkenes, alcohols and carboxylic acids we count mainly the number of carbon atoms in the general formulae as n = 1, 2, 3, 4, 5 etc in the compounds.

1. All names given to any member of **alkanes** must end with a suffix “**ane**”
2. All names given to any member of **alkenes** must end with a suffix “**ene**”
3. All names given to any member of **alcohols** must end with a suffix “**ol**”
4. All names given to any member of **carboxylic acids** must end with a suffix “**oic**”

Foot note: All the names given to any member of **alcohols** and **carboxylic acids** always replace the ending “**e**” of “**ane**” in **alkanes** with a suffix “**ol**” and “**oic**” respectively.

Homologous series Carbon No	Root name	Alkanes “ane”	Alkenes “ene”	Alcohols “ol”	Carboxylic acids. “oic”
1	Meth -	Methane	N. A	Methanol	Methanoic acid
2	Eth -	Ethane	Ethene	Ethanol	Ethanoic acid
3	Prop -	Propane	Propene	Propanol	Propanoic acid
4	But -	Butane	Butene	Butanol	Butanoic acid
5	Pent -	Pentane	Pentene	Pentanol	Pentanoic acid
6	Hex -	Hexane	Hexene	Hexanol	Hexanoic acid

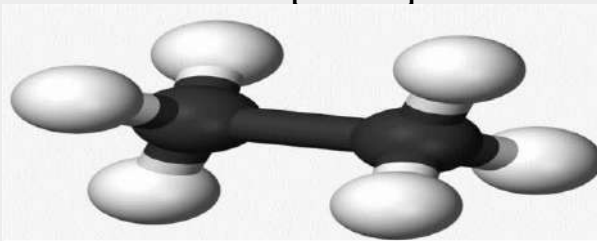
15.9. ALKANES [C_nH_{2n+2}]

Figure 1.8. Model of the structure of ethane

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General characteristics of alkanes as hydrocarbon compounds

Alkanes are saturated hydrocarbons that contain carbon – carbon single bonds. Alkanes are the first (family) group of homologous.

They conform to the general formula C_nH_{2n+2} , where $n = 1, 2, 3, 4, \dots$ etc. They contain only carbon – carbon single bonds, hence they are called saturated hydrocarbons.

Examples of alkanes include: gasoline (petrol), diesel, paraffin, naphtha, natural gas, candle wax. This means the petroleum fuels we use in our everyday activities belong to a group of homologous series called alkanes.

They are found in petroleum and natural gas. Petroleum contains alkanes with up to 70 carbon atoms.

The first four alkanes are gases at room temperature. The next twelve are liquids. The rest are solids at room temperature and pressure.

Their melting and boiling points increase with chain length because attraction between the molecules increases – so it takes more energy to separate them.

- a) Copy and complete the table below correctly to show a model of the structural formulae of the first five alkanes. Remember that each carbon atom has four (4) bonds.

No. carbon atoms	Name of alkane	Molecular formula	Structural formulae	Compressed structure
1	Methane		<pre> H H — C — H H </pre>	CH₄
2	Ethane	C ₂ H ₆	<pre> H H H — C — C — H H H </pre>	CH₃CH₃
3	Propane	C ₃ H ₈	<pre> H H H H — C — C — C — H H H H </pre>	CH₃CH₂CH₃
4	Butane	C ₄ H ₁₀	<pre> H H H H H — C — C — C — C — H H H H H </pre>	CH₃CH₂CH₂CH₃
5	Pentane	C ₅ H ₁₂	<pre> H H H H H H — C — C — C — C — C — H H H H H H </pre>	CH₃CH₂CH₂CH₂CH₃

15.9.1. PHYSICAL PROPERTIES OF ALKANES**a) Boiling points and densities of alkanes**

Boiling point increases with increase in molecular mass of each chain length, increasing the intermolecular forces in the molecules, a lot of heat will be required to break these forces.

Density alkanes increases with increase in molecular mass but decreases with branching.

Alkane	M.pt/ °C	B.pt/°C	Density (g/cm ³)
Methane	-182	-164	0.716
Ethane	-183	-89	0.546
Propane	-190	-42	0.582

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Butane	-138	-0.5	0.601
Pentane	-130	36.3	0.626
Hexane	-95	69	0.659

ACTIVITY: 1.8(b)**What to do**

In groups or individually, discuss how the increase in the number of carbon atoms affect the boiling points of the first four alkanes. Share and compare your findings with other groups in your class. Write them down in your notebook.

Discussion Questions:

- a) Copy and complete following precisely by filling in the missing information using the table below.

Alkane	Methane	Ethane	Propane	Butane	Pentane	Hexane
Molecular formula						
Structural formula						
No. carbon atom						
Physical state (at 25°C)						

- b) State and explain the trend in the:

- boiling points of the first six alkanes.
- density of the first six alkanes.

ACTIVITY: 1.8(d)**What to do**

In groups or individually, discuss on the following physical properties of alkanes. Share compare your findings with the rest of other groups in your class. Use the table below to write your comprehensive work in your note book.

Discussion Question:

- a) Copy and complete the following by explaining how alkanes behave with each physical property. Use the table below.

Physical property	Expected behaviours of alkanes
○ General formula	
○ Functional group	
○ Existence	
○ Physical state at r.t.p	
○ Boiling points	
○ Physical appearance	
○ Solubility in water	
○ Effect on blue and red litmus paper	
○ Density	
○ Electrical conductivity	
○ Ability to react	

a) ISOMERS OF ALKANES.

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Isomers are organic compounds with the same molecular formula, but different structural formulae. The more carbon atoms in a compound, the more isomers it has. Since isomers have different structures, they also have slightly different physical properties. For example, branched isomers have lower boiling points, because the branches make it harder for the molecules to get close. So, the attraction between them is less strong, and less heat is needed to overcome it.

Compare the butane molecules. Both have the same formula, C_4H_{10} , different structures.

Types of isomers in *n* - Butane

The isomers in *n* - Butane can be grouped as:

- (i) **Straight chain isomers** (unbranched chain) and (ii) **branched chain**. Consider the following isomers of *n*-Butane, C_4H_{10} . We can write their structural formula as below.

Butane, C_4H_{10} .	Open structural formula	Condensed structural formula
Isomer: A		$CH_3CH_2CH_2CH_3$ $C_1 \quad C_2 \quad C_3 \quad C_4$ (Boiling point = $0^\circ C$)
Isomer: B		$CH_3CH(CH_3)CH_3$ (Methyl group) $C_1 \quad C_2 \quad C_3$ (Boiling point = $-10^\circ C$)

ACTIVITY: 1.8(e)**What you need:**

Butane, C_4H_{10} ; isomer: A and isomer: B, your notebook.

What to do

In groups or individually, carryout the following activities to find out the different forms of isomers exhibited in alkanes. Use this as a basis to give the meaning of isomers. Share and compare your findings with the rest of the class. Write them down into your notebook

Discussion Questions:

- a) Copy and complete the table below correctly. Fill in the missing information about butane.

Physical properties	Isomer A	Isomer B
Homologous series		
Molecular mass		
Boiling point		
No. carbon on the parent structure		
Type of isomer		
Opened structural formula		
Condensed structural formula		

- b) From your discussion in part (a) above. Suggest:

- (i) What is meant by the term “**Isomer**”
 (ii) With a reason why isomer A and isomer B have different boiling points and yet they both have a general molecular mass of 58.
 c) Suggest the possible **IUPAC** name given to the following isomers of A and B.

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(i) Isomer A: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (ii) Isomer B: $\text{CH}_3\overset{\text{CH}_3}{\text{CH}}\text{CH}_3$

15.9.2. CHEMICAL PROPERTIES OF ALKANES

Generally, alkanes are very unreactive under ordinary conditions. However, under drastic conditions, they react to form new products.

a) Reaction of alkanes with chlorine gas:

Chlorine gas can react with an alkane gas like methane to form chloromethane. This reaction requires Ultra – violet light (U.v light) from sunlight. In the dark, no reaction takes place, hence a **photochemical reaction** just like photosynthesis. The reaction between chlorine gas and methane gas is called **chlorination** or **substitution** reaction.

ACTIVITY:1.8(f)

What to do:

In groups or individually, discuss the following by researching on the chemical reaction of chlorine gas with methane gas under limited chlorine gas using sunlight. Share and compare your work with other groups in your class.

Discussion Questions:

a) Copy and complete the table below precisely by filling in the missing information about the chemical reaction between chlorine gas and methane gas.

No. chlorine atom	Chlorine gas	Methane gas	Major product	Minor product
1	Cl_2	CH_4	CH_3Cl	HCl
2	Cl_2	CH_4	CH_2Cl_2	2HCl
3	Cl_2	CH_4	CHCl_3	3HCl
4	Cl_2	CH_4	CCl_4	4HCl

b) From your discussion, in part (a) above. Write a:

(i) word equation for the reaction between chlorine gas and methane to form chloromethane gas and hydrogen chloride gas. Illustrate with condition for the reaction.

(ii) chemical for the reaction in (b) (i) above with state symbols.

c) Write the **IUPAC** Names of the following compounds:

(i) CH_3Cl (ii) CH_2Cl_2 (iii) CHCl_3 (iv) CCl_4

b) Combustion of alkanes in air.

All alkanes on combustion, undergo **complete combustion** in supply of excess oxygen, forming carbon dioxide and water vapour, and giving out plenty of heat energy. So, they are used as fuels. Methane burns the most easily. If there is not enough oxygen, the alkanes undergo **incomplete combustion**, giving out poisonous carbon monoxide gas and less heat energy.

ACTIVITY: 1.8(g) To prove that water and carbon dioxide gas are formed when candle wax burns in oxygen.

What you need:

Candle wax, glass funnel, delivery tubes, boiling tubes, cold water, calcium hydroxide solution (lime water), anhydrous copper (II) sulphate, petri dish and rubber bungs.

Caution! Take great care on the broken glass materials since they can cut the skin.

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What to do:

In groups:

1. Set the apparatus as shown below in the figure 1.8.5.
2. Light the candle wax and leave it to continue with burning for 15 minutes. Making observations on what takes place in tube A & B.
3. Add a few solid powders of anhydrous copper (II) sulphate into tube A, after the experiment. Make your observation.

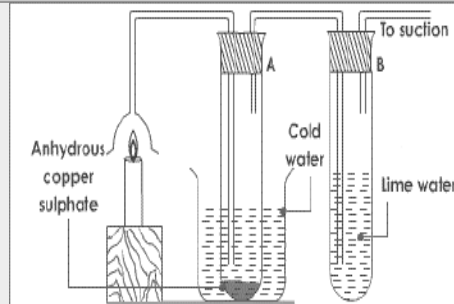


Fig 1.8.5. To prove that carbon dioxide and water are formed when candle wax burns

Discussion Questions:

- a) State what was observed in the boiling tube A and B when candle wax burnt.
 - (i) Tube A _____
 - (ii) Tube B _____
- b) Name the compounds formed in the tube A and B when candle wax burnt.
 - (i) Tube A _____
 - (ii) Tube B _____
- c) Identify the major elements which constitute the candle wax that formed the compounds named in (b) above.
- d) Write:
 - (i) a word equation for the reaction that took place when candle wax burnt in air.
 - (ii) a chemical equation for the reaction in (d) (i) above with state symbols.
- e) Why was ice – cold water used in this experiment?
- f) Write:
 - (i) a word equation for the reaction that took place in the tube B containing calcium hydroxide solution and gaseous product formed.
 - (ii) a chemical equation for the reaction in (f) (i) above with state symbols.

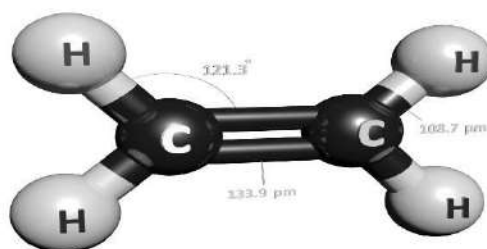
15.10. ALKENES (C_nH_{2n})

Figure 1.9. Model of the structure of ethene

General characteristics of alkenes as hydrocarbon compounds:

Alkenes are unsaturated hydrocarbons with carbon – carbon, ($C = C$) double bonds. They conform to the general formula C_nH_{2n} where $n = 1, 2, 3, 4, \dots$ number of carbon atoms.

They contain only carbon – carbon double bonds, hence they are called unsaturated hydrocarbons.

The first four alkenes are gases at room temperature. The next twelve are liquids. The rest are solids at room temperature and pressure.

Alkenes are known as olefins meaning oil forming organic compounds.

The melting and boiling points of alkenes increase with chain length because attraction between the molecules increases – so it takes more energy to separate them.

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ACTIVITY: 1.9(a) To make models of the structure of alkenes as hydrocarbons.**What you need:**

Access to small raw fruits, toothpicks and your notebook.

What to do:

In groups or individually, conduct the following research to investigate more about alkanes. Share and compare your findings with other groups in your class. Write the down in your notebook.

Discussion Questions:

a) Copy and complete the table below correctly. Use this general formula C_nH_{2n}

No. carbon atoms	Root name	No. Hydrogen atoms	Molecular formular	Add suffix "ene" to complete the name
2	Eth-	4		
3	Prop-	6		
4	But-	8		
5	Pent-	10		
6	Hex-	12		
7	Hept-	14		
8	Oct-	16		
9	Non-	18		
10	Dec-	20		

b) Copy and complete the table below correctly to show a model of the structural formulae of the first five alkenes. Remember that each carbon atom has four (4) bonds.

No. carbon atoms	Name of alkene	Molecular formular C_nH_{2n}	Open structural formula	Condensed structure
2	Ethene	C_2H_4	<pre> H H C = C H H </pre>	
3			<pre> H H H - C - C = C H H H </pre>	
4		C_4H_{10}		$CH_3CH_2CH=CH_2$
5	Pentene			
6	Hexene			

15.10.1. PHYSICAL PROPERTIES OF ALKENES**ACTIVITY: 1.9 (b) To investigate the general Physical properties of alkenes.****What to do**

In groups or individually, discuss on the following physical properties of alkenes. Share compare your findings with the rest of other groups in your class. Use the table below to write your comprehensive work in your note book.

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Discussion Question:

Copy and complete the following by explaining how alkanes behave with each physical property. Use the table below.

Physical property	Expected behaviours of alkanes
1) General formula	
2) Functional group	
3) Existence	
4) Physical state at (r.t.p)	
5) Melting points	
6) Boiling points	
7) Physical appearance	
8) Solubility in water	
9) Effect on blue and red litmus paper	
10) Density	
11) Electrical conductivity	
12) Ability to react	

a) BOILING, MELTING POINTS AND DENSITY OF ALKENES

Melting or Boiling points of alkenes increase with increase in the molecular mass which increases the intermolecular forces of attraction between the molecules hence more heat is required to break these forces leading to increased melting or boiling points.

Alkene	M. Pt / °C	B. Pt / °C	Density (g/cm ³) at 25°C
Ethene	-169.2	-103.7	0.570
Propene	-185.2	-47.6	0.597
But – 1 – ene	-185.2	-6.3	0.617
Pent – 1 – ene	-165.0	30	0.643
Hex – 1 – ene	-140.0	63	0.654

ACTIVITY: 1.9(c)**What to do**

In groups or individually, discuss how the increase in the number of carbon atoms affect the boiling points of the first four alkanes. Share and compare your findings with other groups in your class. Write them down in your notebook.

Discussion Questions:

a) Copy and complete following precisely by filling in the missing information using the table below.

Alkane	Ethene	Propene	But – 1 -ene	Pent – 1 – ene	Hex – 1- ene
No. carbon atom					
Molecular formula					
Structural formula					
Physical state at 25°C					

b) State and explain the trend in the:

- boiling points of the first six alkenes.
- melting points of the first six alkenes.

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(iii) density of the first six alkenes.

b) ISOMERS OF ALKENES.

S No	Structure
1	$\text{CH}_3-\text{CH}=\text{CH}_2$
2	$\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$
3	$\begin{array}{c} \text{CH}_2=\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$
4	$\begin{array}{c} \text{CH}_2=\text{CH}-\text{CH}-\text{CH}_2-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$

IUPAC NAME

Prop -1 - ene

But - 2 - ene

2 -methyl prop-1 - ene

3-methyl pent - 1 - ene

Alkenes exhibit both **chain** isomers and **position** isomers. The positions of the **functional** group of an alkene in its structures can be (-1-ene or - 2 - ene).

ACTIVITY:1.9(c)

What to do:

In groups or individually, discuss the following. Share and compare your findings with the rest in your class.

Discussion Questions:

a) Using the table 1.9.3 above. Copy and complete the table below and answer by filling in the spaces provided precisely.

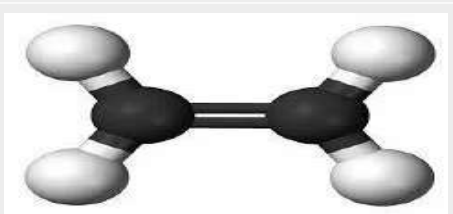
IUPAC name	Prop -1 -ene	But -2-ene	2-methyl prop -1 -ene	3-methyl Pent -1 -ene
Structural formula				
Position of methyl group (-CH ₃)				
Position of C = C double bond				
No. carbon atoms				

b) State **one** similarity and difference between **but - 2 - ene** and **2 - methyl prop - 1 - ene**.

(i) Similarity. (ii) **Difference**

c) What general term is given to the two compounds in (b) above?

15.11. ETHENE GAS:



A model of the structure of Ethene gas

ACTIVITY: 1.10(a) Laboratory Preparation of Ethene Gas by the Dehydration of ethanol.

What you need:

Round bottomed flask, delivery tube, heat source, gas jar, water trough, glass jar stand, ethanol, concentrated Sulphuric acid, bromine water, sands and acidified potassium manganate (VII) solution.

CAUTION!

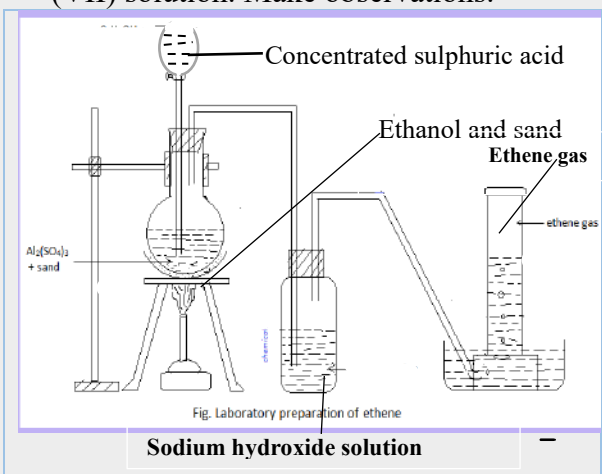
- Perform this experiment in fume chamber, to avoid inhaling poisonous sulphur dioxide gas produced by concentrated Sulphuric acid.
- Concentrated Sulphuric acid is highly corrosive, handle it with care.
- Avoid putting ethanol near fire point, since it is highly flammable.

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- This experiment **MUST** be performed only by a teacher or a technician and **NOT** by learners.

What to do:

- Arrange the setup of the apparatus as shown below.
- Place a few sands into the flask, add ethanol a half – way the flask
- Run concentrated sulphuric acid through a dropping funnel into the flask containing a mixture of sand aluminium sulphate and ethanol to dehydrate off water from ethanol. Aluminium sulphate facilitates dehydration of ethanol, and sand prevents frothing formation in the flask.
- Heat the mixture with great care until bubbles of a colourless gas evolved.
- The gas is then passed into a washed bottle of sodium hydroxide solution to remove any carbon dioxide gas produced and collected over water.
- Bubble the collected gas through bromine water and acidified potassium manganate (VII) solution. Make observations.

**Discussion Questions**

- a) Copy and complete the following by filling in the missing information precisely.

Physical property of ethene gas	Description
○ Colour	<i>Colourless gas</i>
○ Odour (smell)	<i>Has pungent smell</i>
○ Solubility in water	<i>Immiscible with water</i>
○ Density	<i>Less dense than water</i>

- b) State the role of each of the following in the experiment.
- Concentrated Sulphuric acid.
 - Sodium hydroxide solution.
 - Sand.
- c) State the temperature at which the reaction takes place.
- d) Ethene gas was bubbled through the following solutions. State what was observed in each case. Use the table below.

Test	Expected observations
(i) Bromine water	<i>Turns from brown to colourless solution</i>
(ii) Acidified potassium manganate (VII) solution.	<i>Purple solution of manganate (VII) turns to colourless solution</i>
(iii) Acidified potassium dichromate (VI) solution.	<i>Orange solution of dichromate (VI) turns to green solution</i>

- e) State the role of each of the following reagents.
- Bromine water.
 - Acidified potassium manganate (VII) solution.
 - Acidified potassium dichromate (VI) solution
- f) Write:

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- (i) a word equation for the reaction between bromine water and ethene gas to form 1,2-dibromoethane.
- (ii) the chemical equation for the reaction in (f) (i) above with state symbols.
- g) Write:
- (i) a word equation for the reaction leading to the formation of ethene gas.
- (ii) the chemical equation for the reaction in (g) (i) above with state symbols.

15.12. USES OF ALKENES

ACTIVITY:1.11(a) To find out the various applications (uses) of alkenes**What you need:**

Access to computer or smartphone for internet, textbooks, and your notebook.

What to do:

In groups or individually, visit a computer library or using your smartphone, carryout research to explore the various applications (uses) alkenes. Share and compare your findings with the rest of the class. Write them down in your notebook.

Discussion Questions:

- a) Explain the different uses of alkenes of in our day – to – day life activities.

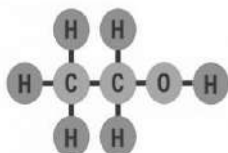
15.13. ALCOHOLS ($C_nH_{2n+1}OH$) What are alcohols?

The alcohols are the family of organic compounds that contain carbon, hydrogen and oxygen atoms. The general formula of alcohols is $C_nH_{2n+1}OH$ where n is the carbon number. The **OH** is the functional group of alcohols, the *Hydroxyl group*.

ACTIVITY: 1.12(a)**Discussion Questions:**

- a) Copy and complete the following by filling in the missing information correctly. Use the formula $C_nH_{2n+1}OH$ in the table below.

No. carbon atom	n=1	n=2	n=3	n=4	n=5
Root Name	Methan-	Ethan-	Propan-	Butan-	Pentan-
Alcohol name					
$C_nH_{2n+1}OH$					
Molecular formula					
Molecular mass	31	45	59	73	87
Structural formula					
Boiling point/ $^{\circ}C$	65	78	87	117	137

a) ETHANOL

Ethanol can be prepared in two methods, one by **biological** method and two by **Chemical** method. **Biological method** of preparing ethanol involves fermentation process.

Fermentation: Refers to the anaerobic break down of carbohydrates such as sugars and starch into ethanol using biological enzymes from yeast or bacteria. It occurs in the absence of oxygen.

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ACTIVITY: 1.12(b) Laboratory preparation of Ethanol by fermentation of glucose**What you need:**

Measuring cylinder, boiling tube, test tube, delivery tube, yeast, saturated glucose solution, rubber bungs.

What to do: In groups or individually,

1. Arrange the setup of the apparatus as shown in fig 1.12(b)
2. Measure 50cm³ of glucose solution into a boiling tube.
3. Leave the setup in a warm place for three days. Make observation.
4. Record the observation after three (3) days.

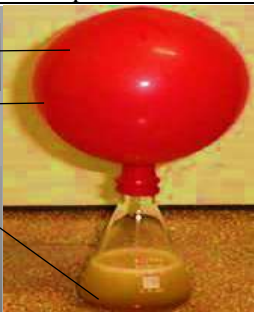


Fig.1.12(b) Fermentation of glucose solution

Discussion Questions:

- a) State what was observed in the test tube containing lime water.
- b) Identify the gas that is responsible for the colour change of the lime water.
- c) What are the conditions necessary for fermentation process to take place?
- d) Write:
 - (i) a word equation for the reaction leading to formation of ethanol and carbon dioxide from glucose solution.
 - (ii) a chemical equation for the reaction that took place in (d) (i) above with state symbols.

Gas X

Inflated
balloonA mixture
of Yeast
and glucose
solution**Discussion Questions:**

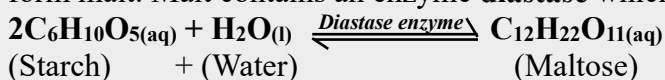
- a) Identify gas X and name one reagent which can be used to identify the gas. State what would be observed if the reagent named above is treated with the gas.
- b) State your observation if the reagent named in (a) above is treated with the gas X.
- c) State the process that led to production of gas X.
- d) What are the necessary conditions for the process stated in (c) above?

16. 2. Preparation of Ethanol from starch

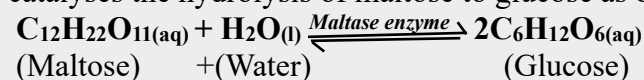
Sources of starch: Millets, sorghum, maize corn, and cassava.

What happens:

When they are grounded and mixed with water. Starch eventually undergoes hydrolysis to form malt. Malt contains an enzyme **diastase** which hydrolyses starch to maltose.



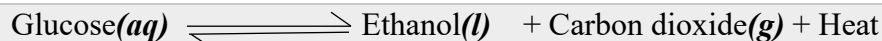
When yeast is added to the **maltose** at room temperature and left to ferment for 3 days, **glucose** molecules are formed. Yeast contains two enzymes, **maltase** and **zymase**. **Maltase** catalyses the hydrolysis of maltose to glucose as below.



Zymase catalyses the breakdown of glucose into ethanol, carbon dioxide, producing heat in the process.



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**ACTIVITY:1.12(c)****What you need:**

Cassava flour, water, millet (sorghum) yeast, polythene bag (banana leaves), distillation pot (flask), copper (laboratory) Liebig condenser, heat source, filter funnel, glass (beakers) bottles.

What to do: In groups

1. Mix the grounded cassava flour with clean water to form a slolid paste.
2. Transfer the paste into the drum or ground and cover it with polythene bag or banana leaves. Leave it to stay for 1 week. Make observation.
3. After 1 week, roast the fermented cassava paste and dry it in the sun.
4. Mix the roasted fermented cassava paste with millet (sorghum) yeast.
5. Transfer the mixture into a big drum and add adequate amounts of water that cover up the mixture. Leave it to stay for 3 days. Observe the heat produced if it is exothermic (heat loss) or endothermic (heat gain).
6. After 3 days, filter the mixture to obtain liquid filtrate.
7. Arrange the set of the apparatus for distillation process of the liquid filtrate.
8. Carryout the distillation process to obtain pure ethanol as distillate.

Discussion Questions:

a) What is the purpose of:

- (i) Leaving the paste in the drum for 1 week before roasting?
- (ii) Mixing the roasted paste with millet (sorghum) yeast?
- (iii) Covering the mixture of roasted cassava paste and yeast for 3 days?

b) Name the:

- (i) process under investigation in this experiment.
- (ii) enzymes found in the cassava paste before adding yeast.
- (iii) enzymes found in the millet (sorghum) yeast that catalyzed the process.

c) Write:

- (i) a word equation for the reaction that took place when starch is hydrolyzed to maltose sugars.
- (ii) a balanced chemical equation for the reaction that took place in (c) (i) with state symbols.

d) Write:

- (i) a word equation for the reaction that took place when maltose is hydrolyzed to glucose sugars.
- (ii) a balanced chemical equation for the reaction that took place in (d) (i) with state symbols.

e) Write:

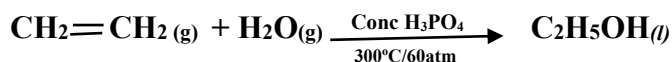
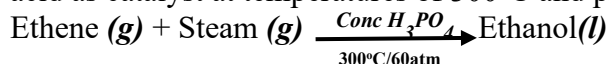
- (i) a word equation for the reaction that took place when glucose is hydrolyzed to ethanol, carbon dioxide gas and heat.
- (ii) a balanced chemical equation for the reaction that took place in (e) (i) with state symbols.

b) Industrial manufacture of Ethanol (Chemical method):

In Uganda, there are local brewing industries such as Nile breweries in Jinja, Uganda breweries limited in Luzira Kampala among others produced ethanol on large (commercial) scale for local and international consumers.

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Most of them obtain their raw materials from starch plant materials to prepare beer (ethanol). However, others produce ethanol on large scale by hydrolysis of Ethene. Ethene can be converted directly to ethanol by steam cracking of long chain alkanes. After cracking, ethene gas is passed through steam in the presence of concentrated phosphoric acid as catalyst at temperatures of 300°C and pressure of 60 atmosphere.

**ACTIVITY:1.12(d): Field study to a nearby chemical plant for brewing Ethanol.**

What to do: In groups

1. Organize a field trip along with your teacher to a nearby brewing industry (factory).
2. Prepare topical questions (questionnaire) that you will use to engage the laboratorians in their respective fields.

Field study task:

- a) Prepare and write down a final report of your findings from the factory. Your final report should include the following:
 - (i) title of the filed study.
 - (ii) raw materials used in the process of making ethanol.
 - (iii) descriptive stages of the entire process of making ethanol.
- b) Present your final report to the rest of the class, HOD Chemistry and the Headteacher. Use word (Power point) document of the computer.

PHYSICAL PROPERTIES OF ETHANOL**ACTIVITY:1.12(e)**

What to do: In groups

Discuss how the physical properties of Ethanol are affected by each of the following. Share and compare your findings with the rest of the class. Write them down in your notebook.

Discussion Questions:

- a) Copy and complete the table below by filling in the missing information precisely.

Property	Expected behaviour of ethanol
o Colour	
o Odour (smell)	
o Density	
o Boiling point.	
o Physical state at r.t.p	
o Effect on litmus paper	
o Solubility in water	

CHEMICAL PROPERTIES OF ETHANOL**a) Combustion of ethanol.**

Alcohols such as ethanol burn in plenty supply of air with a yellow non – sooty flame to form carbon dioxide gas, water and heat energy.

**b) Dehydration reaction of Ethanol**

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When concentrated Sulphuric acid is heated at temperatures between 170 – 180°C, ethanol is dehydrated to form ethene gas and water.

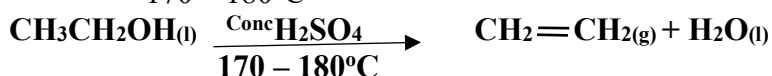
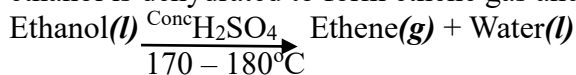


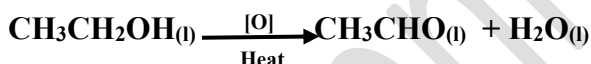
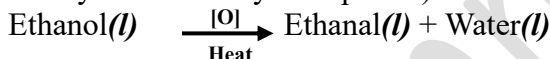
Table showing what happens during the dehydration of ethanol

Dehydration of Ethanol	Expected behaviour
(i) Conditions of the reaction	Concentrated sulphuric acid and temperatures of 170 – 180°C
(ii) Final product(s) formed	Ethene gas and water
(iii) Identification test of the product	Use of liquid bromine: Turns from brown to colourless Acidified potassium manganate (VII) solution: Turns from purple to colourless
(iv) Word equation	Ethanol(l) $\xrightarrow[170-180^{\circ}\text{C}]{\text{ConcH}_2\text{SO}_4}$ Ethene(g) + Water(l)
(v) Chemical equation	$\text{CH}_3\text{CH}_2\text{OH}(l) \xrightarrow[170-180^{\circ}\text{C}]{\text{ConcH}_2\text{SO}_4} \text{CH}_2 = \text{CH}_2(g) + \text{H}_2\text{O}(l)$

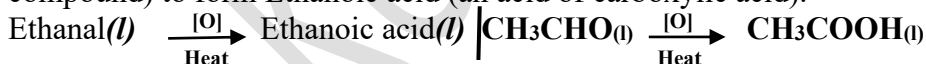
c) Oxidation reaction of Ethanol:

An alcohol such as ethanol can be oxidized using strong oxidizing agent like acidified potassium manganate (VII) solution [O], to form the corresponding carbonyl compound then finally the corresponding carboxylic acid. This reaction occurs in two stages.

The **first stage** involves **partial** oxidation of ethanol on heating to form ethanal (an aldehyde of carbonyl compound) and water.



The **second stage** involves **complete** oxidation of ethanal (an aldehyde of carbonyl compound) to form Ethanoic acid (an acid of carboxylic acid).



USES OF ALCOHOLS

Used as disinfectant such as hand sanitizers for killing germs

Ethanol blended with gasoline serves as fuels in some vehicles to reduce greenhouse gas emissions.

Ethanol is used in cleaning products, to disinfect surfaces, equipment and medical devices.

Ethanol is used as food additive, a solvent and a preservative in the food industry.

Ethanol is used as biofuels reducing total dependence on fossil fuels

Ethanol is used as a solvent and a reagent in the production pharmaceutical products.

Ethanol is used as consumable products in beverages such as beer, spirits, wine.

ETHANOL AND THE SOCIETY.

a) Individual impacts of excessive consumption of ethanol

It leads to total dependence and addiction to an individual.

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Too much consumption can lead to chronic health related problems such as liver disease, heart disease and increase risk of certain cancers.

Excessive consumption of ethanol, exacerbates mental health conditions of a person causing depression and anxiety.

Excessive consumption of ethanol, leads to family and relationship impairments.

b) **Societal impacts of excessive consumption of ethanol**

Drinking while driving cars, leads to fatal accidents in the society.

Excessive consumption of alcohol disrupts family dynamics and community relationship.

Excessive consumption of ethanol, can contribute to social issues such as domestic violence, crime, and homelessness.

1.13. CARBOXYLIC (ALKANOIC) ACIDS [$C_nH_{2n}O_2$]

Carboxylic acids are grouped as organic acids. Alkanoic acids form a homologous series with the general formula $C_nH_{2n}O_2$. The functional group $-COOH$ is also called the carboxyl group. Most organic acid compounds such as citrus fruits (lemon and oranges). Sour milk (yoghurt), Vinegar, tea leaves, Aspirin are examples of carboxylic acids.

The names (Nomenclature) of every member of carboxylic acids in the homologous series come from their corresponding alkane, whereby the ending “e” in alkanes is replaced with the ending “oic acid” in carboxylic acids.

NOMENCLATURE (NAMES) OF CARBOXYLIC ACIDS

No. carbon atoms (n)	Root name	No. Hydrogen atoms	Molecular formula	Add “oic acid” to complete the name.
n = 1	Methan-	2	CH_2O_2	Methanoic acid
n = 2	Ethan-	4	$C_2H_4O_2$	Ethanoic acid
n = 3	Propan-	6	$C_3H_6O_2$	Propanoic acid
n = 4	Butan-	8	$C_4H_8O_2$	Butanoic acid
n = 5	Pentan-	10	$C_5H_{10}O_2$	Pentanoic acid

- b) Copy and complete the table below correctly to show a model of the structural formulae of the first four acids. Remember that each carbon atom has four (4) bonds.

No. carbon atoms	Name of acid	Molecular formula	Structural formulae	Compressed structure
1	Methanoic acid		$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{C}-\text{OH} \end{array}$	HCOOH
2	Ethanoic acid	$C_2H_4O_2$	$\begin{array}{c} \text{H} \quad \text{O} \\ \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{OH} \\ \\ \text{H} \end{array}$	CH_3COOH
3			$\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{OH} \\ \quad \end{array}$	CH_3CH_2COOH

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			H H	
4		C ₄ H ₈ O ₂	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & & \text{O} & \\ & & & & & // & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - & \text{C} & - \text{OH} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & & & \end{array} $	CH ₃ CH ₂ CH ₂ COOH

SOURCES AND COMMON NAMES OF CARBOXYLIC ACIDS

a) Copy and complete the table below precisely by writing the *IUPAC* names and structural formulae of the following carboxylic acids.

Carbon atoms	Common name	Source of the acid	IUPAC Name	Molecular formula	Structural formula
1	Formic acid	Stinger Bees	Methanoic acid	C ₂ H ₂ O ₂	HCOOH
2	Acetic acid	Vinegar	Ethanoic acid	C ₂ H ₄ O ₂	CH ₃ COOH
3	Propionic acid	Sour milk	Propanoic acid	C ₃ H ₆ O ₂	C ₂ H ₅ COOH
4	Butyric acid	Sour milk and animal colon.	Butanoic acid	C ₄ H ₈ O ₂	C ₃ H ₇ COOH
5	Valeric acid	Roots of valerian plants	Pentanoic acid	C ₅ H ₁₀ O ₂	C ₄ H ₉ COOH
6	Caproic acid	Goat's, cow's milk, coconut oil	Hexanoic acid	C ₆ H ₁₂ O ₂	C ₅ H ₁₁ COOH
7	Enanthic acid	Fats and oils	Heptanoic acid	C ₇ H ₁₄ O ₂	C ₆ H ₁₃ COOH
8	Caprylic acid	Coconut or palm oil	Octanoic acid	C ₈ H ₁₆ O ₂	C ₇ H ₁₅ COOH

b) Copy and complete the following by suggesting the possible sources of the following carboxylic acids. Use internet to access possible details.

Plant acids	Source	Animal acids	Source
Citric acid	Citrus fruits	Stearic acid	Animal fats
Oxalic acid	Plant species of oxalis	Palmitic acid	Palm oil and vegetable oils
Malic acid	Wine, fruits, vegetables like carrots, peas, potatoes		
Tannic acid	Nutgalls of oak trees	Oleic acid	Animal and vegetable oils (Olive oil)
Salicylic acid (Aspirin)	Vegetables like cauliflowers and fruits like watermelons, bark and leaves of willow and poplar trees		

PHYSICAL PROPERTIES OF ALKANOIC ACIDS

a) BOILING AND MELTING POINTS OF ALKANOIC ACIDS

The melting or boiling points of carboxylic acids generally increase with increasing molecular mass of the acids. All carboxylic acids associate to each other through hydrogen

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bonding. This increases their melting and boiling points as the number of hydrogen bonds increases with length of the carbon chain and functional group.

Acid	M.pt/ °C	B.pt/°C	Density (g/cm ³)
Methanoic acid	8.4	101	1.220
Ethanoic acid	16.6	118	1.049
Propanoic acid	-20.8	141	0.992
Butanoic acid	-6.5	164	0.964
Pentanoic acid	-34.5	186	0.939
Hexanoic acid	-4.0	205	0.659
Benzoic acid	122	266	1.27

ACTIVITY: 1.13(b)**Discussion Questions:**

a) Copy and complete following precisely by filling in the missing information in the table below.

Carboxylic acid	Methanoic acid	Ethanoic acid	Propanoic acid	Butanoic acid	Pentanoic acid
Molecular formula					
Structural formula					
No. carbon atom					
Physical state (at 25°C)					

b) State and explain the trend in the:

- boiling points of the first five carboxylic acids.
- melting points of the first five carboxylic acids.

ACTIVITY: 1.13(c)**What to do**

In groups or individually, discuss on the following physical properties of Carboxylic acids. Share compare your findings with the rest of other groups in your class. Use the table below to write your comprehensive work in your note book.

Discussion Question:

a) Copy and complete the following by explaining how carboxylic acids behave with each physical property. Use the table below.

Physical property	Expected behaviours of alkanoic acids
○ General formula	
○ Functional group	
○ Existence	
○ Physical state at r.t.p	
○ Melting points	
○ Boiling points	
○ Solubility in water	
○ Effect on blue and red litmus paper	
○ Density	
○ Electrical conductivity	

CHEMICAL PROPERTIES OF CARBOXYLIC ACIDS

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a) Esterification of carboxylic acids.

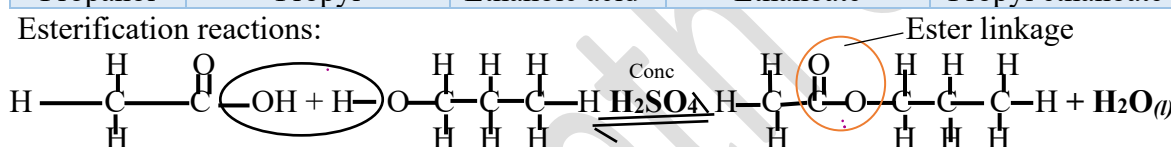
Esterification refers to the reaction between alcohols and carboxylic acids in the presence of concentrated sulphuric acid forming ester and water. Esters have sweet fruity smell in the compounds in which they form. The names of all esters are derived from the alcohols and carboxylic acids which form them. Consider the table below:

Note these points:

- Two molecules join together form a larger molecule called **Ester**, with the loss of a small molecule, water. So, this reaction is called **condensation reaction**.
- The reaction is **reversible**, and **sulfuric acid** acts a catalyst.
- The alcohol part **comes first** in the name – but **second** in the formula.
- Many esters have attractive smells and tastes. So, they are added to shampoos and soaps, Jellies for their smells (Scents), and to ice cream and other foods as flavourings.

Alcohol	Part of ester name from an alcohol	Carboxylic acid	Part of ester name from the acid	Complete name of Ester
Methanol	Methyl	Ethanoic acid	Ethanoate	Methyl ethanoate
Ethanol	Ethyl	Methanoic acid	Methanoate	Ethyl methanoate
Propanol	Propyl	Ethanoic acid	Ethanoate	Propyl ethanoate

Esterification reactions:

**ACTIVITY:1.12(g) To explore the Esterification reaction of alcohols.**

What to do: In groups

Discuss the following chemical properties of the alcohols using the table below. Share and compare your finding with the rest of the other group members.

Discussion Questions:

a) Copy and complete the following using the table below precisely.

Alcohol	Part of ester name from an alcohol	Carboxylic acid	Part of ester name from the acid	Complete name of Ester
i) Methanol		Ethanoic acid		Methyl ethanoate
ii) Ethanol		Methanoic acid		Ethyl methanoate
iii) Ethanol		Ethanoic acid		Ethyl ethanoate
iv) Propanol		Ethanoic acid		Propyl ethanoate

b) Write:

- a word equation for the reaction which took place in (b) (i) up to (iv) to form ester.
 - a chemical equation for the reaction which took place in (c) (i) above.
- c)** Suggest the name of the process that aided the formation of esters.
- d)** Copy and complete the table below precisely.

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Ester	Alcohol		Carboxylic acid	
	Name	Formula	Name	Formula
HCOOCH ₃				
HCOOC ₂ H ₅				
CH ₃ COOC ₂ H ₅				
CH ₃ COOC ₃ H ₇				

APPLICATIONS OF CARBOXYLIC ACIDS

Carboxylic acids are versatile compounds with a wide range of applications due to their various unique properties they exhibited. Below are some of their applications:

(1) Preservatives:

Carboxylic acids like benzoic acid and sorbic acid, are used to preserve food stuffs, beverages and pharmaceuticals to prevent microbial growth and spoilage.

(2) Synthesis of drugs:

They are used as intermediates in the synthesis of drugs such as aspirin, ibuprofen etc.

(3) Synthesis of soaps and detergents:

Fatty acids are used in the production of soaps and detergents through saponification reactions

(4) Production of polymers:

Carboxylic acids are also used as monomers in the production of polymers like polyamides that makes Nylon, polyesters.

(5) Cosmetics

Carboxylic acids such as glycolic acid and lactic acid, are used in skincare products due to their moisturizing and exfoliating properties.

(6) Cleaning agents

They are used in cleaning surfaces and products as disinfectants and degreasers, due to their ability to dissolve and remove all stains. Example include vinegar.

QUICK EXAM STYLE QUESTIONS:

- Ethanoic acid is a member of the homologous series that conform to a general formula $C_nH_{2n}O_2$.
 - Name the series to which ethanoic acid belongs.
 - What is the functional group of the series?
 - Ethanoic acid is a *weak* acid. Explain what this means, using an equation to help you.
 - Ethanoic acid reacts with sodium carbonate.
 - What would you see during the chemical reaction?
 - Write a balanced equation with state symbols for the reaction between ethanoic acid and sodium carbonate.
 - Another member of the series for which $n = 3$ reacts in similar way as ethanoic acid.
 - Name the member and draw its structural formula.
 - Write a balanced equation for the reaction between this compound and sodium hydroxide with state symbols.
- Ethanoic acid reacts with ethanol in the presence of concentrated Sulphuric acid.
 - Name the organic product formed.
 - Which type of compound is it?
 - What is the function of Sulphuric acid?
 - The reaction is *reversible*. What does it mean?
 - Write a chemical equation for the reaction.

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3. Hex-1-ene is an unsaturated hydrocarbon. It melts at 2140°C and boils at 63°C . Its empirical formula is CH_2 . Its relative molecular mass is 84.
- (i) To which family does hex-1-ene belong?
 - (ii) What is its molecular formula?
- b) (i) Hex-1-ene reacts with bromine water. Write an equation to show this reaction.
- (ii) What is this type of reaction called?
 - (iii) What would you see during the reaction?
4. The saturated hydrocarbons form a homologous series with the general formula $\text{C}_n\text{H}_{2n+2}$.
- What is a homologous series?
 - Explain what the term saturated means.
 - Name the series described above.
 - (i) Give the formula and name for a member of this series with two carbon atoms.
 - (ii) Draw its structural formula.
- e) (i) Name a homologous series of unsaturated hydrocarbons, and give its general formula.
- (ii) Give the formula and name for the member of this series with two carbon atoms.
- (iv) Draw the structural formula for the compound.
- f) Copy and complete the following statements correctly by using the words below to fill in the missing words in the passage. The words maybe used once or twice at times.
Fractional distillation; boiling point; crude oil; hydrocarbons; temperatures; lighter; longer; heavier; shorter; high; volatile; cracking.
- (i)is a method of separating the different fractions ofbasing on the principle that each fraction has different..... After..... the complex hydrocarbons are broken down at highinto useful short chainby a process calledin the present of a catalyst.
 - (ii) Theandchain hydrocarbons are lessand haveboiling points compared to theandchain hydrocarbons.
5. When ethanol vapour is passed over heated concentrated Sulphuric acid, a dehydration reaction occurs, and the gas ethene is produced.
- Draw a diagram of suitable apparatus for carrying out this reaction in the lab.
 - What is meant by a dehydration reaction?
 - Write an equation for this reaction, using the structural formulae.
 - (i) What will you see if the gas that forms is bubbled through bromine water?
 - (ii) You will not see this if ethanol vapour is passed through bromine water. Why not?

1.14. POLYMERS AND POLYMERIZATION

			
Wood	Sisal	Nylon	Rubber

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Cotton wool	Paper	Perspex	Polyethylene

A polymer can be natural or synthetic materials. Most of the common materials used in carrying out today's activities are made up of polymers.

Polymers: are large molecules made up of repeated units of monomers joined together to form long chain of complex molecules. **Monomers:** are units of small molecules which chemically combine together to form large and complex molecules called polymers.

Natural polymers: These are complex molecules made up of repeated units of natural monomers. **Synthetic polymers.** These are complex molecules made up of repeated units of man-made monomers.

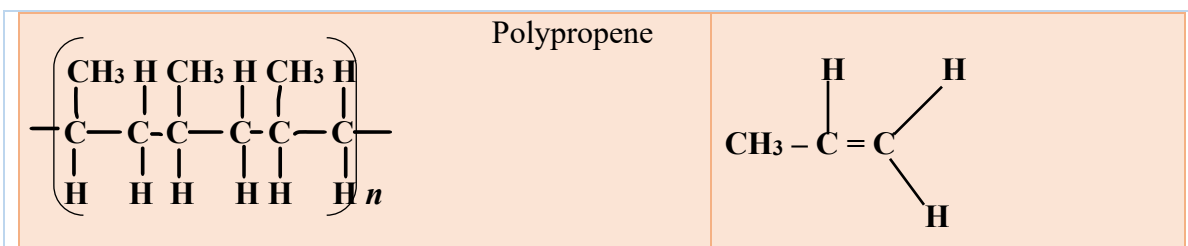
Polymerization reaction is a chemical reaction where several monomers react with each other to form repeated units of large molecules. It involves two types of reactions, addition polymerization and condensation polymerization.

ADDITION POLYMERIZATION OF PLASTICS

Addition polymerization is the process in which unsaturated monomers react with each other to form a polymer without loss of any byproducts. This type of polymerization commonly occurs with unsaturated monomers such as alkenes which contain carbon – carbon double bonds.

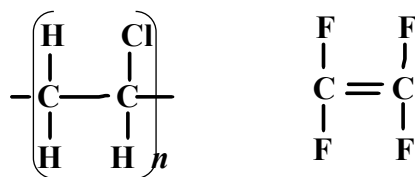
Polymer	Monomer
$\left[\begin{array}{cccccc} \text{H} & \text{Cl} & \text{H} & \text{Cl} & \text{H} & \text{Cl} \\ & & & & & \\ -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C}- \\ & & & & & \\ \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \end{array} \right]_n$ Polychloroethene (Polyvinyl chloride PVC)	$\begin{array}{c} \text{H} \quad \text{Cl} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{H} \quad \text{H} \end{array} \quad (\text{Chloro ethene})$
Polytetrafluoroethene (Teflon) $\left[\begin{array}{cccccc} \text{F} & \text{F} & \text{F} & \text{F} & \text{F} & \text{F} \\ & & & & & \\ -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C}- \\ & & & & & \\ \text{F} & \text{F} & \text{F} & \text{F} & \text{F} & \text{F} \end{array} \right]_n$	$\begin{array}{c} \text{F} \quad \text{F} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{F} \quad \text{F} \end{array} \quad (\text{Tetrafluoro ethene})$
$\left[\begin{array}{cccc} \text{C}_6\text{H}_5 & \text{H} & \text{C}_6\text{H}_5 & \text{H} \\ & & & \\ -\text{C} & -\text{C} & -\text{C} & -\text{C}- \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array} \right]_n$ Polystyrene	$\text{C}_6\text{H}_5 - \begin{array}{c} \text{H} \\ \\ \text{C} \end{array} = \begin{array}{c} \text{H} \\ \\ \text{C} \\ \\ \text{H} \end{array}$ (Phenyl ethene)

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QUICK EXAM STYLE QUESTIONS:

1. a) The polymer 'Teflon' is obtained from the monomer tetrafluoroethene, which has this structure (Y) (X)



- (i) Which feature of the monomer makes polymerization possible in X?
- (ii) Which type of polymerization occurs in structure X?
- (iii) Draw three units in the structure of the macromolecule that forms polymer of X.
- (iv) Give the chemical name for this polymer X
- b) The polymer Polychloroethene Y has been used to make 'cling film', for covering food to keep it fresh.
- (i) What does **n** represent?
- (ii) Name the monomer, and draw its structural formula.
- (iii) Which type of polymerization takes place?
- c) One property of Polychloroethene is its low *permeability* to moisture and gases.
- (i) Explain what is meant by the term **permeability**.
- (ii) That property is important in keeping food fresh. Why?
2. Polychloroethene is *non-biodegradable*.
- a) Explain the term **non -biodegradable** materials.
- b) Describe the major environmental problems caused by the poor disposal of such plastics.

USES OF SYNTHETIC POLYMERS

ACTIVITY: 1.14(b) To find out the different uses of synthetic polymers:

What you need:

Access to computer or using your smartphone for internet, textbook, notebook.

What to do:

In groups or individually, carryout the following research to investigate the different uses of synthetic polymers in our day – today life activities. Share and compare your findings with the rest of the class.

Discussion Questions:

- a) Copy and complete the table below precisely.

Synthetic Polymer	Monomer	Type of polymerization
Polyethene		
Polyvinylchloride (PVC)		
Teflon		
Polypropene		

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Polystyrene		
Nylon (Polyamide)		
Perspex		
Terylene		

PLASTICS: (are here to stay)

Plastics: the problem There were only a few plastics around the world before the 1950s. Since then, dozens of new ones have been developed, and more are on the way. Now it is hard to imagine life without them. They are used everywhere. One big reason for their success is their unreactivity. But this is also a problem. They do not break down or rot away. Most of the plastics thrown out in the last 50 years are still around – and may still be here for another 50 years from now. A mountain of waste plastic is growing.

**Polythene: the biggest problem**

Polythene is the biggest problem. It is the most-used plastic in the world, thanks to its use in plastic bags and food packaging. Around 5 trillion polythene bags are made every year. Most of plastic materials, are used only once or twice, then thrown away.

In many places, rubbish is collected and brought to **landfill sites**. The plastic bags fill up these sites. In other places, rubbish is not collected. So, the plastic bags lie around and cause many problems.

Word problems associated with plastics

- They choke birds, fish including other animals that try to eat them.
- They fill up the animals' stomachs so that they cannot eat proper food, and starve to death. (Animals cannot digest plastics.)
- They clog up drains, and sewers, and cause flooding.
- They collect in rivers, and get in the way of fish. Some river beds now contain a thick layer of plastic, causing death of fish and water pollution.
- They blow into trees and onto beaches. So, the place looks a mess. Tourists are put off – and many places depend on tourists.

Measures and Mitigations to be taken on plastic use:

Because of these problems, plastic bags have been banned in many countries including Bangladesh, Rwanda, Kenya, Morocco, Italy and several states in China and India.

Development of Biodegradable plastics:

Biodegradable polythene is already here in Uganda. Some of the plastics nowadays are **biodegradable**: they contain **additives** such as starch that bacteria can feed on. Some of the plastics are **photodegradable**: it contains **additives** that break them down in the presence of sunlight. In both cases, the result is that the polythene breaks down into tiny flakes. The amount of additive can be varied for different purposes – for example to make rubbish sacks that will break down within weeks.

ACTIVITY: 1.14(c)

In groups or individually, discuss the effects of waste plastics in the society. Share and compare your findings with the rest of the class. present them before the rest of the class.

Discussion Questions:

- Explain the problems caused by waste plastics in the society.

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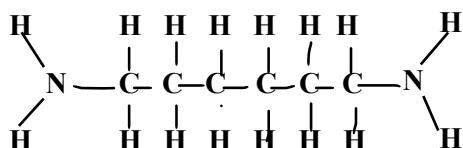
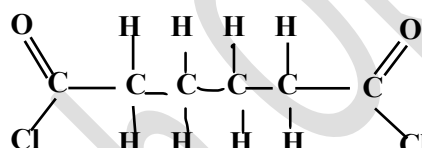
- b) How can the problems of waste plastics be prevented?
 c) Countries such as Rwanda, Kenya Morocco and China are no longer using non – biodegradable plastic bags but rather biodegradable plastic bags. Explain the concept of “*biodegradable plastics*” to the environment.

CONDENSATION POLYMERIZATION:

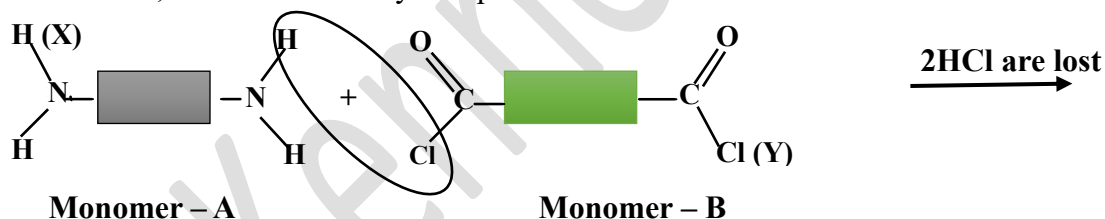
Condensation polymerization is the process whereby monomer molecules join up together to form complex molecules with loss of small molecules. Such as water or hydrogen chloride as smaller byproducts. In condensation polymerization, unlike addition polymerization, no double bonds break. Instead:

- two different monomers join.
- Each different monomer has two but same functional groups that take part in the reaction.
- the monomers join at their functional groups, by getting rid of or **eliminating** small molecules. **Condensation polymerization of Nylon (Polyamide)**

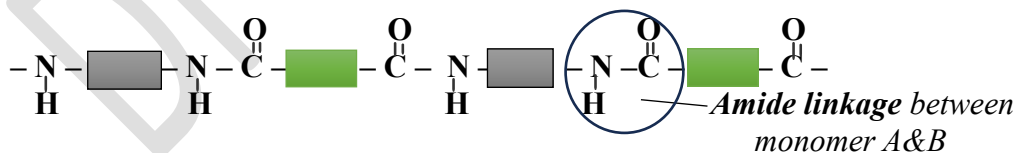
Below are two but different monomers A and B used in making Nylon.

**Monomer – A****Monomer – B**

Monomer – A has two NH_2 group at each end and **Monomer – B** has two COCl group at each end also. Only these functional groups take part in condensation polymerization reaction. The other part of **Monomer – A** and **Monomer – B** can be represented as *blocks* of molecules, to make them easy to represent at this level.

**Monomer – A****Monomer – B**

Monomer – A at part (X) joins another **Monomer – B** while **Monomer B** at part (Y) also joins another **monomer A** eliminating hydrogen chloride molecule. The overall polymer of **Nylon (Polyamide)** is

**ACTIVITY:1.14(f)****What to do:**

In groups or individually, carryout the following research to investigate the different uses of natural polymers in our day – today life activities. Share and compare your findings with the rest of the class.

Discussion Questions:

- a) Copy and complete the table below precisely.

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Natural Polymer	Monomer	Type of polymerization
Polyester		
Nylon (Polyamide)		
Cellulose		
Starch		
Protein		
Rubber		
DNA		
Polypeptide		
Polyester		

1.15. FATS AND OILS

Fats and oils: Are natural polymers called lipids obtained from animals and plants. Both fats and oils are composed majorly by natural monomers called **fatty acids** and **glycerol** obtained from both plant and animal sources.

Examples of **animal fats** are:

i) **Butter:** Obtained from dairy products, **Lard:** Obtained from rendered pork fat.

Examples of **plant fats** are, **coconut oil**, and **palm oil**.

Examples of **plant oils** are **olive oil**, **Avocado oil**, and **sunflower oil**.

Characteristics of fats and oils.

Physical properties	Natural fats	Natural oils
Physical state at r.t.p	Solid at room temperature	Liquid at room temperature
Composition of polymers	Made up of fatty acids	Made up of glycerol
Source	Animal fats	Plant oils

1.16.SOAPS AND DETERGENTS.

SOAPS:

Soap is a sodium or potassium salt of a long chain carboxylic acids or fatty acids. Soap is also known as sodium stearate or potassium stearate. The process of making soaps using sodium (potassium) hydroxide and fatty acid (oil) is known as **saponification**.

Sodium stearate is mainly used as washing soaps while potassium stearate is used as bathing soaps

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**STRUCTURE OF SOAPS**

Soaps are made of two ends, one end is the sodium (potassium) ion, the other end is the long chain carboxylic acids (esters). The end which is made up of sodium (potassium) ions forms the **hydrophilic head** of the soap while the other end of the soap forms **hydrophobic tail** of esters. The hydrophilic head of the soap loves water, hence water loving, forming **polar head** of the soap while the other end hates (repels) water, it forms **non – polar tails** of the soap.

The whole soap is called **Sodium stearate** with the general formula $StCOO^-Na^+$ where **St** is the long chain Hydrocarbon tail (Hydrophobic tail / non – polar tail) while COO^-Na^+ is the (Hydrophilic head / Polar head) of soap

How to make bar soaps**ACTIVITY1.15(a)****What you need:**

Coconut oil, 20% sodium hydroxide, sodium chloride, soap mould board, stirring rod, mixing bucket, weighing balance, measuring cylinder, heat source, glass beaker and hand gloves.

CAUTION!

Avoid getting into contact with sodium hydroxide since it is very corrosive.
 Avoid carrying out this activity with windows and doors closed since sodium hydroxide release strong shocking and suffocating particles in the air, or conduct the activity in a fume cupboard to avoid inhaling these particles.
 Always put on hand gloves and eye goggles to avoid getting into contact with sodium hydroxide.

What to do: In groups

1. Obtain 50cm^3 of coconut oil using a measuring cylinder into a glass beaker.
2. Measure 50cm^3 of 20% sodium hydroxide solution using another measuring cylinder. Transfer it into a beaker containing coconut oil.
3. Transfer the mixture of coconut oil and 20% sodium hydroxide and boil until it forms porridge – like liquid, while stirring using a glass rod.
4. Weigh to obtain 20g of sodium chloride. Transfer the salt into a glass beaker containing a mixture of porridge – like liquid and continue stirring.
5. Transfer the liquid mixture and pour it into a soap mould board, allow it to cool until it solidifies.

6. Present your sample soap to the rest of the class.

Discussion Questions:

- a) Explain what is meant by term “**saponification**”
- b) State the raw materials you have used in the preparation of the above soap.

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- c) Of what importance is sodium chloride during soap preparation?
 d) Draw a flow chart to show how soap is prepared.
 e) What name is given to the process of making soap?
 f) Write a:
 (i) Word equation for the reaction between sodium hydroxide and stearic acid (coconut oil) to form sodium stearate salt.
 (ii) Chemical equation for the reaction in (e) (i) using state symbols.

How to make liquid soaps**ACTIVITY:1.15(b)****What you need:**

Access to internet from a computer or smartphones, and your notebook.

What to do: In groups:

Discuss by carrying out research on how liquid soap is prepared by visiting computer library or using your smartphone for internet. Share and compare your findings with the rest of the class. Write them down in your notebook.

Discussion Questions:

- a) State the raw materials required when preparing liquid soap.
 b) Describe how liquid soap is formed from the materials stated in (a) above.

DETERGENTS

OR



Detergents are also known as **soapy** or **soapless** detergents. Just like soaps, detergents work the same way as soap but they are more effective than soaps. Soapy detergents are soaps obtained from natural fats or oils. Soapless soaps are **synthetic detergents** that have similar properties to soaps but don't contain the usual soaps like sodium salts of fatty acids. Therefore, they are referred to as **soapless soaps**.

Soapless detergents

Soapless detergents include washing powder, washing-up liquids and shampoos. They are called 'soapless' because they contain no soap.

Soapy detergents**How to make soapless detergents****ACTIVITY:1.15(c)****What you need:**

Castor oil, Glass beaker, Dropper, Concentrated Sulphuric acid, cold water, bucket, measuring cylinder and dropping pipette.

What to do: In groups

1. Using a measuring cylinder, obtain 5cm^3 of concentrated sulphuric acid and pour it into a clean glass beaker.

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2. Add 2cm^3 of castor oil using a dropping funnel into a glass beaker containing concentrated acid, swirl gently to mix. What happens to the walls of beaker?
3. Gently warm the mixture and add about 10cm^3 of 4M sodium hydroxide and stir. The mixture gets hot, viscous and dark
4. Add 5cm^3 of distilled water and stir. Then decant to separate the liquid from the solids. The solid is the Soapless detergent which is then washed with distilled water.
5. Remove a small sample of the product above into a clean test tube. Add about 5cm^3 of water and shake well for sometimes. Note what happens.

Discussion Questions:

- a) Suggest what is meant by soapless detergent.
- b) Name the raw materials required in the preparation of soapless detergents.
- c) Draw a flow chart to show how soapless detergents are prepared from the materials named in (b) above.

Effectiveness of soapy and soapless detergents as Cleansing agents**ACTIVITY:1.15(d)****What you need:**

Measuring cylinder, two basins, two pieces of dirty oily cloth, measuring cylinder, 5% soapy detergent solution, 5% soapless detergent solution, 1mol dm^{-3} magnesium sulphate solution and labels.

What to do:

In groups carryout the following procedures to compare which of the two detergents is more effective as a cleaning agent and readily forms lather with water and removes stains with ease.

1. Obtain basins and label them as **A₁** and **A₂** respectively.
2. To basin **A₁**, add one litre of magnesium sulphate solution (hard water).
3. Measure and add 100cm^3 of 5% soapy detergent solution in basin **A₁**.
4. Wash a dirty oily piece of cloth in basin **A₁**. Make your observation after drying the piece of cloth in the sun.
5. Repeat procedures 2, 3 and 4 using basin **A₂** and 5% soapless detergent solution.

Discussion Questions:

- | | |
|--|--|
| a) Identify which of the two detergents | b) Which of the two detergents is: |
| (i) Formed lather readily with hard water. | (i) More efficient at removing stains in a piece of cloth? |
| (ii) Did not formed lather readily with hard water | (ii) Less efficient at removing stains |
| c) What conclusion do you make about the two detergents? | d) Make a poster showing how soap removes (emulsifies) fats or oils. |
| e) Explain your answers in (b) above. | |

Emulsification action of soaps or Soapless detergents in removing greasy stains

Both soaps and Soapless detergents are very effective in removing dirt and disease - causing organisms from our bodies and clothes.

Soaps and Soapless detergents have both polar groups (hydrophilic heads, as water loving parts) and non – polar groups (hydrophobic tails, that attract greasy stains)

The hydrophobic ends of soap molecules move away from the water surface while the hydrophilic ends move towards layer of water molecules. This reduces the surface tension of water, allowing formation of lather with water.

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The hydrophobic ends move around in bunches and attract themselves on dirt from clothes and our bodies through chemical reaction and mechanical action. This helps breaking down the dirt (oily) particles into tiny droplets of particles that settle as a suspension called **emulsion**.

Knowledge Check:

- a) Name the part(s) of soap that:
 - (i) Attracts water molecules to form lather readily with water.
 - (ii) Attracts greasy stain molecules to form emulsions with soap.
- b) Explain the cleansing action of soap on dirty stains.

SOAPLESS DETERGENTS AND ENVIRONMENTAL SAFETY

These are sometimes referred to as synthetic detergents, they have several negative effects on the environment, as follows:

1) Water pollution:

Soapless detergents contain phosphates that can pollute water ways when they are washed down the drains, causing harms to aquatic life and ecosystems.

2) Algal blooms:

Algae grow excessively in the presence of excess phosphates produced from the use of soapless detergents, leading to algal blooms. They cover the surface of the water, blocking sunlight and oxygen from reaching other aquatic organisms, causing them to die.

3) Eutrophication:

This is a process where water bodies become highly enriched with nutrients such as phosphorous and nitrogen leading to excessive growth of algae in the water bodies.

4) Oxygen depletion:

When algae die and decompose, bacteria consume oxygen during the decomposition process. This leads to a net reduction in the amount of oxygen in the water, creating high oxygen demand by aquatic life

How to make use of soapless detergents sustainable

- 1) Encouraging the manufacturers to eliminate or minimize the use of harmful chemicals such as phosphates that can poison and harm aquatic life and disrupt the ecosystems.
- 2) Encouraging the manufacturers to formulate detergents using biodegradable ingredients that can break down more easily without harming the environment, reducing their impact on waterways and ecosystems.
- 3) Encouraging the use of surfactants derived from renewable resources such as bio – based surfactants to reduce on their effect on the environment.

ACTIVITY:1.15(e)**What you need:**

Access to the computer or smartphone, textbooks, notebooks.

What to do: In groups

Carryout **research** by visiting computer library or using your smartphones for internet or textbooks to find out the effects of soapless detergents on the environments such as water bodies. Share and compare your findings with the rest of the other groups in your class. write them in your notebook.

Discussion Questions:

- a) Explain the following terms as used in water pollution.
 - (i) Algal blooms
 - (ii) Eutrophication.
- b) What are the effects of excessive use of soapless detergents into the water body?

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- c) What are the:
- (i) Advantages of normal soaps over soapless detergents?
 - (ii) Disadvantages of soapless detergents?
 - (iii) Differences between normal soaps and soapless detergents?
 - (iv) Similarities between normal soaps and soapless detergents?

SAMPLE OF ACTIVITY OF INTERGRATION

SCENARIO

Today every house hold in Uganda on a daily basis use carbon to support life. They may include food stuffs, drinks (both hard and soft drinks), household materials, laundry materials, petroleum products, most of these organic products have specific uses, classes of the organic compounds and different functional groups.

Your school is organizing academic day where all the parents are invited. The Headteacher of your school, wants the parents and the rest of the entire school to appreciate the values of “**Carbon in life**” in our day – to – day life activities.

TASK:

The head of Chemistry department, selected you to prepare an article that will be presented before the parents and the rest of the school. Your presentation should include:

- a) The class and functional group of the different organic compounds found in the places mentioned above.
- b) The different importance of each material of the different classes of organic compounds mentioned above in our day – to – day life activity.

TOPICAL SUMMARY

In this chapter of “**Carbon in life**”, you learnt that: -

Petroleum oil is also known as crude oil. Petroleum oil is a smelly mixture of hundreds of different alkanes compounds.

Products of petroleum oil can only be obtained by fractional distillation in oil refinery, such products may include natural gas, and different fractions of alkanes.

Natural gas is the first fraction and it contains mainly methane gas. Methane gas is used as fuel gas. Methane is a central raw material used as chemical feedstocks for making other useful organic products.

Biogas is a carbon – based fuel obtained from biomass. It constitutes of natural gas (methane gas) used as fuel gas.

Homologous series Refers to a group of organic compounds with the same general formular, similar structural formulae, similar chemical properties in which the successive members differ from the next by methylene group. (– CH₂ – group).

Hydrocarbons: Are organic compounds that contain carbon and hydrogen atoms only. They are groups as **saturated** and **unsaturated** hydrocarbons.

Alkanes are the first homologous series, with a general formula of C_nH_{2n+2}, followed by alkenes with general formula, C_nH_{2n}, alkynes with general formula, C_nH_n, alcohols with general formula, C_nH_{2n+1}OH, Carboxylic acids with general formula, C_nH_{2n}O₂. Each homologous series have different physical and chemical properties, different functional groups but same root names.

Fermentation and chemical methods are the major brewing methods of preparing ethanol, fermentation is biological since it involves enzymes diastase, maltase and zymase. Excessive consumption of ethanol is not good for your health.

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Polymers: are large molecules made up of repeated units of monomers joined together to form long chain of complex molecules. **Monomers:** are units of small molecules which chemically combine together to form large and complex molecules called polymers. It involves mainly two types natural and synthetic polymers.

. **Polymerization reaction** is a chemical reaction where several monomers react with each other to form repeated units of large molecules. It involves two types of reactions, **addition** polymerization and **condensation** polymerization.

Soap: is a sodium or potassium salt of a long chain carboxylic acids (fats and oils). Soap is also known as sodium stearate or potassium stearate. The process of making soaps using sodium (potassium) hydroxide and fatty acid (oil) is known as **saponification**.

Detergents are mainly Soapless and soapy detergents, they are very effective when used in hard water than ordinary soaps. Some Soapless detergents are non – biodegradable, making them unsustainable to the environment such as water bodies.

S3 New Curriculum chemistry**Theme: Structures and bonds****Chapter 2 – Structures and bonds**

Definitions

An element is a substance which cannot be split or divided up into simpler substances by chemical means. Common elements are oxygen, nitrogen, sulphur and others.

A **compound** is a substance composed of **two or more elements chemically bonded** together.

An atom is the smallest indivisible particle of an element that can take part in a chemical reaction.

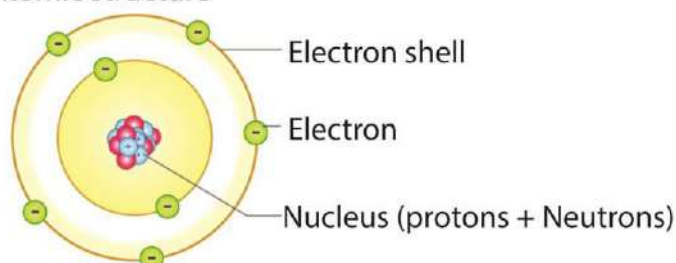
A **molecule** is the smallest particle of an element or compound that can exist in a free and separate state.

Trial 1

Give three examples of each of the following

- (a) Elements
- (b) Compounds
- (c) mixtures

Atomic structure



An atom is composed of three basic subatomic particles, namely the electron, the proton and the neutron; the characteristics of which are given in a table below:

Table: The three main subatomic particles.

Particle	Symbols	Charge	Mass
Electron	e	-1	1/1837
Proton	p	+1	1
Neutron	n	No charge	1

All atoms of the same element contain the **same** number of protons (usually equal to the number of electrons).

Atomic number (Z) is the number of protons in an atom. it is characteristic of an element.

The atomic mass (A) is the sum of the number of protons and neutrons in an atom of an element. **Isotopes** are atoms with the same atomic number but different atomic masses (due to the difference in the number of neutrons in an atom).

Relative atomic mass (Ar) is the **weighted average mass** of an atom of an element compared to **1/12th the mass of a carbon-12 atom**.

Name and chemical symbols of elements

- The chemical symbol(s) is /are a letter or letters that represent an atom of an element.
- If the chemical symbol has one letter then, that letter is a capital letter.
- If the chemical symbol has two letters then, the first letter is capital and the second letter is a small letter and they are written such that they touch each other.

Standard representation of atoms of an element

The element X with atomic mass, A, and atomic number Z is represented as



Trial 1

Write the standard representation of atoms of the following elements:

- Hydrogen
- Oxygen
- nitrogen

A list of the first 20 elements in the periodic table and their chemical symbols

Poem	Name of the element.	Symbol	Atomic numbers (Z)	Atomic mass (A)
Hallo	Hydrogen	H	1	1
Hellene	Helium	He	2	4
Literature	Lithium	Li	3	5
Becomes	Beryllium	Be	4	9
Bore	Boron	B	5	10
Can	Carbon	C	6	12
Not	Nitrogen	N	7	14
Offer	Oxygen	O	8	16
Favourable	Fluorine	F	9	19
Neon	Neon	Ne	10	20
Naye	Sodium	Na	11	23
Mega	Magnesium	Mg	12	25
All	Aluminium	Al	13	26
Sinks	Silicon	Si	14	28
Plus	Phosphorus	P	15	31
Source pans	Sulphur	S	16	32
Clean	Chlorine	Si	17	35.5
Are	Argon	P	18	40
Kept	Potassium	K	19	39
Carefully	Calcium	Ca	20	40

Trial 2

Write electron configurations of the following element

Na, Cl, Ca, O, K and N

Combination of atoms

Atoms combine to form molecules and compound through chemical bonding.

A chemical bond is the force of attraction that binds the atoms within a molecule.

Valence is the combining power of an element or a radical. OR the number of electrons an element loses or acquires to be stable or the number of bonds an atom can form with other atom(s).

Table of the valences of the first twenty elements of the periodic table

Group	1	2	3	4	5	6	7	8
valence	1	2	3	4	3	2	1	0
Elements	H							He
	Li	Be	B	C	N	O	F	Ne
	Na	Mg	Al	Si	P	S	Cl	Ar
	K	Ca						

Types of chemical bonds

A chemical bond may be;

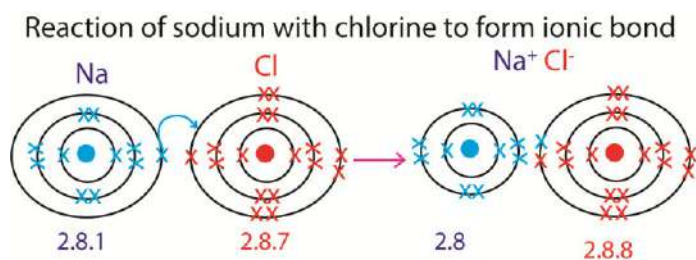
- i) Ionic bond
- ii) Covalent bond
- iii) Metallic bond
- iv) Coordinate/Dative bond
- v) Van de Waal

Noble gases have a full outermost energy level/ shell e.g helium (2), argon (2:8:8), neon (2:8). They are stable and therefore, they do not react with other elements. Elements attain this noble gas configuration by combining with other elements to form a molecule or a compound. They do this in two main ways:

1. Ionic or electrovalent bonding
2. Covalent bonding

Ionic or electrovalent bonding:

An **ionic bond** is a bond formed when one atom donates electron(s) to form a positively charged ion or a **cation** and another accepts electron(s) to become a negatively charged ion or an **anion**. The electrostatic attraction between positive and negative ions constitutes an **ionic bond**. The compound formed is called **ionic compound**.



Ionic bond is the attractive force between positive and negative ions

Properties of ionic compound

- Conduct electricity** when molten or in aqueous solution. This is because ionic compounds consists of positively charged ions which, when in molten state or aqueous state, become mobile and conducts electricity.
- Usually they **have high melting and boiling points**. This is because great energy is needed to break the strong electrostatic force holding the ions together in a regular lattice.
- Usually they are **crystalline solids**.
- Usually they **dissolve in polar solvents such as water**. This is because water molecules are able to bond with both the positive and negative ions. They break up the lattice and keeps the ions apart. If they do not dissolve in polar solvent, it is often because very high energy is required to break up the lattice

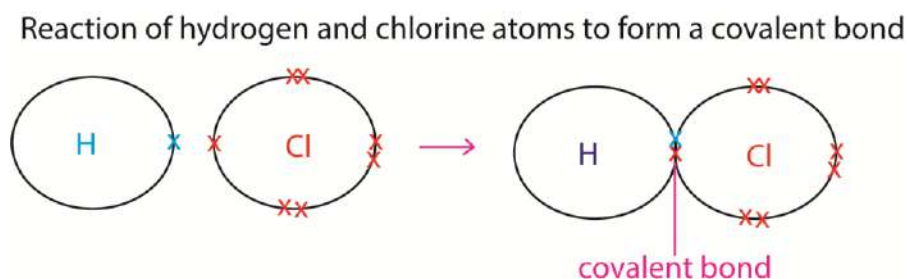
Trial 3

- Electron configurations of calcium and oxygen atoms.
- Draw structural diagrams to show the reaction of calcium and oxygen to form calcium oxide.
- Write electron configuration of calcium and oxygen ions.
- Name the type of bond formed between calcium and oxygen.

Covalent bonding

A **covalent bond** is a bond formed by mutual sharing of one or more electron pairs between atoms of non-metallic elements. For example,

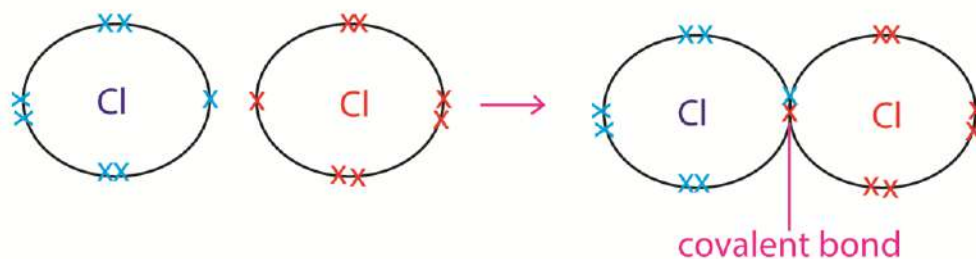
- two hydrogen atoms combine to form a hydrogen molecule,



The shared pair of electrons constitute a covalent bond

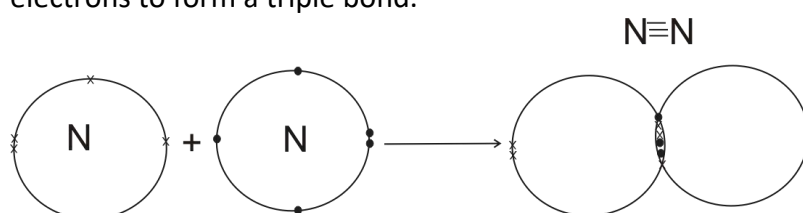
- chlorine atom combines with a hydrogen atom to form a hydrogen chloride molecule.

Reaction of two chlorine atoms to form a covalent bond

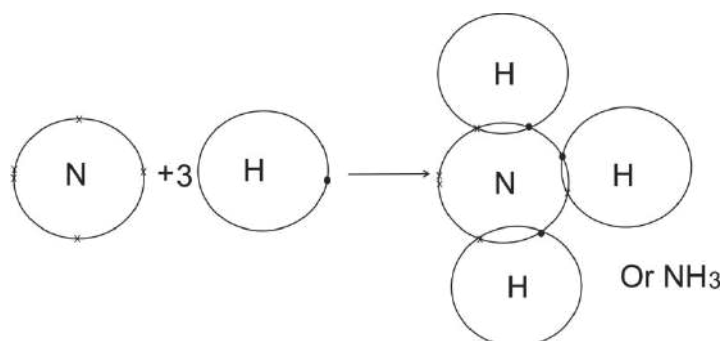


The shared pair of electrons constitute a covalent bond

- (iii) In a covalent bond between two nitrogen atoms involve sharing three pairs of electrons to form a triple bond.



- (iv) In formation of ammonia, Nitrogen uses its unpaired electrons to bond with three hydrogen atoms



Properties of covalent compounds

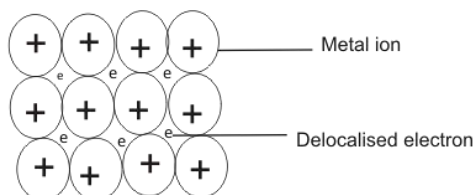
- (i) **Covalent compounds are non-electrolytes.** This is because they consist of molecules with no ions. Non- electrolyte is a solution or a molten compound which does not conduct electricity.
- (ii) They are usually gases, liquids or solids with **low melting points and boiling points** because of the weak inter molecular forces (van der waal forces) of the attraction which exist between simple molecules. Giant molecular substances have higher melting point because the whole structure is held together by strong covalent bonds within the giant molecule as in case of silicon(IV) oxide.
- (iii) They usually **dissolve in non-polar solvents**. Examples of non-polar solvents are tri chloromethane, tetra chloromethane and carbon disulphide. These liquid are covalent compounds and dissolve a great number of covalent compounds. For example tri chloromethane dissolves iodine, carbon disulphide dissolve sulphate. Generally, they do not dissolve in water.

Trial 4

Draw the structure to show the formation of O_2 , CCl_4 , CO_2

Metallic bond

A **metallic bond** is the type of chemical bonding that holds **metal atoms** together in a solid structure. It occurs due to the **delocalization of valence electrons**, meaning that electrons are not bound to a specific atom but instead move freely throughout the metallic structure.

**Properties of metals**

- (i) Due to the presence of delocalised electrons, metals are good conductor of heat and electricity. When the voltage is applied across the metal, the delocalised electrons are able to move freely through the structured conducting an electric current.
- (ii) Metals generally have high densities because the ions are closely packed in the lattice.
- (iii) They usually have high melting and boiling points due to the strong attraction between the positive metals ions and the delocalised electrons. Most non-metals have low melting points and boiling points.
- (iv) They are malleable and ductile. That is, the metals can be pulled out into thin wires.
- (v) Malleable: means that metals can be harmed into different shapes

Trial 5

- (a) Differentiate between an ionic bond and covalent and ionic bond.
- (b) (i) State the type of bonds in ammonia and methane
 - (i) Use diagrams to show how a hydrogen atom combines with a chlorine atom to form hydrogen chloride.
- (c) What type of bond is formed when hydrogen combines with chlorine to form hydrogen chloride? Explain your answer.

THE MOLE CONCEPT AND CALCULATIONS

A large number of calculations that appear in examination papers rely up on the knowledge of the mole concept. This study includes calculations of different kinds which one ought to study.

A. DEFINITIONS.

The **relative atomic mass (RAM)** of an atom is the number of times an atom is heavier than one twelfth of a carbon-12 atom. It is calculated by adding together the number of protons and neutrons in an atom.

The atomic mass is usually a whole number unless where an element occurs in a number of isotopes. Here the atomic mass is the average atomic weight of the isotopes in the ratio of their existence.

Example 1. Chlorine exist in two isotopic forms; ^{35}Cl and ^{37}Cl in percentage compositions of 75% and 25% respectively.

Calculate the relative atomic mass of chlorine.

Every 100 atoms of chlorine there is 75 atoms of ^{35}Cl and 25 atoms of ^{37}Cl

Total mass of 100 atoms
of chlorine

$$= 75 \times 35 + 25 \times 37$$

$$= 3550$$

$$\text{Average atomic mass} = \frac{3550}{100}$$

$$= 35.5$$

$\therefore$ The RAM of chlorine is 35.5

Exercise 1a

1. A sample of ordinary neon is found to consist of ^{20}Ne , ^{21}Ne and ^{22}Ne in percentage $^{20}\text{Ne} = 90.22\%$, $^{21}\text{Ne} = 0.26\%$ and $^{22}\text{Ne} = 8.82\%$. Calculate the actual atomic weight of neon.
2. Copper has a relative atomic mass of 63.55 and consists of two isotopes of mass number 63 and 65. Calculate the percentage composition of the isotopes. (1993/1/4)
3. The two isotopes of hydrogen having mass number 1 and 2 have atomic weight equal to 1.007825 and 2.01473 as determined by mass spectrograph. These isotopes are present in ordinary hydrogen in ratio of 6400 to 1. Determine the mean-atomic weight of hydrogen atoms.
4. 2002/1/2, The mass spectrum of chlorine shows peaks at masses 70, 72, and 74. the heights of the peaks are in ratio of 9:6:1
Calculate
 - a. The average atomic mass of chlorine (2marks)
 - b. The relative abundance of ^{35}Cl and ^{37}Cl . (3 marks)
5. 2001/2/1,
 - a. Define the term relative atomic mass (2mark)
 - b. Explain how the relative atomic mass can be determined by the mass spectrometer (10 marks)
 - c. The mass spectrum of an element, A, contained four lines at mass/charge of 54, 56, 57, and 58 with relative intensities of 5.84, 91.68, 2.17 and 0.31 respectively

2

- i. Explain what the term relative intensities means
- ii. Calculate the relative atomic mass of A

6. 2004/1/03,(b) An element Y has three naturally occurring isotopes with isotopic masses and relative abundances' as shown below:

Isotopic mass	Relative abundance (%)
23.98	78.60
24.98	10.11
25.98	11.29

Calculate the relative atomic mass of Y (2 marks)

Formula masses (relative formula mass or Molecular mass)

This is the number of times a molecule is heavier than one twelfth of a carbon-12 atom. It is calculated by addition of the individual RAM of the constituent atoms in a molecules, Relative formula mass has no units

Example 2.

What is the formula mass of Ozone, O_3 ?

The Ozone molecule contains three oxygen atoms, each of which weighs 16 grams.

$\therefore$ the formula mass of Ozone is $3 \times 16 = 48$

Example 3. *What is the formula mass of Ammonia, NH_3 ?*

RAM of nitrogen atom is 14 and that of hydrogen atom is 1. There are three hydrogen atoms in the molecule and one nitrogen atom;

$\therefore$ the formula mass of ammonia is: $14 + (3 \times 1) = 17$ grams.

Example 4. *What is the formula mass of Glucose, $C_6H_{12}O_6$?* ($C=12$, $H=1$, and $O=16$)

$\therefore$ The formula mass of Glucose is

$$\begin{aligned}
 &= 6 \times 12 + (12 \times 1) + (6 \times 16) \\
 &= 72 + 12 + 96 \\
 &= 180
 \end{aligned}$$

Exercise Ib

- (1) Calculate the formula mass of
 - (a) Ammonium sulphate, $(NH_4)_2SO_4$
 - (b) Sodium carbonate decahydrate, $Na_2CO_3 \cdot 10H_2O$.
 - (c) Oxalic acid, $HOOC-COOH$
 - (d) Maleic acid $HOOC-CH=CH-COOH$

B. THE MOLE

A mole is the AMOUNT of a substance which contain as many elementary units (particles, i.e., molecules, atoms, ions etc) as there are in 12 g of carbon-12 isotope ($^{12}_6C$). A mole of any substance contains 6.02×10^{23} particles.

The number of particles in any mole of a substance (6.02×10^{23}) is called **AVOGADRO'S NUMBER, L.**

2

Molar Mass is the mass in grams of 1 mole of a substance. It is numerically equal to its relative atomic mass or its relative formula mass. e.g., One mole of carbon weighs 12 g, 1 mole of oxygen molecule weighs 32 g, and 1 mole of ammonium sulphate weighs 132 g.

CALCULATIONS USING THE AVOGADRO NUMBER

The Avogadro number, L , is taken as 6.0×10^{23} in the following examples.

Example 5. How many C atoms are there in a carbon rod weighing 8 g ($C=12$).

12 g (RAM of carbon in g or 1 mole) of carbon have

6.0×10^{23} atoms.

$$\therefore 8 \text{ g of carbon has } \frac{8 \times 6.0 \times 10^{23}}{12} \text{ atoms}$$

$$= 4.0 \times 10^{23} \text{ atoms}$$

Example 6. How many grams of copper ($Cu=64$) contain 4.5×10^{23} atoms?

6.0×10^{23} atoms are in 1 mole of copper i.e. 64 g

$$\therefore 4.5 \times 10^{23} \text{ atoms are in } \frac{4.5 \times 10^{23} \times 64}{6.0 \times 10^{23}}$$

$$= 48 \text{ g}$$

Example 7. How many (i) molecules (ii) atoms are there in 35.5 g of gaseous chlorine Cl_2 ?

(i) Mass of 1 mole of chlorine molecule, $Cl_2 = 71$ g

i.e. 71 g of chlorine contain 6.0×10^{23} molecules.

$$\therefore 35.5 \text{ g of chlorine contains } \frac{35.5 \times 6.0 \times 10^{23}}{71}$$

$$= 3.0 \times 10^{23} \text{ molecules}$$

(ii) Each chlorine molecule contains 2 atoms. Hence there are

$2 \times 3.0 \times 10^{23}$ atoms in 35.5 g of gaseous chlorine, Cl_2

$\therefore$ No. of atoms in 35.5 g of chlorine gas = 6.0×10^{23} atoms.

Example 8. How many ions are there in 13.5 g of copper (II) chloride, $CuCl_2$?

Mass of 1 mole of $CuCl_2 = 64 + (2 \times 35.5)$

$$= 135 \text{ g}$$

There are 3 ions (one copper and two chloride ions) in $CuCl_2$ molecule.

i.e. 1 mole or 135 g of $CuCl_2$ contain $3 \times 6.0 \times 10^{23}$ ions.

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$$\therefore 13.5 \text{ g of CuCl}_2 \text{ contain } \frac{4 \times 13.5 \times 36.0 \times 10^{23}}{135}$$

$$= 1.8 \times 10^{23} \text{ ions.}$$

[NB. it contains 0.6×10^{23} copper ions and twice that i.e. 1.2×10^{23} chloride ions.]

The 'mole' concept in calculations

To convert Mass of a substance into moles

$$\text{No. of moles} = \frac{\text{Mass of substance}}{\text{Mass of 1 mole}}$$

To convert moles of a substance in grams

Mass in grams = No. of moles $\times$ mass of 1 mole.

Example 9. How many moles of carbon dioxide molecules are present in 11 g of carbon dioxide, CO_2 ?

Mass of 1 mole of $\text{CO}_2 = 12 + (2 \times 16) = 44 \text{ g}$

$$\begin{aligned} \text{No. moles of CO}_2 &= \frac{\text{Mass in grams}}{\text{Mass of 1 mole}} \\ &= \frac{11}{44} \text{ moles} = 0.25 \text{ moles} \end{aligned}$$

Example 10. What is the mass of 3 mole of carbon monoxide?

The Molar mass of $\text{CO} = (12 + 16) = 28 \text{ g}$

Mass in grams = No. of moles $\times$ mass of 1 mole.

$$= 3 \times 28$$

$$= 84 \text{ g}$$

Example 11. What mass of magnesium ($\text{Mg}=24$) would contain the same number of atoms as 4 g of carbon. ($\text{C}=12$)

$$\begin{aligned} \text{No. of moles of carbon} &= \frac{\text{Mass in grams}}{\text{Mass of 1 mole}} = \frac{4}{12} \text{ moles} \\ &= 0.333 \text{ mol} \end{aligned}$$

A mole of any element contains the same number of atoms.

Hence 0.333 moles of magnesium contain the same number of atoms as 4 g of carbon.

Mass of 0.333 moles magnesium =

$$\begin{aligned} &\text{No. of moles} \times \text{Mass of 1 mole} \\ &= 0.333 \times 24 \end{aligned}$$

$$\frac{5}{8} = 8 \text{ g.}$$

Example 12 . Calculate the mass of
 (i) 1 atom of magnesium.
 (ii) 5 molecule of ethane.

(i) Relative atomic mass of magnesium = 24

$$\begin{aligned} 1 \text{ atom of magnesium} &= \frac{24 \times 1}{6.0 \times 10^{23}} \\ &= 4 \times 10^{-23} \text{ g} \end{aligned}$$

(ii) Molar mass of C_2H_6 = $(2 \times 12) + (6 \times 1)$
 = 30 g

$$1 \text{ molecule of ethane} = \frac{30 \times 1}{6 \times 10^{23}} = 5 \times 10^{-23} \text{ g}$$

$$\therefore 5 \text{ molecules of ethane} = 25 \times 10^{-23} = 2.5 \times 10^{-22} \text{ g}$$

Example 13. Calculate the mole of each element present in 18.4 g of iron (III) sulphate.

$$\begin{aligned} \text{Molar mass of } \text{Fe}_2(\text{SO}_4)_3 &= 2 \times 56 + 3 \times 32 + 12 \times 16 \\ &= 112 + 96 + 192 \\ &= 400 \text{ g} \end{aligned}$$

$$\begin{aligned} \text{No. of moles of } \text{Fe}_2(\text{SO}_4)_3 &= \frac{\text{Mass given}}{\text{Mass of 1 mole}} \\ &= \frac{18.4}{400} = \mathbf{0.046} \text{ mole} \end{aligned}$$

$$\therefore \text{No. of moles of iron} = 0.046 \times 2 = 0.092 \text{ moles.}$$

$$\therefore \text{No. of moles of sulphur} = 0.046 \times 3 = 0.138 \text{ moles}$$

$$\therefore \text{No. of moles of oxygen} = 0.046 \times 12 = 0.552 \text{ moles}$$

Exercise II

1. Using the Avogadro's constant equal to **L**, find the number of atoms in the following masses of calcium: (i) 200 g (ii) 1 kg, (iii) 4 g.

2. Taking Avogadro's constant equal to 6×10^{23} , find the number of molecules in

(a) 3.4 g of ammonia, NH_3 (b) 7.7 g of carbon tetrachloride CCl_4

6

(c) 16 g of sulphur dioxide, SO_2 (d) 1 g of hydrogen, H_2 .3. What mass of carbon dioxide has the same number of molecule as 7 g of nitrogen, N_2 ?4. What is the mass of 0.5 mole of aluminium sulphate, $\text{Al}_2(\text{SO}_4)_3$?

5. Which of these contains the same number of atoms: 16 g of sulphur, 40 g of calcium, 40.5 g of aluminium or 28 grams of iron?

(Ca = 40, N = 14, C = 12, S = 32, O = 16, Al = 27, Fe = 56)

C. Percentage composition: Formulae

Example 14. Calculate the percentage composition by mass of magnesium oxide, ($\text{Mg} = 24$, $\text{O} = 16$).(i) Write formula of the compound: MgO (ii) Find the formula mass of the compound: $24 + 16 = 40$

(iii) Express each atomic mass as a percentage of the formula mass.

$$\text{Percentage of Magnesium} = \frac{24 \times 100\%}{40} = 60\%$$

$$\text{Percentage of oxygen} = \frac{16 \times 100\%}{40} = 40\%$$

Example 15. Calculate the percentage of water of crystallization in hydrated sodium carbonate, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$?

$$\text{Formula mass of } \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} = 46 + 12 + 48 + 20 + 160 = 286$$

Note: The percentage of water should be calculated as a complete, separate unit i.e. ($10 \text{ H}_2\text{O} = 180$) in this case.

$$\begin{aligned} \text{per cent water in } \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} &= \frac{180 \times 100\%}{286} \\ &= 62.94\% \end{aligned}$$

Formulae: Empirical and molecular

The **empirical formula (E.F)** of a compound is the simplest formula which expresses its composition by mass. It does not give the exact number of various atoms present in one formula unit of molecule of a compound.

The **molecular formula (M.F)** of a compound is the one which shows the exact number of each kind of each atom present in a molecule of a compound.

The molecular formula is always a simple multiple of the empirical formula.

i.e. $M.F = n(E.F)$ where n is a whole number e.g., E.F of glucose is CH_2O , whereas its M.F. is $C_6H_{12}O_6$ which $(CH_2O)_6$.

$$\text{OR } \frac{\text{Molecular formula mass}}{\text{Empirical formula mass}} = n, \text{ a whole number.}$$

Example 16. Determine the formula of a lead compound, given that 4.14 g of lead combines with 0.64 g of sulphur and 1.28 g of oxygen.

We shall use the 'mole' concept to find the formula of the compound. Proceed as shown in the table below:

Element symbol	Lead, Pb	Sulphur, S	Oxygen, O
Mass in g	4.14	0.64	1.28
RAM	207	32	16
No. of moles	$\frac{4.14}{207} = 0.02$	$\frac{0.64}{32} = 0.02$	$\frac{1.28}{16} = 0.08$
Ratio of mole	$\frac{0.02}{0.02} = 1$	$\frac{0.02}{0.02} = 1$	$\frac{0.08}{0.02} = 4$
No. of atoms	1	1	4
E.F	$PbSO_4$		

Example 17. The combustion of 0.92 g of copper gave 1.15 g of copper oxide.

Calculate the percentage of copper and oxygen in the sample and hence its formula.

Mass of oxygen in the oxide = $1.15 - 0.92 = 0.23$ g

$$\text{Percentage of copper} = \frac{0.92 \times 100\%}{1.15} = 80\%$$

$$\text{Percentage of oxygen} = \frac{0.23 \times 100\%}{1.15} = 20\%$$

	Copper, Cu = 64	Oxygen, O = 16
Percentage	80	20
No. of moles	$\frac{80}{64} = 1.25$	$\frac{20}{16} = 1.25$
No. of atoms	$\frac{1.25}{1.25} = 1$	$\frac{1.25}{1.25} = 1$
Formula	CuO	

∴ The formula of copper oxide is CuO.

Example 18 3.22 g of hydrated sodium sulphate, $\text{Na}_2\text{SO}_4 \cdot n\text{H}_2\text{O}$ were heated till all the water of crystallization was driven off. The anhydrous salt left had a mass of 1.42 g. Determine the formula of the hydrated salt.

Compound	Na_2SO_4	H_2O
Mass in g	1.42	1.80 (3.22 - 1.42)
R.M.M	142	180
No. of moles	$\frac{1.42}{142} = 0.01$	$\frac{1.80}{18} = 0.1$
Ratio of mole	1	10

Hydrated sodium sulphate is $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$

Example 19. An organic compound was found to contain 12.8% carbon, 2.1% Hydrogen and 85.1% bromine. A rough estimate gave its relative molecular mass between 150 and 200. Determine its molecular formula. Explain your answer.

Element	C (=12)	H (=1)	Br (=80)
No. of mole	$\frac{12.8}{12} = 1.06$	$\frac{2.1}{1} = 2.1$	$\frac{85.1}{80} = 1.06$
/mole ratio/No. of atoms	1	2	1

Empirical formula = CH_2Br

E.F. mass = $12 + 2 + 80 = 94$

Molecular formula mass = $n \times \text{E.F. Mass} = n \times 94$

i.e. M.F. mass can be 94 or 188 or 289.

In this case, it is then 188 and the M.F. of the compound is $\text{C}_2\text{H}_4\text{Br}_2$

Exercise III

- What is the percentage of hydrogen and oxygen in ethanoic acid, CH_3COOH ?
- 24.0 g of a lead oxide was completely reduced by heating it with carbon to 20.8 g of lead. What is the formula of the oxide that was reduced?
- If 0.5 mole of a hydrated salt $\text{X} \cdot n\text{H}_2\text{O}$ contains 63 g of water of crystallization, find the value of n.
- 2.8 g of iron reacted with 5.3 g of chlorine to form the chloride.
 - Determine the simplest formula of the chloride.
 - If the molecular mass of the chloride is 325. Find the molecular formula and structural formula of the chloride.
- When 3.6 g of magnesium was burnt in nitrogen, 5.0 g of magnesium nitride was formed. Determine the simplest

formula of the nitride.

6. A liquid compound X has relative molecular mass of 114 and the ratio of carbon atoms to hydrogen atoms is 4:9. Determine the molecular formula of the compound. To which homologous series of organic compounds does it belong?

7. An organic compound contains by mass 60% carbon, 13.3% hydrogen and 27.7% oxygen. Determine its simplest formula? Find its possible molecular formula if its relative molecular mass is estimated to be between 50 and 80.

8. When 0.29 g of a substance X was burnt in oxygen, 0.66 g of carbon dioxide and 0.27 g of water were formed. Calculate (i) the masses of carbon, hydrogen and oxygen in X.
(ii) Determine the empirical formula of X.

9 (a) A compound X contains carbon, 66.7% and hydrogen 11.1% the rest being oxygen. Determine the empirical formula of X.

(b) The vapor density of X is 36; determine the molecular formula of X.

10. Combustion of 5.4 g of a compound containing C, H and O only gave 7.92 g of CO_2 and 3.24 g of H_2O . Calculate the empirical formula of the compound? The compound has a R.M.M of 180, determine its molecular formula.

1997/2

(a) A compound b contain 92.31% carbon and 6.69% hydrogen

Determine the empirical formula of B (2marks)

(b) B burns with a sooty flame and has a vapour density of 39. Determine molecular formula and write its molecular structure.

2002/2/1(a)

A compound X, vapor density 58, contain carbon 62.07%, hydrogen 10.34%, the rest being oxygen. X does not burn with sooty flame.

- (i) Calculate the empirical formula of X
(C=12, H=1, O=16) (3marks)
- (ii) Determine the molecular formula of X

1993/1/7

(a) When 4.90 g of organic compound X, containing carbon and hydrogen only was burnt in oxygen, 15.78g of carbon dioxide and 5.38 g of water were formed. Calculate the empirical formula of X.

2003/1/7

(a) When 4.90g of organic compound X, containing carbon and hydrogen only was burnt in oxygen, 15.78g of carbon dioxide and 5.38g of water were formed. Calculate the empirical formula of X (2 ½ marks)

(b) X was steam distilled at 80°C and 760mmHg and the distillate was found to contain 90.8% by mass of X. (the vapor pressure of water at 80°C is 240mmHg).

- a. Calculate the formula mass of X.
- b. Deduce the molecular formula of X.

2005/1/16

A compound P contains 52.2% carbon, 13.0% hydrogen, the rest being oxygen

- (i) Determine the empirical formula of P.
- (ii) When vaporized, 0.1g of P occupied 78.8cm^3 at 157°C and pressure of 740mmHg.
 - (i) Calculate the formula mass of P
 - (ii) Determine the molecular formula of P

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- (iii) Write the structural formula of all the possible isomers of P
- (iv) P does not react with sodium metal. Identify P
- (v) Write equation to show how P can be prepared from methanol.

D. Calculation involving Equations, Masses and Volumes

Chemical changes can be described by use of chemical equations, e.g., carbon dioxide reacts with oxygen to form carbon dioxide. This reaction is written as:



The plus sign (+) indicates addition of carbon to oxygen (vice versa) OR a mixture of the two. The arrow ($\rightarrow$) is often read 'yields'. Substances on the left of the arrow are **reactants, or starting materials**. Those on the right are the **products** of the reaction.

Chemical equations have meaning on the atomic and molecular level. The equation (1) means that one atom of carbon (C) reacts with one molecule of oxygen (O_2) to produce one molecule of carbon dioxide, CO_2 . One carbon atom weighs 12 grams. One oxygen molecule weighs 32 grams, (2 x 16 grams per oxygen atom).

Similarly, in the above equation 12 grams of carbon react with 32 grams of oxygen. i.e., 12 mass units of carbon react with 32 mass units of oxygen. The ratio of the masses of the individual reactants is 12:32.

It is not convenient to weigh gases. They are normally measure by volumes. The volume occupied at s.t.p. by one mole of oxygen or carbon dioxide is 22.4 dm^3 or 22400 cm^3 . This volume is called the molar volume.

Equal volumes of all gases, under the same conditions of temperature and pressure, contain the same number of molecules (Avogadro's Law)

It follows that 1 mole of any gas occupies a volume of 22.4 dm^3 at s.t.p. and a volume of 24.0 dm^3 at r.t.p. (room temperature and pressure).

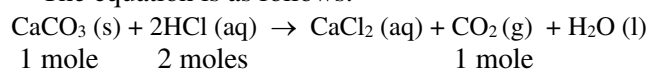
These are the values we will be using in all calculations.

Gay-Lussac's Law

When gases react together, they do so in volumes which are related to each other in simple ratio and to the product if gaseous, provided all measurements are made at the same temperature and pressure.

Example 20 *How much marble (CaCO_3) remains undissolved and what volume of carbon dioxide (at r.t.p.) formed when all 14.6 g of hydrochloric acid are allowed to react with 50g marble?*

The equation is as follows:



The question can be solved using mole concept.

Molar mass mole of HCl = 36.5 g.

$$\text{Moles of HCl used} = \frac{14.6}{36.5} = 0.4 \text{ mole}$$

Mass of CaCO_3 that reacted (thus dissolved)

From the equation,

2 moles of HCl react with 1 mole of CaCO_3 (=100)

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$$\begin{aligned}
 &0.4 \text{ mole of HCl react with } \frac{0.4}{2} = 0.2 \text{ mole of CaCO}_3. \\
 &= (0.2 \times 100)\text{g} \\
 &= 20 \text{ g of CaCO}_3
 \end{aligned}$$

$\therefore$ Mass of undissolved $\text{CaCO}_3 = (50 - 20) = \mathbf{30 \text{ g}}$.

Volume of CO_2

2 moles of HCl liberate 1 mole or 24 dm^3 of CO_2 at r.t.p.

$$\therefore 0.4 \text{ mole of HCl will liberate } \frac{0.4 \times 24}{2} = 4.8 \text{ dm}^3 \text{ of CO}_2.$$

Example 21. According to the equation: $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$

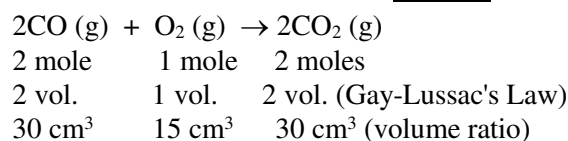
How many moles of carbon which react with 9.6 dm^3 of oxygen at r.t.p?

1 mole or 24 dm^3 (at r.t.p.) of O_2 react with 2 mole of carbon.

1 dm^3 of O_2 reacts with $\frac{2}{24}$ moles of carbon

$$\therefore 9.6 \text{ dm}^3 \text{ react with } \frac{2 \times 9.6}{24} = 0.8 \text{ moles of carbon.}$$

Example 22. 40 cm^3 of oxygen was added to 30 cm^3 of carbon monoxide and the mixture ignited. What is the volume and composition of the resulting mixture? (All volumes measured at the same temperature and pressure).



The volume of CO_2 produced is the same as that of CO used up, i.e. 30 cm^3 of CO_2 are produced.

15 cm^3 of the 40 cm^3 of oxygen are used up in the reaction, i.e. 25 cm^3 of oxygen

30 cm^3 of carbon dioxide produced and thus has a volume of 55 cm^3 .

EXERCISE IV

1. Calculate the mass of magnesium oxide obtained by burning 2.4g of magnesium in excess oxygen.
2. 0.2 g of copper (II) oxide is heated in a stream of hydrogen. Calculate the mass of copper obtained if the reduction is 100 percent.
3. What mass and volume of oxygen (at r.t.p.) would be formed when 4.9 g of potassium chlorate (KClO_3) are heated and completely decomposed into potassium chloride and oxygen?

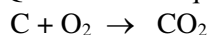
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4. Find the mass of sodium carbonate produced and the volume of carbon dioxide gas evolved at r.t.p. when 42 g of sodium hydrogen carbonate (NaHCO_3) is heated and completely decomposed?
5. Write the equation of the reaction between sodium and water. 4.6 g of sodium are completely dissolved in water. Calculate the mass and volume (at r.t.p.) of hydrogen produced.
6. When 3.1 g of a carbon dioxide MCO_3 are heated to constant mass, the volume of carbon dioxide evolved during the heating is 600 cm^3 at r.t.p. What is the relative atomic mass of M?
7. 2.8 g of iron is reacted completely with chloride according to the reaction: $2 \text{Fe (s)} + 3 \text{Cl}_2 \text{ (g)} \rightarrow 2 \text{FeCl}_3 \text{ (s)}$. Find the mass of iron (III) chloride formed and the minimum volume of chloride at r.t.p. needed for this reaction.
8. Find the mass of copper (II) sulphide that can be precipitated when hydrogen sulphide is bubbled through a solution of copper (II) sulphate containing 4.0 g of copper (II) sulphate.
9. The equation for the reduction process taking in the blast furnace is $\text{Fe}_2\text{O}_3\text{(s)} + 3\text{CO (g)} \rightarrow 2\text{Fe(s)} + 3 \text{CO}_2 \text{ (g)}$
What is the maximum mass of iron which could be extracted from 40 tones of iron (III) oxide? What is the mass and volume (at r.t.p) of carbon dioxide produced?
10. What mass of 90 percent pure ammonium chloride would be needed just to react completely with 14.8 g of slaked lime, Ca(OH)_2 ? What volume at s.t.p of ammonia will be produced?
11. What volume of oxygen is need for complete combustion of 20 litres of propane (C_3H_8) to carbon dioxide and water, measured at r.t.p.?
12. Ethene reacts with oxygen according to the equation

$$\text{C}_2\text{H}_4 \text{ (g)} + 3\text{O}_2 \text{ (g)} \rightarrow 2\text{CO}_2 \text{ (g)} + 2\text{H}_2\text{O (g)}$$
 15 cm^3 of ethene were mixed with 60 cm^3 of oxygen and the mixture was sparked to complete the reaction. All volumes were measured at a pressure of 1 atm. and (i) at a temperature of 120°C (ii) at room temperature.
Find the volume and composition of the gas mixture left at the end of the experiment in each case.
- 13*. 10 cm^3 of a gaseous hydrocarbon were mixed with 90 cm^3 of oxygen and sparked. The resulting volume at r.t.p. was 70 cm^3 , which reduced to 30 cm^3 on shaking with sodium hydroxide. Find the empirical formula of the hydrocarbon.
- 14 According to the equation,

$$\text{C}_3\text{H}_8 \text{ (g)} + 5\text{O}_2 \text{ (g)} \rightarrow 3\text{CO}_2 \text{ (g)} + 4\text{H}_2\text{O (l)}$$
 (a) How many moles of CO_2 will be produced in the reaction of 1 Mole of O_2 ?
 (b) How many grams of CO_2 are produced by 1 mole O_2 ?
 (c) How many grams of propane (C_3H_8) will react with 3.2 g of O_2 ?
 (d) How many moles of propane (C_3H_8) will produce 9 g of H_2O ?
 (e) How many moles of CO_2 will be produced with 36 g of water?
 *(f) What are the maximum moles of CO_2 that will be produced from a mixture of 12 g of C_3H_8 and 8 g of O_2 ?
 Assume complete reaction.

**Q15 From the equation



What will be the partial pressure of CO_2 in the product mixture formed from 20 g of O_2 and 3 g of CO_2 , assuming complete reaction.

1992/1/2

When 20 g of a mixture containing anhydrous sodium carbonate and sodium hydrogen carbonate was heated and cooled, the mass of the mixtures changed to 13.8 g.

- (a) Write equation for the reaction that took place during the heating (1 ½ marks)
- (b) Calculate the percentage of sodium carbonate in the mixture

1995/1/7

1.18g of compound P on evaporation occupied 300cm³ at s.t.p.

- (a) Calculate the relative molecular mass of P.
- (b) The empirical formula of P is C₂H₄O. Determine its molecular formula
- (c) Compound P reacts with sodium carbonate to produce a gas that turns lime water milky. Write the structural formula of P.

2000/1/17

When 0.55g of organic compound T was burned in oxygen 0.220g of carbon dioxide and 0.135g of water were formed.

- (i) Determine empirical formula of T
- (ii) When 0.225g of T was vaporized at 127°C and 760mmHG it occupied a volume of 119.11cm³. (Molar volume of a gas at s.t.p is 22.4dm³). calculate the formula mass of T and determine molecular formula of T
- (iii) T reacts with acidified potassium dichromate to form ethane-1,2-dioic acid. Write the formula and the IUPAC name of T.

2001/1/8

When 142cm³ of a hydrocarbon Q, of molecular mass 58 was exploded with excess oxygen and cooled to room temperature, the volume of the residual gas was 694cm³. After addition of concentrated potassium hydroxide, the volume decreased to 126cm³

- (a) determine the molecular formula of Q
- (b) write the names and the structural formula of all isomers of Q

2004

- (a) When 8.8g of a hydrocarbon, Z, was burn in excess air, 14.4g of water and 13.44dm³ of carbon dioxide were obtained at s.t.p. determine the empirical formula of Z.
- (b) The vapor density of Z is 22. Write the name and molecular formula of Z.
- (c) Write equations to show how Z can be synthesized from alcohol
- (d) Indicate a mechanism for the first stage of the reaction in (c)(i).

2005/1/16

A compound P contains 52.2% carbon, 13.0% hydrogen, the rest being oxygen

- (i) Determine the empirical formula of P.
- (ii) When vaporized, 0.1g of P occupied 78.8cm³ at 157°C and pressure of 740mmHg.
- (vi) Calculate the formula mass of P
- (vii) Determine the molecular formula of P
- (viii) Write the structural formula of all the possible isomers of P
- (ix) P does not react with sodium metal. Identify P
- (x) Write equation to show how P can be prepared from methanol.

Calculations involving Concentration of Solution.

Definitions

1. The **concentration** of a substance is the number of moles or the mass of a solute dissolved or contained in a known volume of solution. Usually the concentration of a solution is expressed in either number of grams or number of moles of solute per litre of solution.

2. A **standard substance/primary standard** is a pure substance whose mass can easily be weighed accurately and is used to standardize others.

Properties of primary standard

- Should have high molecular mass to be weighed accurately with little errors
- Should form stable solutions
- It should be stable in air, i.e., it shouldn't be hygroscopic, deliquescent, react with air, easily lose its water of crystallization and/or it should not be volatile.

3. A **Standard solution** is a solution whose concentration is known, usually expressed in terms of molarity, mole/litre, moles/dm³, or grams/litre.

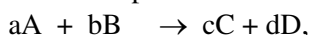
4. The **Molarity** of solution is the number of moles of the solute contained in 1 litre, 1000 ml or 1000 cm³ of solution.

$$\text{Molarity} = \frac{\text{mass of solute dissolved in one litre}}{\text{molar (formula) mass}}$$

$$\text{or} = \frac{\text{concentration in g/l}}{\text{molar (formula) mass}}$$

$$\text{Hence, No. of moles} = \frac{\text{Molarity} \times \text{volume}}{1000 \text{ ml of solution}}$$

If solutions of compounds A and B react according to the equation



$$\text{then } \frac{V_A \times M_A}{V_B \times M_B} = \frac{a}{b}$$

where, V_A and V_B : volumes of solution
and m_A and m_B : molarities of solutions

5. A **molar solution** is a solution containing 1 mole of a solute/substance per litre.

Example 23. Calculate the molarities of (a) 7g of potassium hydroxide KOH, in 250cm³ solution (b) 9.8g of sulphuric acid in 2 dm³ solution.

(a) RMM of KOH = 39 + 16 + 1 = 56 g

7g of KOH in 250cm³ solution $\equiv$ 28g KOH in 1000cm³
or 1 dm³ solution

$$\text{Molarity of KOH} = \frac{g / dm^3}{RFM} = \frac{28}{56} = 0.5M$$

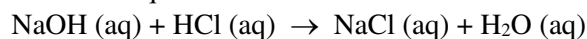
b) RMM of H₂SO₄ = 2 + 32 + 64 = 98g

9.8g of H₂SO₄ in 2dm³ solution = 4.9g in 1 dm³ solution.

$$\text{Molarity of H}_2\text{SO}_4 = \frac{g / dm^3}{RFM} = \frac{4.9}{98} = 0.05M$$

Example 24. 25.0 cm³ of a solution of sodium hydroxide of concentration 0.1 mole dm⁻³ were exactly neutralized by 20.0 cm³ of a solution of hydrochloric acid. Calculate the concentration of the acid solution (a) as a molarity (b) in g/dm³.

Reaction equation:



Mole of NaOH

1000 m³ contain 0.1 moles

$$25 \text{ cm}^3 \text{ will contain } \frac{25 \times 0.1}{1000} = 0.0025 \text{ moles}$$

Moles of HCl

From the equation

1mole of NaOH react with 1 mole of HCl

$$\Rightarrow \text{moles of HCl} = \text{moles of NaOH} \\ = 0.0025 \text{ moles}$$

Molarity of HCl

20cm³ contain 0.0025mole

$$1000\text{cm}^3 \text{ will contain } \frac{0.0025 \times 1000}{20} = 0.125 \text{ moles}$$

$$\Rightarrow \text{molarity of HCl} = \underline{0.125M}$$

Concentration = moles x RFM

$$= 0.125 \times 36.5$$

$$= 4.58 \text{ g/dm}^3$$

Example 25. In a titration 30 cm³ of 0.4M sodium hydroxide require 40cm³ of 0.15 M phosphoric acid, H₃PO₄.

- (i) How many moles of sodium hydroxide are present in 30cm^3 of solution?
 (ii) How many moles of phosphoric acid are present in 40cm^3 of solution?
 (iii) Find the number of moles of sodium hydroxide required to neutralize one mole of phosphoric acid?
 (iv) Write the formula of the sodium salt formed and the equation for the reaction between the acid and sodium hydroxide?

Answer

$$(i) \text{ moles of NaOH} = \frac{30 \times 0.4}{1000} = 0.012 \text{ moles}$$

$$(ii) \text{ moles of H}_3\text{PO}_4 = \frac{40 \times 0.15}{1000} = 0.006 \text{ moles}$$

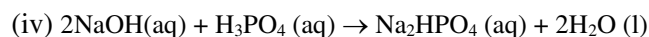
$$(iii) \text{ NaOH} : \text{H}_3\text{PO}_4$$

$$0.012 : 0.006$$

$$0.006 : 0.006$$

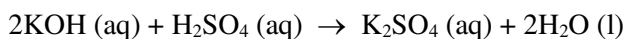
$$2 : 1$$

2 moles of NaOH required neutralizing one of phosphate acid.



The formula of sodium salt formed is Na_2HPO_4

Example 26. 24.0cm^3 of sodium of 0.1M KOH were exactly neutralized by 30.0cm^3 of sodium of χM H_2SO_4 . Find the concentration of the acid solution (a) as the molarity χ (b) in g/dm^3



Mole of KOH

1000 cm^3 contain 0.1 moles

$$24.0\text{ cm}^3 \text{ will contain } \frac{24 \times 0.1}{1000} = 0.0024 \text{ moles}$$

Moles of H_2SO_4

From the equation

2mole of KOH react with 1 mole of H_2SO_4

$\Rightarrow$ moles of $\text{H}_2\text{SO}_4 = \frac{1}{2}$ moles of KOH

$$= \frac{0.0024}{2} = 0.0012 \text{ moles}$$

Molarity of χ of H_2SO_4

30cm^3 contain 0.0012mole

$$1000\text{cm}^3 \text{ will contains } \frac{0.0012 \times 1000}{30} = 0.04 \text{ moles}$$

$$\therefore \chi = 0.04\text{M}$$

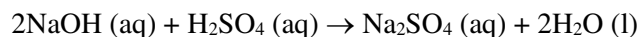
Thus g/dm^3 of $\text{H}_2\text{SO}_4 = 0.04 \times 98 = 3.92\text{g/dm}^3$

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Example 27. In an experiment to determine the concentration dilute sulphuric acid in moles per litre, 25 cm³ of 0.2M sodium hydroxide solution required 24.6 of the acid. Calculate the molarity of the acid.

Equation.



Moles of the NaOH that reacted

1000 cm³ contains 0.2 mole

$$\begin{aligned} 1\text{ cm}^3 & \text{ contains } \frac{0.2}{1000} \text{ moles} \\ 25\text{ cm}^3 & \text{ contain } \frac{0.2 \times 25}{1000} = 0.005 \text{ moles} \end{aligned}$$

Moles of H₂SO₄

From the equation: 2 mole of NaOH reacts with 1 mole H₂SO₄.

0.05 mole of NaOH react with (½ x 0.005) mole of acid

Molarity

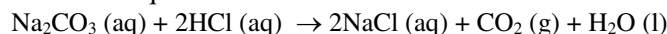
∴ Mole of H₂SO₄ in 24.6cm³ that reacted = 0.0025mole

$$\begin{aligned} 1\text{ cm}^3 & \text{ contain } \frac{0.0025}{24.6} \\ \Rightarrow 1000\text{ cm}^3 & \text{ contain } \frac{0.0025 \times 1000}{24.6} = 0.102 \text{ mole} \end{aligned}$$

Therefore, molarity of H₂SO₄ solution is **0.102M**

Example 28. Find the volume of 0.25M HCl required to exactly react with 20 cm³ of 0.1M sodium carbonate solution.

Reaction equation



Moles of the Na₂CO₃ that reacted

1000 cm³ contains 0.1 moles

$$\begin{aligned} 1\text{ cm}^3 & \text{ contains } \frac{0.1}{1000} \text{ moles} \\ 20\text{ cm}^3 & \text{ contain } \frac{0.1 \times 20}{1000} = 0.002 \text{ moles} \end{aligned}$$

Moles of HCl

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20

From the equation; 1 mole of Na_2CO_3 reacts with 2 mole HCl

0.02 mole of Na_2CO_3 react with (0.002×2) mole of HCl
 $= 0.004$ moles

Volume of hydrochloric acid required
 0.25 moles are contained in 1000cm^3

$$\Rightarrow 0.004 \text{ moles will be in } \frac{1000 \times 0.004}{0.25}$$

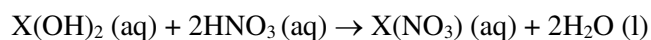
$$= 16 \text{ cm}^3$$

The volume required is **16 cm^3**

Example 29. 20 cm^3 of 0.1 M nitric acid was completely neutralized by 10 cm^3 of a hydroxide of a divalent metal $\text{X}(\text{OH})_2$ which was made by dissolving 5.8g of the hydroxide in 1 litre of solution. ($H=1$, $O=16$).

(a) Calculate the molarity of the hydroxide.

Reaction equation



No. of moles of the nitric acid that reacted

$$\begin{array}{ll} 1000 \text{ cm}^3 \text{ contains} & 0.1 \text{ moles} \\ 1 \text{ cm}^3 \text{ contains} & \frac{0.1}{1000} \text{ moles} \\ \Rightarrow 20 \text{ cm}^3 \text{ contain} & \frac{20 \times 0.1}{1000} \\ & = 0.002 \text{ moles} \end{array}$$

(b) Calculate the molar mass of $\text{X}(\text{OH})_2$ and hence determine the atomic mass of X.

From the reaction equation 2 moles of nitric acid react with 1 mole of the hydroxide.

1 mole of nitric acid react with $\frac{1}{2}$ moles of $\text{X}(\text{OH})_2$

$$\Rightarrow \text{moles of the hydroxide in reacted } 0.002 \times \frac{1}{2}$$

$$= 0.001 \text{ mole}$$

Molarity of $\text{X}(\text{OH})_2$

$\therefore 10 \text{ cm}^3$ of $\text{X}(\text{OH})_2$ contain 0.001 mole

$$\begin{array}{ll} 1 \text{ cm}^3 \text{ contains} & \frac{0.001}{1000} \text{ moles} \\ \Rightarrow 1000 \text{ cm}^3 \text{ contains} & \frac{0.001 \times 1000}{10} \\ & = 0.1 \text{ moles} \end{array}$$

Therefore, molarity of the hydroxide = **0.1M.**

$$\begin{aligned} \text{(b) Molar mass of } X(\text{OH})_2 &= \frac{\text{concentration g/l}}{\text{molarity}} \\ &= \frac{5.8}{0.1} = 58\text{g} \end{aligned}$$

The atomic mass of X is obtained from using the molar mass, i.e.

$$\begin{aligned} \text{Molar mass of } X(\text{OH})_2 &= X + 2(16+1) = 58 \\ &= X + 34 = 58 \\ \Rightarrow X &= 24\text{g} \end{aligned}$$

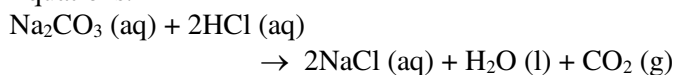
Thus, the hydroxide of compound X is magnesium hydroxide $\text{Mg}(\text{OH})_2$.

Example 29. In determination of water of crystallization of $\text{Na}_2\text{CO}_3 \cdot n\text{H}_2\text{O}$, 14 cm^3 of 0.2M HCl was required to neutralize 25 cm^3 of sodium carbonate prepared by dissolving 4g of $\text{Na}_2\text{CO}_3 \cdot n\text{H}_2\text{O}$ in 250 cm^3 of solution.

(a) Calculate the molarity of sodium carbonate.

(b) Calculate the molar mass of the hydrated sodium carbonate and hence calculate the number of moles n of water of crystallization.

Equations.



Moles of the acid that reacted

$$\begin{aligned} 1000\text{ cm}^3 &\text{ contain } 0.2\text{ mole} \\ 1\text{ cm}^3 &\text{ contain } \frac{0.2}{1000}\text{ moles} \\ \therefore 14\text{ cm}^3 &\text{ contain } \frac{0.2 \times 14}{1000} = 0.0028\text{ moles} \end{aligned}$$

From the reaction equation, 2 moles of HCl react with 1 mole of the base.

$$\begin{aligned} \text{Thus, moles } \text{Na}_2\text{CO}_3 \text{ in } 25\text{ cm}^3 \text{ that reacted} &= \frac{0.0028}{2} \\ &= 0.0014\text{ moles} \end{aligned}$$

$$\begin{aligned} 1\text{ cm}^3 &\text{ contain } = \frac{0.28}{2 \times 25} \\ \Rightarrow 1000\text{ cm}^3 &\text{ contain } \frac{0.28 \times 1000}{2 \times 25} \end{aligned}$$

$$22$$

$$= 0.056 \text{ M.}$$

thus, molarity of sodium carbonate solution is 0.056M.

(b) Molar mass of hydrated sodium carbonate

$$\text{Molar mass} = \frac{\text{Concentration g/l}}{\text{Molarity}} = \frac{2 \times 4}{0.056} = 286 \text{ g}$$

Value of n in the formula $\text{Na}_2\text{CO}_3 \cdot n\text{H}_2\text{O}$ is calculated using the molar mass.

$$\text{thus, } (23 \times 2) + 12 + (16 \times 3) + 18n = 286$$

$$18n = 286 - 106$$

$$n = 10$$

Example 29. During determination of Basicity of an acid H_nA , 25 cm^3 of KOH prepared by dissolving 11.2 g in 1 litre of solution was required to neutralize 25 cm^3 of 0.1M H_nA .

(a) Calculate the molarity of the hydroxide.

(b) Calculate the basicity of the acid.

a) Molarity of the hydroxide.

$$\text{Molarity} = \frac{\text{concentration (g/l)}}{\text{Formula mass}} \quad \text{Formula mass of KOH} = 39 + 16 + 1$$

$$= 56 \text{ g}$$

$$\therefore \text{Molarity} = \frac{11.2}{56} = 0.2 \text{ M}$$

(b) Basicity of the acid.



Reaction ratio n:1

Moles of the KOH that reacted

$$= \frac{0.2 \times 25}{1000} = 0.005 \text{ moles}$$

Moles of the acid that reacted

$$= \frac{0.1 \times 25}{1000} = 0.0025 \text{ moles}$$

Thus
$$n = \frac{0.005}{0.0025} = 2$$

Example 31. What volume of 5M H_2SO_4 is required to prepare 250 cm^3 of 0.25M of the acid.

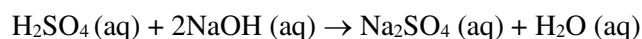
$$\text{Molarity}^1 \times \text{Volume}^1 = \text{Molarity}^2 \times \text{Volume}^2$$

$$5 \times \text{volume conc. acid} = 0.25 \times 250$$

$$\begin{aligned} \text{Volume of the conc. acid} &= \frac{0.25 \times 250}{5} \\ &= 12.5 \text{ cm}^3 \end{aligned}$$

Example 32. 10 cm^3 of concentrated sulphuric acid were diluted to 1000 cm^3 . 25 cm^3 of this diluted acid were neutralized by 30 cm^3 of 0.3 M sodium hydroxide. Calculate the mass of the acid in 1 dm^3 of the concentrated acid.

Equation.



Moles of NaOH that reacted.

$$1000 \text{ cm}^3 \text{ contain } 0.3 \text{ mole}$$

$$1 \text{ cm}^3 \rightarrow \frac{0.3}{1000}$$

$$\therefore 25 \text{ cm}^3 \rightarrow \frac{0.3 \times 25}{1000}$$

$$= 0.0075 \text{ moles}$$

but, 2 moles of NaOH react with 1 mole of the acid.

$\therefore$ moles of H_2SO_4 in 25 ml of the dilute acid

$$= 0.0075 \times \frac{1}{2}$$

$$1000 \text{ cm}^3 \text{ contain } \frac{0.0075 \times 1000}{25} = 0.3 \text{ moles}$$

$\therefore$ molarity of the dilute solution is 0.3M

Since 10 cm^3 of the concentrated acid was diluted to 1000 cm^3 to make the dilute solution, then.

$$\text{Molarity}^1 \times \text{Volume}^1 = \text{Molarity}^2 \times \text{Volume}^2$$

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$$10X = 1000 \times 0.18$$

where X is the molarity of the concentrated acid.

thus, $X = 18M$

The mass of the acid in $1dm^3$ of concentrated acid = Molarity $\times$ Molar mass

$$= 18 \times 98$$

$$= 1764 \text{ g.}$$

Example 33. In the determination of the percentage purity of NH_4Cl ; 5.0 g of an impure salt of NH_4Cl was boiled with $100cm^3$ of $2M$ $NaOH$ for about 10 minutes. The residual solution was made up to $250 cm^3$ in a volumetric flask with distilled water. $25 cm^3$ portion of the resultant solution required $22.4 cm^3$ of $0.5M$ HCl for neutralization using methyl orange indicator.

(a) Calculate the percentage of NH_4Cl in the sample

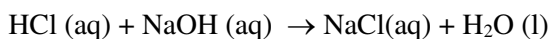
(b) Hence determine the percentage of ammonia in the pure sample.

Equations

(i) Sodium hydroxide react with NH_4Cl to release ammonium gas which is lost on heating as per the equation.



(ii) Excess sodium hydroxide reacts with HCl as per the equation.



Moles of the acid that reacted with excess base

$$\begin{aligned} 1000 \text{ cm}^3 &\text{ contain } 0.5 \text{ moles} \\ \therefore 22.4 \text{ cm}^3 &\text{ contain } \frac{0.5 \times 22.4}{1000} \\ &= 0.0112 \text{ moles} \end{aligned}$$

But, 1 mole of the acid reacts with 1 mole of the base

$\therefore$ moles of excess $NaOH$ in $25 cm^3$ of the solution

$$250 \text{ cm}^3 \text{ will contain } \frac{0.0112 \times 250}{25} = 0.0112 \text{ moles}$$

25

Moles of NaOH added to impure NH_4Cl 1000 cm^3 contain 2 moles

$$\therefore 100 \text{ cm}^3 \text{ contain } \frac{2 \times 100}{1000} = 0.2 \text{ moles}$$

Mole of NaOH that reacted with NH_4Cl is equivalent to the difference between moles NaOH added and that found in excess.

$$= 0.2 - 0.112 = 0.088 \text{ moles}$$

Since 1 mole of NaOH reacts with 1 mole NH_4Cl , then mole of NH_4Cl that reacted
 $= 0.088$

$$\begin{aligned} \text{Mass of } 0.088 \text{ moles of } \text{NH}_4\text{Cl} &= \text{molarity} \times \text{molar mass} \\ &= 0.088 (14+4+35.5) \\ &= 4.708 \text{ g} \end{aligned}$$

$$\text{Percentage purity} = \frac{\text{Mass of pure substance} \times 100}{\text{Total mass of impure sample}}$$

$$\therefore \text{Percentage purity of } \text{NH}_4\text{Cl} = \frac{4.708 \times 100}{5.0}$$

$$= 94.2\%$$

(b) Percentage of ammonia in the pure sample

Every 53.5 g of NH_4Cl contain 17g of NH_3

$$\therefore 4.708 \text{ contain } \frac{17 \times 4.708}{53.5} = 1.496 \text{ g}$$

$$\begin{aligned} \text{Percentage of } \text{NH}_3 \text{ in pure sample} &= \frac{1.496 \times 100}{4.708} \\ &= 31.77\% \end{aligned}$$

Exercise V

- What is the molarity of 21.2 g of sodium carbonate dissolved in 10 litres of solution (Na=23, C=12, O=16).
- Find the volume of 0.01M sodium hydroxide which is required to react exactly with 25 cm^3 of 0.02M HCl.
- 25 cm^3 of 0.1M sodium carbonate was found to react with 23.3 cm^3 of HCl to complete neutralization. Calculate the molarity of hydrochloric acid.
- A solution of 0.1M HCl and volume 25 cm^3 required 12.5 cm^3 of a metal hydroxide $\text{M}(\text{OH})_2$ which was prepared by dissolving 1.45 g of the $\text{M}(\text{OH})_2$ in 250 cm^3 of solution. (H=1, O=16)
 - Calculate the molarity of $\text{M}(\text{OH})_2$

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- (b) Calculate the molar mass of $M(OH)_2$
(c) Calculate the atomic mass of M.
5. Calculate the molarity and concentration in grams per litre of a sodium hydroxide solution, if 12.5 cm^3 of $0.2M$ H_2SO_4 acid required 25 cm^3 of NaOH for complete neutralization.
6. 25 cm^3 of oxalic ($H_2C_2O_4 \cdot nH_2O$) acid solution prepared by dissolving 2.52 g of the acid into 250 cm^3 required 20 cm^3 of sodium hydroxide solution prepared by dissolving 8 g of the hydroxide in one litre of solution.
- (a) Calculate the molarity of sodium hydroxide solution.
(b) Calculate the molarity of oxalic ($H_2C_2O_4 \cdot nH_2O$) acid solution
(c) Calculate the value of n (the mole of molecules of water of crystallization ($C=12, H=1, O=16, N=23$))
7. Calculate the molarities/concentrations of the following solutions.
(a) 20.0 g of sodium hydroxide, NaOH, in 250 cm^3 of solution;
(b) 3.8 g of iron (II) sulphate, $FeSO_4$, in 100 cm^3 solution.
8. Calculate the number of moles of the named substance in
(a) 50 cm^3 of $0.1 M$ nitric acid.
(b) 100 cm^3 of $1.5 M$ sodium carbonate.
9. Calculate the masses of dissolved substances contained in
(a) 100 cm^3 of $2 M$ sodium sulphate solution, Na_2SO_4 .
(b) 50 cm^3 of $0.2 M$ potassium carbonate, K_2CO_3 .
10. 8.48 grams of a mixture of sodium chloride and anhydrous sodium carbonate were dissolved in 100 cm^3 of solution. 5 cm^3 of this solution required 40 cm^3 of $0.1M$ HCl for complete neutralization. Calculate the percentage composition of the original mixture.
11. Concentrated hydrochloric acid containing 36% by mass of the acid has a density of 1.18 g/cm^3 . Calculate the volume of this acid required to neutralize 8.4 grams of sodium hydrogen carbonate.
12. 20 cm^3 of $0.9M$ sodium hydroxide solution were added to 50 cm^3 of nitric acid. The excess acid neutralized 20 cm^3 of $1.1M$ alkali. Calculate the molarity of the original acid solution and the number of grams of nitric acid per dm^3 of this solution.
13. 2 grams of impure calcium carbonate were added to 80 cm^3 of $1.25M$ hydrochloric acid. The resulting solution was made up to 100 cm^3 and 10 cm^3 of this neutralized 12.5 cm^3 of sodium hydroxide solution containing 20 g/dm^3 . Determine the percentage purity of the calcium carbonate.
14. 0.15 gram of an ammonium salt was heated with excess sodium hydroxide. The ammonia given off was absorbed in 25 cm^3 of $0.16M$ HCl. The excess acid required 12 cm^3 of $0.2M$ sodium hydroxide for neutralization. Calculate the percentage of ammonia in the salt.
15. 24 cm^3 of solution of tribasic acid containing 12 grams of the acid per dm^3 of solution just neutralized 48 cm^3 of $0.09M$ sodium hydroxide. Calculate the molar mass of the acid.

1985/2/6

- (a) (i) Explain the term primary standard solution (3marks)
(ii) Why is sodium hydroxide NOT recommended as a primary standard for volumetric analysis? (2 marks)
- (b) (i) Why does the salt of sodium carbonate behaves as an alkali in acid-base titration?
(ii) 21.40g of hydrated sodium carbonate ($\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$) was dissolved in water to make a litre of solution. 25 cm³ of this solution required 18.70cm³ of 0.2M standard hydrochloric acid solution for complete neutralization. Calculate the value of X.

1991/1/11

A hydrated compound Q contains the following ions, iron, sulphate and ammonium.

- (a) When 9.64g of Q was heated, 4.32g of water was evolved. Determine the number of moles of water of crystallization in Q. (1mark).

- (b) When a solution containing 9.64g of Q was heated strongly with excess sodium hydroxide solution, 2.14g of hydroxide of iron was formed together with ammonia. The ammonia required 100cm³ of 0.2M hydrochloric acid for complete neutralization.

Calculate

- (i) the number of moles of ammonium ions in Q
(2marks)
- (ii) the number of moles of irons in Q (2marks)
- (c) A solution containing 9.64g of Q was treated with excess barium chloride solution gave 9.32g of barium sulphate. Determine the number of moles of sulphate ions in Q (2marks)
- (d) Determine the molecular formula of Q. (2marks)

1991/1/1

Concentrated sulphuric acid is 18M. Calculate the volume, in cm³ of the acid that would be required to prepare 200cm³ of 2.0M sulphuric acid (3 marks)

1992/2/8

- (b) Concentrated nitric acid is 70% (w/w) and has a density of 1.42gcm⁻³.
(c) 12.68 cm³ of the acid in (b) was dissolved in water and the solution made up to 250 cm³ with distilled water. Calculate the volume that would react completely with 25.0cm³ of 0.2M sodium carbonate (4 marks)

1994/1/15

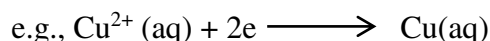
1.55g of an acid $\text{C}_n\text{H}_{2n}(\text{COOH})_2$ was dissolved in water and the solution made up to 250cm³. 25.0cm³ of this solution required 23.5cm³ of 0.1M aqueous sodium hydroxide solution for complete neutralization. Calculate n (9 marks)

2003/2/8c

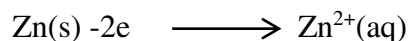
1.07g of a nitrogen containing compound was boiled with excess sodium hydroxide solution to produce ammonia. The ammonia produced neutralized 200cm³ of a 0.1M monobasic acid. Calculate the percentage by mass of nitrogen in the compound (3)

Redox titration

Redox reaction is a reaction in which both reduction and oxidation take place simultaneously. A reduction reaction is where there is addition of electrons to an ion species;



whereas, oxidation is loss of electrons from a compound.



The species which accepts electrons such as Cu^{2+} is reduced but is referred to as an oxidizing agent.

The species which donates electrons such as Zn^{2+} is oxidized but is referred to as a reducing agent

Oxidation Number

This is a number used to express the oxidation state of an element. Atoms of elements are given oxidation number zero. Then when two elements are combined the atoms or ions of more electronegative element are regarded as being in negative oxidation state, and those of a more electropositive element in a positive state. If we arbitrarily assume that the bonds between the atoms are

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all ionic, the oxidation number of an element is simply the number of electrons given up or received by an atom of an element. In sodium chloride the oxidation number of sodium is 1, while that of chlorine is -1. hydrogen in HCl has oxidation number 1 but in sodium hydride, NaH is -1. Some elements have several oxidation states. Sulphur has oxidation number -2 in H_2S , 4 in SO_2 and VI in SO_3 . chlorine shows a wide variety of oxidation number, e.g. -1 in HCl, 0 in Cl_2 , 1 in ClO^- , 3 in ClF_3 , 4 in ClO_2 , 5 in ClO_3^- , 6 in ClO_3 and 7 in ClO_4^- .

An increase in oxidation number of an element during a reaction means that an element has been oxidized. conversely, a decrease in oxidation number means that an element has been reduced.

Calculating oxidation number of ions of an element in complex ions.

Unless otherwise stated or implied in these calculations, the oxidation number of a metal is equal to group number, e.g., Na is 1, Mg is 2 Al is 3. Oxidation number of O is -2 and that of any group 7 element is -1.

Example 34. calculate the oxidation number of manganese ion in KMnO_4 .

Here the oxidation number of K = 1 and that of oxygen is implied to be -2. if the oxidation number of Mn is X

Then, $1 + X + -2 \times 4 = 0$ (overall charge ion the molecule)

$$X = 7$$

Exercise VI

Calculate the oxidation numbers of the Bolden atoms in molecules or atoms below

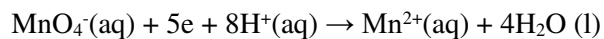
1. (i) KIO_3 (ii) SO_4^{2-} (iii) $\text{Cr}_2\text{O}_7^{2-}$
 (iv) H_2O_2 (v) KClO_4 (vi) $\text{Na}_2\text{S}_2\text{O}_6$

Half equations

These are balanced based on the number of electrons lost or accepted by a central atom.

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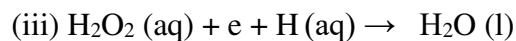
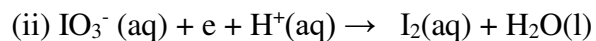
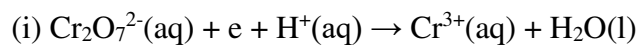
Example, equation for reduction of manganate (VII) in acidic medium is written as follows:



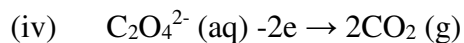
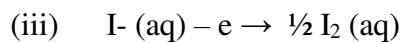
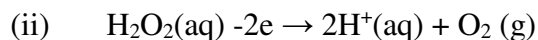
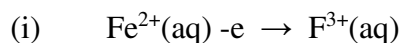
Five electrons are added because the oxidation number of Mn is reduced from 7 to 2.

Exercise VI I

Balance the following reduction equations



The following are common oxidation half equation reactions:

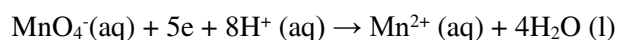


Overall equation

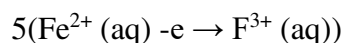
These are constructed from half equation in such a way that the number electron received by an oxidizing agent is equal to the number of electron lost from a reducing agent.

For example a reaction between MnO_4^- and Fe^{2+} in acidic medium, the half equation of Fe^{2+} is multiplied by 5 to provide the 5 electron necessary to reduce MnO_4^- .

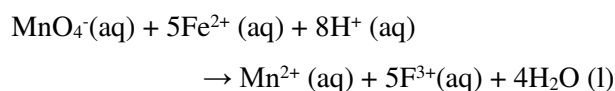
That is,



Plus



Equals

**Exercise VI**

- Write overall equations between the following pairs of ions or compounds, assume acidic medium where necessary
 - MnO_4^- and I^-
 - MnO_4^- and H_2O_2
 - KIO_3 and KI
 - MnO_4^- and $\text{C}_2\text{O}_4^{2-}$
 - $\text{K}_2\text{Cr}_2\text{O}_7$ and FeSO_4
- 0.9875g of an impure potassium manganate (VII) was dissolved in water to make 250cm^3 of solution. When 20.0cm^3 of this solution was acidified with dilute sulphuric acid, warmed and titrated against sodium ethanedioate (oxalate) solution, made by dissolving 1.675g of anhydrous sodium ethanedioate to make 250cm^3 , 24.40cm^3 of sodium ethanedioate solution was used. ($\text{Na}_2\text{C}_2\text{O}_4 = 134$ and $\text{KMnO}_4 = 158$).

Calculate percentage purity of potassium permanganate (VII)

- 10cm^3 of concentrated hydrogen peroxide was diluted to 250cm^3 . 20cm^3 of the dilute solution required 28.30cm^3 of 0.02M potassium manganate (VII) solution for complete reaction. Find the volume strength of the concentrated hydrogen peroxide solution.

1990/1/5

Potassium permanganate is commonly used as a reagent in volumetric analysis. However it is not a primary standard.

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- (a) State two disadvantage of using potassium manganate (VII) as a reagent in volumetric analysis.
- (b) Give two reasons why potassium manganate (VII) is not used as a primary standard
- (c) Write a half equation for the reduction of manganate (VII) ions in
 - (i) acid solution
 - (ii) alkaline solution

1994/1/17

- (a) In volumetric estimation of reducing agent, potassium dichromate (VI) is preferred to potassium manganate (VII) as an oxidant
Explain why potassium dichromate (VI) is preferred as an oxidant. (2 marks)
- (b) 3.8g of solder containing tin was dissolved in excess hydrochloric acid. The solution was made up to 250cm^3 . 25.0cm^3 of this solution required 23.5cm^3 of a 0.01M potassium dichromate (VI) solution for complete reaction.
Calculate the percentage, by mass, of tin in the solder.

- (d) Explain why chlorine is more soluble in dilute sodium hydroxide than in water (4 marks)

1999/1/1

- (a) Determine the oxidation number of
 - i. Nitrogen in HNO_3
 - ii. Sulphur in S_2O_8 (1 marks)
- (b) Write half equation for the conversion of $\text{S}_2\text{O}_8^{2-}$ to SO_4^{2-} ions
- (c) Complete the following and balance the equations
 - i. $\text{S}_2\text{O}_8^{2-} + \text{I}^- \rightarrow$
 - ii. $\text{Sn}^{2+}(\text{aq}) + \text{S}_2\text{O}_8^{2-}(\text{aq}) \rightarrow$

2001/1/14

Iron (II) sulphate is normally used to standardize a solution of potassium manganate (VII) acidified with sulphuric acid.

- (i) write equation for the reaction between potassium manganate (VII) and iron (II) sulphate
- (ii) State why hydrochloric acid is not used to acidify potassium manganate (VII) solution
- (b) 25 cm^3 of acidified solution of 0.02M potassium manganate (II) reacted exactly with 24.95 cm^3 of sodium nitrite. Potassium manganate (VII) reacts with sodium nitrite according to the following equation
$$2\text{MnO}_4^-(\text{aq}) + 5\text{NO}_2^-(\text{aq}) + 6\text{H}^+(\text{aq}) \rightarrow 2\text{Mn}^{2+}(\text{aq}) + 5\text{NO}_3^-(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$$

Calculate the concentration of sodium nitrite in moles per litre.

Iodometry

Here a substance is reacted with excess potassium iodide in redox equation to liberate iodine; the amount of iodine liberated is determined by titration with sodium thiosulphate solution to determine the concentration of the substance.

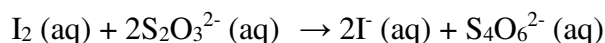
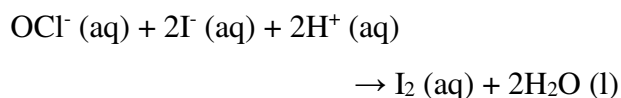
Example 35

Find concentration of NaOCl in gram per litre given that 20cm³ of NaOCl solution was reacted with excess potassium iodide, the liberated iodine required 25 cm³ of 0.1M sodium thiosulphate solution.

Solution

In acid medium, sodium chlorate reacts with iodide ions to liberate iodine, which oxidizes thiosulphate ions.

The reactions take place according to the following equations



Moles of sodium thiosulphate that reacted
100cm³ contain 0.1moles

$$25\text{cm}^3 \text{ contain } \frac{0.1 \times 25}{1000} = 0.0025 \text{mols}$$

$$\begin{aligned} \text{Moles of iodine produced from equation 2} \\ = \frac{0.0025}{2} = 0.00125 \text{moles} \end{aligned}$$

Moles NaOCl

From equation 1, moles of NaOCl is equal to the moles of iodine = 0.0015 moles

Molarity of NaOCl

20cm³ of NaOCl solution contain 0.00125 moles

$$\begin{aligned} 1000\text{cm}^3 \text{ contain } & \frac{0.00125 \times 1000}{20} \\ & = 0.0625\text{M} \end{aligned}$$

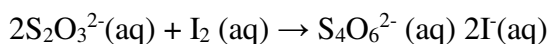
Concentration in grams per litre

$$\begin{aligned} 1 \text{ mole of NaOCl weighs} & (40 + 16 + 35.5) \\ & = 91.5\text{g} \\ 0.0625 \text{ moles weigh} & = 91.5 \times 0.0625 \\ & = 5.7\text{g/l} \end{aligned}$$

Exercise VII**1986/2/8e**

6.53 g of impure copper was dissolved in excess nitric acid. The solution obtained made up to 250cm³ of solution with water. To 25cm³ of this solution excess aqueous potassium iodide was added. The reaction that took place is given by the equation
 $2\text{Cu}^{2+} (\text{aq}) + 4\text{I}^- (\text{aq}) \rightarrow \text{Cu}_2\text{I}_2 (\text{s}) + \text{I}_2 (\text{aq})$

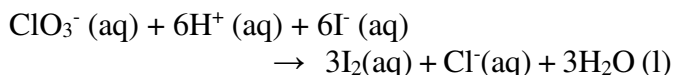
The iodine liberated reacted with 20cm³ of 0.5M aqueous sodium thiosulphate according to the equation



Calculate the percentage of by mass the purity of copper in the sample (Cu =64) (6marks)

1997/2/7c

2.0g of a mixture of potassium chloride and potassium chlorate were dissolved in 250cm³ of water. 10cm³ of the solution was mixed with excess potassium iodide. The iodine liberated required 8.0cm³ of a 0.2M sodium thiosulphate solution for complete reaction. Potassium chlorate and potassium chloride react according to the equation



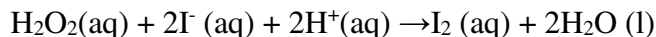
Calculate the percentage of potassium chlorate in the mixture (6marks)

Question

During the determination of concentration of hydrogen peroxide in g/dm³ of the medical hydrogen peroxide, 12 cm³ of medical hydrogen peroxide was diluted to 250cm³ with distilled water. 20cm³ of the

dilute solution was acidified required and excess KI added. The liberated iodine required 22cm³ of sodium thiosulphate solution containing 9.3g Na₂S₂O₃.5H₂O in 250cm³ of solution.

In acid medium, hydrogen peroxide reacts with iodide ions follows”



Calculate the volume strength of medical hydrogen peroxide.

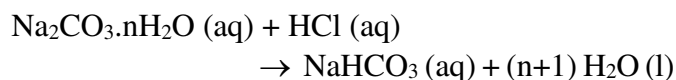
Double indicator titration.

Titration requiring use of two indicators is employed to determine the concentrations and proportions of any two of the following base in a mixture, i.e. NaOH, Na₂CO₃ and NaHCO₃. Ideally, the two indicators referred to here are the phenolphthalein and methyl orange indicator. The basis of these titrations is that, NaOH, and Na₂CO₃ are titrated to complete neutralization with methyl orange, whereas, NaHCO₃ is acidic to phenolphthalein indicator and thus Na₂CO₃ is only half neutralized when titrated with an acid in presence of phenolphthalein indicator.

Example 36. 7.15 g sodium carbonate Na₂CO₃.nH₂O was dissolved in water to make 250 ml of solution. 25.0 cm³ of this solution required 12.50 cm³ of 0.2 M HCl using phenolphthalein indicator. Calculate the value if n.

Equation

In presence of phenolphthalein indicator sodium carbonate is only half neutralized as per the equation



Moles of the acid that reacted:

1000 cm³ contain 0.2 mole

$$\therefore 12.5 \text{ cm}^3 \text{ contain } \frac{0.2 \times 12.5}{1000}$$

$$= 0.0025 \text{ mole}$$

From the reaction equation, 1 mole of Na₂CO₃.nH₂O reacts with 1 mole of HCl.

$\therefore$ 0.0025 mole of the acid react with 0.0025 mole of Na₂CO₃.nH₂O.

Molarity of Na₂CO₃.nH₂O

$\Rightarrow$ 25.0 cm³ of Na₂CO₃.nH₂O solution contain 0.0025 mole of Na₂CO₃.nH₂O.

$$1000 \text{ cm}^3 \text{ contain } \frac{0.0025 \times 1000}{25}$$

$$= 0.1\text{M}$$

$\therefore$ Molarity of Na₂CO₃.nH₂O solution is 0.1M

$$\text{RMM of Na}_2\text{CO}_3.n\text{H}_2\text{O} = \frac{\text{Concentration g/l}}{\text{Molarity}}$$

$$= \frac{7.15 \times 4}{0.1}$$

$$= 286$$

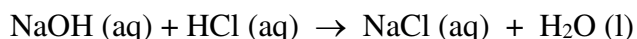
$$\text{Thus, } (23 \times 2) + 12 + (16 \times 3) + n(2+16) = 286$$

$$106 + 18n = 286$$

$$n = 10$$

Example 35. 25.0 cm³ of a solution containing a mixture of sodium hydroxide and sodium hydrogen carbonate require 15.0 cm³ of 0.1M HCl using phenolphthalein indicator. 25.0 cm³ of the same mixture required 28 cm³ of the acid using methyl orange indicator. Calculate the proportion by mass of the two bases in the mixture.

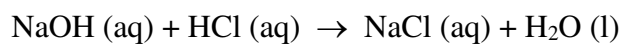
Equation for the reaction that took place using phenolphthalein



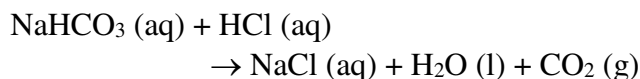
35

$\therefore 15 \text{ cm}^3$ of 0.1M HCl was required to neutralize NaOH is 25 cm^3 of the mixture.

Equations using methyl orange indicator



and



$\therefore$ The difference between the volume ($28 - 15 = 13 \text{ cm}^3$) of the acid was required to neutralize NaHCO_3 .

No. of moles of the acid that reacted with NaOH

1000 cm^3 contain 0.1 mole

$$\therefore 15 \text{ cm}^3 \text{ contain } \frac{0.1 \times 15}{1000} = 0.0015 \text{ mole}$$

From the reaction ratio of 1:1, 1 mole of HCl reacts with 1 mole of NaOH.

$$\Rightarrow \text{moles of NaOH in } 25 \text{ cm}^3 = 0.0015 \text{ mole}$$

$$\begin{aligned} 1000 \text{ cm}^3 \text{ will contain } & \frac{0.0015 \times 1000}{25} \\ & = 0.06 \text{ M} \end{aligned}$$

Concentration of NaOH in g/L = RMM x Molarity

$$= 40 \times 0.06$$

$$= 2.4 \text{ g/dm}^3$$

No. of moles of the acid that reacted with NaHCO_3 .

1000 cm^3 contain 0.1 mole

$$\therefore 13 \text{ cm}^3 \text{ contain } \frac{0.1 \times 13}{1000} = 0.0013 \text{ mole}$$

From the reaction ratio of 1:1,
1mole of HCl reacts with 1 mole of the NaHCO_3 .

$$\Rightarrow \text{moles of NaHCO}_3 \text{ in } 25 \text{ cm}^3 = 0.0013 \text{ mole}$$

$$\begin{aligned} 1000 \text{ cm}^3 \text{ will contain } & \frac{0.0013 \times 1000}{25} \\ & = 0.052 \text{ M} \end{aligned}$$

Concentration of NaHCO_3 in g/L = RMM x Molarity

$$= 84 \times 0.052$$

$$= 4.368 \text{ g/dm}^3$$

The ratio of masses of NaOH : NaHCO_3 is 2.4 : 4.368

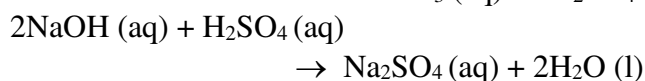
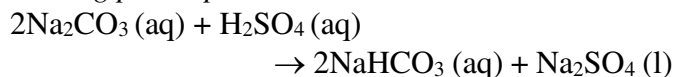
$$= 1 : 1.82$$

Example 36. (1990/1/13) 25.0 cm^3 of a solution containing a mixture of sodium carbonate and sodium hydroxide require 30.0 cm^3 of 0.1M H_2SO_4 using phenolphthalein indicator. Another 25.0 cm^3 of the same mixture required 48.0 cm^3 of the acid using methyl orange indicator. Calculate the molar concentration of the mixture with respect to:

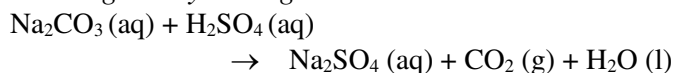
(i) Sodium carbonate (ii) sodium hydroxide. (9Marks)

Reaction Equations.

1. Using phenolphthalein indicator.

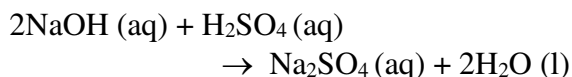


2. Using methyl orange indicator.



35

36



From the reactions above, the extra volume ($48 - 30 = 18 \text{ cm}^3$) of acid under methyl orange indicator is used to half neutralization of Na_2CO_3 . This also is equal to the volume of the acid that converts Na_2CO_3 to NaHCO_3 .

$\therefore$ the volume of the acid that completely neutralized Na_2CO_3 in 25 cm^3 of solution $= 18 \times 2 = 36 \text{ cm}^3$.

And volume of the acid that reacted with NaOH $= 30 - 18 = 12 \text{ cm}^3$.

No. of moles of the acid that reacted with NaOH.

1000 cm^3 contain $\frac{0.1 \text{ mole}}{1000}$

$\therefore 12 \text{ cm}^3$ contain $\frac{0.1 \times 12}{1000} = 0.0012 \text{ mole}$

From the reaction ratio of 1:2, 1 mole of H_2SO_4 reacts with 2 moles of NaOH.

$\Rightarrow$ moles of NaOH in $25 \text{ cm}^3 = 0.0012 \times 2 \text{ mole}$

1000 cm^3 will contain $\frac{0.0012 \times 2 \times 1000}{25}$

$= 0.048 \text{ M}$

$\therefore$ Molarity of solution in respect to NaOH $= 0.048 \text{ M}$.

No. of moles of the acid that reacted with Na_2CO_3 .

1000 cm^3 contain 0.1 mole

$\therefore 36 \text{ cm}^3$ contain $\frac{0.1 \times 36}{1000} = 0.0036 \text{ mole}$

From the reaction ratio of 1:1, 1 mole of H_2SO_4 reacts with 1 mole of the Na_2CO_3 .

$\Rightarrow$ moles of Na_2CO_3 in $25 \text{ cm}^3 = 0.0036 \text{ mole}$

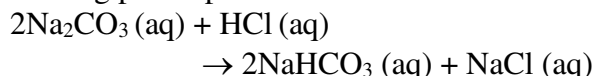
1000 cm^3 will contain $\frac{0.0036 \times 1000}{25} = 0.144 \text{ M}$

$\therefore$ the molarity of solution in respect to Na_2CO_3 is **0.144M**

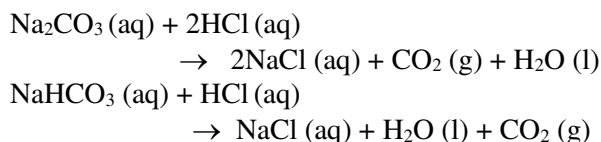
Example 37. 25.0 cm^3 of a solution containing a mixture of sodium carbonate and sodium hydrogen carbonate require 15.0 cm^3 of 0.5 M HCl using phenolphthalein indicator. Another 25.0 cm^3 of the same mixture required 34.5 cm^3 of the acid using methyl orange indicator. Calculate the concentration of the mixture with respect to: (i) sodium carbonate (ii) sodium hydrogen carbonate per dm^3 of solution.

Reaction Equations.

1. Using phenolphthalein indicator.



2. Using methyl orange indicator.



From the reactions above, 15.0 cm^3 of the acid was required for half neutralization of Na_2CO_3 .

$\therefore$ the volume of the acid that completely neutralized Na_2CO_3 in 25 cm^3 of solution $= 15 \times 2 = 30 \text{ cm}^3$.

and volume of the acid that reacted with NaHCO_3 initially in 25.0 cm^3 of the mixture $= 34.5 - 30 = 4.5 \text{ cm}^3$.

No. of moles of the acid that reacted with Na_2CO_3 .

1000 cm^3 contain 0.5 mole

$$\therefore 30 \text{ cm}^3 \text{ contain } \frac{0.5 \times 30}{1000} = 0.015 \text{ mole}$$

From the reaction ratio of 2:1, 2 mole of HCl react with 1 mole of the Na_2CO_3

$$\Rightarrow \text{moles of NaOH in } 25 \text{ cm}^3 = \frac{0.015}{2} = 0.0075 \text{ moles}$$

$$1000 \text{ cm}^3 \text{ will contain } = \frac{0.0075 \times 1000}{25} = 0.3 \text{ M}$$

Concentration of Na_2CO_3 in g/L = RMM x Molarity

$$= 106 \times 0.3$$

$$= 31.8 \text{ g/dm}^3$$

No. of moles of the acid that reacted with NaHCO_3

$$1000 \text{ cm}^3 \text{ contain } 0.5 \text{ mole}$$

$$\therefore 4.5 \text{ cm}^3 \text{ contain } \frac{0.5 \times 4.5}{1000} = 0.00225 \text{ mole}$$

From the reaction ratio of 1:1, 1 mole of HCl reacts with 1 mole of NaHCO_3 .

$$\Rightarrow \text{moles of NaHCO}_3 \text{ in } 25 \text{ cm}^3 = 0.00225 \text{ mole}$$

$$1000 \text{ cm}^3 \text{ will contain } \frac{0.00225 \times 1000}{25}$$

$$= 0.09 \text{ M}$$

Concentration of NaHCO_3 in g/L

$$= \text{RMM} \times \text{Molarity}$$

$$= 84 \times 0.09$$

$$= 7.56 \text{ g/dm}^3$$

Exercise IV

1. 25.0 cm^3 of a solution containing a mixture of sodium carbonate and sodium hydroxide require 27.2 cm^3 of $0.05 \text{ M H}_2\text{SO}_4$ using phenolphthalein indicator. Another 25.0 cm^3 of the same mixture required 37.6 cm^3 of the acid using methyl orange indicator. Calculate the mass of each base in 250 cm^3 of the mixture.

2. 25.0 cm^3 of a solution containing a mixture of sodium carbonate and sodium hydrogen carbonate require 11.2 cm^3 of 0.1 M HCl using phenolphthalein indicator and a further 28.6 cm^3 of the acid using methyl orange indicator. Calculate the percentage (w/v) of sodium carbonate in the mixture.

1993/2/7c

25.0 cm^3 of a solution containing a mixture of sodium carbonate and sodium hydrogen carbonate required 15.00 cm^3 of 0.5 M hydrochloric acid for complete reaction using phenolphthalein indicator.

25.0 cm^3 of solution of the mixture required 34.50 cm^3 of the acid using methyl orange indicator. Calculate the mass of sodium carbonate and sodium hydrogen carbonate in the solution in grams per litre. (6marks)

Answers**Exercise I**

- i. (a) 132, (b) 286, (c) 90 (d) 116
 ii. $^{63}\text{Cu} = 72.5\%$, $^{65}\text{Cu} = 27.5\%$

Exercise II

1. (i) 5L, (ii) 25L (ii) 0.1L.
 2. (a) 1.2×10^{23} (b) 1.2×10^{22} (c) 3×10^{23} (d) 3×10^{23}
 3. 11; 4. 171; 5. 16 g of sulphur, 28 g of iron.

Exercise III

1. H = 6.7%, O = 53.3%; 2. PbO_2 ; 3. 7; 4. FeCl_3 ; 5. Mg_3N_2
 6. C_8H_{18} , alkane; 7. $\text{C}_3\text{H}_8\text{O}$; 8. (i) H=0.03 g, C=0.18 g, O = 0.08 g (ii) $\text{C}_3\text{H}_6\text{O}$; 9. $\text{C}_4\text{H}_8\text{O}$, $\text{C}_4\text{H}_8\text{O}$; 10. CH_2O , $\text{C}_6\text{H}_{12}\text{O}_6$

Exercise IV

1. 33.33 g; 2. 0.16 g Cu; 3. $1.44 \text{ dm}^3 \text{ O}_2$, 1.92 g oxygen
 4. 26.5 g Na_2CO_3 , 6 dm^3 of CO_2 ; 5. 0.2 g of H_2 , 24 dm^3 of H_2 .
 6. 64; 7. 8.125 g FeCl_3 , 1.8 dm^3 of Cl_2 ; 8. 1.536 g
 9. 28 tonnes of iron, 33 g of CO_2 , 18 dm^3 of CO_2 .
 10. 23.8 g of NH_4Cl , 8.96 dm^3 of NH_3 11. 100 litres of C_3H_8
 12. (i) 59.3 cm^3 , (ii) 45 cm^3 at r.t.p; 13. C_4H_8 14. C_4H_8
 15. (a) 0.6 mole of CO_2 , (b) 26.4 g of CO_2 (c) 0.88 g of C_3H_8 , (d) 0.125 mole of C_3H_8 , (e) 66 g of CO_2 , (f) 0.75;
 16. 0.4

Exercise IV

1. 0.02M; 2. 50 cm^3 ; 3. 0.2146M; 4. (a) 0.05, (b) 74, (c) 40
 5. 0.2M; 6. (a) 0.2M, (b) 0.1M, (c) n=2; 7. (a) 2M, (b) 0.25M
 8. (a) 0.005 mole, (b) 0.15 mole; 9. (a) 24.4 g, (b) 1.38 g
 10. 50% of each; 11. 8.59 cm^3 ; 12. 0.8M, 50.4 g/dm^3 , 13. 96.9%;
 14. 18.1 per cent; 15. 200.

Exercise IV : 1. Na_2CO_3 , 4.41; NaOH, 2.688 2. 0.445%

STRUCTURE OF COMPOUNDS/SUBSTANCES

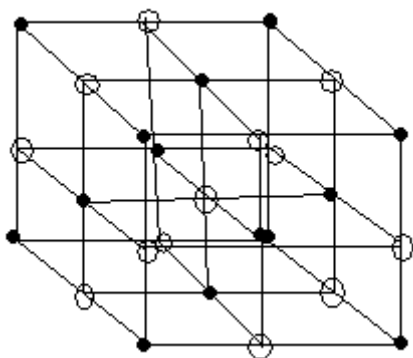
Different types of bonds give rise to different types of structures. This is illustrated in the table below

Bonding	Structure
Ionic/ electrovalent	Giant ionic structure
Covalent	Simple molecular structure
Covalent	Giant molecular structure
Metallic	Giant metallic

Giant ionic structure

This is a giant structure consisting of a regular arrangement of a large number of oppositely charged ions resulting into a three dimensional crystal. Each ion is surrounded by a number of oppositely charged ions and they are held together by strong electrostatic forces of attraction. The number of ions surrounding a given oppositely charged ion is known as the co-ordination number. The examples of structures include: sodium chloride and magnesium chloride.

Structure of sodium chloride



Each sodium ion is surrounded by six chloride ions and each chloride ion is also surrounded by six sodium ions. The co-ordination number is therefore 6:6.

Properties of ionic compounds

1. They are solids with a regular shape. This is because of the strong electrostatic forces of attraction keeping the ions closely packed.
2. They have high melting points. This is due to the strong electrostatic force of attraction between oppositely charged ions.

3. They do not conduct electricity in solid states but do conduct in their molten states or solution form. This is because in solid states, the ions are localized and the electrons are not mobile but in molten form, the ions and electrons are free to move (electrons become delocalized) and thus conduct electricity.

4. They have high density as the ions are closely packed.

5. Ionic compounds are soluble in water and other polar solvents but they are insoluble in organic solvents like benzene. Ionic compounds dissolve in polar solvents like water as the ions are attracted by the polar molecule.

A polar compound is a covalent compound in which charge separation exists between atoms. This is due to the differences in electronegativity between the atoms.

Electronegativity is the tendency of an atom to attract bonding electrons towards itself. Electropositivity is the tendency of an atom to push away bonding electrons from itself.

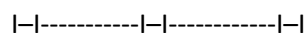
For example, in a water molecule oxygen is more electronegative and attracts the bonding electrons towards itself giving it a partial negative charge. Hydrogen attains a partial positive charge.

Structure of a water molecule

Simple molecular structures

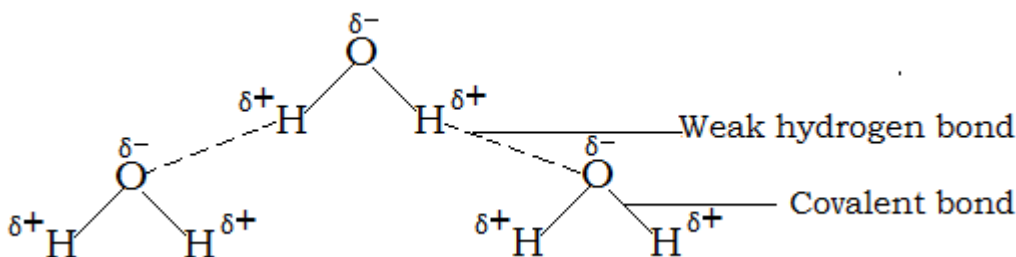
The structure consists of simple independent molecules joined together by weak intermolecular forces of attraction. E.g. Vander Waal's force of attraction. The atoms in the molecule are held together by strong covalent bonds. Compounds with this structure exist as gases, liquids or solids with low melting points. Examples include iodine, carbon dioxide, ammonia

Structure of iodine



Strong covalent bond holding iodine atoms together

Weak Vander Waal's force holding iodine molecules together



Properties of simple molecular structures

1. They have low melting and boiling points because the molecules are held by weak intermolecular forces of attraction.
2. They are non conductors of electricity because they do not have mobile electrons/ ions.
3. Most of them are gases and liquids. Very few are solids.
4. They have low densities as the molecules are not closely packed.
5. They dissolve in organic solvents.

Giant atomic/molecular structure

Compounds with such structures consist of molecules /atoms linked together by strong covalent bonds resulting into a giant three dimensional or three dimensional structures like in graphite and diamond respectively (see details under Carbon and its compounds)

Properties of giant molecular structures

1. They have high melting and boiling points because of great energy needed to break the bonds.
2. They do not conduct electricity except graphite which has delocalized electrons
3. They are insoluble in water.

Giant metallic structure

Giant metallic structure consists of very many metal ions surrounded by a sea /cloud of electrons. These electrons are released by metal atoms.

Properties of giant metallic structures

1. They are solids with high melting and boiling points except mercury which is a liquid.
2. They conduct electricity in both liquid and solid states due to the presence of mobile electrons. NB Most metals are malleable and ductile. This is because, since the bonding agent in a metal is a moving electron cloud, the ions of the metal; usually slide relative to one another under stress (without shattering the lattice and produce a new position of stability).

Sample questions on bonding and structure

Bonding and structure

1. Describe briefly what is meant by the following types of bonding: electrovalent, covalent, coordinate and metallic. Use examples to illustrate how the above bonds are formed between any two named atoms or molecules.

2. Explain what is meant by metallic bond. Explain why a typical metal is (a) a good conductor of electricity (b) a good conductor of heat (c) in some conditions at least malleable and ductile.
3. The compounds named below are all covalent compounds. With the aid of the table of electron structures, give a diagram for a molecule of each of these compounds showing the outer most electron shells only: (i) tetra chloromethane CCl_4 , phosphorus(III) chloride PCl_3 (iii) silane SiH_4 (iv) tri chloromethane CHCl_3 (v) phosphine PH_3 and (vi) dichloromethane CH_2Cl_2
4. Explain briefly why ammonia and oxygen molecules participate readily in coordinate bonding. Give an example for the formation of an ion by ammonia by this means. Show by means of electronic diagram the formation of a coordinate linkage between phosphorus tri chloride and oxygen.
5. By means of electronic diagrams, show the structure of (a) calcium atom (b) chlorine atom (c) calcium chloride. State the differences between electrovalent and covalent compounds.
6. Describe how structures of the following compounds account for their property in term of electrical conduction (a) copper (b) graphite (c) diamond
7. With the aid of well labeled drawings explain the structure of the following: Graphite, diamond, sodium chloride and chlorine molecule.

FOSSIL FUELS

A fuel is a substance consumed to provide energy through combustion or chemical reaction or nuclear reaction.

Fossils are geologically altered remains of a once-living organism and / or its behaviour.

Fossil fuels are fuels that were formed as a result of a gradual decomposition/ decay of the remains of plants and animals that lived millions of years ago.

Or these are natural substances formed from anaerobic decay of dead plants and animals that lived millions of years ago and are used as a source of fuel.

Types of Fossil Fuels

There are mainly three types of fossil fuels, namely:

- 1) Coal; it is a solid at room temperature.
- 2) Crude oil; thick dark brown or black liquid.
- 3) Natural gas; colourless gas.

These fossil fuels differ in the way they were formed. However, they have many physical and chemical properties in common. All are (1) natural resources (2) sources of energy and (3) non-renewable.

Uganda has large deposits of fossil fuels especially crude oil and natural gas. Coal is not found in Uganda.

How Was Each of These Fuels Formed?

(1) Coal

Coal was formed by the action of heat and pressure on the remains of mainly terrestrial /land plants over millions of years. Coal is formed by slow decomposition of dead plant materials. This process is called carbonisation. Plants contain a high amount of carbon; this is why coal also contain high amounts of carbon.

(2) Crude oil and natural gas

Crude oil and natural gas were formed by the action of heat and pressure on the remains of sea plants and animals over millions of years.

Fossil fuels contain carbon because they are formed from the remains of plants and animals which are made up of carbon. These fuels are known as fossil fuels because they were formed from fossils which were formed millions of years ago which were decomposed under high pressure and temperature.

Which of The Fossil Fuels Does Not Pollute Our Environment?

Natural gas. This is mainly methane and is transported as CNG or LNG. The advantage of using natural gas is that it is less polluting than crude oil and so is described as clean fuel.

Unlike natural gas, coal consists of the elements Sulphur, nitrogen, hydrogen and oxygen. These elements are very reactive. This is why coal is highly combustible and also a heavy pollutant.

How Are Fossil Fuels Extracted?

There are two main methods for extracting fossil fuels from the ground. These are (1) mining (2) drilling. Fossil fuels are obtained from the earth by mining and drilling.

Mining is the extraction of solid fossil fuels from the earth. For example, coal is extracted by digging underground or scrapping from the surface.

Drilling is a cutting process in which a drill bit is used to cut or enlarge a circular hole in a solid material. Crude oil and natural gas can easily be forced to flow to the surface using drilling method.

Why Are Fossil Fuels Used as Sources of Energy?

- ✓ This is because of their ability to release a lot of energy when burned.
- ✓ Fossil fuels are cheap and easy to use.

In What Ways Are Fossil Fuels Used as Energy Sources?

- Generation of electricity.
- In transportation as fuel for automobiles and jets.
- For cooking in gas stoves.
- Heating homes and buildings.
- Heating water.

Other Uses of Fossil Fuels

By-products of crude oil are used to produce various chemicals like;

- Plastics.
- Paint.
- Lubricants for vehicles and machines.
- Tar for tarmacking roads.
- Soap and dyes.
- Medicines like aspirin.
- Fertilizers and pesticides.

Why Are Fossil Fuels Described as Non-Renewable?

This is because they cannot be regenerated or produced any natural process. The formation of fossil fuels happened millions of year ago and such process may not happen again in the near future. In other words, fossil fuels are described as non-renewable natural resources because they cannot easily be replaced. Fossil fuels take a long time to be formed or renewed.

There is need for sustainable use of these fuels. Sustainability means using a resource to meet our own current needs without compromising the ability of future generations to meet their own needs.

End

S3 New Curriculum chemistry Theme: Fuels and energy Chapter 5 –**Fossil fuels**

Fossil fuels are **natural energy sources** formed from the remains of ancient plants and animals that were buried and compressed over millions of years. The three main types of fossil fuels are:

- **Coal** – A solid fuel made primarily of carbon, used for electricity generation and industrial processes.
- **Oil (Petroleum)** – A liquid fuel refined into gasoline, diesel, and other products for transportation and industry.
- **Natural Gas** – A gaseous fuel composed mostly of methane, used for heating, electricity, and fuel for vehicles.

Fossil fuels are **non-renewable** because they take millions of years to form and are being consumed much faster than they can replenish. Fossil fuels are used as fuel because they produce heat when burned. They are cheap and easy to use.

Extraction of fossil fuels

The extraction of fossil fuels involves several processes to **locate, drill, mine, and refine** these natural energy sources. The method used depends on the type of fossil fuel:

1. Coal Extraction

- **Surface Mining** (Open-pit or strip mining) – Used when coal deposits are close to the surface. Large machines remove layers of soil and rock to expose the coal.
- **Underground Mining** – Used for deeper deposits. Miners dig tunnels to reach the coal seam and transport it to the surface.

2. Oil Extraction

- **Drilling** – Wells are drilled into underground reservoirs where crude oil is trapped.
- **Offshore Drilling** – Platforms are built to extract oil from underwater reserves, often in oceans.
- **Hydraulic Fracturing (Fracking)** – High-pressure water, sand, and chemicals are injected into rock layers to release trapped oil.



3. Natural Gas Extraction

- **Conventional Drilling** – Similar to oil extraction, wells are drilled into gas reservoirs.
- **Fracking** – Used to extract gas from shale rock by breaking it apart with high-pressure fluid.
- **Liquefaction** – Natural gas is sometimes cooled into liquid form for easier transport.

After extraction, fossil fuels are processed and refined to remove impurities and produce usable fuels like gasoline, diesel, and electricity.

Ways of refining crude fossil fuel

Refining crude fossil fuels involves separating and processing the raw mixture into useful products like gasoline, diesel, and jet fuel. The key refining processes include:

1. Fractional Distillation: Crude oil is heated in a large distillation column, where different components separate based on their boiling points. Lighter fractions like gasoline rise to the top, while heavier ones like asphalt settle at the bottom.

2. Cracking This process **breaks down** large hydrocarbon molecules into smaller, more valuable ones:

- **Thermal Cracking** – Uses heat to break down heavy molecules.
- **Catalytic Cracking** – Uses a catalyst to speed up the process and produce more gasoline.

3. Hydroprocessing

- **Hydrotreating** – Removes impurities like sulfur, nitrogen, and metals, making fuels cleaner.
- **Hydrocracking** – Uses hydrogen to break down heavier fractions into lighter fuels.

4. Reforming: This process modifies the molecular structure of hydrocarbons to improve fuel quality. It's used to **increase octane levels** in gasoline, making it more efficient.

5. Alkylation: Combines smaller hydrocarbons to produce high-quality gasoline with better combustion properties.

6. Blending: Refined fuels are mixed to meet specific requirements, like different octane levels for gasoline or winter vs. summer fuel blends.

7. Coking: Converts the heaviest fractions into **petroleum coke**, a carbon-rich byproduct used in industrial processes.

Each of these refining steps helps optimize crude fossil fuels for different uses, from vehicle fuel to plastics production.

Uses of fossil fuel

Fossil fuels play a crucial role in modern society, powering various industries and everyday life. Here are some of their key uses:

1. Energy Generation

- **Electricity Production** – Power plants burn coal, oil, or natural gas to generate electricity.
- **Heating** – Homes and industries use natural gas for heating and cooking.

- **Transportation** – Cars, trucks, airplanes, and ships rely on gasoline, diesel, and jet fuel.

2. Industrial Applications

- **Manufacturing** – Factories use fossil fuels to power machines and production lines.
- **Chemical Production** – Used to make plastics, fertilizers, synthetic fibers, and pharmaceuticals.



Plastic products

- **Steel & Cement Making** – Coal and oil help in high-temperature processes like steel refining and cement production.

3. Agriculture

- **Fertilizers & Pesticides** – Many are derived from petroleum-based chemicals.
- **Machinery Fuel** – Tractors and other farming equipment run on fossil fuel-powered engines.

4. Commercial & Domestic Uses

- **Cooking & Heating** – Natural gas and propane are commonly used in homes and restaurants.
- **Backup Power** – Diesel generators provide emergency electricity for hospitals, data centers, and remote areas.

5. Lubrication & Maintenance

- **Engine Oils & Greases** – Petroleum-based lubricants keep machinery running smoothly.

6. Transportation Infrastructure

- **Road Construction** – Asphalt (a petroleum product) is essential for paving roads.

S3 New Curriculum chemistry

Theme: Reactants and products

Chapter 6 – Chemical reactions

Chemical reactions

A chemical reaction is a process in which one or more chemical substances called reactants combine or breakdown to form new substances called products. Substances are either chemicals elements or compounds.

Rate of a chemical reaction:

The **rate of a chemical reaction** refers to how quickly reactants are converted into products.

Ways of measuring the rate of reaction

Scientists use various methods to measure reaction rates depending on the type of reaction. Here are some common techniques:

1. Measuring Change in Mass

- If a reaction produces a gas, the mass of the system will decrease over time.
- A **balance (scale)** can be used to record mass loss.
- Example: **Decomposition of calcium carbonate (CaCO_3) producing CO_2 gas.**

2. Measuring Volume of Gas Produced

- A **gas syringe** or an **inverted measuring cylinder** in water can measure gas volume over time.
- Useful for reactions that generate gases like hydrogen (H_2) or carbon dioxide (CO_2).
- Example: **Reaction of magnesium (Mg) with hydrochloric acid (HCl) producing hydrogen gas.**

3. Monitoring Color Change (Spectrophotometry)

- If the reaction involves a color change, a **spectrophotometer** measures absorbance changes.
- Example: **Reaction of iodine with starch forming a blue-black color.**

4. Tracking pH Changes

- If a reaction produces or consumes acidic or basic substances, a **pH meter** can monitor pH over time.
- Example: **Neutralization of acids and bases.**

5. Measuring Change in Conductivity

- Conductivity changes if ions are formed or consumed during a reaction.
- A **conductivity meter** can track this change.
- Example: **Dissolution of salts in water affecting ionic concentration.**

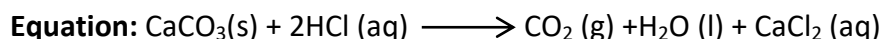
6. Measuring Precipitate Formation

- Some reactions form insoluble solids (precipitates).
- **Cloudiness/turbidity** of the solution can be measured.
- Example: Reaction **acid with sodium thiosulphate to form sulphur**

7. Tracking Temperature Changes

- Exothermic or endothermic reactions alter the temperature.
- A **thermometer** or **calorimeter** records temperature change.
- Example: **Combustion reactions or dissolution of salts in water.**

Experiment to determine the rate of reaction of calcium carbonate with dilute hydrochloric acid by measuring the rate production carbon dioxide



Setup

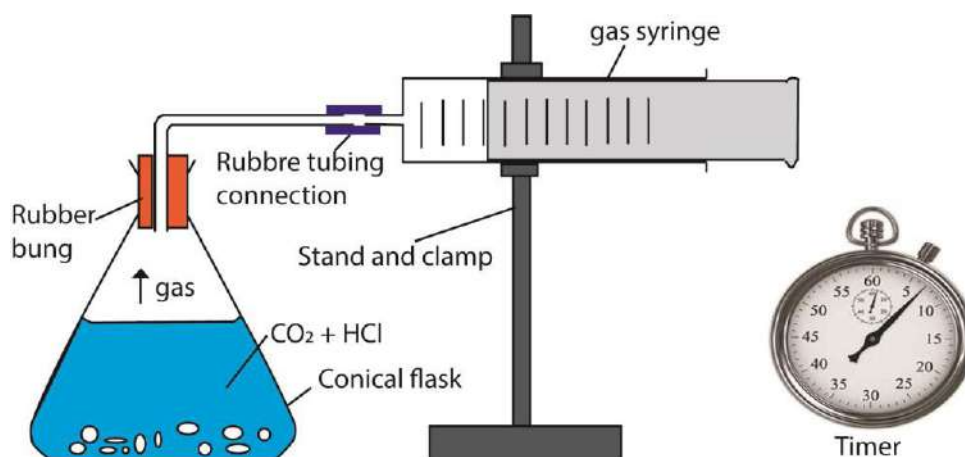


Figure 1

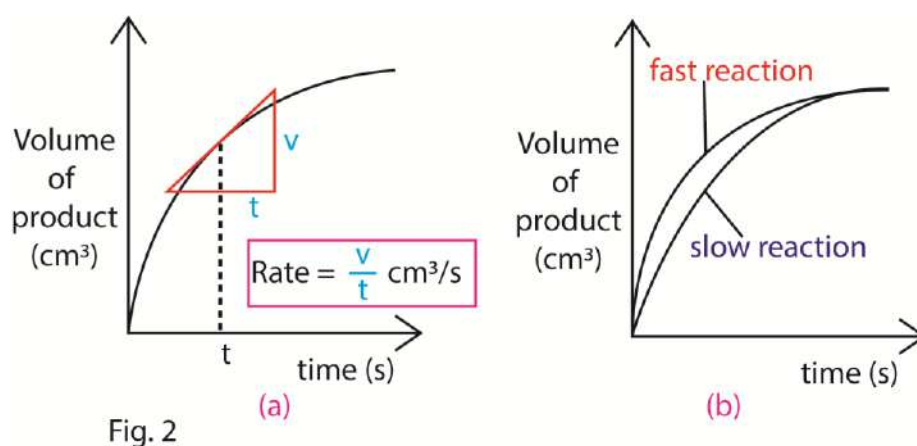
Procedure

- Measure 50cm³ of dilute hydrochloric acid using a measuring cylinder into the conical flask.
- Add 10g of calcium carbonate to the acid close the rubber bung and immediately start a stop clock
- Read and record the volume of carbon dioxide gas produced in the syringe every five second and record your results in the table below

Time in second	5	10	15	20	25	30	35	40
Volume of CO ₂ produced (cm ₃)								

- Plot a graph showing how the volume of carbon dioxide changes with time.
- The rate of reaction at any time is determined from the slope of the gradient at that particular instance.

Figure 2 (a) Shows how the rate of reaction at time t is determined by drawing a tangent at that time while (b) shows curves for the reactions with higher rates (or fast reaction) and for slow reactions.

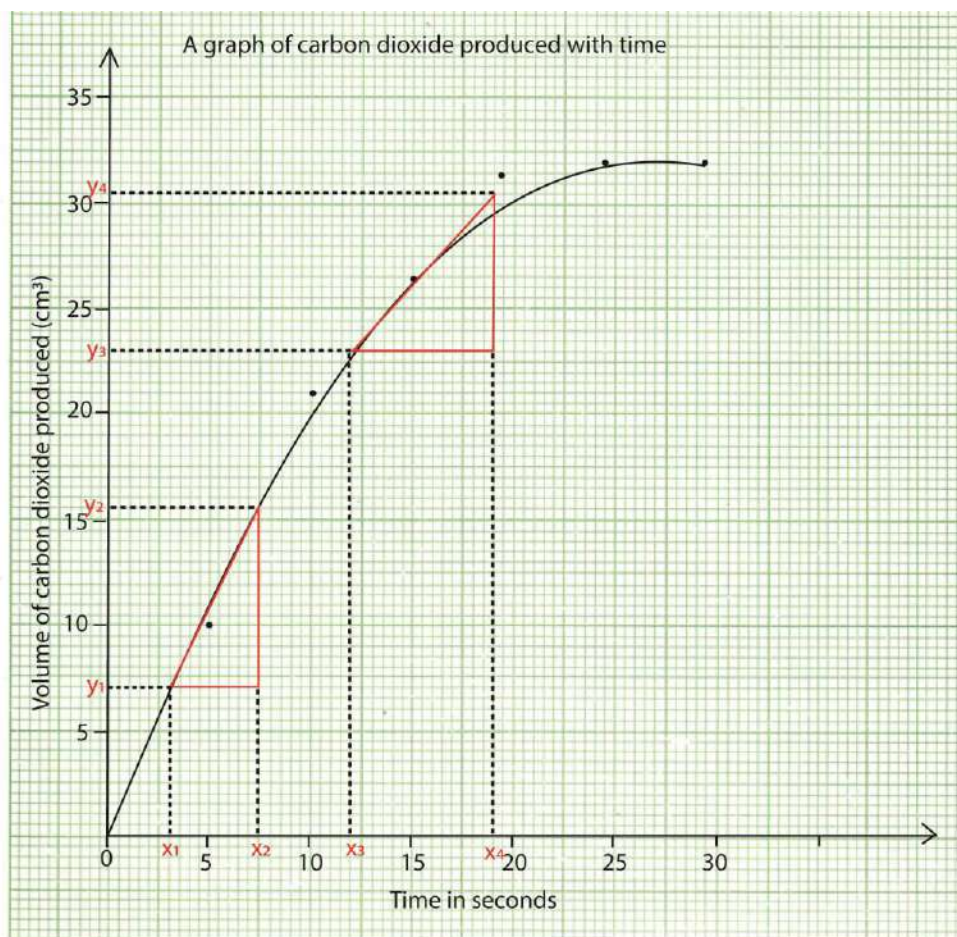


Example 1

The table below shows the volume of carbon dioxide liberated with time when dilute hydrochloric acid was reacted with calcium carbonate at 25⁰C.

Time in second	0	5	10	15	20	25	30
Volume of CO ₂ produced (cm ₃)	0	10	21	26.5	31.5	32	32

- Plot the graph of volume of hydrogen evolved (vertical) against time (horizontal axis)



- (ii) Determine the rate of reaction at 5 seconds.

Rate at 5 seconds = $\frac{y_1}{x_1} = \frac{10}{5} = 2.0 \text{ cm}^3/\text{s}$

- (iii) Determine the rate of reaction at 20 seconds

Rate at 20 seconds = $\frac{y_4 - y_3}{x_4 - x_3} = \frac{30 - 23}{20 - 15} = 1.4 \text{ cm}^3/\text{s}$

- (iv) Explain the difference between the rates at 12 seconds and 20 seconds

The rate is higher at 5 seconds than at 20 seconds because the concentrations of the reactants at 5 seconds are higher than those at 20 seconds

- (v) Explain why the rate does not change between 25 and 30 seconds

Because one of the reagents is used up

Factors that affect a chemical reaction

There are about four factors that affect a chemical change/reaction. These are:

1. A catalyst
2. Temperature
3. Concentration
4. Pressure
5. Surface area of the reactants

1. Concentration:

Increase in concentration increases the rate of reaction due to increase in the frequency of molecular collision.

Experiment: Investigating the Effect of Concentration on Reaction Rate

Aim: To determine how changing the concentration of **hydrochloric acid (HCl)** affects the rate of reaction with **magnesium (Mg) strips**, measured by the time taken for a given length of magnesium strip to react with equal volume of acid of different concentration.

Materials Needed:

- Hydrochloric acid (HCl) solutions of different concentrations (e.g., 0.5 M, 1 M, 2 M)
- Magnesium strips (Mg)
- Measuring cylinder
- Stopwatch
- Beaker
- Gas syringe (or an inverted measuring cylinder in water to collect gas)
- Safety goggles and gloves

Procedure:

1. **Prepare Different HCl Solutions** – Pour 50 mL of HCl solution into separate beakers with varying concentrations (0.5 M, 1 M, 2 M, 3M).
2. **Add Magnesium Strip** – Drop a magnesium strip into the first beaker containing 0.5 M HCl.
3. **Measure Time and Gas Volume** – Start the stopwatch and record the time taken for the reaction to **stop fizzing** (when hydrogen gas stops forming).
4. **Repeat for Different Concentrations** – Perform the experiment using magnesium strips of equal length with 1 M and 2 M HCl, recording reaction time and gas volume produced.
5. **Compare Results** – Observe how higher concentrations of HCl cause faster reactions and **more rapid gas production**.

Sample results table

HCl Concentration (M)	0.5	1	2	2
Time for Reaction to Stop (s)	90	60	42	25

Observation

The **3 M HCl solution** reacts the fastest, completing in **25 seconds**, while **0.5 M HCl** takes **90 seconds**.

Conclusion:

Higher concentration of HCl → Faster reaction, i.e. Increasing HCl concentration results in **more frequent collisions** between acid molecules and magnesium atoms, speeding up the reaction.

Experiment: Investigating the Effect of Concentration on Reaction Rate

Aim: To determine how the concentration of **sodium thiosulphate** affects the rate of reaction with **hydrochloric acid**, measured by how quickly the solution turns **opaque**.

Materials Needed:

- Sodium thiosulphate solution (different concentrations: **0.1 M, 0.2 M, 0.5 M, 1 M**)
- Hydrochloric acid (HCl, constant concentration)
- Conical flask
- Stopwatch
- White paper with a black "X" mark
- Measuring cylinders
- Safety goggles and gloves

Procedure:

- Prepare different sodium thiosulphate solutions** – Pour 50 mL into separate 100 cm³ flasks with varying concentrations (**0.1 M, 0.2 M, 0.5 M, 1 M**).
- Place flask containing 0.1M sodium thiosulphate solution on the marked paper** – The black "X" helps measure how quickly the solution becomes cloudy.
- Add hydrochloric acid** – Pour **10 mL of HCl** into the flask and immediately start the stopwatch.
- Observe and record time** – Stir the mixture and note the time when the "X" **completely disappears** due to the formation of sulfur precipitate.
- Repeat the experiment** – Perform the reaction for each concentration of sodium thiosulphate.

Sample Results Table:

Sodium thiosulphate solution	0.1M	0.2M	0.5M	1M
Time taken for mark "X" to disappear (seconds)	90	65	40	24

Observation

1 M sodium thiosulphate reacted in **25 seconds** (Shorter time), while **0.1 M** took **90 seconds** (longer time)

Conclusion:

- **Higher concentration → Faster reaction**
- **More concentrated solutions lead to a quicker disappearance of the "X"**
because more particles are available for reaction, increasing the frequency of collisions.

2. Temperature:

Increasing the temperature increases reaction rates because of the disproportionately large **increase** in the number of high energy collisions. It is only these collisions (possessing at least the activation energy for the **reaction**) which result in a **reaction**.

Experiment: Investigating the Effect of Temperature on Reaction Rate

Aim: To determine how changing the temperature of **hydrochloric acid** affects the rate of its reaction with **magnesium**, measured by the volume of hydrogen gas produced.

Materials Needed:

- Hydrochloric acid (1 M)
- Magnesium strips (same size for consistency)
- Beaker (100 mL)
- Water bath or hot plate (for heating solutions)
- Thermometer
- Gas syringe or measuring cylinder (for collecting gas)
- Stopwatch
- Safety goggles and gloves

Procedure:

- Pour **50 mL** of the acid into 100ml beaker.
- Warm the acid to **30°C**
- Drop a magnesium strip into the beaker.
- Start stopwatch and begin timing the reaction immediately.
- (v)** Measure the time taken for the reaction to stop.
- (vi)** Repeat the experiment with the acid warmed 35°C, 45°C and 60°C.

Sample result Table

Temperature of the acid (°C)	30	35	45	60
Time taken for the reaction to stop (seconds)	120	80	50	30

Observations

The **60°C acid** reacted the fastest, completing in **30 seconds**, while the **30°C acid** took **120 seconds**.

Conclusion:

Higher temperature → Faster reaction because at higher temperature the reaction took a shorter time. At higher temperature, particles move **faster**, increasing the **frequency of collisions** and **energy for successful reactions**.

3. Surface area:

Increasing the surface area of a solid reactant exposes more of its particles to attack. This results in an **increased** chance of collisions between reactant particles, so there **are** more collisions in any given time and the **rate of reaction increases**. The surface of a solid can be increased by crushing a substance into a powder or by reducing its particle size.

Experiment: Investigating the Effect of Surface Area on Reaction Rate

Aim: To determine how the surface area of **calcium carbonate (CaCO_3)** affects the rate of reaction with **hydrochloric acid (HCl)**, measured by the time taken for equal mass of chips and powder to react with equal volumes of the acid.

Materials Needed:

- Calcium carbonate in different forms:
 - **Large marble chips** (low surface area)
 - **Powdered calcium carbonate** (high surface area)
- Hydrochloric acid (1 M)
- Measuring cylinder (50 mL)
- Conical flask
- Stopwatch
- Safety goggles and gloves

Procedure:

- Measure hydrochloric acid** – Pour **50 mL of HCl** into a conical flask.
- Add marble chips** – Drop **5 g** of large calcium carbonate chips into the acid.
- Start stopwatch** – Immediately begin timing the reaction.
- Measure time taken for the reaction to complete.**
- Repeat for powder** – Perform the experiment keeping all other conditions constant.

Sample Results Table:

Calcium Carbonate form	Large Chips	Powder
Time for Reaction to Stop (seconds)	120	42

Observations

The powdered form reacts the fastest (**40s**), while the marble chips react the slowest (**115s**).

Conclusion:

- **Smaller particle size (or large surface area) → Faster reaction**
- **Higher surface area the shorter the time taken for the reaction** because more particles are exposed for reaction.

4. Pressure:

This affects gaseous reactions since gases unlike solids are compressible.

Increasing pressure on gases brings reactant particles close to each other increasing the frequency of collision and hence the rate of reaction. Pressure can be increased by decreasing the volume of the container. For example, in the Haber process, a large yield of ammonia is obtained from high pressure as per the following reaction.

5. **Catalyst:** increases the rate of reaction by lowering activation energy or energy barrier to the reaction.

By definition

A catalyst is substance that increases the rate of a chemical reaction without itself undergoing any permanent chemical change.

To investigate the effect of a catalyst on the production of oxygen

You are provided with three test tubes labelled A, B, C; to test tube

A put hydrogen peroxide

B put manganese dioxide

C put both manganese dioxide first followed with hydrogen peroxide

Allow the experiment to stand for 10 minutes.

Insert a glowing split in test tube A, B C respectively

Record your observation and conclusion.

Observations

In **A** oxygen was given off after a long period. The glowing splint kept on glowing.

In **B** no oxygen was given off.

In **C** oxygen was given off after a short time. The glowing splint busted into a flame

Conclusion:

The manganese dioxide speeds up the production of oxygen from hydrogen peroxide.

Reversible reactions

Reversible reactions are chemical reactions where the reactants can form products, but the products can also revert back to the original reactants under certain conditions.

A reversible reaction where A and B react to form C and D can be represented as follows

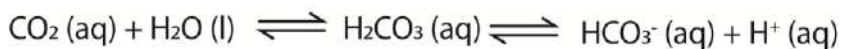


At equilibrium the rate of the formation of the product is equal to the rate of formation of the reactants.

Chemical equilibrium

Chemical equilibrium is a state in which the rate of the forward reaction equals the rate of the backward reaction. In other words, there is no net change in the concentrations of the reactants and the product. This kind of equilibrium is also called dynamic equilibrium.

An example of reversible reactions are Dissolution of carbon dioxide in soda at high pressure.



At high pressure carbon dioxide dissolves in water to form carbonic acid that hydrolyse in water to form hydrogen carbonate ion and hydrogen ions. When the bottle is opened, pressure decreases and this reaction reverses.

Le Chatelier's Principle

This principle states that when a system at equilibrium is disturbed, it **adjusts to counteract the change** and restore equilibrium.

Factors that affect reversible reactions

According to **Le Chatelier's Principle**, several factors influence reversible reactions and can shift the equilibrium between reactants and products.

1. Concentration of Reactants and Products

- **Increasing reactant concentration** → Shifts equilibrium forward, producing more products.
- **Increasing product concentration** → Shifts equilibrium backward, favoring reactants.
- Example: In the **Haber process** ($\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$), adding more nitrogen or hydrogen increases ammonia production.

2. Temperature

- **For exothermic reactions** (heat is released): Increasing temperature shifts equilibrium **backward**, favoring reactants.
- **For endothermic reactions** (heat is absorbed): Increasing temperature shifts equilibrium **forward**, producing more products.
- Example: **Decomposition of calcium carbonate** ($\text{CaCO}_3 \rightleftharpoons \text{CaO} + \text{CO}_2$) is endothermic, so heating favors decomposition.

3. Pressure (For Gaseous Reactions)

- **Increasing pressure** → Shifts equilibrium toward the side with fewer gas molecules.
- **Decreasing pressure** → Shifts equilibrium toward the side with more gas molecules.
- **Example:** In the Haber process, increasing pressure favors ammonia formation because fewer gas molecules exist on the product side.

4. Presence of a Catalyst

- A **catalyst** speeds up both forward and backward reactions **without shifting equilibrium**.
- It **reduces activation energy**, allowing equilibrium to be reached faster.
- Example: Iron is used as a catalyst in the Haber process but does not change the equilibrium position.

5. Nature of the Reaction

- Some reactions naturally **favor one direction** based on their energy requirements or stability of products.
- Example: **Weak acid dissociation** ($\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$) favors mostly reactants because acetic acid only partially dissociates.

Application of reversible reaction

Manufacture of ammonia

By reaction nitrogen and hydrogen



Conditions for production of ammonia

1. Pressure:

Production of ammonia is accompanied with a decrease in the number of moles of a gas and as such high yield of ammonia is favored by high pressure.

2. Temperature

Production of ammonia produces heat; high yields are thus expected at low temperatures.

However, at low temperature, the reaction is slow that a compromise temperature of 450°C - 500°C is used.

3. A catalyst is iron

Uses of ammonia

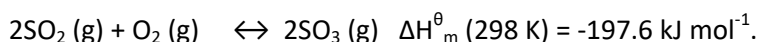
- Disinfectant
- For preparation of liquid soap
- Fertilizers

Note that prolonged use of ammonia nitrate fertilizer may make the soil acidic because ammonium nitrate in soil reacts with water forming ammonium hydroxide and nitric acid plants absorb ammonium ions leaving nitric acid in solution.



The contact process

In sulphur dioxide is reacted with oxygen in presence of Vanadium (V) oxide (catalyst) to form sulphur trioxide. The conversion of sulphur dioxide and oxygen into sulphur trioxide (sulphur (VI) oxide) is an exothermic reaction with a decrease in gas volume.



Condition for production of sulphur trioxide

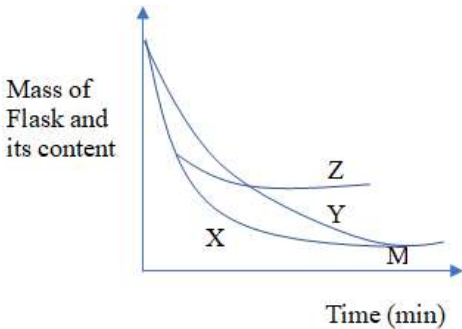
Temperature: 420°C

Catalyst : vanadium (V) Oxide

Pressure:

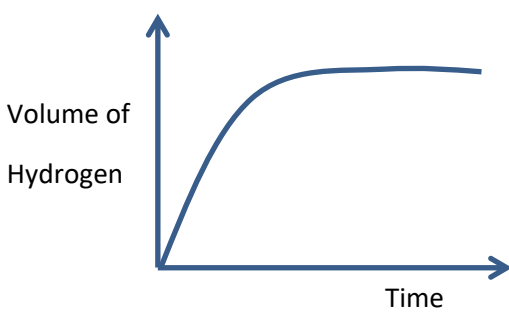
An increase in pressure would increase the yield of the reaction because there are fewer number of moles of gas on the right hand side. However, a pressure of 1 atm is enough to obtain a very good yield. Thus it would be a waste of money and resources to increase the pressure

Revision exercise

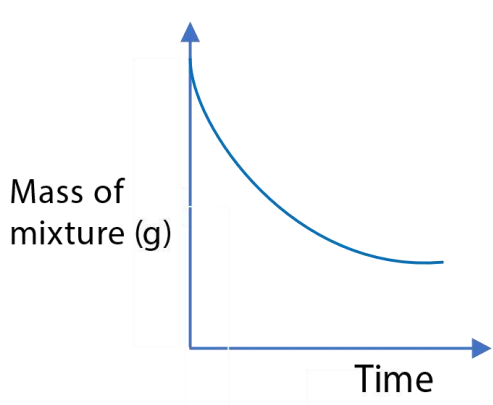
1.		Curve Y in diagram shows the results that were obtained during the investigation of the rate of the reaction between iron and dilute hydrochloric acid under normal conditions. Curve X and Z were obtained when some conditions of the experiment were changed. 		
	(a)	(i)	List three condition that were changed to obtain curve X	(3marks)
		(ii)	State what point M represent	(½ mark)
	(b)		Some conditions you have listed in (a)(i) were changed to obtained curve Z.	
		(i)	State the condition changed	(01mark)
		(ii)	Give a reason for your answer	
2.			Sodium thiosulphate reacts with hydrochloric acid according to the following equation $\text{S}_2\text{O}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \longrightarrow \text{H}_2\text{O}(\text{l}) + \text{SO}_2(\text{g}) + \text{S}(\text{s})$	
	(a)		State what would be observed if dilute hydrochloric acid is added to sodium thiosulphate solution	(½ mark)
	(b)		The rate of reaction with dilute acids is affected by the concentration of sodium thiosulphate.	
		(i)	State one other factor other than concentration that can affect the rate of reaction	(½ mark)
		(ii)	Briefly explain the effect of the factor you have state in (b)(i) on the rate of reaction.	(02mark)
		(iii)	Describe an experiment can be carried out in the laboratory to show the effect of the factor on have stated in (b)(i) on the rate of reaction. Diagram not required)	(6 ½ marks)
3.			State and explain the effect of each of the following conditions on the rate of chemical reaction.	
	(a)		Particle size	(04marks)

	(b)		Concentration of reactants	(05 msrks)																		
	(c)		Temperature	(06marks)																		
4.			The table shows the variation in the concentration of sodium thiosulphate with time. <table><tr><td>Time (s)</td><td>200</td><td>100</td><td>40</td><td>20</td><td>10</td></tr><tr><td>Concentration of thiosulphate (moldm⁻³)</td><td>0.05</td><td>0.09</td><td>0.15</td><td>0.20</td><td>0.25</td></tr><tr><td>$\frac{1}{\text{concentration}}$ of thiosulphate (mold⁻¹m⁻³)</td><td></td><td></td><td></td><td></td><td></td></tr></table>	Time (s)	200	100	40	20	10	Concentration of thiosulphate (moldm ⁻³)	0.05	0.09	0.15	0.20	0.25	$\frac{1}{\text{concentration}}$ of thiosulphate (mold ⁻¹ m ⁻³)						
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Concentration of thiosulphate (moldm ⁻³)	0.05	0.09	0.15	0.20	0.25																	
$\frac{1}{\text{concentration}}$ of thiosulphate (mold ⁻¹ m ⁻³)																						
		(i)	Determine the values of $\frac{1}{\text{concentration}}$ of thiosulphate, copy the table and enter your answer in the space provided in the table.	(01mark)																		
		(ii)	Plot a graph of $\frac{1}{\text{concentration}}$ of thiosulphate (vertical axis) against time (horizontal axis.	(03marks)																		
		(iii)	State any conclusion that can be drawn from the shape of the graph.	(1 ½ marks)																		
5.	(a)	(i)	Write an equation for the reaction between dilute nitric acid and calcium carbonate	(1 ½ mark)																		
		(ii)	State how temperature can affect the rate of reaction in (a)(i) above	(01mark)																		
		(iii)	Give a reason why a large surface area of calcium carbonate can speed up rate of reaction in (a)(i)	(2 marks)																		
	(b)		Magnesium can react with hydrochloric acid to form hydrogen. State the conditions and write equation for the reaction	(2 ½ marks)																		
	(c)		The table below shows the volume of hydrogen evolved when various lengths of magnesium ribbon were reacted with fixed volume of hydrochloric acid <table><tr><td>Length of ribbon (cm)</td><td>1.0</td><td>2.0</td><td>3.0</td><td>5.0</td><td>6.0</td></tr><tr><td>Volume of hydrogen (cm³/min)</td><td>2.2</td><td>3.6</td><td>5.2</td><td>9.2</td><td>10.8</td></tr></table>	Length of ribbon (cm)	1.0	2.0	3.0	5.0	6.0	Volume of hydrogen (cm ³ /min)	2.2	3.6	5.2	9.2	10.8							
Length of ribbon (cm)	1.0	2.0	3.0	5.0	6.0																	
Volume of hydrogen (cm ³ /min)	2.2	3.6	5.2	9.2	10.8																	
		(i)	Plot a graph of volume of hydrogen (vertical axis) against length of magnesium ribbon (horizontal axis)	(5marks)																		

		(ii)	Explain the shape of the graph you have drawn.	(2marks)																		
		(iii)	Using your graph, determine the rate of reaction if 4.0cm of magnesium ribbon was used	(01mark)																		
6.			Oxygen is formed from hydrogen peroxide in the presence of Manganese dioxide according to the following equation $2\text{H}_2\text{O}_2(\text{aq}) \xrightarrow{\text{MnO}_2} 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$																			
	(a)		In an experiment, a certain volume of hydrogen peroxide was used to prepare oxygen at room temperature. With the aid of a suitable diagram describe how the following can be determined.																			
		(i)	The volume of oxygen evolved	(5 ½ marks)																		
		(ii)	The rate of evolution of oxygen	(03marks)																		
	(b)		In another experiment one half of volume of hydrogen peroxide in (a) was diluted with equal volume of water. On the same axes draw graphs to show the variation of volume of oxygen with time in (a) and (b).	(03marks)																		
	(c)		Oxygen produced from 200cm ³ of 0.5M hydrogen peroxide solution was reacted completely with magnesium. Calculate the mass of magnesium that reacted	(3 ½ marks)																		
7.	(a)		Define the term rate of reaction	(02marks)																		
	(b)		The table below shows variation of hydrogen evolved with time when dilute hydrochloric acid was added to excess zinc. <table border="1"><tr><td>Volume of hydrogen (cm³)</td><td>0</td><td>20</td><td>35</td><td>46</td><td>56</td><td>72</td><td>79</td><td>79</td></tr><tr><td>Time (s)</td><td>0</td><td>10</td><td>20</td><td>30</td><td>40</td><td>60</td><td>80</td><td>90</td></tr></table> Plot a graph of volume hydrogen evolved against time	Volume of hydrogen (cm ³)	0	20	35	46	56	72	79	79	Time (s)	0	10	20	30	40	60	80	90	(05mrks)
Volume of hydrogen (cm ³)	0	20	35	46	56	72	79	79														
Time (s)	0	10	20	30	40	60	80	90														
	(c)		Using the graph determine the time taken to collect 60cm ³ of hydrogen gas	(01mark)																		
	(d)	(i)	Draw tangents on your graph at points when the time is 20 and 60 seconds and determine the gradient of each tangent.	(04marks)																		

		(ii)	Compare the rate of reaction at 20 and 60seconds. Explain your answer	(03marks)
8.			The graph in figure below shows the variation of volume of hydrogen evolved with time when excess magnesium was added to 100cm³ of 1.0M sulphuric at room temperature. 	
	(a)		Calculate the number of moles of hydrogen ions that is contained in 100cm ³ of	
		(i)	A 0.5M H ₂ SO ₄	(01mark)
		(ii)	A 1.0M H ₂ SO ₄	(01mark)
	(b)	(i)	Sketch on the same axes of the graph in figure above the graph that would be obtained if the same mass of magnesium was added to 100cm ³ of a 0.5M sulphuric acid at room temperature.	(01mark)
		(ii)	Mark on the graph the times the two reactions have reached completion.	(01mark)
		(iii)	Compare the time the reaction took to reach completion when 0.5M sulphuric acid was used to that when 1.0M sulphuric acid was used	(01mark)
9.	(a)		Describe an experiment to show how surface area can affect the rate of reaction between calcium carbonate and 2M hydrochloric acid. Your answer must include: • A labelled diagram of apparatus • Sketch of expected graph • Mention how the graph can be used to reach a conclusion.	11 ½
	(b)		Briefly explain why, when 4M hydrochloric acid was used instead of the 2M acid, the rate of reaction was faster. Explain this observation.	(2½ marks)
	(c)		State one other factor than mentioned above that can affect the rate of reaction between hydrochloric acid and calcium carbonate	(01mmark)

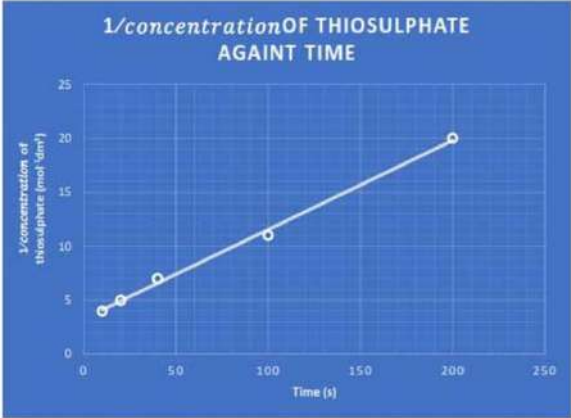
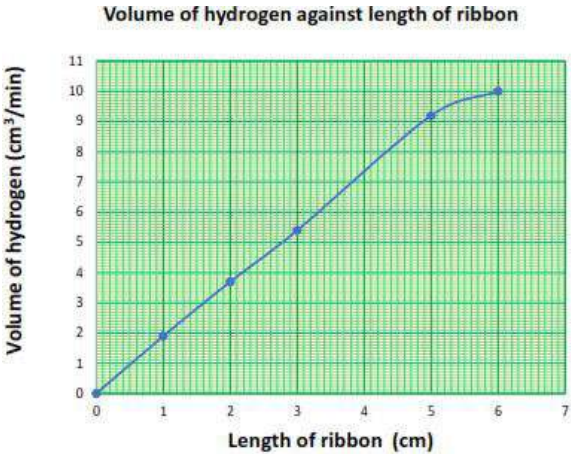
10.	(a)		What is meant by the rate of reaction?	(02marks)																		
	(b)		State how the following factors affect the rate of a chemical reaction:																			
		(i)	Temperature																			
		(ii)	Surface area of the reactants																			
	(c)		The table below shows the volume of hydrogen collected at various time intervals when magnesium was reacted with a 2M hydrochloric acid <table><tr><td>Time (s)</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>Volume of hydrogen collected (cm³)</td><td>0</td><td>25</td><td>45</td><td>60</td><td>70</td><td>75</td><td>77</td><td>77</td></tr></table>	Time (s)	0	1	2	3	4	5	6	7	Volume of hydrogen collected (cm ³)	0	25	45	60	70	75	77	77	
Time (s)	0	1	2	3	4	5	6	7														
Volume of hydrogen collected (cm ³)	0	25	45	60	70	75	77	77														
		(i)	Plot a graph of volume of hydrogen versus time	(04 marks)																		
		(ii)	Determine the rate of reaction at 3 seconds	(02marks)																		
		(iii)	Determine the volume of hydrogen evolved at 3.5 seconds	(01mark)																		
	(d)		State how the rate of reaction at 3 seconds would be affected if a 1M hydrochloric acid was used	(02marks)																		
11.			When a certain volume of 0.1M hydrochloric acid was reacted at room temperature with excess iron fillings, 120cm ³ of the gas were produced.																			
	(a)		Draw a labelled diagram to show how the rate of reaction was determined																			
	(b)		Write equation for the reaction that took place																			
	(c)		Calculate the																			
		(i)	Volume of hydrochloric acid required to produce 120cm ³ of the gas																			
		(ii)	The mass of iron filling that reacted																			
	(d)		Draw a sketch graph of the volume of the gas against time																			
	(e)		State how the rate of reaction would change if the reaction was carried out at a temperature above room temperature																			
12.	(a)		State the factors that can affect the rate of chemical reaction.																			
	(b)		A mixture of a known mass of magnesium and a certain volume of 2M hydrochloric acid were put in a conical flask and the mass of the mixture was recorded at various intervals. The																			

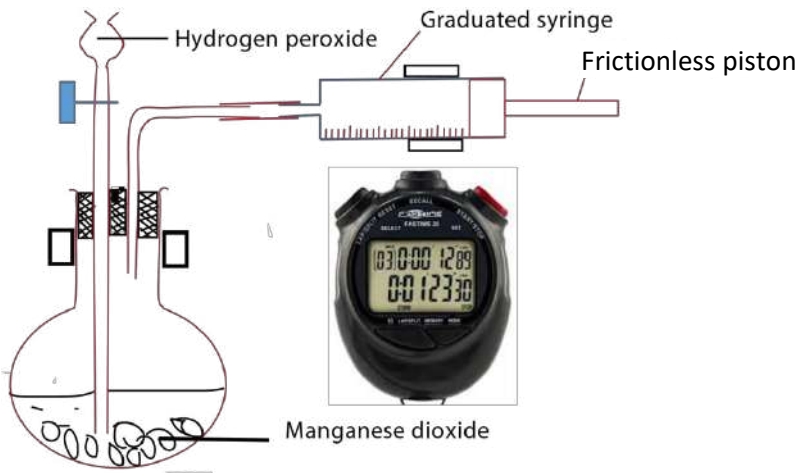
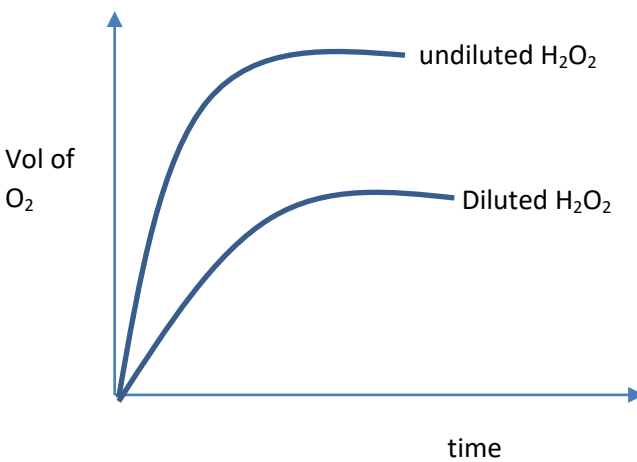
			results of the experiment are shown in the graph below.  On the same axes, draw a graph that would be obtained when same mass of magnesium was reacted with the same volume of 1M hydrochloric acid.																			
	(c)		5.0g of calcium carbonate was reacted with 20cm ³ of 2M hydrochloric acid.																			
		(i)	Write equation for the reaction between hydrochloric acid and calcium carbonate																			
		(ii)	The mass of calcium carbonate that was left. (Ca = 40, C = 12, O =16)																			
13.	(a)	(i)	What is the rate of reaction?																			
		(ii)	How does the particle size affect rate of reaction? Explain your answer																			
	(b)		The table of results shows the time taken for sulphur to form when various concentrations of sodium thiosulphate were used. <table border="1"><tr><td>Concentration of S₂O₃²⁻ (M)</td><td>0.2</td><td>0.6</td><td>0.8</td><td>1.2</td><td>1.6</td></tr><tr><td>Time for sulphur to form (s)</td><td>60</td><td>20</td><td>15</td><td>10</td><td>7.5</td></tr><tr><td>1/t (s⁻¹)</td><td>0.017</td><td>0.05</td><td>0.07</td><td>0.10</td><td>0.13</td></tr></table> Plot a graph of 1/t on vertical axis against concentration of thiosulphate.	Concentration of S ₂ O ₃ ²⁻ (M)	0.2	0.6	0.8	1.2	1.6	Time for sulphur to form (s)	60	20	15	10	7.5	1/t (s ⁻¹)	0.017	0.05	0.07	0.10	0.13	
Concentration of S ₂ O ₃ ²⁻ (M)	0.2	0.6	0.8	1.2	1.6																	
Time for sulphur to form (s)	60	20	15	10	7.5																	
1/t (s ⁻¹)	0.017	0.05	0.07	0.10	0.13																	
	(c)	(i)	Explain the relationship between the rate of the reaction and 1/t.																			
		(ii)	Deduce from the graph, how the rate of reaction varies with																			

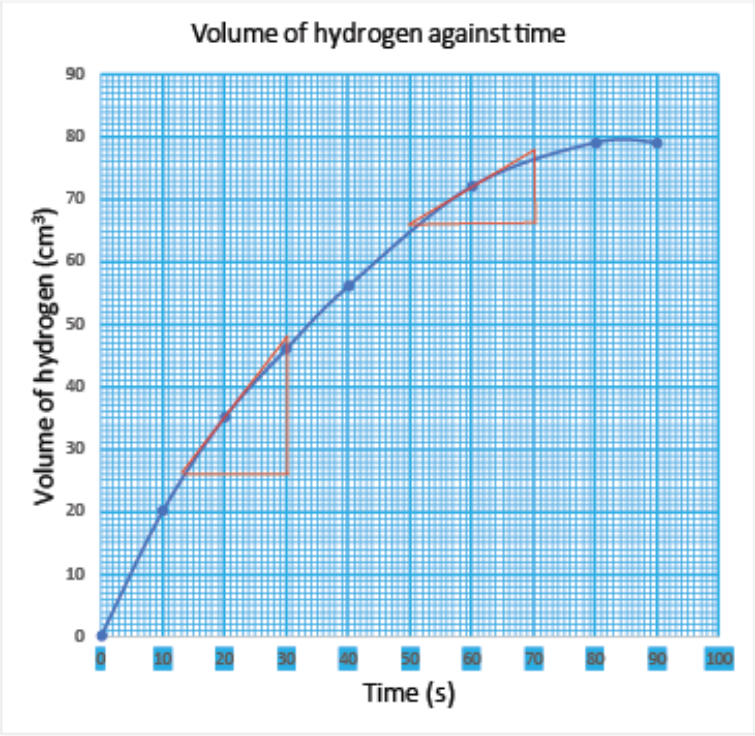
			the concentration of thiosulphate.																	
	(d)		Name one reagent that you would use to test for sulphur dioxide and state what would be observed if the reagent was used.																	
14.			In an experiment to determine the rate of reaction between zinc and sulphuric acid, dilute sulphuric acid was reacted with zinc granules to which some copper II sulphate solution was added. The volume of hydrogen gas evolved at various <table><tr><td>Time in minutes</td><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td></tr><tr><td>Vol. of gas in cm³</td><td>0</td><td>10</td><td>21</td><td>26.5</td><td>31.5</td><td>32</td><td>32</td></tr></table>	Time in minutes	0	5	10	15	20	25	30	Vol. of gas in cm ³	0	10	21	26.5	31.5	32	32	
Time in minutes	0	5	10	15	20	25	30													
Vol. of gas in cm ³	0	10	21	26.5	31.5	32	32													
	(a)	(i)	What is the role copper (II) sulphate solution?																	
		(ii)	Write ionic equation for the reaction above																	
		(iii)	Explain what would happen to the reaction if zinc granules were replaced with zinc powder.																	
	(b)	(i)	Plot the graph of volume of hydrogen evolved (vertical) against time (horizontal axis)																	
		(ii)	Describe how you would determine the rate of reaction at 12 minutes.																	
		(iii)	Compare the rate of reaction at 12 minutes with that at 20 minutes. Give reason for your answer																	
		(iv)	What happens to the shape of the after 24 minutes? Explain your answer																	

Suggested answers

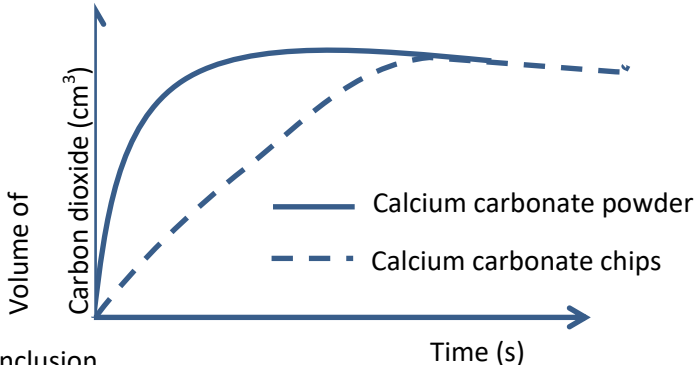
1.	(a)	(i)	Note that - the rate of reaction in curve X is higher than in curve Y, - The initial and final mass in X and Y is the same implying that the mass of iron was not changed Thus, to increase the rate of reaction - temperature was increased - particle size of iron was reduced - the concentration acid was increased					
	(b)	(i)	The concentration of the acid was reduce					
		(ii)	The initial mass is the same but final mass in Z is higher suggesting that not all the iron reacted due to insufficient acid in the volume added.					
2.	(a)		Yellow solid and chocking gas					
	(b)	(i)	Temperature					
		(ii)	The temperature increases the rate of reaction by increasing the rate of collision high energetic molecules.					
		(iii)	Two experiments are curried out using equal volumes of thiosulphate and acid of the same concentration. In one experiment the temperature of thiosulphate is kept at room temperature while for the other the temperature is raised to 60°C. When acid is added to thiosulphate, the yellow suspension form earlier in thiosulphate solution at 60°C that that at room temperature. This indicates that the reaction is faster at higher temperature than at low temperature.					
3.			Refer to page 5-9					
4.		(i)	Time (s)	200	100	40	20	10
			Concentration of thiosulphate (mol dm ⁻³)	0.05	0.09	0.15	0.20	0.25
			$\frac{1}{c}$ of thiosulphate (mol ⁻¹ m ⁻³)	20	11	7	5	4

		(ii)	
		(ii)	Rate of reaction increases as concentration increase (Note that the recipricol of concentration is dilution, and dilute solution takes long to form cloudness)
5.	(a)	(i)	$\text{CaCO}_3(\text{s}) + 2\text{HNO}_3(\text{aq}) \longrightarrow \text{Ca}(\text{NO}_3)_2(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
		(ii)	Temperature increases the rate of reaction
		(iii)	Increase in surface area increases the rate of reaction by increasin the probability of contact between the liquid and the solid
	(b)	(i)	Heat Contentration of hydrogen
	(c)	(i)	
		(ii)	The rate of production of hydrogen increases with the lengnt of magnesium ribblon up to 5cm when the acid become insufficient that increase in rate decreases.
		(iii)	At 4.0 cm the rate is $7.3\text{cm}^3/\text{minute}$

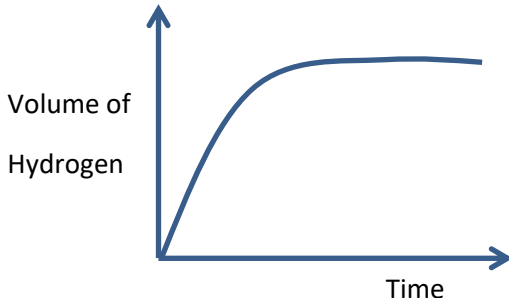
6.	(a)	The setup is as shown below  Hydrogen peroxide Graduated syringe Frictionless piston Manganese dioxide Hydrogen peroxide is added to manganese (IV) oxide and immediately stop clock started. The amount of oxygen produced in graduated syringe in a given time is noted.
	(ii)	Rate of production of oxygen = _____
(b)		 Vol of O_2 undiluted H_2O_2 Diluted H_2O_2 time Note that dilution of hydrogen peroxide reduces the rate of production of oxygen and the maximum volume of oxygen produced.
(c)		Mole of hydrogen peroxide = _____ Moles of hydrogen produced 2mole of hydrogen peroxide produce 1 mole of oxygen $\Rightarrow$ 0.1 mole of hydrogen peroxide produce _____ Mole of magnesium that reacted $2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$ From equation

			2 mole of O_2 react with ($2 \times 24 = 48g$ of magnesium $\Rightarrow$ 0.05mole of O_2 react with _____
7	(a)		Rate of reaction is the amount of products produced per unit time
	(b)		
	(d)	(i)	At 20 second Rate = _____ At 60 seconds Rate = _____
		(ii)	The rate at 20 seconds is higher than at 60 seconds because the concentration of the reactants decrease as time goes on.
8	(a)	(i)	1000cm^3 contain 0.5 moles $\Rightarrow$ 100 cm^3 cotains _____
		(ii)	1000cm^3 contain 1 moles 100 cm^3 cotains _____

		(i) & (ii)	
		(i)	The reaction with 0.5M sulphuric acid takes longer because the reaction is slower. Since the concentration of acid is lower.
9			Experiment set up is shown below Two experiments are carried out using equal masses of calcium carbonate chips and calcium carbonate powder and equal volumes of acid. In each case the volume of carbon dioxide produced at intervals of time is measured. A plot of the graph of the volume of carbon dioxide by calcium carbonate chips and calcium carbonate powder with time is given below

			 Conclusion The graphs show that the rate of reaction is higher with calcium carbonate powder than with chips.
	(b)		The rate of reaction is higher with 4M hydrochloric acid than 2M hydrochloric acid because high concentration of the acid increases the rate of collision between calcium carbonate molecules and acid molecules.
	(c)		Catalyst
10	(a)		The rate of reaction is how fast the reactants are converted to the product with time
	(b)	(i)	Temperature Increasing the temperature increases reaction rates because of the disproportionately large increase in the number of high energy collisions. It is only these collisions (possessing at least the activation energy for the reaction) which result in a reaction.
		(ii)	Surface area: Increasing the surface area of a solid reactant exposes more of its particles to attack. This results in an increased chance of collisions between reactant particles, so there are more collisions in any given time and the rate of reaction increases. The surface of a solid can be increased by crushing it into a powder.

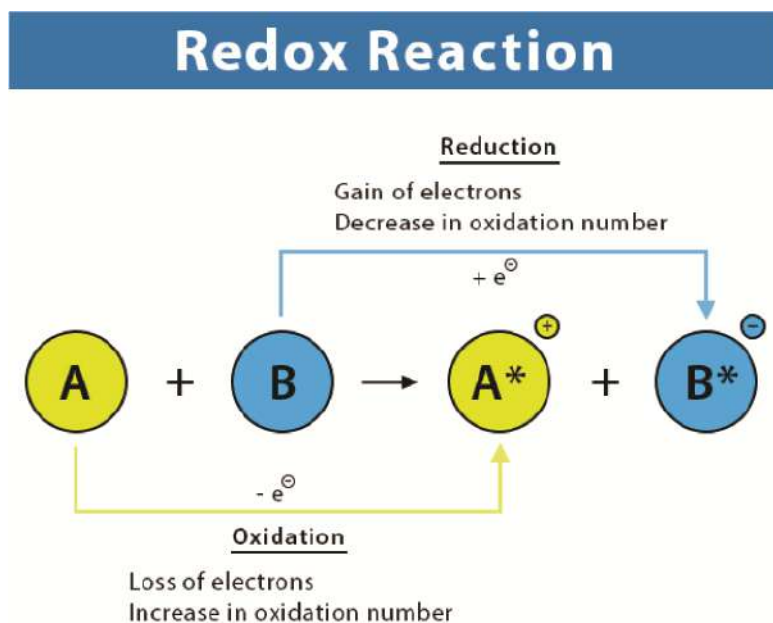
	(c)	(i)	Volume of hydrogen against time
		(ii)	Rate at 3.0 s = _____
	(d)		Would decrease
11	(a)		Experiment set up is shown below Hydrochloric acid Graduated syringe Frictionless piston Iron fillings Hydrochloric acid is added to iron fillings and immediately a stop clock is started and the volume (V) of hydrogen is recorded in a given time (t) Rate = _____
	(b)		$\text{Fe(s)} + 2\text{HCl(aq)} \longrightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
	(c)	(i)	Moles of HCL that reacted 24000cm³ at room temperature of hydrogen require 2 moles of hydrogen 120cm³ of hydrogen will require _____ Volume of hydrochloric acid Remember that molarity is the moles in 1000 cm³

			⇒ 0.1moles of HCl are contained in 1000cm^3 And 0.01mole of HCl will be in ————
		(ii)	Mass of iron fillings 2moles of HCl react with 56g ⇒ 0.01 mole of HCl react with ————
	(d)		A graph of volume of hydrogen liberated against time 
	(e)		The rate increases

S4 New Curriculum chemistry

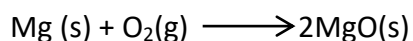
Theme: REDOX Reactions

Chapter 1 – Oxidation and Reduction Reactions

**Oxidation**

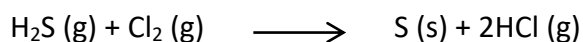
Oxidation can be defined as as

- Gaining of oxygen, for example, reaction of magnesium with oxygen



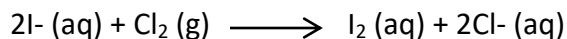
When magnesium is burnt in oxygen there is increase in weight because oxygen reacts with oxygen to form heavier magnesium oxide

- Loss of hydrogen for example sulphur is oxidised when hydrogen sulphide reacts with chlorine



In this case hydrogen atoms are removed from sulphur to chlorine

- Loss of electrons or increase in oxidation state for example, during the reaction potassium iodide with chlorine gas to form iodine and potassium chloride



In this reaction, the oxidation state of iodine increases from -1 to 0; hence iodide ions are oxidised.

Reduction

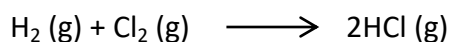
Reduction can be defined as

- (i) Loss of oxygen for example when copper oxide is reacted with magnesium, copper is reduced because magnesium removes oxygen from copper oxide

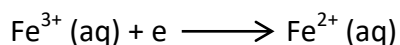


The color of copper oxide changes from black to brown copper crystals

- (ii) Addition of hydrogen, for example, chlorine is reduced when it reacts with hydrogen to form hydrogen chloride

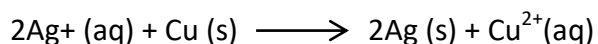


- (iii) Addition of electron, and oxidation state decrease, for example when sulphur dioxide turns brown solution of iron (III) to a green solution of iron II



Redox reaction

In a reaction both oxidation and reduction occur concurrently with one reactant oxidised while another is reduced, thus redox reaction. For example, when copper is added to solution of silver nitrate solution, a green solution of copper nitrate is formed with black ppt of silver because copper is oxidised to copper II ions while silver ions are reduced to black silver particles



Oxidation state

Oxidation state refers to the charge an atom would have if electrons were completely transferred in a bond, rather than shared.

	Rule	Example	Oxidation state
1.	Oxidation state of free element or atoms is zero	Cu, S, Fe, Na, K	0
2	Oxidation state of a simple ion is equal the charge on the ion	Na ⁺ Cl ⁻ Cu ²⁺ Al ³⁺	+1 -1 +2 +3
3	In compound metal atoms have positive oxidation states while nonmetals have negative oxidation states but the compound has zero oxidation state for example MgO	Mg ²⁺ O ²⁻	+2 -2

Trial 1

(a) Determine the oxidation states of the elements in the following compounds

- (i) CaO
- (ii) HCl
- (iii) NaI

(b) Determine the oxidation state of copper in the following oxides

- (i) CuO
- (ii) Cu₂O

Oxidizing agent

An oxidizing agent is a substance that gains electrons and causes another substance to lose electrons in a redox reaction. It is reduced in the process while the other substance is oxidized.

Common oxidizing agents include:

- **Oxygen (O_2)** – Used in combustion and respiration.
- **Chlorine (Cl_2)**
- **Potassium dichromate** – is an oxidizing agent in the laboratory analytical methods.
- **Hydrogen peroxide (H_2O_2)** – A powerful oxidizer used in bleaching and disinfecting.
- **Potassium permanganate ($KMnO_4$)** – Common in redox titrations and water purification.
- **Chlorine (Cl_2)** – Used for disinfecting and in chemical synthesis.
- **Nitric acid (HNO_3)** – Acts as an oxidizing agent in reactions involving metals.
- **Iron(III) (Fe^{3+})**

Reducing agents

A **reducing agent** is a substance that **loses electrons** and **causes another substance to gain electrons** in a redox reaction. It is **oxidized** in the process while the other substance is **reduced**.

Common reducing agents include:

- **Hydrogen (H_2)** – Used in many industrial reduction processes.
- **Carbon (C)** – Found in coke and charcoal, used in metal extraction.
- **Sulfur dioxide (SO_2)** – Acts as a reducing agent in various chemical reactions.
- **Iodide ions (I^-)** – Can reduce iron(III) (Fe^{3+}) to iron(II) (Fe^{2+}).
- **Tin (II) chloride ($SnCl_2$)** – Often used in analytical chemistry.
- **Ascorbic acid (Vitamin C)** – A mild reducing agent in biological and chemical processes.

Electrolysis

This is the decomposition of an ionic compound in molten or solution form into its constituent elements by electricity.

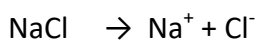
Note that

1. An electrolyte is an ionic substance that conducts electricity in solid or molten form.
2. Solid electrolyte do not conduct electricity because the ions cannot move because they are held together by strong ionic bonds.

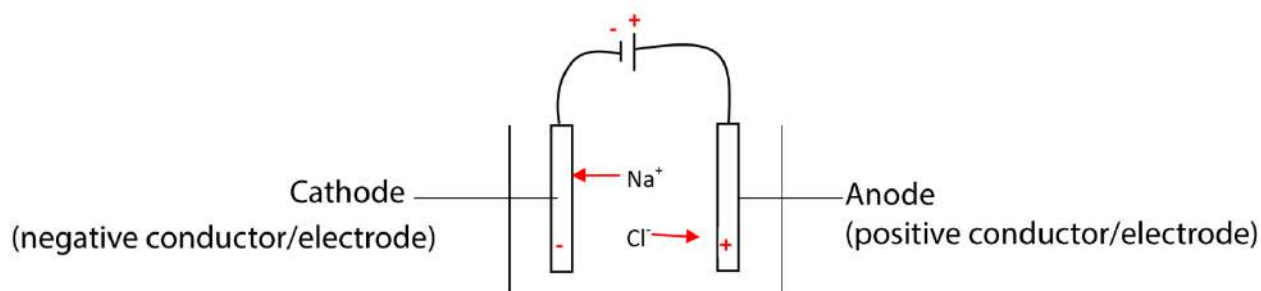
Mechanism of electrolysis

Consider electrolysis of molten sodium

In molten form sodium chloride ionizes as follows



In an electrolytic cell, unlike charges attract; Na^+ migrate to the cathode while Cl^- migrate to the anode.



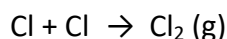
At the cathode, Na^+ ion acquires an electron to become sodium atom.



At the anode the Cl^- loses an electron to form a Cl atom



The chlorine atoms combine to form chlorine gas.

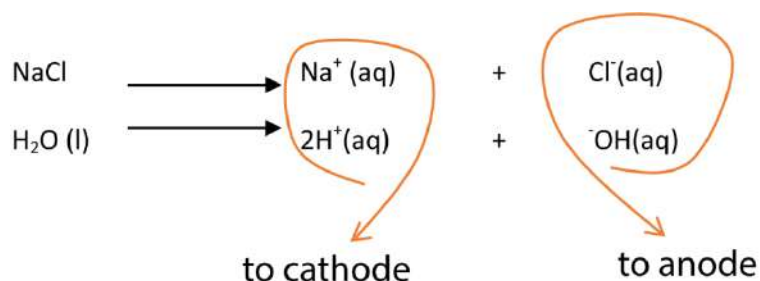


Ultimately, sodium chloride in molten form is decomposed by electrolysis to sodium metal and chlorine gas.

Selective discharge

Consider electrolysis of sodium chloride solution

Both sodium chloride and water ionize to form ions



The positively charged ions migrate to the cathode while negatively charged ions migrate to the anode.

Factors that decide the ion to be eliminated or discharged first.

(i) Position in electro-chemical series:

Series for cations (**high to low**) K^+ , Ca^{2+} , Na^+ , Mg^{2+} , Al^{3+} , [C], Zn^{2+} , Fe^{2+} , Pb^{2+} , Cu^{2+} , Ag^+

Series for anions (**high to low**) SO_4^{2-} , NO_3^- , Cl^- , Br^- , I^- , OH^-

The ion lower in electro-chemical series is discharged first. Consequently, during electrolysis of dilute sodium chloride, H^+ which is low in the reactivity series than Na^+ is discharged at the cathode whereas, OH^- ions are discharged at the anode.

At the cathode
 $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$

At the anode
 $4\text{OH}^-(\text{aq}) - 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

(ii) Concentration:

When Cl^- , Br^- or I^- are concentrated, then they will be discharged with respect to OH^- .

In this case electrolysis of concentrated sodium chloride solution liberates chlorine at the anode and hydrogen gas at the cathode (concentration does not affect Na^+)

At the cathode
 $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$

At the anode
 $2\text{Cl}^-(\text{aq}) - 2\text{e}^- \rightarrow \text{Cl}_2(\text{g})$

(iii) Nature of electrode

(a) When mercury cathode is used Na^+ is discharged with preference to H^+ .
 Therefore, electrolysis concentrated sodium chloride using mercury cathode liberate Na at the cathode and chlorine gas at the anode (due to high concentration of Cl^-).

At the cathode
 $\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}$

At the anode
 $2\text{Cl}^-(\text{aq}) - 2\text{e}^- \rightarrow \text{Cl}_2(\text{g})$

Application of electrolysis

1. Industrial preparation of gases; Cl_2 , H_2 and O_2 .
2. Extraction of metal: metals above carbon in the reactivity series are almost all extracted by electrolysis.

Example Extraction of sodium.

Ore: molten NaCl

Anode: iron cylinder is cheap and has a melting point above the melting point of NaCl.

Cathode: carbon because it does not react with chloride.

Calcium chloride added to the mixture to

- lower the melting point of sodium chloride from 800°C to 600°C ,
- reduce solubility of sodium in molten sodium chloride,
- Lower the corrosive vapor of sodium chloride.

Sodium is collected in dry nitrogen to protecting it from reacting with air.

At the cathode

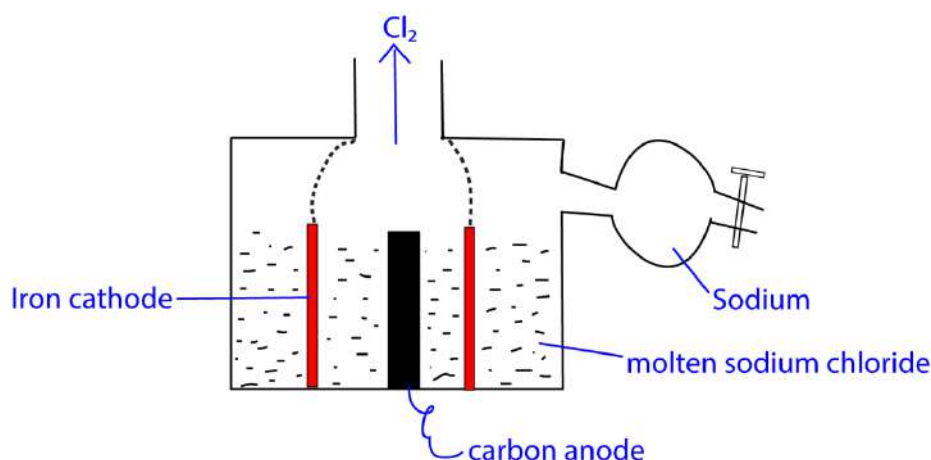
$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$

At the anode

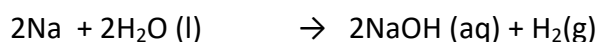
$2\text{Cl}^- - 2\text{e}^- \rightarrow \text{Cl}_2(\text{g})$

Setup

The setup is such chlorine produced in a different compartment not react with sodium



When sodium amalgam is dropped in water, sodium reacts to form sodium hydroxide and hydrogen gas.



Sodium hydroxide is concentrated to form pellets.

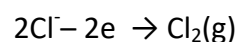
Preparation of sodium hydroxide

By electrolysis of concentrated solution of sodium chloride using carbon anode and mercury cathode. At the anode chlorine is liberated and at mercury cathode Na^+ instead of H^+ is discharged and dissolved in mercury to form mercury amalgam.

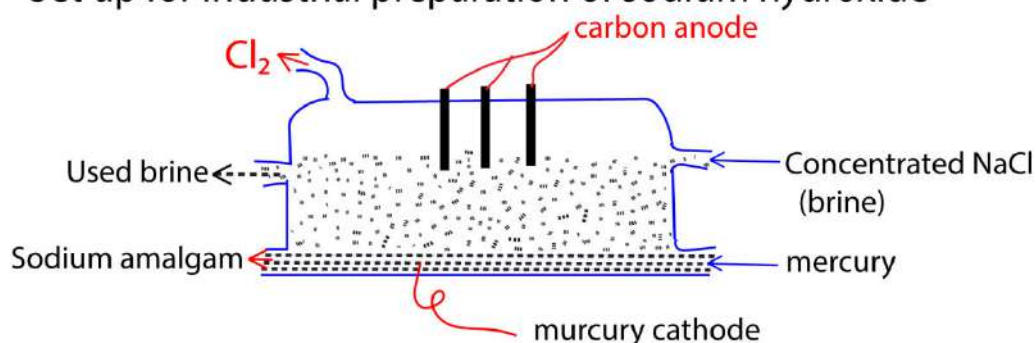
At the cathode



At the anode



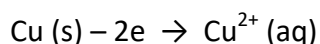
Set up for industrial preparation of sodium hydroxide



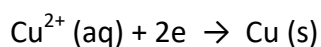
This process is disadvantageous because it releases poisonous mercury into the environment

Purification of copper

Anode: impure copper (dissolves)



Cathode (copper is deposited)



Electrolyte: copper sulphate solution

Copper migrated from the anode to the cathode.

Faraday's Laws of electrolysis

1. The mass of a substance liberated at an electrode is proportional to the quantity of electricity used.

The quantity of electricity Q in coulombs = It

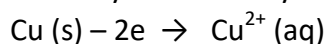
(I = current of electricity in amperes, t = time in second)

2. The moles of electricity required to liberate one mole of an element is proportion to the charge on its ions. (1mol of electricity = 1Faraday = $1F = 96500 \text{ C}$)

Example

Calculate the mass of copper liberated by a current of 1A for 1hour. ($\text{Cu} = 63.5$)

Quantity of electricity $Q = It = 1 \times (1 \times 60 \times 60) = 3600\text{C}$

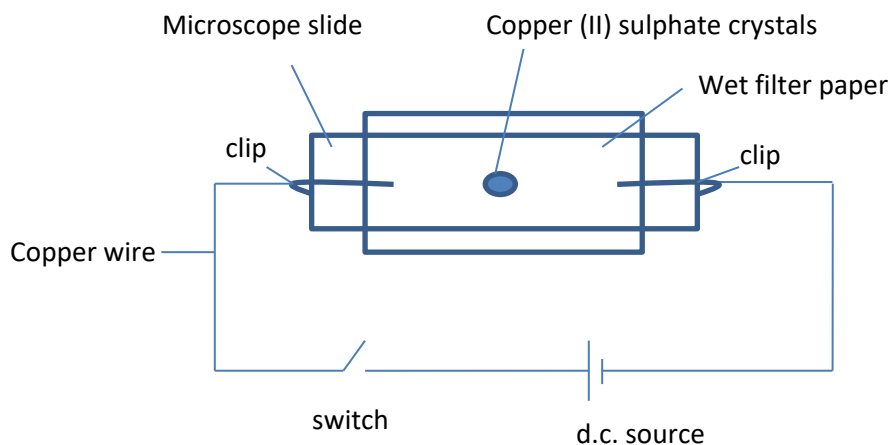


It implies that $(96500 \times 2) \text{ C}$ is required to liberate 63.5 g of copper

Therefore, 3600C will liberate ————— = 1.18g

Revision questions

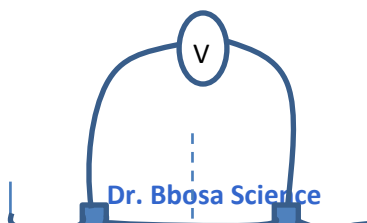
1. Dilute sulphuric acid was electrolyzed using carbon electrodes.
 - (a) State what was observed at the
 - (i) Anode (1mark)
 - (ii) Cathode (1mark)
 - (b) Write an equation for the reaction at the anode (2marks)
 - (c) Dilute copper (II) was electrolyzed using copper electrodes. State what was observed at the anode (1mark)
2. (a) Both copper wire and copper (II) sulphate conduct electric current. Name the particles which conduct electricity current in
 - (i) Copper wire
 - (ii) Aqueous copper (II) ions
- (b) The set up apparatus in the diagram below was used to find out what happens when an electrolyte is connected to a source of electric current.



State what was observed

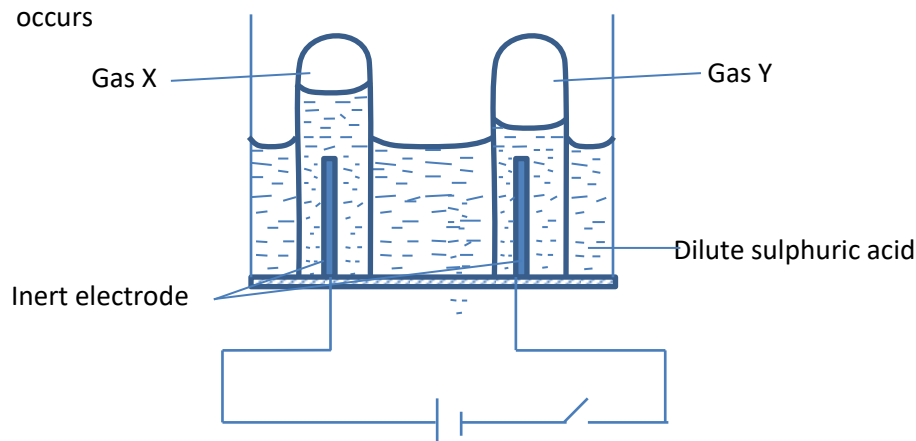
- (i) When the switch was closed
 - (ii) If copper sulphate crystal was replaced with potassium manganate (VII) crystal and the switch closed once again.
 - (c)
 - (i) Give a reason for the observation you have made in (b)(i) and (ii)
 - (ii) State any general conclusion that can be drawn following the reason you have given in (c)(i).
3. An aqueous solution of potassium iodide was electrolyzed between carbon electrode
 - (a) State what was observed at the anode
 - (b)
 - (i) Name the product formed at the anode
 - (ii) Describe the test that can be carried out to identify the product at the cathode
 - (c) Litmus paper was dropped into the solution around the cathode at the end of the experiment.
 - (i) State what was observed
 - (ii) Give a reason for your answer in (c)(i).
 - (d) Draw a labelled diagram of a setup of an apparatus that can be used to prepare oxygen by electrolysis of water.

4. (a) State two factors that can determine the product formed at an electrode during electrolysis
(b) Explain why aqueous solution of copper (II) chloride conducts electric current whereas solid copper (II) chloride does not.
(c) a dilute solution of copper (II) chloride was electrolyzed using graphite electrodes
(i) state what was observed at the cathode and write equation for the reaction that took place.
(ii) Name the substance that was produced at the anode.
(iii) Explain how the product you have named in (c)(ii) above is formed at the anode and write equation to illustrate your answer.
(d) The electrolysis of copper (II) chloride was repeated using copper electrodes. State what was observed at the anode and briefly explain your answer.
5. (a) An aqueous solution of copper (II) sulphate was electrolyzed between graphite electrode.
(i) state what was observed at the cathode
(ii) write equation for the reaction that took place at the anode.
(b) The solution that remained after electrolysis in (a) was tested with litmus solution
(i) state what was observed
(ii) Give a reason for your answer in (b)(i)
(c) The electrolysis in (a) was repeated using copper electrode that had been weighed before the experiment. State the change in mass of the electrode that took place after electrolysis.
6. Acidified water was electrolyzed using platinum electrode
(a) Write an equation for the reaction that took place at then
(i) anode (1½ marks)
(ii) cathode (1½ marks)
(b) Name one other substance that can be used as electrode in the electrolysis of acidified water. (1marks)
7. (a) (i) using an example state the difference between an electrolyte and an electrode.(04marks)
(ii) Explain why aqueous solution of sodium chloride conducts electricity whereas solid sodium chloride does not. (02mark)
(b) The diagram below shows a set-up of an electrochemical cell which can be used to compare the reactivity of zinc and copper.



- (i) Identify the rod that is positively charged (01)
- (ii) Identify R and state its purpose (1 ½ mark)
- (iii) Write equation for the reaction taking place at the copper and zinc rods (03 marks)
- (iv) Write equation for the overall reaction in the cell. (1 ½ marks)
- (v) State what would happen if zinc metal is dropped in a solution containing copper (II) ions (02)

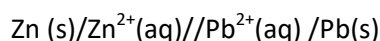
8. The diagram below shows an electrolytic cell in which electrolysis of dilute sulphuric acid occurs



- (i) Name the gases X and Y that are evolved during electrolysis (02marks)
- (ii) Give equation for the reaction occurring at the anode. (02marks)
- (iii) Indicate the direction of electrons in the circuit. (01mark)
- (iv) Calculate the volume of gas X produced when the current of one ampere flows for 10 minutes through the electrolyte. [1 faraday = 9.6×10^4 coulombs, 1 mole of a gas occupies $2.4 \times 10^4 \text{ cm}^3$ at room temperature and pressure]
- (v) State two industrial applications of electrolysis other than manufacture of sodium hydroxide

(b) sodium hydroxide can be manufactured using a mercury cell. How would the manufacturing process affect the environment? (3 ½ marks)

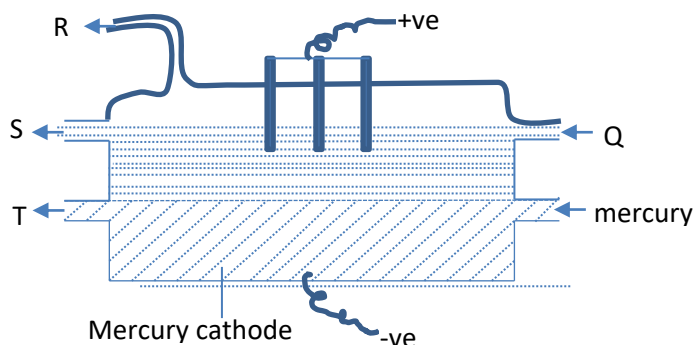
9. A notation for an electromotive cell is given below



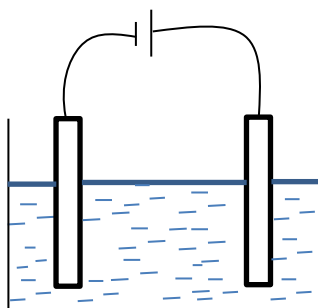
- (a) Name two substances that can be used as electrolytes (2marks)
- (b) State which of the electrodes is the anode
- (c) Write equation for the reaction at
 - (i) The anode
 - (ii) The cathode

(d) Write the overall equation for the cell reaction

10. During the manufacture of sodium hydroxide, concentrated sodium chloride solution (brine) is electrolyzed using a mercury cathode as shown in the diagram below

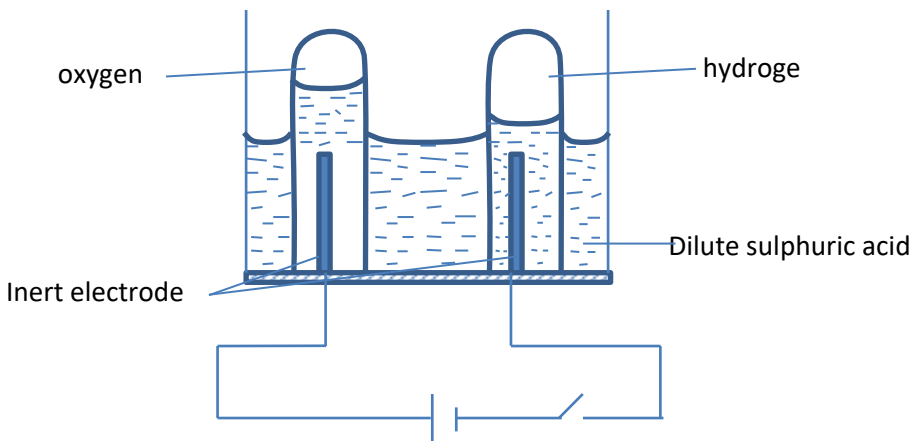


- (a) Name the substance used as the anode
 - (b) Name the substance
 - (i) Taken in at Q
 - (ii) Taken out through R, S, T.
 - (c) Name one other substance formed during the manufacture of sodium hydroxide
 - (d) Describe briefly how solid sodium hydroxide can be obtained from the products of electrolysis.
11. The diagram below shows an arrangement of apparatus used for purification of copper



- (a) Name the substance used as
 - (i) Anode
 - (ii) Cathode
 - (iii) Electrolyte
- (b) Write equation for the reaction that took place at
 - (i) Anode
 - (ii) cathode

Suggested answers

1.0	(a)	(i)	Colorless gas
		(ii)	colorless gas
	(b)		$4\text{OH}^-(\text{aq}) - 4\text{e}^- \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
	(c)		Anode decrease in size
2	(a)	(i)	Electrons
		(ii)	Ions
	(b)	(i)	Blue color moves towards the cathode
		(ii)	Purple color moves towards the anode
	(c)	(i)	Blue Cu^{2+} migrate to the cathode
	(d)		During electrolysis cations migrate to the cathode whereas anions migrate to the anode
3.	(a)		Brown color
	(b)	(i)	Iodine
		(ii)	A blue color forms when a drop of iodine solution is added
	(c)	(i)	Color of litmus turns blue
		(ii)	Removal of hydrogen ion and iodide ion leaves an alkaline solution of KOH
	(d)		
4	(a)		- Position in the reactivity series - Concentration - Nature of electrode
	(b)		In aqueous solution copper (II) chloride ions are free to move whereas in solid state ions do not move
	(c)	(i)	Brown coat formed on graphite

			$\text{Cu}^{2+}(\text{aq}) + 2\text{e} \longrightarrow \text{Cu}(\text{s})$ (copper ions are lower than hydrogen ions in reactivity series)
		(ii)	oxygen
		(iii)	$\text{OH}^{-} \text{ ions are lower than } \text{Cl}^{-} \text{ in the reactivity series and is preferentially discharged}$ $4\text{OH}^{-}(\text{aq}) - 4\text{e} \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
	(d)		The anode decrease in size due to oxidation $\text{Cu}(\text{s}) - 2\text{e} \longrightarrow \text{Cu}^{2+}(\text{aq})$
5	(a)	(i)	Brown coat formed on graphite $\text{Cu}^{2+}(\text{aq}) + 2\text{e} \longrightarrow \text{Cu}(\text{s})$ (copper ions are lower than hydrogen ions in reactivity series)
	(b)	(i)	Turned litmus solution red
		(ii)	Removal of copper ions from solution leaves acid H^{+}
	(c)		The anode decrease in size due to oxidation $\text{Cu}(\text{s}) - 2\text{e} \longrightarrow \text{Cu}^{2+}(\text{aq})$ The cathode increase in size due to deposition $\text{Cu}^{2+}(\text{aq}) + 2\text{e} \longrightarrow \text{Cu}(\text{s})$
6	(a)	(i)	Anode : $4\text{OH}^{-}(\text{aq}) - 4\text{e} \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
		(ii)	Cathode: $2\text{H}^{+}(\text{aq}) + 2\text{e} \longrightarrow \text{H}_2(\text{g})$
	(b)		Graphite
7	(a)	(i)	An electrolyte is an ionic compound that conducts electricity in solution of molten form while electrodes are conductors dissolved in electrolytic solution
		(ii)	Aqueous sodium chloride conducts electricity because it has free moving ions whereas solid sodium chloride has ions that do not move.
	(b)	(i)	Copper rod
		(ii)	R is porous partition that allows exchange of charges without mixing the solutions or completes the circuit
		(iii)	At zinc rod $\text{Zn} - 2\text{e} \longrightarrow \text{Zn}^{2+}(\text{aq})$ Copper rod $\text{Cu}^{2+}(\text{aq}) + 2\text{e} \longrightarrow \text{Cu}(\text{s})$

		(iv)	$\text{Zn (s)} + \text{Cu}^{2+}\text{(aq)} \longrightarrow \text{Zn}^{2+}\text{(aq)} + \text{Cu(s)}$
8.	(a)	(i)	X= oxygen and Y = hydrogen
		(ii)	$4\text{OH}^{-}\text{(aq)} - 4\text{e}^{-} \longrightarrow 2\text{H}_2\text{O (l)} + \text{O}_2\text{(g)}$
		(iii)	From anode to cathode
		(iv)	$Q = it = (1 \times 10 \times 60) = 600\text{C}$ (96000 x 4) produce 24000cm ³ of oxygen 600C produce _____ = 37.5 cm ³
		(v)	Electroplating Extraction of metals Purification of copper
	(b)		Mercury when released in the environment is very poisonous, it accumulates through food chains
9	(a)		Zinc sulphate and copper sulphate
	(b)		The anode is zinc
	(c)	(i)	$\text{Zn (s)} - 2\text{e}^{-} \longrightarrow \text{Zn}^{2+}\text{(aq)}$
		(ii)	$\text{Cu}^{2+}\text{(aq)} + 2\text{e}^{-} \longrightarrow \text{Cu(s)}$
		(iv)	$\text{Zn (s)} + \text{Cu}^{2+}\text{(aq)} \longrightarrow \text{Zn}^{2+}\text{(aq)} + \text{Cu(s)}$
10.	(a)		Carbon
	(b)	(i)	Brine/concentrated sodium chloride solution
		(ii)	R – chloride S- used brine T- sodium amalgam
	(c)		hydrogen
	(d)		Sodium amalgam dissolves in water. Sodium reacts with water to form sodium hydroxide and hydrogen $2\text{Na (s)} + 2\text{H}_2\text{O(l)} \longrightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$ Sodium hydroxide solution is concentrated by evaporation to form pellets.

11	(a)	(i)	Anode is Impure copper sulphate
		(ii)	Cathode is pure copper sulphate
	(b)	(i)	$\text{Cu (s)} - 2\text{e}^- \rightarrow \text{Cu}^{2+}(\text{aq})$
		(ii)	$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$

INDUSTRIAL PROCESSES DETAILED NOTES AND SAMPLE

1. *Industrial processes*
2. *Some sample questions on the above topic*
3. *Try so hard to answer the sample questions and look for more qns.*

Don't say tomorrow, it will be too late for chemistry revision, and yesterday is gone forever, you have got today to revise your chemistry! "Revise as if tomorrow is not there"
May god bless you

Industrial processes

These are the procedures involved in manufacturing of items.

Or;

An industrial process is a set of procedures that convert material resources into finished commodities.

Industries do a great work of processing raw materials into finished products ready for use by the final consumer.

There is a variety of natural resources in Uganda, ranging from minerals, stones, plants and animals.

These act as a starting point for many manufacturing industries within your locality.

Examples of some industries, their products and uses

Industry	Product	Name of product	Use of some products
Textile industry		clothes	Covering the body
Steel making industry		Steel bars, iron nails, iron sheets	Construction
Battery industry		Battery	Providing electric energy
Mining industry		Cement	Building houses
Plastics industry		Plastic chairs, plastic buckets, plastic cups, jerrycans etc	Sitting on, carrying water, taking water/tea, fetching water.
Fertilisers industry		Fertilisers	Restoring soil fertility
Beverage's industry/Soda drinks industry/food industry		soda	Human consumption
Water industry/food industry		water	Human consumption
Sugar industry/food industry		sugar	Human consumption

Soap making industry.		Soaps and detergents	Washing
Soap making industry		Sanitizer	Killing micro-organisms
Cosmetics industry		Jelly	Human beautification

Describe how industries relate to each other

Industries are interdependent on the other so that the economy as a whole function properly and efficiently. The primary sector is where the materials for the secondary industry are gathered. In the secondary sector, the product is then made into consumable product which is then distributed by the tertiary industry.

Types/Classification of industries.

Industries can be classified as;

- Primary
- Secondary
- Tertiary

Primary industries

These are industries that produce or create products that are sold to the public.

Examples of primary industries.

- Lumbering
- Mining
- Farming
- fishing

Secondary industries

These are industries that process raw materials from the primary sector into finished products.

The products manufactured under secondary sector are either consumed by the consumers or used as raw materials for other processing industries.

Examples of secondary industries.

- Oil refinery
- Cement industry

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Tertiary industries

These are industries that provide services and facilitate a smooth flow of goods and services in the market.

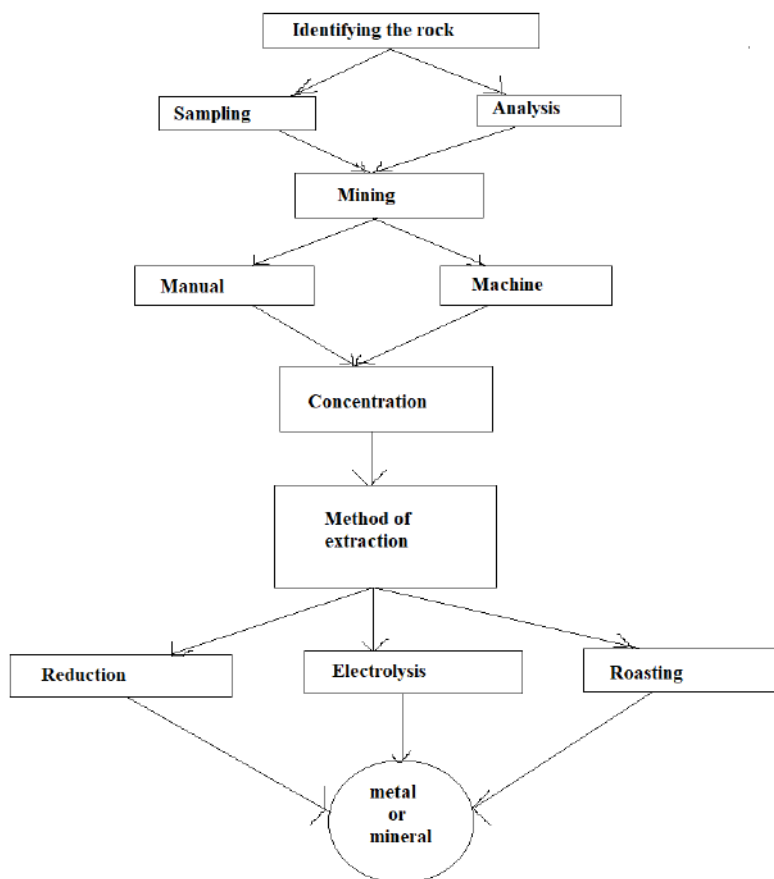
Examples of tertiary industries.

- Warehousing
- Insurance
- Banking
- Transport
- Advertising

Some industries, corresponding products their products

Industry	Product	Class of industry
Mining	Limestone	Primary
Processing	Cars	Secondary
Service	telecommunication	Tertiary

Processes of obtaining useful chemicals from rocks**Summary of the processes**



Sample questions and answers

- Why are sampling and analysis of the rock necessary before mining?
 - Sampling helps in obtaining information which is necessary for the economic assessment of the ores.
 - Analysis gives information about the ore before massive mining can begin
- State at least two ways in which sampling is done.
 - Hand sampling
 - Machine sampling or excavation
- Compare manual and excavation (machinery) methods of mining.
 - The manual method consumes more time than the excavation method
 - The manual method is also riskier in terms of accidents than the excavation method.
- State three properties of the ore considered before concentration of the ore is done.
 - Chemical composition of the ore
 - Conductivity of the ore

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- Solubility of the ore
 - Density of the ore
5. Outline three ways in which the ore is concentrated.
- Froth flotation
 - Roasting
 - Gravity separation
6. Complete the table by giving examples of metals extracted by each of the methods indicated.

Method of extraction	Examples of metals extracted
Electrolysis	Sodium, potassium
Chemical reduction	Iron, aluminium
Roasting in air	Zinc, copper
No extraction process method needed	Gold, silver

Useful chemicals from rocks

Most rocks are made up of naturally occurring inorganic solids that have a crystalline structure and a distinct chemical composition.

These are known as minerals and are chemical in nature.

Examples of elements that make up the minerals (and their uses) found in the Earth's rocks

Note (research about the image of each mineral element from baroque)

Mineral	Chemical Symbol/ formula	Image	Use
Silver	Ag		➤ Making jewellery ➤ Making trophies ➤ Making mirrors
Copper	Cu		➤ Making electric wires ➤ Used in construction such as roofing ➤ Making jewellery
Graphite	C		➤ Making lead pencils ➤ Making electronic products, such as electrodes. ➤ Making lubricants
Cinnabar	HgS		➤ Making colouring paints

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THE KISJO CONTENT (INDUSTRIAL PROCESSES)

			➤ Providing mercury
Galena	PbS		➤ Manufacturing lead ➤ Making lead pipes and sheets
Pyrite	FeS ₂		➤ Used in the production of sulphur dioxide. ➤ Used in the production of sulphuric acid.
Halite or rock salt	NaCl		➤ It is added in food to make it tasty. ➤ Used to preserve food such as fish. ➤ It maintains blood pressure.
Corundum	Al ₂ O ₃		➤ Used as an abrasive ➤ Grinding optical metals ➤ Polishing metals
Cuprite	Cu ₂ O		➤ Making jewellery
Haematite	Fe ₂ O ₃		➤ Producing pigments. ➤ Radiation shielding
Calcite	CaCO ₃		➤ Used as a construction material ➤ For agricultural soil treatment.
Dolomite	CaMg(CO ₃) ₂		➤ As an ingredient in the production of glass, bricks, and ceramics.
Gypsum	CaSO ₄ .2H ₂ O		➤ Manufacture of wallboard, cement.
Anhydrite	CaSO ₄		➤ Used as a drying agent ➤ in plaster, paint, and varnish.
Albite	NaAlSi ₃ O ₈		➤ Manufacture of glass and ceramics.
Olivine	(Mg,Fe) ₂ SiO ₄		➤ Added to blast furnaces to remove impurities from steel and to form a slag
Orthoclase	KAlSi ₃ O ₈		➤ Manufacture of glass and ceramics.

THE KISJO CONTENT (INDUSTRIAL PROCESSES)

Quartz	SiO ₂		➤ Used to make oscillators for watches, clocks, radios, televisions, electronic games, computers.
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Extraction and purification of metals

Metals are electropositive, they ionize by loss of electrons to form positively charged ions. They are therefore reducing agents.

Occurrence of metals.

Metals occur in the earth's crust as **ores**.

An ore is a rock which contains metal compounds from which metals can be extracted. Therefore, ores are the major source of metals on earth.

General idea on extraction of metals

For a metal to be extracted, its concentration in a given ore must be high.

Therefore, it is necessary to concentrate (purify) the ore before extracting the metal.

The method of metal extraction normally depends on the position of the metal in the electrochemical/reactivity series.

K	These are very reactive metals according to their order and therefore form very stable compounds. They are not found as free elements and can only be extracted from their compounds by electrolysis
Na	
Ca	
Mg	
Al	
Zn	These are moderately reactive metals. They usually occur as ores in form of oxides, sulphides, carbonates and more other forms. They are extracted by reduction of the ores using either carbon or carbon monoxide
Fe	
Pb	
Cu	These are not very reactive. They are therefore found as free elements. They are the easiest to extract since they form unstable compounds. They can be extracted by mere heating of the ore
Hg	
Ag	
Au	

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Note;

We only look at the extraction of only three metals, one metal from each category, that is to say;

- Aluminium, Al
- Iron, Fe
- Copper, Cu

The processes involved in extracting and purifying metals especially intermediate metals in Uganda.

The metals are extracted by reduction and there have four basic stages in the process after obtaining the ore/mining;

1. Concentration of the ore
2. Roasting of the ore to convert the carbonates and sulphides to oxides and remove water vapour. It is easier to reduce the oxide than the carbonate, that's why ores are heated directly in air before electrolysis.
3. Reduction of the ore
4. Purification/refining of the metal extracted

Concentration of the ore

This process involves removing impurities.

Impurities can be separated by physical means like, picking by hand, washing, using a magnet or solvent extraction.

In the case of copper, the ore is first crushed and then mixed with water.

Air is then blown through the mixture and the clean ore separates and collects at the surface as froth. This is referred to as froth floatation.

Roasting

The concentrated (pure) ore is roasted in air at high temperatures to produce oxides which are easier to reduce than carbonates or sulphides.

Sulphur is removed as sulphur dioxide and carbon dioxide driven off from the carbonates.

e.g.,

zinc sulphide + oxygen $\longrightarrow$ sulphur dioxide + zinc oxide

Reduction of the ore

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The ore is usually heated in a furnace in the presence of a suitable reducing agent, usually coke (carbon) or carbon monoxide which converts the ore into the required metal.

e.g.,

Iron(III) oxide + carbon dioxide $\longrightarrow$ iron + carbon dioxide

Purification/refining of the metal

The metal obtained is often impure. Purification is normally done by electrolysis in the case of copper and zinc. In other cases, the impure metal is heated in a hearth open to air where the impurities oxidize and rise to the surface as a scum and can be removed.

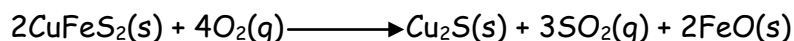
COPPER

The principle/ chief ores of copper are:

- copper pyrites (CuFeS_2);
- cuprite (Cu_2O);
- copper (I) sulphide (Cu_2S) or glance and
- malachite (CuCO_3 , $\text{Cu}(\text{OH})_2$).

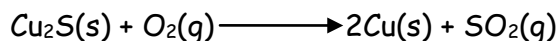
Extraction of copper from copper pyrites

The ore obtained from mining is first concentrated by a process of froth floatation and then it is roasted in air to produce copper(I) sulphide.



By adding silicon dioxide (sand) and heating the mixture in the absence of air, the solid iron (II) oxide impurity is converted into slag which is poured off with the reaction mixture leaving behind only copper (II) sulphide.

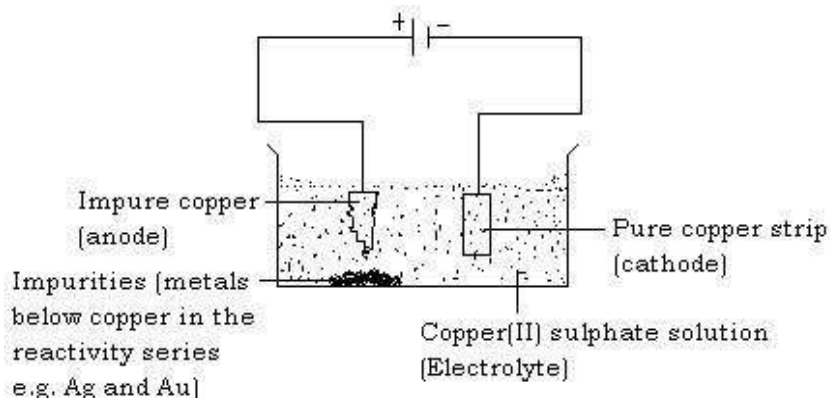
The copper (II) sulphide is reduced to metallic copper by heating in a regulated supply of air (oxygen).



(Impure copper)

The metallic copper produced is impure copper (blister copper) and has to be purified. Purification of the copper is done by the process of electrolysis.

Purification of blister copper



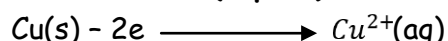
The impure copper/blister copper is made the anode and strip of pure copper serves as the cathode.

The electrolyte is acidified solution of copper (II) sulphate.

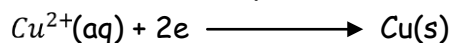
During electrolysis, pure copper is transferred from the impure copper anode to the pure copper cathode.

Therefore, the anode dissolves and decreases in size as the cathode grows bigger.

At the anode (impure)



At the cathode (pure)



(Pure copper)

Ions above copper in the reactivity series remain in the solution.

Uses of copper

- For wiring electrical circuits since it is a good conductor of electricity and is relatively cheap. The copper used for this purpose must be very pure since impurities increase electrical resistance.
- It is used for making ornaments like ear rings and pins, bangles e.t.c. being little attacked by air.
- It is used for making alloys like bronze (copper and tin) and brass (copper and zinc), copper coinage (copper and tin), German silver (copper, zinc and nickel) - Used for making water pipes and boilers.
- Copper is used as a roofing material because it weathers to acquire a coating of green basic copper carbonate, $\text{CuCO}_3 \cdot \text{Cu(OH)}_2 \cdot n\text{H}_2\text{O}$, which lends a colourful touch to the building.

Sample Questions on copper extraction

(kindly look for more questions from the library books **#dont wait for tomorrow**)

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1. Identify the principle ores of copper

Diagram	Name
	Copper pyrites
	Copper glance
	Malachite
	Cuprite

2. Mention where in Uganda can such ores be found

- Kilembe mines in Kasese

3. State the main steps involved in extracting copper from its ores

- Mining,
- Concentration,
- Roasting, and
- Purification.

4. Describe how copper mining affects the environment

- Erosion
- Formation of sink holes
- Destruction of ecosystems and habitats
- Contamination of soil and water by chemicals from mining processes

5. State the uses of copper in everyday life.

- Check the notes above

IRON

The principle/chief ores of iron are:

Structure	Name
	Haematite, Fe_2O_3
	Magnetite, Fe_3O_4
	Iron pyrite, FeS_2

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	Siderite or Spathic iron ore (FeCO_3)
	Limonite ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$).

These ores in Uganda can be found in

- Muko in Kabale district
- Kisoro district
- Sukulu and Bukusu in Tororo district

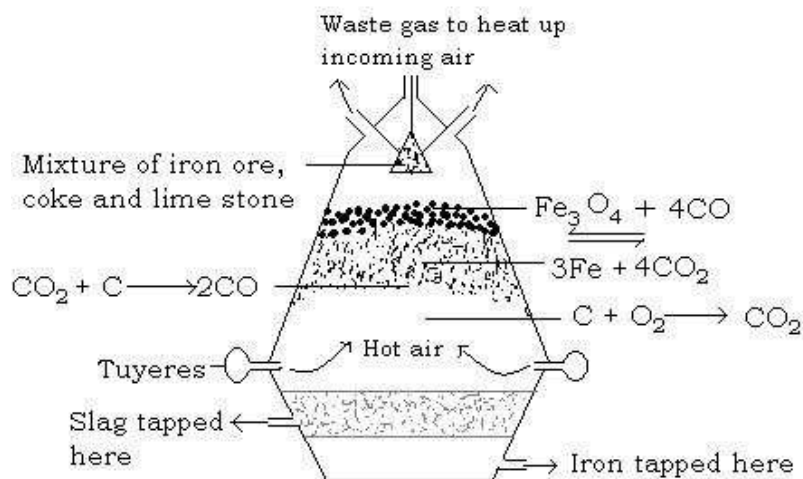
Extraction process (iron from magnetite, Fe_3O_4).

The iron ore is crushed and roasted in air to remove water and other non-metallic impurities especially sulphur and phosphorus which are oxidized away as gaseous oxides.

The roasted ore is now mainly Fe_3O_4 . When an ore is roasted in air and Fe_3O_4 is the main product, it is known as sintering

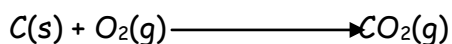
The roasted ore is mixed with coke (carbon) and limestone (calcium carbonate) and introduced into the blast furnace where the reduction of the ore takes place.

The blast furnace



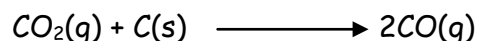
Very hot air is introduced from low down into the blast furnace.

As the hot air passes through the mixture of roasted ore, coke and limestone, the coke burns to form carbon dioxide (is oxidized) in an exothermic process.

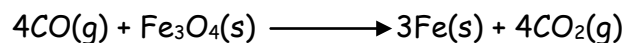


As the carbon dioxide produced rises through the furnace, it is reduced by the excess hot coke to produce carbon monoxide.

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The carbon monoxide formed reduces the iron ore at a high temperature (about 1000°C) to form iron metal.

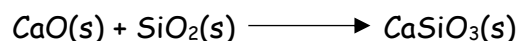
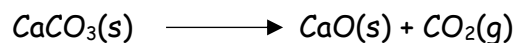


The molten iron formed sinks to the bottom of the blast furnace where it is tapped and solidified into blocks of pig iron.

Note; The procedures are the same when using Haematite ore (Fe_2O_3)

The role of limestone

Limestone removes Silicon (IV) oxide which is the main impurity in the iron ore. Limestone at high temperature decomposes to form calcium oxide and carbon dioxide. The calcium oxide formed combines with silicon (IV) oxide (impurity) to form molten calcium silicate (slag).



The slag being less dense than iron forms a separate layer above iron and thus are tapped separately.

The slag protects the molten iron against any further oxidation by oxygen in the hot air in the blast furnace.

The carbon dioxide produced as a bi-product in the furnace is quickly reduced by hot coke to carbon monoxide which is required as a reducing agent.

By products from the blast furnace include:

- Calcium silicate/slag (used for making roads, manufacturing cement, manufacturing glass);
- Waste gases like Carbon dioxide, carbon monoxide, steam and unreacted nitrogen;
- Calcium phosphate (used as an inorganic fertilizer).

Types of Iron

There are three types of iron which are classified based on their percentage purity. The percentage purity also determines the strength and use of the iron.

The types of iron are;

- Cast iron (pig iron)
- Wrought iron
- Steel.

Cast iron

This is an impure iron which contains relatively high proportions of carbon (4%) and small proportions of other substances such as silicon, phosphorus and sulphur.

Such impurities make cast iron to be hard, brittle and to have a lower melting point than pure iron. It cannot be welded and has little tensile strength.

Cast iron can be used to make hot water pipes, Bunsen burner bases, cookers, in railings and other purposes where little strain is imposed.

Wrought iron

This is the purest form of iron (contains about 0.3% carbon) and is obtained from cast iron by heating it with iron (III) oxide in a furnace by a process known as —puddling.

The oxygen of the iron oxide oxidizes carbon and sulphur to their respective gaseous oxides, phosphorus to Phosphates(V) and silicon to silicates which form slag.

The semi molten mass is then hammered and rolled so that the slag is squeezed out and a mass of almost pure iron remains.

It is very tough, malleable and ductile and is therefore used to make iron nails, sheeting, ornamental work, horse shoes and agricultural implements.

Wrought iron is sometimes referred to as low carbon steel.

Steel

Steel is an alloy of mainly iron with carbon and other elements like manganese, chromium, silicon, cobalt and sometime tungsten.

The quality of steel depends on the amount of carbon present and this in turn determines its intended use.

Steel is generally used in the construction of buildings, bridges, ships, car bodies, cutting and boring tools, crushing machines and stainless cutlery such as knives, forks e.t.c.

Uses of iron

- Used in the manufacture of steel
- Used as a medical supplement
- Used to make bridges, bicycles, chains and cutting tools.

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- Iron catalysts are used in the industrial production of ammonia

ALUMINIUM

The principle ore of aluminium is; bauxite.

Sample questions on Aluminium extraction.

Study the figure below and answer the questions that follow

1. Name the processes taking place at;

- **A**-Mining
- **X**-Concentration
- **R**- Roasting
- **Z**-Melting
- **M**- Electrolysis

2. Describe what takes place in the processes X, Z, M, and R.

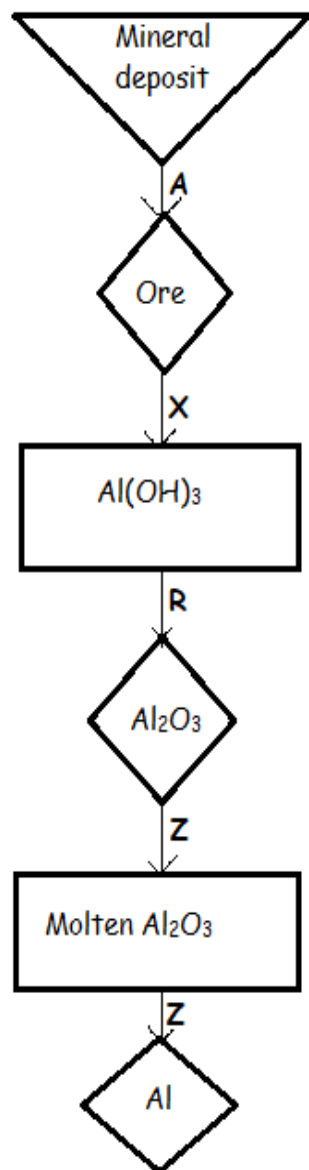
- At X, the ore is crushed and mixed with concentrated sodium hydroxide solution.
- At R, pure aluminium hydroxide is heated to form pure aluminium oxide.

- At Z, aluminium oxide is dissolved in molten cryolite to lower its melting point.
- At M, the mixture of aluminium oxide in molten cryolite is electrolyzed between graphite electrodes.

3. Mention the uses of aluminium.

- Aluminium is used in the manufacture of food packing materials, aircrafts, source plans and cars.

Note; aluminium is a widely used metal because of its light weight.



FERTILISERS

A fertilizer is a substance containing essential elements for the healthy growth of plants. When added to the soil, fertilisers normally improve soil fertility.

The most important elements are nitrogen, potassium, and phosphorus and sometimes sulphur.

These elements are absorbed by plants as compounds especially as nitrates phosphates.

Nitrates as fertilisers

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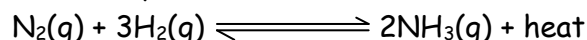
Nitrates are obtained from nitrogen as one the major elements to form nitrate fertilisers.

Uses of nitrogen

- Used in the manufacture of fertilisers
- Used in the manufacture of ammonia

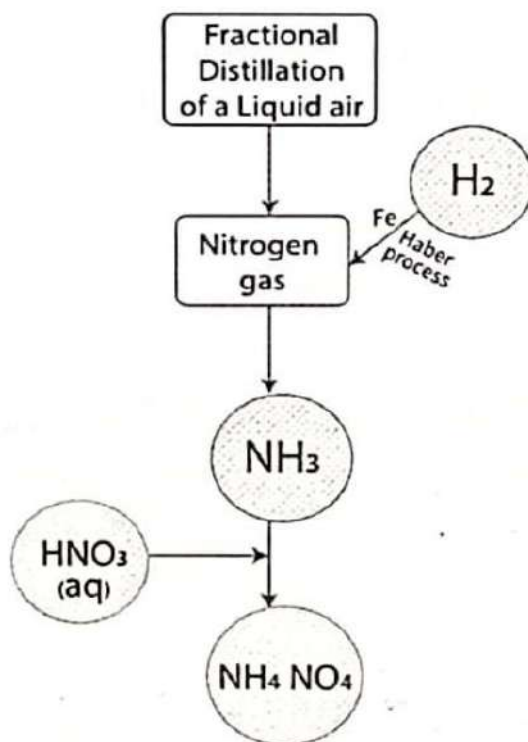
Obtaining nitrates from nitrogen starting from atmospheric air

- The atmospheric air is trapped and condensed to form liquid air.
- The liquid air is pumped into the plant where it goes through various stages until it reaches the fractionating column where it is slowly warmed up and the air components(gases) boil off one by one according to their boiling point. Nitrogen having the lowest boiling point (-196°C), it is obtained first.
- Hydrogen gas is then mixed with nitrogen in the presence of finely divided iron catalyst and the mixture is heated to form ammonia gas.



- The ammonia gas is then mixed with nitric acid to form ammonium nitrate
 $\text{HNO}_3(\text{aq}) + \text{NH}_3(\text{g}) \longrightarrow \text{NH}_4\text{NO}_3(\text{aq})$
- The ammonium nitrate obtained acts as the fertilizer containing nitrogen.

Summary using a flow chat



Conversion of nitrates into fertilisers

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Ammonia.

On daily basis, our bodies produce urine on daily basis.

The smell of the urine is due to ammonia vapour residues in urea.

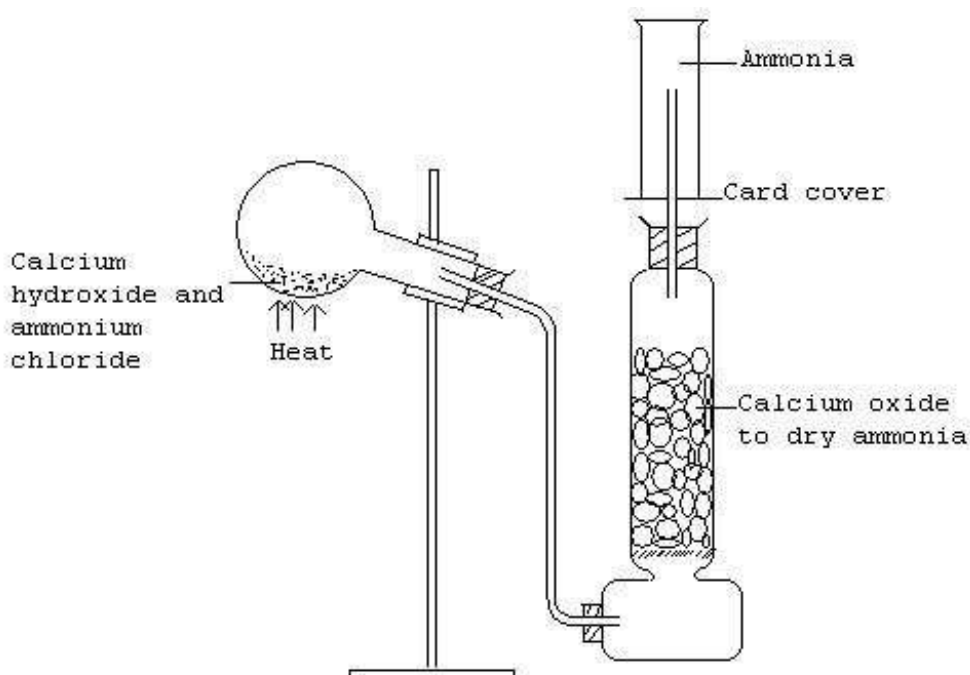
Ammonia naturally occurs in many parts of the environment like in soil, air, and vegetation.

Ammonia is an essential compound of nitrogen and hydrogen.

Ammonia can be prepared in the laboratory.

Laboratory preparation of ammonia gas

- Round bottomed flask
- Gas jar
- Source of heat
- Cork
- Retort stands and clamps
- Delivery tube
- Ammonium chloride
- Calcium hydroxide
- Calcium oxide (quick lime) to dry ammonia
- Litmus paper

Setup of apparatus

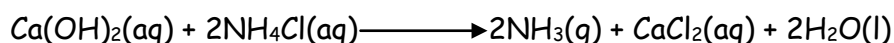
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Procedure

- Assemble the apparatus as shown in the diagram above
- Grind a mixture of ammonium chloride and calcium hydroxide and place it in a round bottom flask of a hard glass

NB. The neck of the flask should bend down wards and the flask should be in a slanting position because the formed water vapour will condense and if allowed to run back on the hot flask causes breakage

- Heat the mixture in the flask to evolve ammonia gas

Equation

- The gas is then dried by passing it over lumps of calcium oxide.

Haber process (industrial manufacture of ammonia)

Most of the ammonia used today is manufactured by reacting nitrogen and hydrogen at the temperatures of 400-500°C and a pressure of 200 atmospheres.

Finely divided iron is used as a catalyst in the industrial manufacture of ammonia.

Sample questions and answers

1. On testing ammonia with moist litmus paper, what did you observe?
 - The red litmus paper turned blue and the blue litmus paper remained blue.
2. Write an equation for the formation of ammonia from calcium hydroxide and ammonium chloride.

$$\text{Ca(OH)}_2(\text{aq}) + 2\text{NH}_4\text{Cl}(\text{aq}) \longrightarrow 2\text{NH}_3(\text{g}) + \text{CaCl}_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$$
3. Explain the role of calcium oxide in the preparation of ammonia.
 - To dry ammonia gas
4. Why is ammonia gas collected by upward delivery?
 - Ammonia gas is less dense than air.
5. How important is ammonia gas?
 - Used as a refrigerant gas.
 - Used in the manufacture of fertilisers.
 - Used in the manufacture of plastics, pesticides, dyes and other chemicals.
 - Used as an ingredient in many cleaning products.

Fertilisers from ammonia gas

Ammonia is converted into a variety of fertilisers, such as potassium nitrate, ammonium nitrate, ammonium sulphate and diammonium phosphate (DAP), calcium ammonium nitrate (CAN).

These fertilisers are better than each other depending on the nitrogen composition.

The best fertiliser is the one with the highest composition of nitrogen.

Structure/photo of the above fertilisers

These fertilisers are known as **inorganic fertilisers**

Inorganic fertilisers are therefore synthetic fertilisers that contain a high percentage of essential elements like nitrogen, phosphorus and potassium. They are relatively cheap and soluble in water.

Sample questions and answers

Consider the fertilisers in the table below

Fertiliser	Chemical formula
Ammonium nitrate	NH_4NO_3
Potassium nitrate	KNO_3
Urea	NH_2CONH_2

- a) Calculate the molar mass of each compound in the table above. (RAM; N=14, H=1, O=16, K=39.1, C=12).

Molar mass of NH_4NO_3

$$\begin{aligned} &= (2 \times 14) + (4 \times 1) + (3 \times 16) \\ &= 80 \end{aligned}$$

Molar mass of KNO_3

$$\begin{aligned} &= (1 \times 39.1) + (1 \times 14) + (3 \times 16) \\ &= 101.1 \end{aligned}$$

Molar mass of NH_2CONH_2

$$\begin{aligned} &= (2 \times 14) + (4 \times 1) + (1 \times 12) + (1 \times 16) \\ &= 60 \end{aligned}$$

- b) Calculate the percentage composition of nitrogen in each of the compounds.

Ammonium nitrate

Mass of nitrogen in NH_4NO_3

$$\begin{aligned} &= (2 \times 14) \\ &= 28\text{g} \end{aligned}$$

Percentage composition of nitrogen in NH_4NO_3

$$\begin{aligned} &= \left(\frac{28}{80} \times 100 \right) \% \\ &= 35\% \end{aligned}$$

Potassium nitrate

Mass of nitrogen in KNO_3

$$\begin{aligned} &= (1 \times 14) \\ &= 14\text{g} \end{aligned}$$

Percentage composition of nitrogen in KNO_3

$$\begin{aligned} &= \left(\frac{14}{101.1} \times 100 \right) \% \\ &= 13.8477\% \end{aligned}$$

Urea

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Mass of nitrogen in NH_2CONH_2

$$= (2 \times 14)$$

$$= 28\text{g}$$

Percentage composition of nitrogen NH_2CONH_2

$$= \left(\frac{28}{60} \times 100\right)\%$$

$$= 46.667\%$$

c) Write the compounds in descending order, with respect to the percentage compositions of nitrogen.

➤ Urea, Ammonium nitrate, Potassium nitrate

d) Identify the best fertiliser out of the three fertilisers. Give a reason for your answer.

➤ The best fertiliser is urea, since it has the highest percentage of nitrogen.

Preparation of ammonium nitrate fertiliser

- Ammonium nitrate is produced by reacting nitric acid with ammonia
- The resulting solution is concentrated to 97.5% - 98% in the final concentrator.
- The concentrated solution is fed to a prilling tower, and some part of the solution is fed to a slurry tank.
- In the slurry tank, the filter is dispersed, the oversize and small fines are melted.
- The 97.5% ammonium nitrate solution from the concentrator is discharged and the moisture content of the mixture is adjusted by the addition of scrubbing liquor.

Health awareness

Nitrates and nitrites are widespread contaminants of vegetables, fruits, and waters. Fertilisers applied to crops can be washed into rivers and leached into soil and ground water.

The levels of these compounds are increased as a result of wastes from chemical industries, effluents, nitrogenous fertilisers, and herbicides in agriculture.

Therefore, determining the nitrate and nitrite levels in biological food and food samples is important for human health and the environment.

Organic fertilisers such as cow dung, plant materials (humus) and human waste have less harm to the environment, and therefore can be used in preference to inorganic fertilisers.

Industrial processes that make use of natural resources

- Power generation
- Fish processing
- Turning wind turbines
- Cement manufacturing
- Textile production

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THE KISJO CONTENT (INDUSTRIAL PROCESSES)

- Processes in the oil and gas industry

Industries that use natural resources

Natural resource	Processes that make use of the natural resource	Industry in which the resource is used
Lakes	Fishing	Fish processing industry
Rivers	Generation of hydro-electric energy.	Hydropower industry
Swamps	Disposal of wastes from the industries.	Water purifying industries
Forests	Carpentry	Wood and pulp industry
Air	Signal transfer	communication

Technology and environment.

Technology has improved transport, industry, and communications in everyday life.

Technology has also improved the quality of the environment. For example, energy used at home can be generated from different sources, such as the sun, water and biomass. Explain how technology has improved the use of natural resources in the industrial sector.

➤

Importance of industrial processes in utilizing natural resources to make useful chemicals

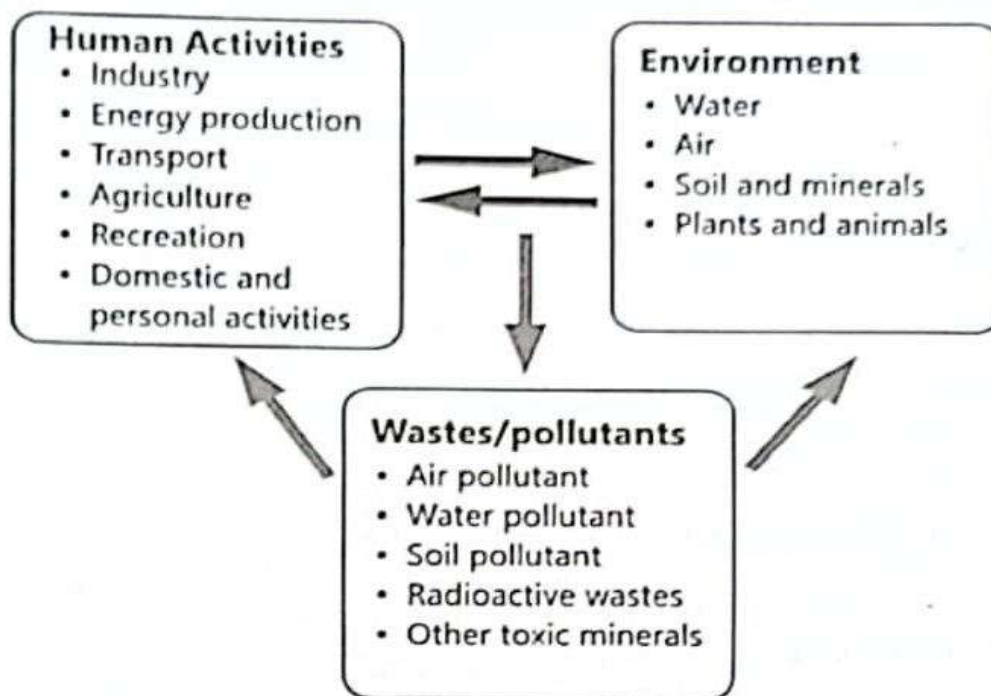
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Dangers arising from industrial processes

- Water pollution
- Global warming
- Degradation of land quality

Ways of minimizing/reducing the dangers of industrial processes

- Proper treatment of industrial wastes before disposal.
- Recycling
- Planting more trees to absorb harmful gases like carbon-dioxide.
- Etc (look for more)

The relationship between human activities, the environment and wastes

Recycling, reusing and reducing.

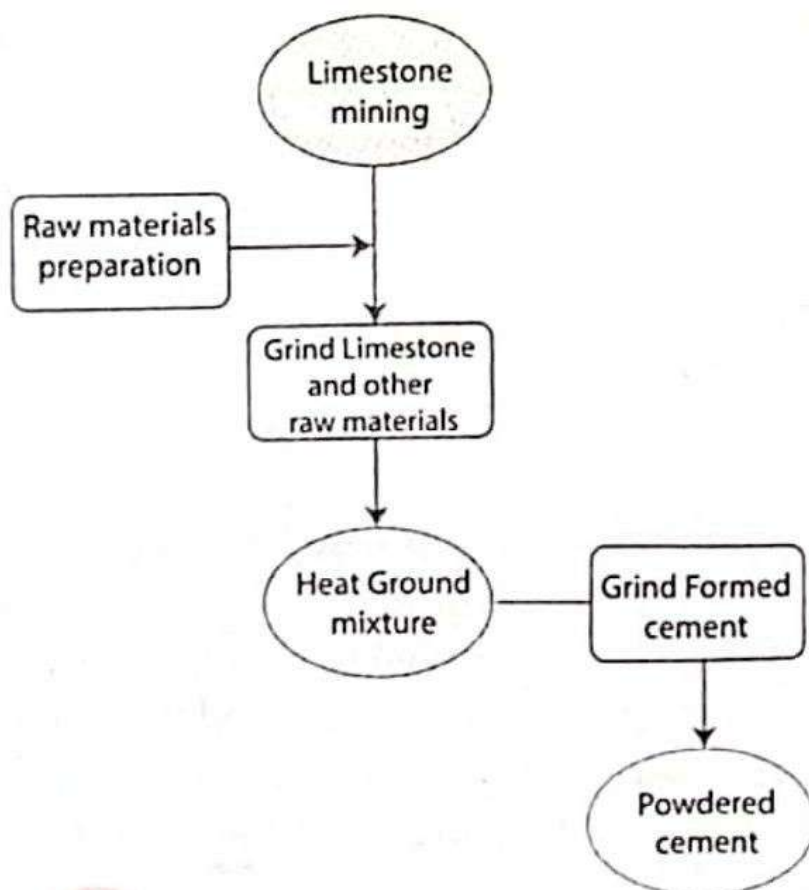
Importance of recycling, reusing and reducing.



Manufacture of lime and cement

The process of cement manufacture occurs in the following stages;

- Mining of limestone.
- Other raw materials (alumina and clay) are added to limestone and grinded.
- The mixture is then heated to form cement.
- The formed cement is then grinded again after cooling to form powdered cement.

Summary of the process**Benefits of manufacturing lime and cement**

- Provides employment opportunities.
- Construction of safer structures for shelters.
- Earns foreign exchange for the government.
- It's for study purposes.

Dangers arising from manufacture of lime and cement.

Please, explain the following points and add on if you can;

- Air pollution
- Water pollution
- Global warming
- Accidents
- Difficulty in breathing.

How to minimizing dangers arising from manufacture of lime and cement.

- Using suitable respiratory gears like nose masks, etc
- Wearing suitable eye protection wear like safety glasses.
- Using alternatives like fly ash instead of cement.
- Using better alternatives to make cement.

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Production of sodium hydroxide and chlorine gas by electrolysis of sodium chloride solution/brine.

What you need.

- Batteries
- Electrical wires
- Carbon rods
- Spatula
- Glass beaker
- 10M sodium chloride solution
- Distilled water
- Test tube
- Burning splint
- Measuring cylinder.

Safety precautions

- Chlorine gas is poisonous; therefore, the experiment should be carried out in a fume cupboard or be carried out by the teacher in small amounts.
- Sometimes alkali solution is corrosive and it should be handled with care.

What to do/steps taken.

- Measure 100cm^3 of distilled water using a measuring cylinder and pour into a glass beaker.
- Measure 4 spatula endful of sodium chloride and add them to the water in the beaker.
- Stir the mixture to form a solution.
- Introduce the carbon rods in the solution and attach them to the wires from the battery.
- Insert the test tubes at each carbon rod.

Observation.

- When tested with moist blue litmus paper, it turned red and got bleached.
- The gas discharged burnt with a pop sound when tested, indicating hydrogen gas.

Products, conclusion and application.

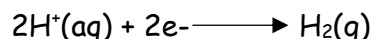
Chlorine gas is produced at the anode.

Hydrogen gas is produced at the cathode.

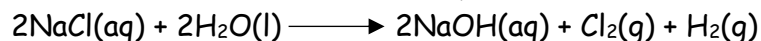
Equation at the anode.



Equation at the cathode.



Overall equation for the electrolysis of brine



Uses of the products from electrolysis of brine.

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THE KISJO CONTENT (INDUSTRIAL PROCESSES)

Product	Use
Chlorine	➤ Used for making paints and dye stuffs ➤ Used for making solvents for degreasing and dry cleaning ➤ Used in the manufacture of hydrochloric acid. ➤ Used for making medical drugs, pesticides and weed killers.
Hydrogen	➤ Used to harden vegetable oils ➤ Used as a fuel in hydrogen fuel cells ➤ Used in making nylon.

Uses of sodium hydroxide solution as a product also;



Polymers

A polymer is a complex molecule with a large molecular mass formed by the combination of many small molecules with relatively small molecular mass.

Types of polymers

There are mainly two types of polymers;

- Natural polymers
- Synthetic polymers

Uses of polymers

Natural polymers

Polymer	Monomer	Use
Starch	Glucose	Source of energy
Proteins	Amino acids	Repair of worn-out tissues
Cellulose	Glucose	Cell walls
Glycogen	Glucose	Source of energy
Lipids (fats and oil)	Fatty acids and glycerol	Source of energy,
Natural rubber	Isoprene	Making foot wears
Nylon (was first produced by an American chemist Wallace Hume Carothers at the DuPont on 28 th February, 1935)		Manufacture of car tyres
Cotton		Used in making textiles.

Synthetic (artificial) polymers

Polymer	Monomer	Use
Polyethene	ethene	➤ Making containers, eg plastics bowls and dust bins ➤ used as wrapping materials ➤ manufacture of polythene bags
Polyvinyl chloride (PVC)	Chloro ethene	➤ Making rain coats ➤ Electrical insulation ➤ making pipes and films
Polystyrene	styrene	➤ making packing materials ➤ making house hold items such as combs, plastic cups and a common lining in refrigerators
Artificial rubber		➤ making of shoes
Melamine		➤ making of melamine glasses
Teflon		➤ used in insulation of wirings in aerospace and computer applications

Dangers of synthetic polymers

- They prevent water from penetrating into the soil hence reducing soil fertility.
- They cause air pollution due to the emission of gases(outgassing) resulting from exposure of the polymer to heat.
- They also contribute to global warming when burnt to produce greenhouse gases like carbon-monoxide.

Important chapter summary concepts

- Uganda has a large number of natural resources, which can be used in various industrial processes.
- Metals can be extracted from their ores.
- The method used to extract a metal from an ore depends on the reactivity of the metal.
- Metals which are high in the reactivity series are extracted by electrolysis but those averagely reactive are extracted by chemical reduction.
- In the extraction of sodium, molten sodium chloride is used as the electrolyte. Calcium chloride is added to lower its melting point.
- The main ore of aluminium is bauxite.
- Iron is extracted by heating a mixture of the ore, limestone and coke in a blast furnace.
- Copper is extracted from cuprite or malachite. It can be refined by electrolysis of copper(II) sulphate using a copper cathode.
- Ammonia is a useful gas, especially in the manufacture of fertilisers and is obtained on a large scale by the haber process.
- Ammonium sulphate, ammonium nitrate, urea and DAP are some of the fertilisers from ammonia.
- Fertilisers have many advantages but also some disadvantages.
- Organic fertilisers are useful substitutes for industrial (inorganic) fertilisers but may lack sufficient quantities of key elements for plant growth.
- Cement is made by heating a mixture of limestone, alumina and clay. It is a useful material in construction.
- Sodium hydroxide and chlorine are prepared by electrolysis of concentrated sodium chloride (brine).

Sample activity of integration

Over time, Ugandans have used risky processes in mining expensive minerals like iron, tin and gold. It was reported by investigative journalists of different media houses in Uganda that people in the gold mining business in Mubende district and Karamoja region were using mercury, a hard metal, to separate gold from earthly materials.

Task

As a chemistry learner, write a short message to the leaders in the aforementioned regions, stating the precautions that people can employ for safe mining.

End of chapter questions

1. a) Write the names and locations of three important chemical industries in Uganda.
b) What are the major sources of metals on Earth?
2. a) List the physical and chemical methods employed to concentrate an ore.
b) Describe the electromagnetic separation method of ore concentration.
3. a) Write four reducing agents used in the reduction of metal oxides to metals.
b) What are the three common purification procedures for metal extraction?
4. a) Describe how sodium metal is extracted from sodium chloride.
b) Explain why metals cannot be extracted from the electrolysis of sodium chloride solution.
5. a) What is the role of nitrogen in food packaging?
b) Write a balanced chemical equation for the laboratory preparation of ammonia.
6. a) Explain why drying agents like concentrated sulphuric acid and calcium chloride cannot be used to dry ammonia.
b) Explain how sulphur dioxide is converted to sulphur trioxide in the industrial manufacture of sulphuric acid.
7. a) Give two examples of fertilisers manufactured from ammonia.
b) Explain the environmental impact of fertilisers.
c) Write some examples of naturally occurring organic fertilisers.
d) What are the advantages of organic fertilisers over chemical fertilisers? Explain.
8. Describe the process of manufacturing cement.

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UCE

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margin1st ELEMENT OF CONSTRUCT;THE LEARNER APPRECIATES CONTRIBUTION OF
CHEMISTRY TO OUR ECONOMY. Areas of emphasis;

- ① Manufacture of Oxygen
- ② Manufacture of Chlorine
- ③ Extraction of Metals $\leftarrow \begin{matrix} Fe \\ Cu \\ Al \end{matrix}$
- ④ Manufacture of fertilizers
- ⑤ Manufacture of detergents
- ⑥ Manufacture of sodium hydroxide
- ⑦ Manufacture of sulphuric acid
- ⑧ Manufacture of cement
- ⑨ Manufacture of ethanol
- ⑩ Manufacture of Biogas.

Each Process;

V - Vessel

Cp - chemical processes

Cd - Conversion to
desired product

Ch - Coherence

Pr - Purification

Key points in assessment

⇒ Raw materials

⇒ Process of production.

⇒ Side effects of the process of production and mitigation

⇒ Social benefits

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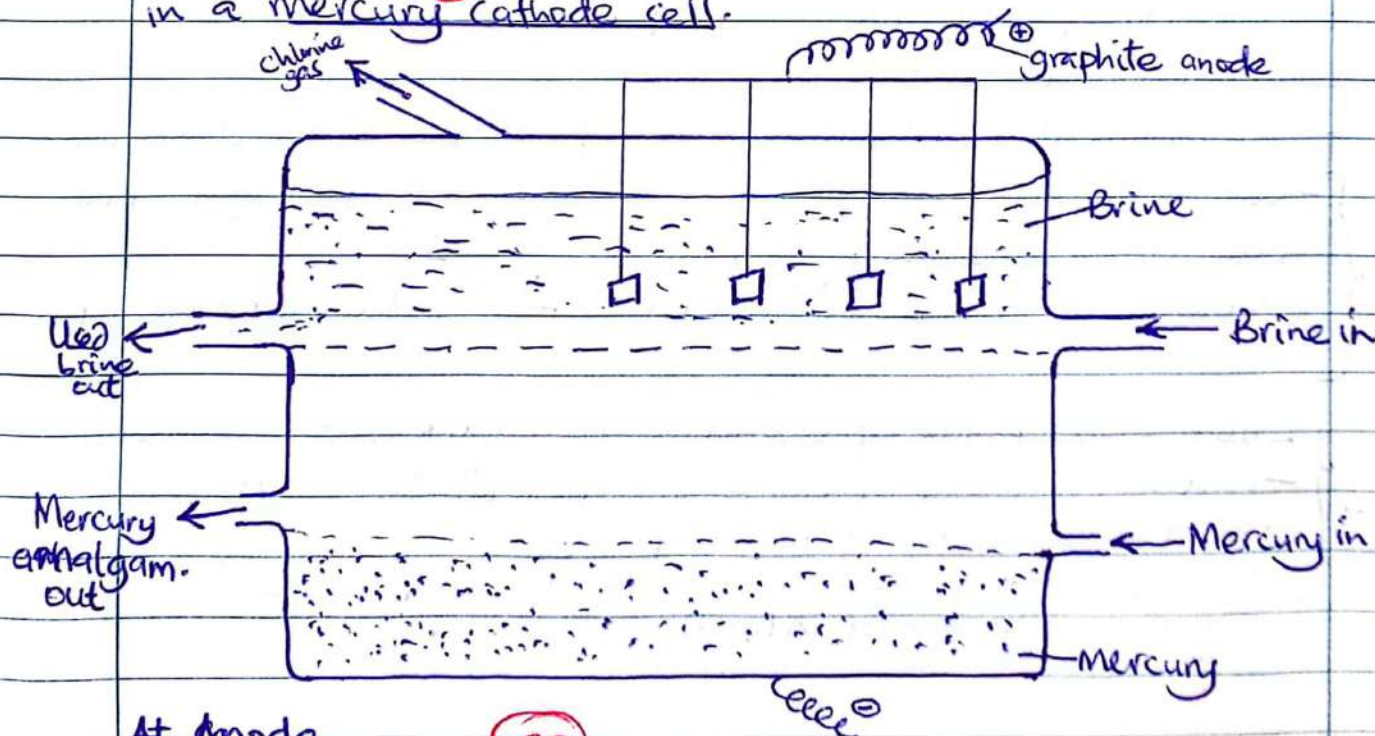
LEARNER APPRECIATES CONTRIBUTION OF CHEMISTRY.
TO OUR ECONOMY.

ASSESSABLE AREAS.

1 and 2. MANUFACTURE OF CHLORINE; - Using Mercury Cathode cell.
OR SODIUM HYDROXIDE SOLUTION.

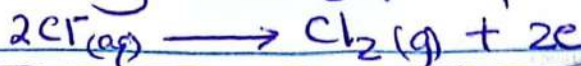
⇒ Raw materials; Concentrated sodium chloride solution, Mercury.
(Brine)

⇒ Process of production;
Chlorine is manufactured by electrolysis of concentrated sodium chloride solution (Brine) using graphite^{as} anode and mercury as cathode in a mercury cathode cell.



At Anode

Chloride and hydroxide ions migrate to the anode and chloride ions are preferentially discharged to form chlorine gas.



For NaOH; At Anode; $\text{Na}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Na}(\text{s})$ The Sodium then mixes with mercury
 $\text{Na}(\text{s}) \longrightarrow \text{Na}(\text{Hg})$

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Sodium amalgam reacts with water forming sodium hydroxide solution

$$\text{Na/Hg (s)} + \text{H}_2\text{O (l)} \rightarrow \text{NaOH (aq)} + \text{H}_2 \text{ (g)} + \text{Hg (l)}$$

⇒ Side effects of the process and mitigation.

- Suffocation due to release of other gases like hydrogen in ^{confined} spaces
Proper Use of PPEs
- Exposure to high concentrations of chlorine leading to death or itching of eyes, noses
Proper Use of PPEs
- Exposure to excessive noise from mechanical equipment/cylinders
Proper Use of Personal Protective Equipment
- Exposure to mercury which is highly toxic and may cause damage to nervous system, reproductive system on long time-exposure.
Proper Use of PPEs

⇒ Social benefits

- Manufacturing plants are source of revenue to the Govt hence improved infrastructures.
 - Employment opportunities hence improving the standards of living
- for Chlorine {
- Chlorine in water is a disinfectant hence kills pathogens that would cause diseases to man.
 - Chlorine is used ^{in manufacture of} as a weed killers, ^{weeds} that reduce crop production.
- for NaOH {
- used in manufacture of soap which is a detergent in homes.
 - Used in manufacture of drugs eg pain ^{relievers} killers - eg Aspirin
 - Used in manufacture of baking powder
 - In paper making industry, it is used to clean the ^{wood} paper
 - Used in extraction of Aluminium.
 - In textile industry to make dyes,

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marginMANUFACTURE OF OXYGEN⇒ Raw materials ; Air ^{or} (Atmospheric air)⇒ Process of productionAtmospheric air is pumped into ^{tanks} ~~plant~~ and the process involves the following stages.① Filtration

Atmospheric air is filtered using air filters to remove dust particles.

② Drying:^{ch} Air is cooled until the water vapour condenses. It is passed through a bed of silica gel. This is to remove water vapour.③ Removal of pollutantsThe remaining air components are passed through beds of charcoal to remove air pollutants (NO_2 , SO_2 , CO , etc)④ Removal of Carbon dioxideIt is then passed through concentrated potassium hydroxide solution to remove Carbon dioxide gas. ^{CP}⑤CompressionAir is repeatedly compressed at high pressure of 200 atm and ^{allowed to} cooled at -200°C to form liquid air. ^{cd}⑥Fractional distillationThe liquid air is pumped into fractionating column where nitrogen at with lowest boiling point (-196°C) comes off and later on ^{further} warming Oxygen is collected at -183°C .⇒ Side effects :- Noise pollution from compressors - ^{proper} Use of appropriate PPE's
- Cold burning of oxygen cylinders⇒ Social benefits :- ~~for air~~ - For breathing by sea divers, astronauts
→ Employment - salary - ^{improved} ~~improved~~ ^{standard of living} ~~improved~~ - Manufacture of steel.
→ Govt revenue - ^{improved} ~~improved~~ ^{infrastructure} ~~improved~~ - Used in ~~acet~~ oxy-acetylene flames for cutting and welding of metals.

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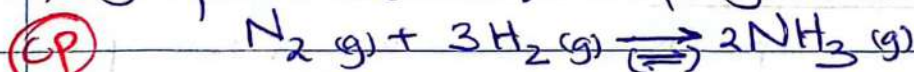
A. AMMONIUM FERTILIZERS

- ← Urea (H_2NCONH_2)
- ← Ammonium nitrate
- ← Ammonium sulphate
- ← Ammonium phosphate
- ← Calcium Ammonium Nitrate (CAN)

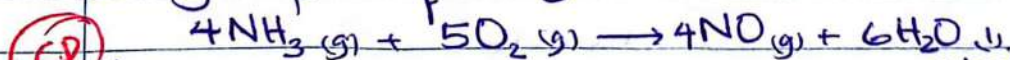
⇒ Raw materials; Nitrogen gas, Hydrogen gas.

⇒ Process of production:

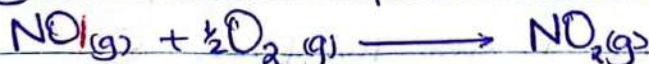
Nitrogen from fractional distillation of liquid air is reacted with hydrogen from natural gas in a ratio of 1:3 respectively to form ammonia by Haber process. The reaction requires low temperature (450-500°C), high pressure (200 atm) and finely divided Iron Catalyst.



Ammonia produced is heated in air (oxygen) in presence of platinum catalyst forming nitrogen monoxide and water. All these in a tank.

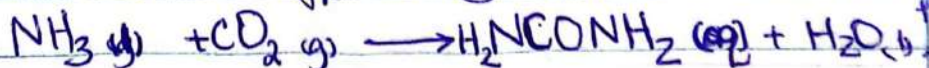


Nitrogen monoxide is further oxidised to nitrogen dioxide

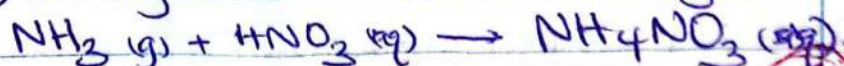


Nitrogen dioxide is dissolved in water in presence of oxygen, in a tank, forming nitric acid.

Urea; Obtained by reacting liquid ammonia gas with carbon dioxide at high pressure.



Ammonium Nitrate; By reacting nitric acid with ammonia gas.



Ammonium Sulphate;
$$\text{NH}_3 (\text{g}) + \text{H}_2\text{SO}_4 (\text{l}) \rightarrow (\text{NH}_4)_2\text{SO}_4 (\text{aq})$$
 (Cd)

The fertilizers are further concentrated and converted to solid form.

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⇒ Side effects + Mitigation:

- Runoff into water bodies promoting ^{increased} algae growth thus oxygen supply is cut
- Some when dissolved in water form acidic solution that alter soil pH ^{off leading to suffocation of aquatic animals} hence low crop production.

⇒ Social benefits:

- Source of employment as residents get salaries hence improved standards of living.
- High levels of nutrients supplied ^{leading} by ~~to~~ to improved crop production hence better standards of living ^{to farmers}
- Some when dissolved in water form acidic solutions that alter soil pH, hence low crop pro

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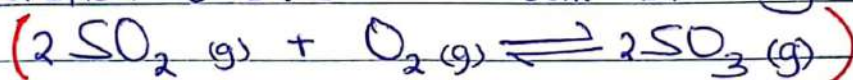
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Do not
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marginMANUFACTURE OF SULPHURIC ACID

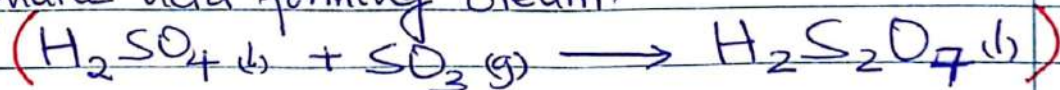
⇒ Raw materials ; - Sulphur dioxide gas.
- Oxygen gas.

⇒ Process of production;

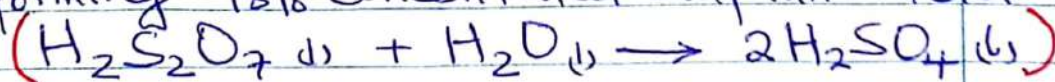
Dry Sulphur dioxide free from impurities is heated with dry pure Oxygen gas at ^{low} temperature (of about 450-500°C), high pressure (of about 1-3 atmospheres) in presence of vanadium(V) oxide catalyst to form sulphur trioxide. The reaction occurs in a combustion cylinder.



(CP) Sulphur trioxide is dissolved in with little concentrated Sulphuric acid forming oleum.



(CD) Oleum is added to moderated ^{volumes} ~~amounts~~ of distilled water forming 98% concentrated Sulphuric acid.



⇒ Side effects and mitigation

- Toxic misty fumes from oleum ^{which} ~~are~~ when inhaled may cause breathing problems and many side effects.
- Acid spills on the floor/surfaces leading to accidents

⇒ Social benefits

- Employment $\equiv$ salaries $\equiv$ improved standards of living
- ~~Use~~ Manufacture of fertilizer $\equiv$ increased farm yields $\equiv$ improved standards
- used in car batteries as electrolyte - - -

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MANUFACTURE OF CEMENT.

⇒ Raw materials: Limestone, clay, gypsum.

⇒ Process of production:

A mixture of Limestone and clay is crushed and milled into a fine powder.

The fine powder is then mixed with water and allowed to flow down a rotating drum (cylinder) in which it is strongly heated to about 1500°C.

Limestone decomposes to calcium oxide and carbon dioxide

$$(\text{CaCO}_3(s) \longrightarrow \text{CaO}(s) + \text{CO}_2(g))$$

(CP) Calcium oxide reacts with aluminium oxide and silicon dioxide in clay form lumps of calcium aluminate and calcium silicate

(PF) The lumps are crushed to form cement as a fine powder
 Gypsum is added during the grinding process to moderate the reaction between cement and water (setting of cement)
 Cement is packed in bags ready for use.

⇒ Side effects and mitigation

- Dust, noise during crushing
- Poisonous fumes during crushing
- Falling objects
- Hot surface burns
- Slips and falls during lifting, overloading

} use of
Appropriate PPE's

⇒ Social benefits.

- Employment opportunity to residents to improve their standard of living.
- Revenue to government of Uganda to increase improve on infrastructure

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MANUFACTURE OF ETHANOL

⇒ Raw materials ; - Starch containing material eg Maize, Cassava, sweet
or glucose
millet, Sorghum, Potato
- Malt, Yeast.

⇒ Process of production.

The starch containing material is crushed and roasted (heated) to extract starch. Malt is then added to starch in a container and it is then covered. Malt contains an enzyme, diastase that catalyses hydrolysis of starch to maltose.

(CP) Yeast is then added to maltose after about 3 days at room temperature. Maltase in yeast catalyses the hydrolysis of maltose to glucose.

(Cd) Zymase enzyme in yeast catalyses the decomposition of glucose to ^{Crude} ethanol and carbon dioxide.

Crude ethanol is converted to pure ethanol by fractional distillation. (Pr)

⇒ Side effects and mitigation

- Ethanol spills on surface leading to falls and accidents.
 - Hot surface burns during distillation. — Appropriate PPEs
- (Add more points)

⇒ Social benefits.

- Employment opportunity, salary hence improved standards of living.
 - Ethanol is used as a fuel in motor engines.
 - As a solvent for some paints.
 - Manufacture of drugs.
 - Manufacture of perfumes.
- build the point from there.

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MANUFACTURE OF DETERGENTS

(A) NON-SOAPY / SOAPLESS DETERGENTS.

From an alcohol;

→ Raw materials; Duodecan-1-ol, Concentrated sulphuric acid, Sodium hydroxide solution.

Process of production;

Duodecan-1-ol is reacted with cold concentrated sulphuric acid in a plastic ^(v) container forming duodecyl hydrogen sulphate which is reacted with sodium hydroxide solution to form a detergent. This mixture can be added to minimum volumes of water to form liquid detergent.

Some additives such as whitening agents, biological enzymes, fragrance, fillers, stabilizers may be added during the process.

Side effects and mitigation.

- Acid spills on surfaces
- Burns from acids when in contact with skin.
-

Social benefits.

- Production of cleansing agents that one can use to clean surfaces to improve on hygiene hence improved health living.
- Employment ⇒ Salaries ⇒ improved standards of living.

(B) SOAPY DETERGENTS

→ Take them through Saponification.

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MANUFACTURE OF BIO-GAS.

⇒ Raw materials; Organic waste, Water.

⇒ Process of production

Animal or plant (organic) wastes are put in a Container and mixed with some little water.

(CP) The container is covered to prevent aerial oxidation.

The container and contents are maintained at a temperature between 25-30°C.

(Cd) Anaerobic bacteria breakdown the organic matter to finally methane, ammonia, hydrogen sulphide, Carbon dioxide and nitrogen gases.

⇒ Side effects and mitigation

- Fire outbreaks during leakages on pipes

- Poisonous fumes in case of any leakages on pipes

By sealing off all pipes.
Proper use of PPE's

Maintain taps closed.

⇒ Social benefits

- Biogas is used for cooking and lighting since it is a renewable gas.

- Bio-slurry is used as an organic fertilizer and animal feeds to increase farmer's farm production hence better standards of living.

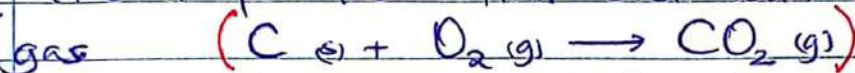
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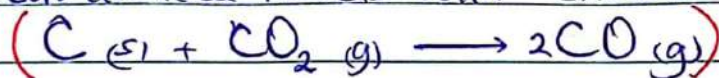
Subject 100

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marginEXTRACTION OF IRON⇒ Raw materials: Ore (Haematite), Limestone, Coke.⇒ Process of production:The raw materials, i.e. haematite, limestone and coke are fed into a blast furnace.Bottom
part
of furnaceStage 1: Oxidation of Coke.Hot air from the bottom ^(oxidises) reacts with coke forming carbon dioxide gasMiddle
part
of furnaceStage 2: Reduction of Carbon dioxide

Carbon dioxide reacts with excess coke forming carbon monoxide

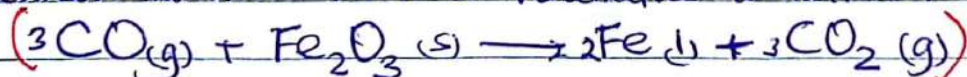


(cp)

Stage 3: Reduction of Iron(III) oxide (haematite)

(cd)

Carbon monoxide reduces haematite to molten iron and carbon dioxide

Role of Limestone

Calcium carbonate decomposes to calcium oxide and carbon dioxide. The calcium oxide reacts with silicon dioxide and aluminium oxide impurities forming calcium silicate and calcium aluminate that are tapped off.

(Pr)

Purification: Pure iron (wrought iron) is obtained by passing air through molten iron to remove non-metal impurities.⇒ Side effects + Mitigation

- Poisonous fumes

- Proper use of PPE

- Excessive noise from the plant

- PPE

- Too much heat from furnace to workers

- Chemical spills on molten iron

⇒ Social benefits

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EXTRACTION OF COPPER.

⇒ Raw materials; Ore (Copper pyrites)
 Silicon dioxide

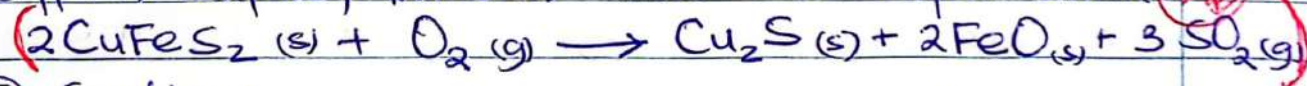
⇒ Production process;

① Concentrating the ore by froth flotation.

The ore is crushed and ^{mixed} ~~mixture~~ with water containing a frothing agent such as palm oil in a container. Air is blown in the mixture to agitate as ore particles go to the surface ^{with oil} while impurities sink down with water. The froth is skimmed off and dilute sulphuric acid is added to break the froth. The froth is filtered off and dried.

② Roasting.

The dried ore is roasted in air in a ^{blast} ~~furnace~~ [✓] to form Copper(I) sulphide, Iron(II) oxide and Sulphur dioxide ^{cd}.



③ Smelting

^{cd} Silicon dioxide is added to the above resultant mixture ^{in absence} ~~in~~ of ^{air} ~~extra~~ all in a furnace to remove Iron(II) oxide.

④ Reduction;

The copper(I) sulphide is heated in controlled amount of air to form Impure copper. ^{cd}

Purification; Impure copper is purified by electrolysis using
^{Pr} Impure copper as anode and pure copper as cathode
 using acidified Copper(II) sulphate solution electrolyte

⇒ Side effects and Mitigation

- Toxic fumes from production process can lead to suffocation and death
- Sulphur dioxide produced as a by-product reacts with water forming acidic rains that affect walls of houses

⇒ Social benefit

- Employment ^{opportunities} - salary - Improved ^{standards of living}

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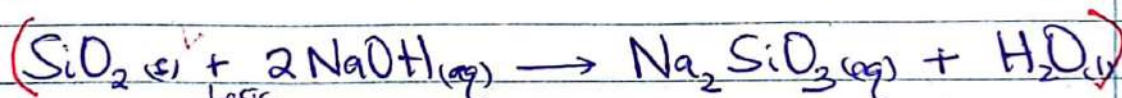
Do not
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⇒ Raw materials; Ore (Bauxite), sodium hydroxide

⇒ Process of production;
Concentrating the ore

Bauxite is ground to powder and heated to convert any Iron(II) oxide impurity present to Iron(III) oxide and also to remove water of crystallisation.

The powder is then boiled with hot concentrated sodium hydroxide solution that dissolves the amphoteric aluminium oxide and acidic silicon dioxide (✓)

(CP) in the ore forming sodium aluminate and sodium silicate respectively (✓) $(\text{Al}_2\text{O}_3(s) + 2\text{NaOH}(aq) \rightarrow 2\text{NaAlO}_2(aq) + \text{H}_2\text{O}(l))$ - The undissolved ^{basic} Iron(III) oxide and titanium(IV) oxide are filtered off

- Carbon dioxide is bubbled through the filtrate to precipitate aluminium hydroxide leaving silicate ions in the solution.

(Cd) - Aluminium hydroxide is washed, dried and heated strongly to produce aluminium oxide.

Electrolysis of aluminium oxide

Aluminium oxide is dissolved in molten cryolite to lower its melting point to 800°C. Electrolysis of molten aluminium oxide occurs using graphite electrodes in an Iron bath lined with graphite.

(Pr) At Cathode; $(\text{Al}^{3+}_{(l)} + 3e \rightarrow \text{Al}_{(l)})$ (At Anode $2\text{O}^{2-}_{(l)} \rightarrow \text{O}_{2(g)} + 4e$)⇒ Side effects and mitigation

- Pollution from poisonous fumes

- Burns caused by contact with hot surfaces of

⇒ Social benefits.

- chemical spills

- Employment opportunities to residents hence improved standards of living

- Source of revenue to government hence improved infrastructures

EXTRACTION OF IRON

⇒ Raw materials: Ore (Haematite), Limestone, Coke.

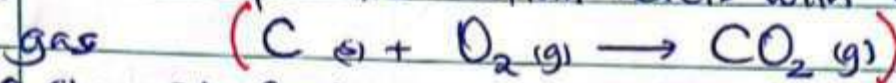
⇒ Process of production:

The raw materials, i.e. haematite, limestone and coke are fed into a blast furnace.

Bottom part of furnace

Stage 1: Oxidation of coke.

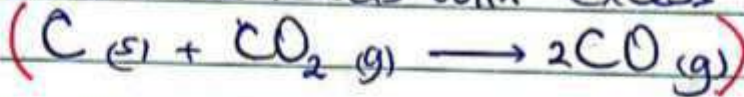
Hot air from the bottom reacts with coke forming carbon dioxide gas (oxidises)



Middle part of furnace

Stage 2: Reduction of carbon dioxide

Carbon dioxide reacts with excess coke forming carbon monoxide

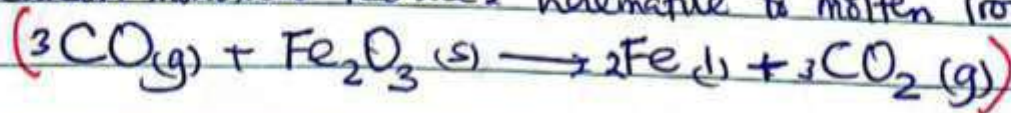


cp

Stage 3: Reduction of Iron(III) oxide (haematite)

cd

Carbon monoxide reduces haematite to molten iron and carbon dioxide



Role of limestone

Calcium carbonate decomposes to calcium oxide and carbon dioxide. The calcium oxide reacts with carbon dioxide and aluminium oxide impurities forming calcium silicate and calcium aluminate that are tapped off.

Pr

Purification: Pure iron (wrought iron) is obtained by passing air through molten iron to remove non-metal impurities.

⇒ Side effects + Mitigation

- Poisonous fumes

- Proper use of PPE

- Excessive noise from the plant

- PPE

- Too much heat from furnace to workers

- Chemical spills of molten slag

⇒ Social benefits

EXTRACTION OF ZINC

⇒ Raw materials; Ore (Zinc blende), Coke

⇒ Process of production;

Concentration of the ore by froth flotation.

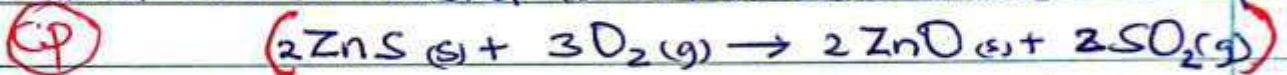
The ore is crushed into powder and mixed with water containing a frothing agent such as palm oil in a tank. Containers.

Compressed air is blown through the mixture to agitate it, making air bubbles rise to the surface with froth. The impurities sink at the bottom since they are wetted by water and ore float on surface since they are wetted by oil.

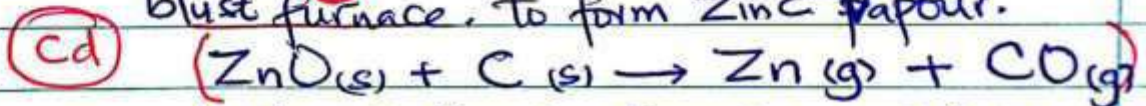
The froth is skimmed off, and dilute sulphuric acid is added to it to break the froth. It is filtered off and dried.

Roasting

The dried ore is roasted to obtain zinc oxide and sulphur dioxide.



Reduction; Zinc oxide is reduced by heating with coke in a blast furnace. to form zinc vapour.



The zinc vapour distils off from the furnace, cooled by a spray of molten lead.

(Pr) Purification; This is by Redistillation.

⇒ Side effects and Mitigation — Proper use of PPE

- Poisonous fumes that are toxic to man leading to deaths.
- Excessive heat and noise from rotating cylinders — Proper use of PPE

⇒ Social benefits

- Employment opportunity to the residents hence improving their standard of living.
- Source of government revenue to improve infrastructure in the region.

that are ornaments Product formed is used in manufacture of alloys such as Brass.

EXTRACTION OF COPPER.

⇒ Raw materials; Ore (Copper pyrites)
Silicon dioxide

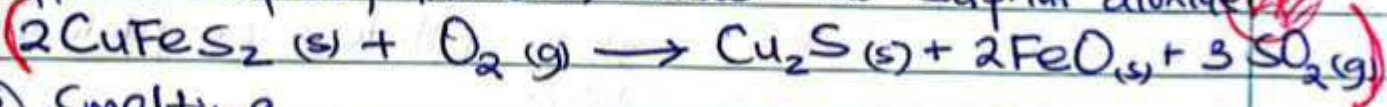
⇒ Production process;

① Concentrating the ore by froth flotation.

The ore is crushed and ^{mixed} ~~mixture~~ with water containing a frothing agent such as palm oil in a container. Air is blown in the mixture to agitate as ore particles go to the surface ^{with oil} while impurities sink down with water. The froth is skimmed off and dilute sulphuric acid is added to break the froth. The froth is filtered off and dried.

② Roasting.

The dried ore is roasted in air in a ^{blast} furnace to form Copper(I) sulphide, Iron(II) oxide and Sulphur dioxide. (cd) ✓



③ Smelting

^{done in a closed} Silicon The above resultant mixture ^{in absence} ~~is re~~ of ~~extra~~ air in a furnace to remove Iron(II) oxide.

④ Reduction;

The copper(I) sulphide is heated in controlled amount of air to form Impure copper. (cd)

Purification; Impure copper is purified by electrolysis using
(P) } Impure copper as anode and pure copper as cathode
using acidified Copper(II) sulphate solution electrolyte

⇒ Side effects and Mitigation

- Toxic fumes from production process can lead to suffocation and death
- Sulphur dioxide produced as a by-product reacts with water forming acidic rains that affect walls of houses

→ Proper control of waste
Use of PPE

⇒ Social benefit

- Employment opportunity - salary - Improved standards of living

EXTRACTION OF SODIUM

⇒ Raw materials ; - Sodium chloride (Rock salt)
- Calcium chloride.

⇒ Production process.

Sodium is extracted by electrolysis of molten sodium chloride using Iron as cathode and graphite as anode in a DOWN'S CELL ✓

(CP) During the process, calcium chloride is added to sodium chloride to lower its melting point from 800°C to 600°C

At the Cathode: $\text{Na}^+ + e \longrightarrow \text{Na} \downarrow$ (cd)

(Pr) The sodium ^{discharge} is tapped off and collected into iron storage tanks containing dry nitrogen.

⇒ Side effects and mitigation

- Poisonous fumes that if inhaled may cause death (chlorine by-products) - Proper use of PPE
- Burns caused ~~with~~ ^{by} contact with hot surfaces on ~~fin~~ ^{Down cell} furnace. - Proper use of PPE

⇒ Social benefits

- Employment opportunities to residents hence improving their standards of living.
- Source of revenue to government hence improving infrastructure.
- Sodium produced is a coolant in nuclear weapons.

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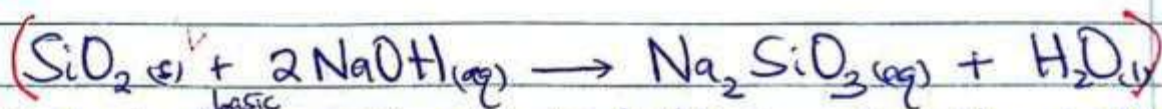
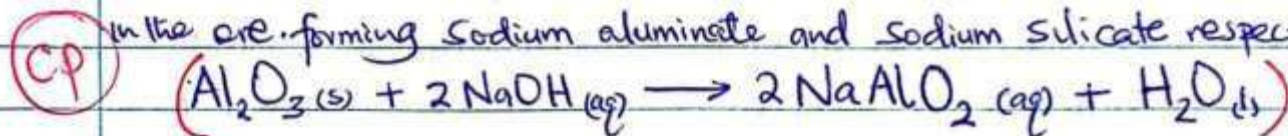
EXTRACTION OF ALUMINIUM.

⇒ Raw materials; Ore (Bauxite), Sodium hydroxide

⇒ Process of production;
Concentrating the Ore

Bauxite is ground to powder and heated to convert any Iron(II) oxide impurity present to Iron(III) oxide and also to remove water of crystallisation.

The powder is then boiled with hot concentrated sodium hydroxide solution that dissolves the amphoteric aluminium oxide and acidic silicon dioxide (✓) in the ore forming sodium aluminate and sodium silicate respectively (✓) in a container.



- The Undissolved ^{basic} Iron(III) oxide and titanium(IV) oxide are filtered off.
- Carbon dioxide is bubbled through the filtrate to precipitate aluminium hydroxide leaving silicate ions in the solution.

(Cd) - Aluminium hydroxide is washed, dried and heated strongly to produce aluminium oxide.

Electrolysis of aluminium oxide

Aluminium oxide is dissolved in molten cryolite to lower its melting point to 800°C. Electrolysis of molten aluminium oxide occurs using graphite electrodes in an Iron bath lined with graphite.

At Cathode; $\text{Al}^{3+}(l) + 3e \rightarrow \text{Al}(l)$ (At Anode $2\text{O}^{2-}(l) \rightarrow \text{O}_2(g) + 4e$)⇒ Side effects and mitigation

- Pollution from poisonous fumes
- Burns caused by contact with hot surfaces of
- chemical spills

Proper use of PPEs

Proper use of PPE

⇒ Social benefits.

- Employment opportunities to residents hence improved standards of living
- Source of revenue to government hence improved infrastructure

- (c) **Effective Nuclear Charge (Z_{eff})** refers to the net positive charge experienced by an atom's outermost electrons due to the nucleus's attraction, considering the shielding effect of inner electrons.

Factors Affecting effective nuclear charge

1. **Nuclear Charge (Z)** – The total number of protons in the nucleus increases attraction.
2. **Electron Shielding** – Inner electrons repel outer electrons, reducing nuclear attraction.
3. **Atomic Size** – Smaller atoms experience stronger nuclear pull on valence electrons.

Trend of effective nuclear charge in the Periodic Table:

- **Across a Period:** Effective nuclear charge increases because more protons are added while shielding remains nearly constant.
- **Down a Group:** effective nuclear charge decreases slightly due to increased shielding from additional electron shells.

- (d) **Atomic radius** is the distance from the center of the nucleus to the furthest electron under the influence of a nuclear charge of an atom.

Trend of atomic radii in the periodic Table

- (i) **Atomic Radius decreases across a Period (Left to Right)**

Reason: Increasing nuclear charge pulls electrons closer to the nucleus, reducing the atomic size. Example: Sodium (Na) has a larger atomic radius than chlorine (Cl) in Period 3.

- (ii) **Atomic Radius increases down a Group (Top to Bottom)**

Reason: Additional electron shells (energy levels) increase the atomic size, even though nuclear charge also rises. Example: Lithium (Li) is smaller than cesium (Cs) in Group 1

- (e) **Metallic character** refers to how easily an element can lose electrons and exhibit typical metallic properties such as conductivity, malleability, and luster. It is influenced by an element's ability to form **positive ions (cations)** and its interaction with electrons.

Trend in the Periodic Table:

- (i) **Metallic character increases down a group ↓** – As atomic size increases, outer electrons are held less tightly, making electron loss easier.
- (ii) **Decreases across a period ➡** – As nuclear charge increases, electrons are more strongly attracted, reducing metallic behavior.

Properties of High Metallic Character:

- **Good electrical and thermal conductivity**
- **High malleability and ductility**
- **Low ionization energy** (easy loss of electrons).
- **Shiny appearance (lustrous)**

Examples:

- **High metallic character:** Cesium (Cs), Francium (Fr), Iron (Fe).
- **Low metallic character:** Nonmetals like Oxygen (O) and Sulfur (S).

Melting points

The melting point of a substance or element is the measure of energy required to break down the regular arrangement of atoms, ions or molecules in the substance (solid state). Therefore, the melting point indicates the strength of the forces knitting atoms, ions or molecules together in the crystal. The strength of those forces in elements varies according to the type of crystals formed and due to this; the melting points of elements in a period do not change uniformly,

(a) Across a Period (Left to Right)

- **Metals (Groups 1- 3)** generally have **high melting and boiling points** due to strong **metallic bonding**.
- **Group 4 elements** (like carbon and silicon) have the **highest melting and boiling points** due to strong **covalent networks**.
- **Group 8 (Noble Gases)** have **low melting and boiling points** because they exist as individual atoms with weak Van der Waals forces.

(b) Down a Group (Top to Bottom) ↓

- **Metals (e.g., Alkali & Alkaline Earth Metals)** experience a **decrease** in melting and boiling points because metallic bonds weaken as atomic size increases.
- **Nonmetals (e.g., Halogens)** experience an **increase** in melting and boiling points due to stronger intermolecular forces in larger molecules.

Chemistry of period 3 elements

Each element in a period provides representative Chemistry of the group into which it exists.

The elements of period 3 are Na, Mg, Al, Si, P, S, Cl and Ar.

Electronic structure

The table below shows electron configuration of period 3 elements

Elements	Na	Mg	Al	Si	P	S	Cl	Ar
Electron configuration	2.8.1	2.8.2	2.8.3	2.8.4	2.8.5	2.8.6	2.8.7	2.8.8

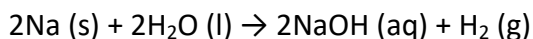
Physical properties of period 3 element

Sodium, magnesium and aluminium are metals whose melting points increase in that order due to increase in the number of electrons contributed to form metallic bonds. Silicon, phosphorus, sulphur, chlorine and argon are non-metals. Silicon, phosphorus and sulphur are solids while chlorine and argon are gases at room temperature.

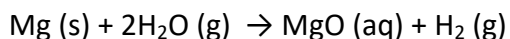
Reaction period 3 elements with water

Reactivity with water decreases across the period because there is a decrease in metallic character of these elements across the period.

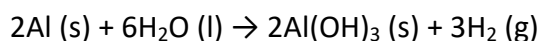
- i) Sodium reacts vigorously with cold water to form sodium hydroxide and hydrogen gas.



- ii) Magnesium does not react with cold water but reacts with steam to form magnesium oxide and hydrogen gas.



- iii) Aluminium reacts with steam very slowly to form aluminium hydroxide and hydrogen gas.



- iv) Silicon, phosphorus, sulphur and argon have no reaction with water.

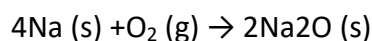
- v) Chlorine reacts with water to form chloric (I) acid and hydrochloric acid.



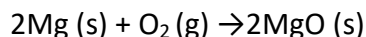
Reaction period 3 elements with oxygen

Period 3 elements (sodium to argon) react differently with oxygen, forming oxides with varying properties.

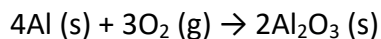
- (i) **Sodium (Na)** reacts violently with a yellow flame to form sodium oxide



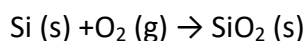
- (ii) **Magnesium (Mg)** Burns in oxygen with a **bright white flame** to form white powder of magnesium oxide.



- (iii) **Aluminum (Al)** forms aluminum oxide, which has a protective layer preventing further reaction:

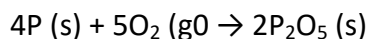


- (iv) **Silicon (Si)** reacts at high temperatures to form silicon dioxide:

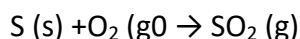


Silicon dioxide is a major component of sand.

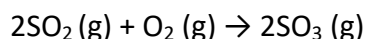
- (v) **Phosphorus (P)** burns with a **bright white flame** to form phosphorus pentoxide when burned in excess oxygen:



- (vi) **Sulfur (S)** burn in oxygen with a blue flame to form sulphur dioxide



In excess oxygen, sulfur trioxide may also form:



- (vii) **Chlorine (Cl₂)** does not react directly with oxygen under normal conditions.

- (viii) **Argon (Ar)** is **inert** and does not react with oxygen.

The table below shows the properties of oxides of period elements

ELEMENTS	Na	Mg	Al	Si	P	S	Cl
OXIDES	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₂ O ₅	SO ₃	Cl ₂ O ₇
PRINCIPAL BONDING	Ionic	Ionic	Ionic-covalent	Covalent	Covalent	Covalent	Covalent
CHARACTER	Basic	Basic	Amphoteric	Acidic	Acidic	Acidic	Acidic
MP\°C	1193	3075	2300	1728	563	30	-91

Physical properties of oxides of period 3 elements

i) Across the period there is a gradual change in character of oxides from strongly basic to strongly acidic. Al_2O_3 is amphoteric oxide, i.e., possesses both basic and acidic properties.

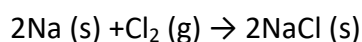
ii) Na_2O , MgO and Al_2O_3 have high melting points due to the strength of ionic bonding. SiO_2 , although covalent, has a high melting point because the crystal of SiO_2 is a giant structure in which silicon and oxygen atoms are bonded together by strong single covalent bonds.

P_2O_5 (white solid) consists of P_4O_{10} molecules; these contain electric dipoles due to electron displacements in their bonds and are quite strongly attracted to each other. This is why its melting point is high compared with that of SO_3 which consists of simple SO_3 molecules.

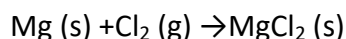
Cl_2O_7 has a low melting point because the molecules are held together by the weak van der Waals forces.

Reaction of Period 3 elements with chlorine

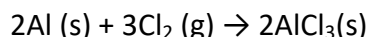
(i) **Sodium (Na)** reacts **vigorously** with chlorine, forming white solid of **sodium chloride**:



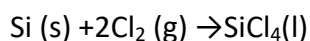
(ii) **Magnesium (Mg)** burns in chlorine to form white crystals of **magnesium chloride**:



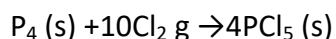
(iii) **Aluminum (Al)** reacts with chlorine gas at high temperatures to form **aluminum chloride**:



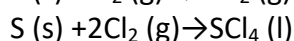
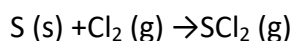
(iv) **Silicon (Si)** forms a liquid **silicon tetrachloride**:



(v) **Phosphorus (P)** reacts with excess chlorine to form a yellow crystalline solid of **phosphorus pentachloride**:



(vi) **Sulfur (S)** reacts with chlorine to form **sulfur dichloride or redish-brown liquid of sulfur tetrachloride**:



(vii) **Chlorine (Cl₂)** No reaction since it is already chlorine.

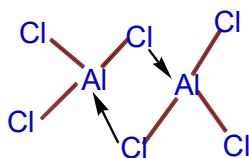
(viii) **Argon (Ar)** Does not react (inert noble gas).

The table below provides properties of chlorides of period 3 elements

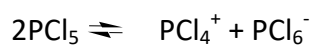
ELEMENTS	Na	Mg	Al	Si	P	S	Cl
CHLORIDES	NaCl	MgCl ₂	AlCl ₃	SiCl ₄	PCl ₅	S ₂ Cl ₂	Cl ₂
BONDING	Ionic	Ionic	Covalent	Covalent	Covalent	Covalent	Covalent
STATES	Solid	Solid	Solid	Liquid	Solid	Liquid	Gas
MP\°C	808	714	192	-68	160	-76	

Physical properties of period 3 chlorides.

NaCl and MgCl₂ have high melting points due to the strength of the ionic bonding. AlCl₃ has a fairly high melting point because in the solid state, it consists of Al₂Cl₆ molecules and not simple AlCl₃. These molecules are produced through dative bonding between Al and Cl in the Al₂Cl₆ molecules.



SiCl₄ and S₂Cl₂ are liquids and consist of respective simple SiCl₄ and S₂Cl₂ molecules and in the solid state the molecules are held by weak van der Waals force which explains their very low melting points. PCl₅ a pale yellow solid has a fairly high melting point because the solid undergoes partial ionisation.



Group 1: The alkali metals

The elements of group 1, lithium, sodium, potassium, rubidium, caesium and francium are called the **alkali metals**. The alkali metals are soft and silvery coloured: They are good-conductors of heat and electricity. Because they rapidly tarnish in air, they are generally stored under oil.

Some physical properties of group 1A elements are given in the table 8.1 below:

Table 8.1 Some physical properties of alkaline metals

Element s	Atomic number	Electronic configuration	M.P. / $^{\circ}\text{C}$	B.P. / $^{\circ}\text{C}$	Atomic radius (nm)
Li	3	2.1	180	1326	0.133
Na	11	2.8.1	98	883	0.157
K	19	2.8.8.1	64	756	0.203
Rb	37	2.8.18.8.1	39	688	0.216
Cs	55	2.8.18.18.8.1	29	690	0.235

Trend of physical properties of group 1 elements

Hardness of group 1 elements decreases down the group due to decrease in strength of metallic bonds as atomic size increases

Density of group 1 elements generally increases down the group (except for potassium) due to increase atomic masses

Melting points of group 1 elements decrease down the group as atomic size increases. Larger atoms have **weaker metallic bonds**, making them easier to melt.

Chemical properties of group 1 elements

The alkali metals are extremely reactive and electropositive metals. In chemical combinations, this single electron is very readily transferred, giving a metal ion of single positive charge with a stable electronic configuration of a noble gas. Compounds of the alkali metals are therefore ionic and exist as high melting point solids. Li^+ due to its smallest size, polarises anions most strongly and thus its compounds are highly covalent which is why they are soluble in organic solvents.

All the alkali metal compounds give characteristic flame colourations that can be used to identify them e.g. lithium (red), sodium (yellow), potassium (lilac), rubidium (red) and caesium (blue).

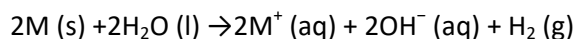
Reaction of group 1 elements with oxygen

Element	Reaction with Oxygen	Equation
Lithium (Li)	Burns red to form lithium oxide	$4\text{Li (s)} + \text{O}_2\text{(g)} \rightarrow 2\text{Li}_2\text{O (s)}$
Sodium (Na)	Burns yellow to form sodium peroxide	$2\text{Na (s)} + \text{O}_2\text{(g)} \rightarrow \text{Na}_2\text{O}_2$
Potassium (K)	Burns purple to form potassium superoxide	$\text{K (s)} + \text{O}_2\text{(g)} \rightarrow \text{KO}_2\text{(s)}$
Rubidium (Rb)	Reacts violently to form rubidium superoxide	$\text{Rb (s)} + \text{O}_2\text{(g)} \rightarrow \text{RbO}_2\text{(s)}$
Cesium (Cs)	Reacts explosively to form cesium superoxide	$\text{Cs (s)} + \text{O}_2\text{(g)} \rightarrow \text{CsO}_2\text{(s)}$

Reaction of group 1 elements with water

Generally group 1 elements react with water to form hydroxides and hydrogen

General Reaction



(where **M** represents the alkali metal)

Specific Reactions

Element	Reaction with Water	Observations
Lithium (Li)	Reacts gently , producing lithium hydroxide (LiOH) and hydrogen (H ₂)	Bubbles form; slow reaction
Sodium (Na)	Reacts quickly , forming sodium hydroxide (NaOH) and hydrogen gas	Fizzes, moves on water surface, melts
Potassium (K)	Reacts violently , forming potassium hydroxide (KOH) and hydrogen gas	Ignites with a purple flame
Rubidium (Rb)	Reacts explosively , forming rubidium hydroxide (RbOH) and hydrogen gas	Very rapid, sparks may appear
Cesium (Cs)	Reacts extremely explosively , forming cesium hydroxide (CsOH) and hydrogen gas	Instant explosion, shatters container

Revision questions

1. Explain why metallic character increases down any group of the Periodic Table.
2. Explain why element in the same group of the Periodic Table show similar chemical properties
3. Which side of the periodic table has non-metals?
4. Explain why the boiling points of halogens increase down the group
5. Describe the trend in reactivity of groups 1 and 7 on moving down the groups.

Theme: Thermochemistry**Topic 18: ENERGY CHANGES DURING CHEMICAL REACTIONS**

Energy changes normally occur every time chemical changes take place.

Energy is neither created nor destroyed but it can be transformed from one form to another.

For example

1. In electrochemical cells, chemical energy is transformed to electrical energy
2. Fuel like coal stores chemical energy which is transformed into heat energy when the fuel is burnt.
3. The chemical energy in food is transformed into heat energy during metabolism to facilitate body processes.

The most common form of energy change in chemical reactions is the **heat changes**. All chemical substances possess chemical energy stored in bonds that connect atoms in a compound.

The energy content (heat content) of a substance which is stored in its bonds is referred to as **Enthalpy**

Also, energy has to be supplied if bonds are to be broken.

The enthalpy of a substance is denoted by **H**.

Changes in enthalpy are denoted by **ΔH**

Enthalpy changes occur in a reaction when some old bonds in the reactants are broken and new bonds are formed in the products

ACTIVITY

You are provided with the following chemicals and apparatus

- Dilute hydrochloric acid (2M HCl)
- Zinc granules
- Calcium oxide
- Ammonium chloride crystals
- Sodium hydroxide pellets
- Distilled water
- Thermometer
- 4 Beakers labeled A, B, C and D
- Spatula
- 4 test tubes

Procedure

- Put **dilute hydrochloric acid** in a beaker labeled **A**.
- Put **distilled water** in a beaker labeled **B**
- Read and record the **initial temperatures** of both solutions
- In beaker **A**, put **zinc granules** using a spatula
- In beaker **B**, put **calcium oxide** using a spatula.
- Read and Record the **final temperatures** of both solutions
- Put 100cm³ of **distilled water** in beaker labeled **C**.
- Put 100cm³ of **distilled water** in beaker labeled **D**.
- In beaker **C**, put one spatula endful of **ammonium chloride crystals**
- In beakers **D**, put one spatula endful of **sodium hydroxide pellets** and stir
- Read and record the **final temperatures** of both solutions.
- Record your results in the table below.

<i>Substance</i>	<i>Initial temperature (°C)</i>	<i>Final temperature (°C)</i>
A		
B		
C		
D		

TASKS:

1. Which reactions produced heat?

2. Which reactions absorbed heat?

3. List two uses of heat in everyday life.

Reactions are categorized into two depending on whether there is release of heat to the surrounding or absorption of heat from the surrounding.

The reaction during which heat is liberated to the surrounding is called **exothermic reaction**.

The reaction during which heat is absorbed from the surrounding is called

Endothermic reaction.

EXOTHERMIC REACTIONS

An exothermic reaction is one during which heat is liberated to the surrounding.

During exothermic reaction, heat is given out and therefore, the temperature of the products rises above room temperature, but with time, it drops to room temperature as heat is lost to the surrounding

At the end of the reaction, ***the heat content of the products is less than that of the reactants.***

Therefore, the enthalpy (heat) change of an exothermic reaction is negative

$$\Delta H = \text{Heat content of products} - \text{Heat content of reactants.}$$

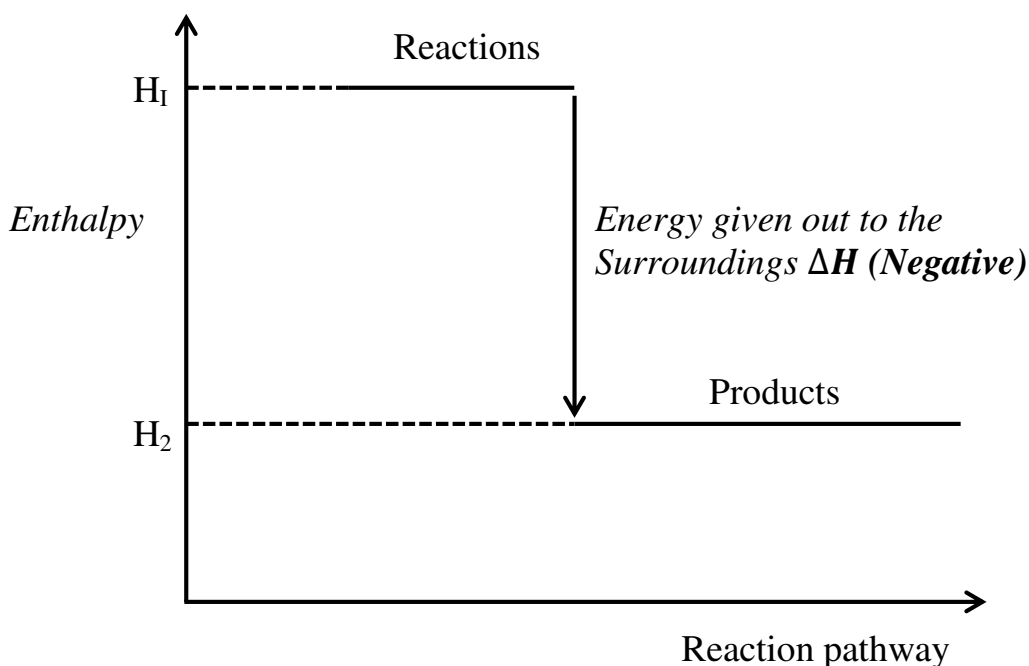
If: H_1 = Is the heat content of reactants

H_2 = Is the heat content of products

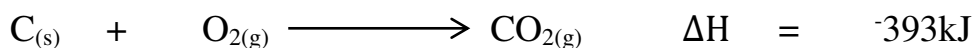
$$\text{Then; } \Delta H = H_1 - H_2$$

Since the heat content of products (H_2) is less than the heat content of reactants (H_1), that is to say $H_2 < H_1$, then it means that **ΔH is negative.**

This can be represented on an energy level diagram.

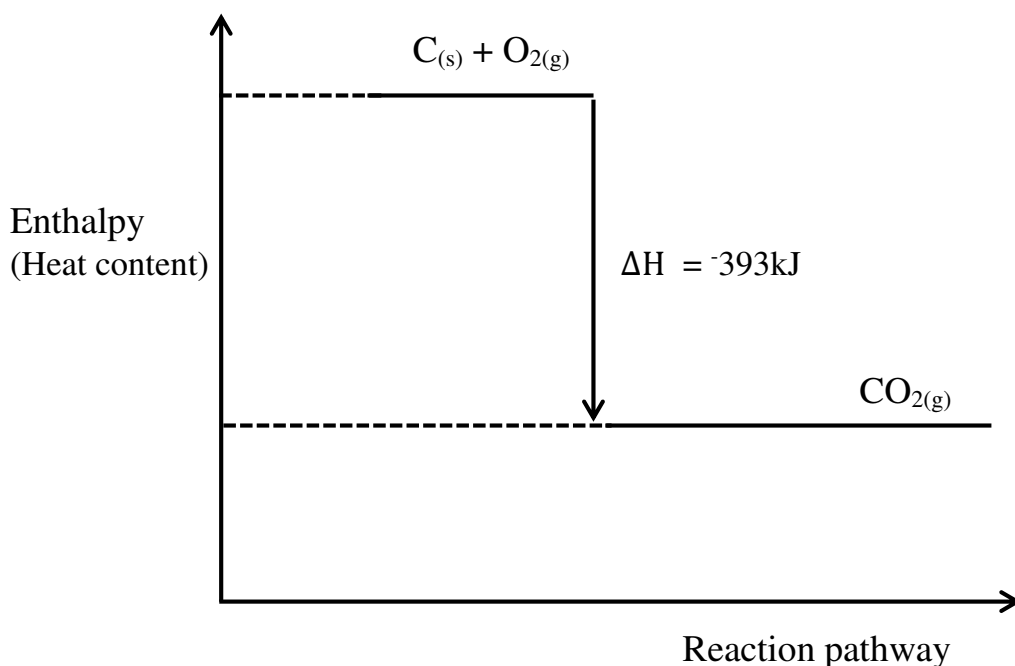


An example of exothermic reaction is when carbon reacts with oxygen. In such a reaction, heat is evolved



The chemical energy in carbon and oxygen is partly transferred to chemical energy in carbon dioxide and partly evolved as heat. This means that carbon dioxide has less energy than the starting material that is carbon and oxygen; thus the value of enthalpy change is negative.

Energy level diagram to represent the reaction.



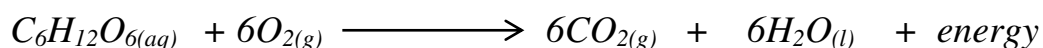
NOTE:

Bond making involves the *liberation of heat to the surroundings* implying that it is an *exothermic process*.

Examples of Exothermic Reactions or Processes In Everyday Life.

1) Cellular respiration

This is a vital exothermic reaction that occurs in living cells of the body. This process provides energy to the body cells to maintain the other vital processes of the body. During this process, glucose molecules are broken down in the body living cells into carbon dioxide, water molecules and energy is released.



The energy released during the break down of sugar is used in the synthesis of Adenosine triphosphate (ATP) which is the energy carrier molecule of the cells.

2) *Rusting of iron*

In the process, heat is released when **iron** comes into contact with **oxygen** and **water** to form Hydrated iron(III) oxide.

3) *Fermentation*

Fermentation is the process by which sugars (for example **glucose**) are decomposed by enzymes (for example **zymase**) produced by microorganism (for example **yeast**) forming **ethanol**, **carbon dioxide** and **energy**.

Fermentation can also mean incomplete oxidation of glucose in the absence of oxygen.

Fermentation is an exothermic reaction, meaning that the process releases energy in form of heat.

4) *Making an ice cube*

This is the process of a liquid changing its state to a solid. Changing water into ice cube is an exothermic reaction, since the water slowly loses heat and starts to cool down to form ice cubes.

5) *Combustion*

This refers to a chemical reaction in which a substance reacts with oxygen

In this kind of reaction, heat energy is released to the surrounding. For example, when wood is burnt in oxygen, large amount of heat energy is produced.

6) *Snow formation of clouds*

Clouds come into existence from the condensation of water vapour. This process is an exothermic reaction.

7) *Burning of candle*

The candle is a hydrocarbon which burns in oxygen to form carbon dioxide and water. This process evolves heat to the surrounding.

Other examples of exothermic reactions include:-

- Laundry detergent
- Decomposition of vegetables
- Neutralization reaction
- Nuclear fission
- Cold packs to heal an injury
- Reaction of water and calcium chloride
- Dissolving a strong acid in water.

ENDOTHERMIC REACTIONS

Endothermic reaction is one during which heat is absorbed from the surroundings.

When an endothermic reaction occurs, the heat required for the reaction is taken from the reacting materials and the temperature of the products falls below the initial temperature.

Eventually, the temperature of the products raises to room temperature again as heat is absorbed from the surrounding.

In this case, the heat content of the product is greater than that of the reactants and the enthalpy change is **positive**.

$$\Delta H = \text{Heat content of products} - \text{Heat content of reactants.}$$

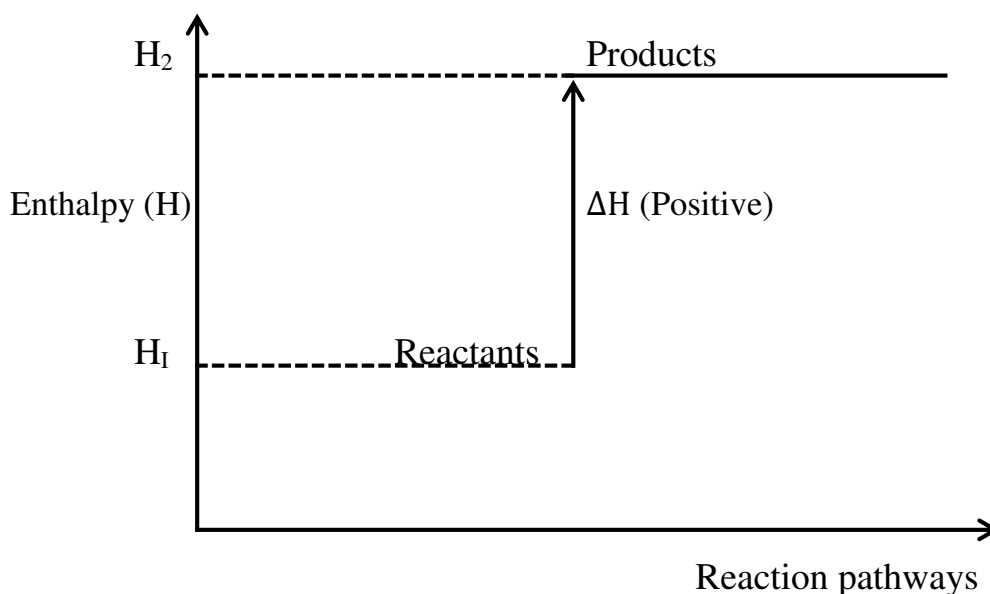
If: H_1 = Is the heat content of reactants

H_2 = Is the heat content of products

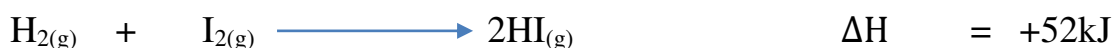
Then; $\Delta H = H_1 - H_2$

Since the heat content of products (H_2) is greater than the heat content of reactants (H_1), that is to say $H_2 > H_1$ then the ΔH is **positive**

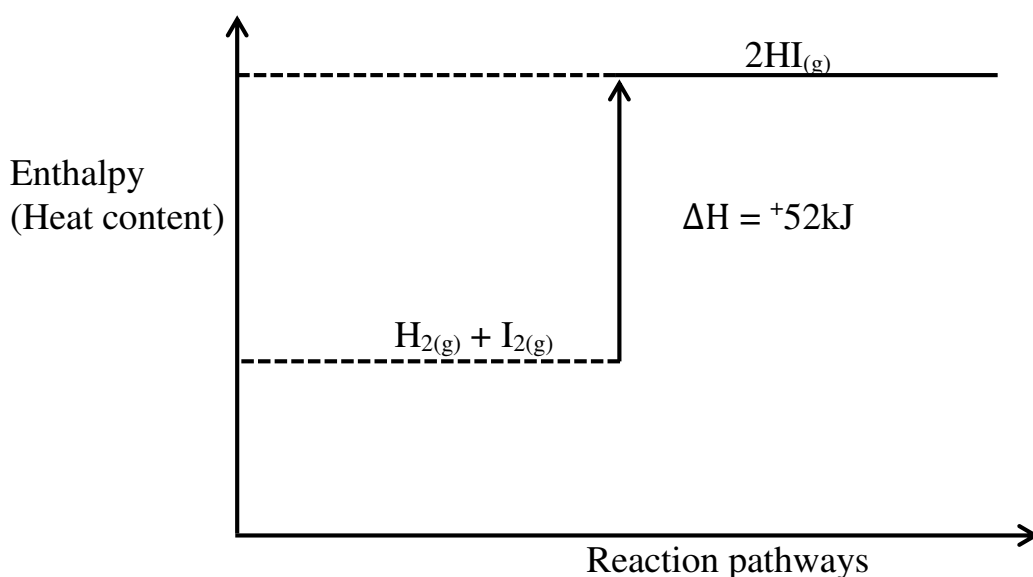
This can be represented on an energy level diagram



For example, when hydrogen reacts with iodine, heat is absorbed from the surroundings.



Hydrogen iodide has more energy than the starting materials, Hydrogen and iodine. Therefore the value of enthalpy change is positive.



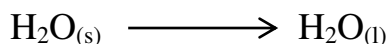
NOTE:

Bond breaking requires the *absorption of heat from the surroundings* implying that it is an **endothermic process**.

Examples of Endothermic Reactions or Processes in Everyday Life

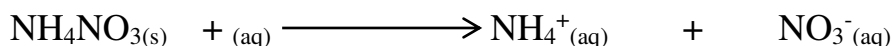
1) *Melting of ice*

Heat is absorbed to convert ice into liquid water



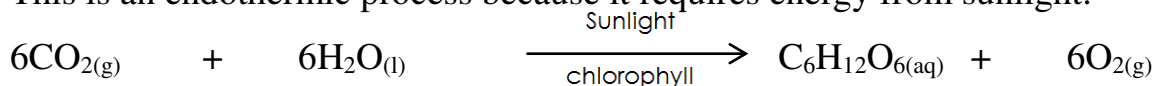
2) *Dissolving ammonium nitrate in water.*

This dissolution process absorbs heat from the surrounding.



3) *Photosynthesis*

This is an endothermic process because it requires energy from sunlight.



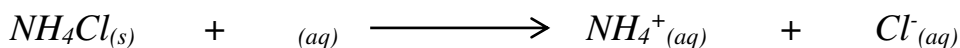
4) *Thermal decomposition of calcium carbonate.*

Heat is needed to break down calcium carbonate into calcium oxide and carbon dioxide



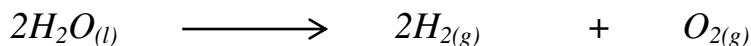
5. Dissolving ammonium chloride in water.

This process absorbs heat from the surrounding, thus it is an endothermic process.



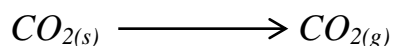
6. Electrolysis of water

This process requires an input of electrical energy, making it an endothermic process.



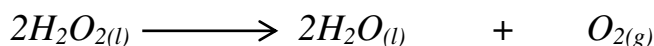
7. Sublimation of dry ice (Solid carbon dioxide).

Dry ice sublimates at a temperature below room temperature, absorbing heat.



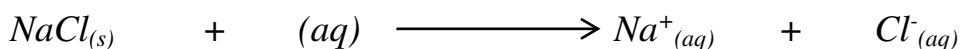
8. Thermal decomposition of hydrogen peroxide.

Breaking down hydrogen peroxide into water and oxygen requires intake of heat



9. Dissolving sodium chloride (common salt) in water.

This dissolution process absorbs heat energy from the surrounding, resulting into temperature fall.



10. Cooking an egg

The process of cooking is an endothermic reaction.

For example, during the process of frying eggs, heat energy from the hot frying pan is absorbing by the raw eggs, hence making the process to be endothermic.

Other examples of endothermic processes include:-

- Reacting ethanoic acid and sodium carbonate.
- Reacting baking soda (sodium bicarbonate) and vinegar (acetic acid)
- Melting of ice to water

Differences between Endothermic and exothermic Reactions

<i>Endothermic reactions</i>	<i>Exothermic reactions</i>
These are reactions during which heat is absorbed from the surroundings	These are reactions during which heat is liberated to the surroundings
Enthalpy changes of endothermic reactions are positive	Enthalpy changes of exothermic reactions are negative
Endothermic reactions result from bond breaking	Exothermic reactions results from bond making

Activity One

Exploring the reactions or processes in which energy is given out or absorbed.

a) Complete the table below by stating whether the reaction given is endothermic or exothermic.

<i>Reaction</i>	<i>Is energy absorbed?</i>	<i>Endothermic/ Exothermic?</i>
Fermentation		
Respiration		
Cooking an egg		
Burning salt in water		
Melting ice cubes		
Neutralization		
Decomposition		

b) State any other five reactions in which energy is

<i>i) absorbed</i>	<i>ii) given out</i>

Activity two**Exploring changes in temperature as different substances dissolve in water.**

What you are required to do:

- i) Label small plastic cups with
 - ✓ *Potassium chloride*
 - ✓ *Calcium chloride*
 - ✓ *Sodium carbonate*
 - ✓ *Sodium hydrogen carbonate*
 - ✓ *Ammonium nitrate*
 - ✓ *Sodium hydroxide*
- ii) Add **150cm³** of water into the cup labeled '*potassium chloride*' and measure its temperature using a thermometer. Record the temperature in the table as **initial temperature (T₀)**
- iii) Weigh **2g** of potassium chloride and add it to the cup in step (ii) above. Stir the mixture until the temperature stops changing and record the **final temperature (T_f)** in the table.
- iv) Repeat steps (ii) to (iii) using the cups labeled '*calcium chloride*', '*sodium carbonate*', '*sodium hydrogencarbonate*', '*ammonium nitrate*' and '*sodium hydroxide*'. Weigh and add the solutes to their corresponding cups.

Table: Temperature changes in some chemical reactions

Experiment	Solid	T₀ (°C)	T_f (°C)
1	Potassium chloride		
2	Calcium chloride		
3	Sodium carbonate		
4	Sodium hydrogen carbonate		
5	Ammonium nitrate		
6	Sodium hydroxide		

Observation and Analysis

1. Which solutes dissolved endothermically?

2. Which solutes dissolved exothermically?

3. Which solute dissolved most exothermically?

4. Which solute dissolved most endothermically?

Importance of Endothermic Reactions in Everyday Life

The significance of endothermic reactions in our everyday life stretch from *household applications* to *industrial processes* and *medical treatments*.

Endothermic reactions have significant importances in our everyday life, and these include the following:

1) Cooling systems:

Refrigeration and air conditioning rely on endothermic reactions to absorb heat and cool the surroundings

2) Food preparation:

Cooking and baking involve endothermic reactions to break down ingredients and create desired textures and flavors.

3) Energy storage:

Batteries and fuel cells use endothermic reactions to store energy for later use.

4) Medical applications:

Some medical treatments, for example cancer therapy, utilize endothermic reactions to target and destroy cancer cells.

5) Environmental remediation:

Endothermic reactions can help to clean pollutants from soil and water.

6) Material synthesis

Endothermic reactions are used to create materials like cement, glass and steel.

7) Water treatments

Endothermic reactions help to purify water by removing impurities and contaminants.

8) Pharmaceuticals:

Endothermic reactions are involved in the productions of some medication.

9) *Food preservation:*

Endothermic reactions help to preserve food by inactivating enzymes and microorganisms.

10) *Scientific research;*

Endothermic reactions are essential in various scientific experiments and investigations.

Importances of Exothermic Reactions in Everyday Life

The significance of exothermic reactions in our everyday life, stretch from *energy production* and *industrial processes* to *cooking* and *transportation*.

Exothermic reactions have significant importance in our everyday life, and these include the following;

1) *Energy production*

Combustion reactions, for example burning fossil fuels, provide energy for transportation, heating and electricity generation.

2) *Cooking and heating*

Exothermic reactions are used in cooking, heating of water and space heating

3) *Industrial process:*

Exothermic reactions are used in cement production, steel manufacturing and chemical synthesis.

4) *Transportation:*

Combustion reactions power internal combustion engines in cars, trucks and airplanes causing them to move.

5) *Food preparation:*

Exothermic reactions are used in baking, grilling and frying.

6) *Water treatments*

Exothermic reactions are used to disinfect water and break down organic matter.

7) *Medicine:*

Exothermic reactions are used in some medical treatments, like cancer therapy.

8) *Welding and cutting*

Exothermic reactions are used in welding and cutting of metals.

9) *Catalytic converters*

Exothermic reactions reduce emissions in the vehicle exhaust systems

10) *Fuel cells:*

Exothermic reactions generate electricity in fuel cells

ENERGY TRANSFORMATIONS

These are processes that convert energy from one form into another.

First law of thermodynamics (law of conservation of energy)

Energy is neither created nor destroyed but it can be transformed from one form to another.

Every useful process transforms energy from one form to another.

Examples of everyday energy transformations include the following:

1. Electricity to light:

Electric lamps transform electrical energy into light and heat energy.

2. Chemical energy to kinetic energy:

Gasoline is transformed into motion (kinetic) energy in cars.

3. Thermal energy to kinetic energy:

Heat from a stove is transformed into motion energy to cook food.

4. Kinetic energy to thermal energy

Brakes on a bike transform motion energy into heat energy.

5. Electrical energy to sound energy:

Speakers transform electrical energy into sound waves.

6. Chemical energy to thermal energy:

Food is transformed into energy for body use through digestion.

7. Solar energy to electrical energy:

Solar panels transform sunlight energy into electrical energy.

8. Kinetic energy to electrical energy

Generators transform motion energy into electrical energy.

9. Thermal energy to electrical energy;

Power plants transform heat energy into electrical energy

10. Mechanical energy to thermal energy:

Friction transforms motion energy into heat energy

ENERGY TRANSFORMATIONS IN BURNING ETHANOL

Ethanol is a *fuel* that has chemical energy stored in carbon – carbon (C-C) bonds, carbon – hydrogen (C-H) bonds and oxygen – hydrogen bonds (O-H)

Ethanol being a combustible fuel, it releases heat energy during combustion.

The *chemical energy* in ethanol is converted into *heat energy* when burnt and this is an example of an exothermic reaction.

Burning of ethanol releases heat energy and the enthalpy change (ΔH) value of this reaction is negative.

The chemical energy in ethanol is also transformed into *light energy* during combustion.

ENERGY TRANSFORMATION IN BURNING PARAFFIN

Paraffin is a group of alkanes. It is a mixture of different types of hydrocarbons with the general formula C_nH_{2n+2} .

Paraffin is a *by-product* of crude oil, which is a fossil fuel.

In fossil fuels, energy is stored in form of chemical energy, which is converted into heat and light energies when burnt.

ENERGY TRANSFORMATIONS IN BURNING WOOD

The chemical energy in wood is released as heat and light because of the chemical reaction between chemicals in wood (for example cellulose) and oxygen in air. This type of chemical reaction is called *combustion* and it requires oxygen.

Combustion converts the stored chemical energy in wood into heat and light energy.

THE FLOW OF ENERGY THROUGH AN ECOSYSTEM

Energy flow in an ecosystem involves reactions which are endothermic (photosynthesis) and those which are exothermic (respiration)

Endothermic reactions occur during photosynthesis, while exothermic reactions occur during consumption and decomposition. Energy flows from one trophic level to another, with some energy lost as heat at each level.

For example;

1) Producers (Endothermic):

Plants absorb light energy from the sun and convert it into chemical energy through photosynthesis (endothermic reaction). This energy is stored in organic compounds like glucose.

2) Herbivores (Exothermic)

Herbivores consume plants and breakdown the organic compounds into simpler molecules, releasing energy (exothermic reaction). The energy released is used for growth, maintenance, and reproduction.

3) Carnivores (Exothermic)

Carnivores consume herbivores and break down their organic compound, releasing energy (exothermic reaction). This energy is used for growth, maintenance, and reproduction.

4) Decomposers (Exothermic)

Decomposers break down dead organic matter, releasing energy (exothermic reaction). This energy is used by other organism in the ecosystem.

5) Energy loss (Exothermic):

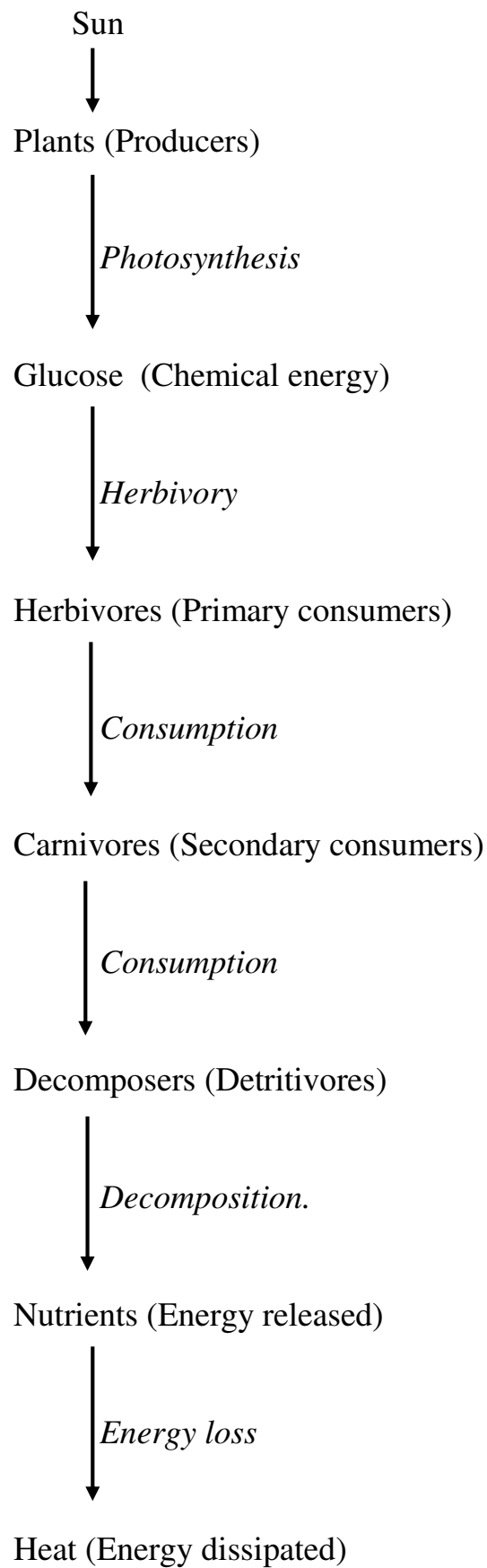
Energy is lost as heat at each tropic level due to metabolic processes (Exothermic reactions)

6) Energy input (Endothermic)

The ecosystem receives energy input from the sun through photosynthesis (endothermic reaction), which supports the entire food web.

NOTE:

An ecosystem is a community of organisms and their physical environment interacting together to form a self-sustaining unit.

ENERGY FLOW DIAGRAM

An Investigation to Compare the Heat Given Out When Different Fuels Burn

(For example: *Peanuts, Oils, paper and Wood*)

Title:	<i>Comparative Investigation of Heat released from Burning Peanuts, oils, paper and Wood.</i>
Aim:	To determine and compare the heat released when peanuts, oils, paper and wood are burnt
Hypothesis:	Wood releases the most heat per unit mass, compared to peanuts, oils and paper.
Variables:	Dependent variable: <i>Temperature</i>
	Control variable: <i>Mass of fuel materials</i>
Materials:	✓ Peanuts ✓ vegetable oil ✓ paper (Newspaper) ✓ Wood ✓ Thermometer ✓ Heat – resistant container or crucible ✓ Weighing balance ✓ Stop watch or timer
Procedure	1. The initial temperature (T_o) of the thermometer is recorded. 2. A fixed mass (for example: 10g) of each fuel material (Peanuts, oil, paper and wood) is weighed. 3. Each material is placed in the heat – resistant crucible. 4. Each material is ignited and its temperature is measured using a thermometer 5. The maximum temperature (T_f) reached and the time taken to reach it are measured and recorded. 6. Steps (3) to (5) for each fuel materials are repeated. 7. The heat released per unit mass is calculated using the formula below 8. <i>Heat released</i> = $\frac{\text{Mass}}{(m)} \times \frac{\text{Specific heat}}{\text{Capacity } (C)} \times \text{change in temperature}$ 9. Per unit mass $Q = MC\Delta\theta$
Risks:	✓ Working with open flames and combustible materials increase the risk of fires and burns. ✓ Handling hot materials and equipment can cause heat - related injuries. ✓ Burning materials can release harmful chemicals, such as volatile organic compounds which can be harmful if inhaled ✓ Inhaling fumes from burning materials can cause respiratory problems. ✓ Exposure to smoke, heat or chemicals can cause eye and skin irritation ✓ Improper use of equipment such as thermometers can lead to damage or breakage ✓ Measure errors or equipment malfunctions can lead to inaccurate data, which can affect the validity of the conclusions of the investigation.

Mitigations	✓ Follow proper laboratory guidelines ✓ Wear appropriate persona; protective equipment (PPE) such as gloves, goggles and lab coats ✓ Ensure proper ventilation and exhaust systems are in place ✓ Use equipment correctly and follow manufacturer's instructions. ✓ Monitor temperature and heat output carefully ✓ Keep a fire extinguisher nearby ✓ Supervision of students during the investigation ✓ Proper disposal of fuel materials
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Data presentation

Table of results. (*These are assumed values*)

Material	T _o (°C)	T _f (°C)	Time (s)	C (Jg ⁻¹ °C ⁻¹)	Heat released
Peanut	25.0	450.0	120	0.59	2500
Vegetable oil	25.0	380.0	90	0.51	1800
Paper	25.0	320.0	60	0.41	1200
Wood	25.0	500.0	150	0.65	3100

Data interpretation

- ✓ Wood released the most heat per unit mass followed by peanuts, oil and paper
- ✓ The release of heat by fuel materials is in the order of wood >Peanuts > Oil>paper
- ✓ The high heat released from wood is due to its high carbon content and combustion efficiency.
- ✓ Peanuts have a higher heat release than oil and paper due to their high protein and fat content, which burn more efficiently.
- ✓ Paper has the lowest heat release due to its low carbon content and higher water content, which reduces combustion.

Conclusion

- ✓ The investigation compared the heat released when peanuts, oils, paper and wood are burnt.
- ✓ The results show that wood releases the most heat per unit mass, followed by peanuts, oil and paper.

INVESTIGATION OF EXOTHERMIC REACTION

(Using the Scientific Method of Dissolving Substances)

(For example: Sodium hydroxide dissolved in water)

Title:	Investigation of Exothermic reactions that occur when Sodium Hydroxide Dissolves in water.
Aim:	To investigate an exothermic reaction that occurs when sodium hydroxide dissolves in water
Hypothesis:	Dissolving sodium hydroxide pellets in water will result in an exothermic reaction, causing a temperature increase, and increasing the mass of sodium hydroxide will increase the temperature change.
Variables:	<i>Dependent variable:</i> Temperature change of the mixture <i>Independent variable:</i> Mass of sodium hydroxide pellets. <i>Control variable:</i> Initial temperature of water
Materials:	✓ Sodium hydroxide pellets. ✓ Distilled water ✓ Thermometer ✓ Stirring rod ✓ Containers (beakers or cups) ✓ Weighing balance
Procedure:	1. The initial temperature (T_i) of the water in a plastic beaker is measured and recorded. 2. A fixed mass (m) of sodium hydroxide (for example 1g) is weighed and recorded. 3. The sodium hydroxide is slowly added to water while stirring 4. Stirring is continued until the sodium hydroxide is fully dissolved. 5. The final temperature (T_f) of the mixture is measured and recorded. 6. Steps (2) to (5) are repeated with different masses of sodium hydroxide (for example 2g, 3g, 4g and 5g) 7. The temperature change (ΔT) is calculated using the formula: $\Delta T = T_f - T_i$ 8. A graph of temperature change (ΔT) against mass of NaOH is plotted.

Risks:	✓ Splashes of sodium hydroxide solution cause skin and eye irritation ✓ Sodium hydroxide solution formed in the investigation can burn if not handled carefully. ✓ Inhaling sodium hydroxide dust can cause respiratory problems for example bronchitis ✓ The exothermic reactions can generate heat, potentially causing burns ✓ Improper use of equipment such as thermometers can lead to breakage. ✓ Measurement errors can lead to inaccurate data, affecting the validity of the investigation.
Mitigations:	✓ Wear appropriate personal protective equipment (PPE) including gloves, goggles and a lab coat ✓ Handle sodium hydroxide solution formed with care, to avoid splashes and spills ✓ Ensure proper ventilation and exhaust systems are in place ✓ Use equipment correctly and follow the instructions of the manufacturer ✓ Monitor temperature and heat output carefully ✓ Keep a fire extinguisher nearby ✓ Dispose the materials properly

Data presentation, Data analysis and Interpretation

Data presentation

Table of results (*These are assumed values*)

m(g)	T _i (°C)	T _f °C)	ΔT (°C)
1.0	20.0	35.0	15.0
2.0	20.0	40.0	20.0
3.0	20.0	45.0	25.0
4.0	20.0	50.0	30.0
5.0	20.0	55.0	35.0

Data analysis

Plot a graph of ΔT against m.

Data Interpretation

The results show a consistent temperature increase with increasing mass of sodium hydroxide, indicating an exothermic reaction.

The temperature change (ΔT) is directly proportional to the mass of sodium of sodium hydroxide

The graph shows a linear relationship between ΔT and concentration of sodium hydroxide, allowing for predictions of temperature changes at different concentrations.

Conclusion:

This investigation demonstrates the exothermic reaction between sodium hydroxide and water, with a clear dependence on the mass of sodium hydroxide.

The results support the hypothesis and provide a quantitative understanding of the temperature change.

INVESTIGATION OF ENDOTHERMIC REACTION

(Using the Scientific Method Of Dissolving Substances)

(For example: Sodium hydrogen carbonate dissolved in water)

Title:	Investigation of Endothermic reaction that occur when <i>sodium Hydrogencarbonate</i> dissolves in water.
Aim:	To investigate an endothermic reaction that occurs when <i>sodium hydrogencarbonate</i> dissolves in water
Hypothesis:	Dissolving <i>sodium hydrogencarbonate</i> in water will result in an endothermic reaction, causing a temperature decrease, and increasing the mass of sodium hydrogen carbonate will increase the temperature decrease.
Variables:	<i>Dependent variable:</i> Temperature change of the mixture <i>Independent variable:</i> Mass of <i>sodiumhydrogen carbonate</i> <i>Control variable:</i> Initial temperature of water
Materials:	✓ Sodium hydrogen carbonate powder ✓ Distilled water ✓ Thermometer ✓ Stirring rod ✓ Containers (beakers or cups) ✓ Weighing balance
Procedure:	1. The initial temperature of water (T_i) in a beaker is measured and recorded 2. A fixed mass (m) of <i>sodium hydrogencarbonate</i> (for example: 1g) is weighed and recorded. 3. NaHCO₃ is slowly added to the water while stirring 4. Stirring is continued until the <i>sodium hydrogencarbonate</i> is fully dissolved 5. The final temperature (T_f) of the mixture is measured and recorded 6. Steps (2) to (5) are repeated with different masses of <i>sodium hydrogencarbonate</i> (for example: 2g, 3g, 4g and 5g) 7. The temperature change (ΔT) is calculated using the formula: $\Delta T = T_f - T_i$ 8. A graph of temperature change (ΔT) against mass of NaHCO₃ is plotted.

Risks:	✓ <i>Sodium hydrogencarbonate</i> is a mild irritant whose splashes can cause skin and eye irritation if not handled carefully. ✓ Inhaling <i>sodium hydrogencarbonate</i> dust can cause respiratory problems such as coughing ✓ Direct contact of the skin with <i>sodium hydrogencarbonate</i> can cause irritation and discomfort ✓ Improper use of equipments such as thermometers can lead to inaccurate data, affecting the validity of the investigation ✓ Some individuals may be allergic to NaHCO_3, which can cause an allergic reaction.
Mitigations:	✓ Wear appropriate personal protective equipment (PPE) including gloves, goggles and a lab coat ✓ Handle <i>sodium hydrogencarbonate</i> with care, to avoid splashes and spills ✓ Ensure proper ventilation and exhaust systems are in place ✓ Use equipment correctly following instructions of the manufacturer ✓ Monitor temperature and heat output carefully ✓ Keep a first aid kit nearby ✓ Dispose off the materials properly ✓ Be aware of any allergies or sensitivities

Data presentation, Data analysis and Interpretation

Data presentation

Table of results (*These are assumed values*)

m(g)	$T_i(^{\circ}\text{C})$	$T_f(^{\circ}\text{C})$	$\Delta T(^{\circ}\text{C})$
1.0	20.0	18.0	-2.0
2.0	20.0	16.0	- 4.0
3.0	20.0	14.0	- 6.0
4.0	20.0	12.0	- 8.0
5.0	20.0	10.0	- 10.0

Data analysis

Plot a graph of ΔT against M

Data interpretation

The results show a consistent temperature decrease with increasing mass of NaHCO_3 , indicating an endothermic reactions.

The temperature change (ΔT) is directly proportional to the mass of NaHCO_3 , supporting the hypothesis.

The graph shows a linear relationship between (ΔT) and the mass of NaHCO_3 , allowing for predictions of temperature changes at different concentrations.

Conclusion:

This investigation demonstrates the endothermic reaction between *sodium hydrogencarbonate* and water, with a clear dependence on NaHCO_3 concentration

The results support the hypothesis and provide a quantitative understanding of the temperature change

OBTAINING USEFUL ENERGY FROM BURNING FUELS

Energy can be released in chemical reactions in various forms for example light and sound energy, although it is usually released as heat energy.

AMOUNT OF ENRGY RELEASED FROM BURNING FUELS

Amount of energy released from a burning fuel (for example ethanol) is got from the expression below.

Energy released (Q) = Mass(m) x specific heat capacity (c) x change in temperature

$$Q = MC\Delta\theta \quad \text{———— (eqn i)}$$

Change in temperature ($\Delta\theta$) = Final temperature (θ_f) – Initial temperature (θ_i)

$$\Delta\theta = \theta_f - \theta_i \quad \text{———— eqn(ii)}$$

Putting equation (ii) into equation (i)

$Q = MC(\theta_f - \theta_i)$

Where Q - Is the energy released by the fuel.

M - Is the mass of the liquid heated

C - Specific heat capacity of the liquid heated

θ_f - Final temperature

θ_i - Initial Temperature

Example

1. 3.5g of a fuel is burned to heat 50cm³ of water. The temperature of water increased from 22°C to 71°C. Calculate the:-

- Energy released by the fuel
- Energy released per gram of the fuel

Solution (i)

Heat released = Mass of water X specific heat capacity X change in temperature

$$\begin{aligned}
 \text{From density} &= \frac{\text{mass}}{\text{volume}} \\
 \text{Density of water} &= 1\text{gcm}^{-3} \\
 \text{Volume of water} &= 50\text{cm}^3 \\
 \text{Mass of water} &= \text{Density X volume} \\
 &= 1\text{gcm}^{-3} \times 50\text{cm}^3 \\
 \text{Mass of water} &= 50\text{g}
 \end{aligned}$$

$$\begin{aligned}
 \text{Heat released by the fuel} &= 50 \times 4.2 \times (71 - 22) \\
 &= 50 \times 4.2 \times 49
 \end{aligned}$$

$$\text{Heat released by the fuel} = \mathbf{10,290\text{Joules}}$$

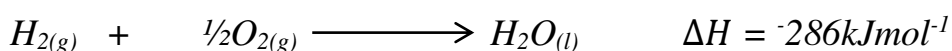
Solution (ii)

$$\begin{aligned}
 \text{Energy released per gram of the fuel} &= \frac{\text{Energy released by the fuel}}{\text{mass of the fuel}} \\
 &= \frac{10,290\text{Joules}}{3.5\text{g}}
 \end{aligned}$$

$$\text{Energy released per gram of the fuel} = \mathbf{2,940\text{Joules per gram}}$$

INTERPRETATION OF CHEMICAL EQUATIONS FOR EXOTHERMIC REACTIONS

Consider the equation below



Interpretation of the equation

When *one mole of hydrogen* is *completely burnt in oxygen*, it *evolves* or produces *286kJ of heat*

Consider the equation below



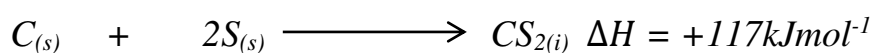
Interpretation of the equation:

When **one mole of carbon** is **completely burnt in oxygen**, it produces **406 kilo joules of heat**.

Note: The negative sign on the value of ΔH means that heat is produced or evolved or liberated to the surrounding implying that is an exothermic reaction

INTERPRETATION OF CHEMICAL EQUATIONS FOR ENDOTHERMIC REACTIONS

Consider the equation below



Interpretation of the equation

When **one mole of carbon** reacts with **2 moles of sulphur**, **117 kilo joules of heat** are **absorbed** from the surrounding.

Note: The positive sign on the value of ΔH means that heat is absorbed from the surrounding, implying that it is an endothermic reaction.

TYPES OF HEAT CHANGES

Heat of combustion (Enthalpy of combustion)

Heat of combustion is the heat change that occurs when one mole of a substance is completely burnt in excess oxygen.

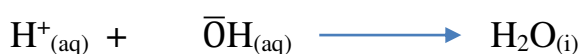
Heat (enthalpy) of combustion is usually **negative**; that is to say ΔH is negative

This implies that heat combustion is an **exothermic process** or reaction, which **evolves heat to the surrounding**.

Heat of neutralization.

This is the heat change that occurs when one mole of hydrogen ions from an acid react with one mole of hydroxyl ions from an alkali to form one mole of water.

Ionic equation:



Heat of neutralization is usually **negative**, that is to say ΔH is negative

This implies that heat neutralization is an **exothermic process** or reaction, which liberates heat to the surrounding.

INTERPRETATION ENERGY PROFILES OF CHEMICAL REACTIONS

There is transfer of energy during a chemical reaction as reactants are converted into products. The idea of transfer of energy during a chemical reaction is illustrated using energy profile diagram.

ENERGY PROFILE DIAGRAM

An energy profile diagram show the theoretical energy pathway of the reaction as it progresses from reactants to products.

The energy profile diagram compares the energy of the reactants to that of the products

The difference between the energy of reactants and that of products represents the amount of energy transferred to the surroundings or absorbed from the surroundings.

Potential energy

This is the energy stored in the bonds of a compound or substance.

Activation energy

This is the minimum energy needed for the reaction to proceed.

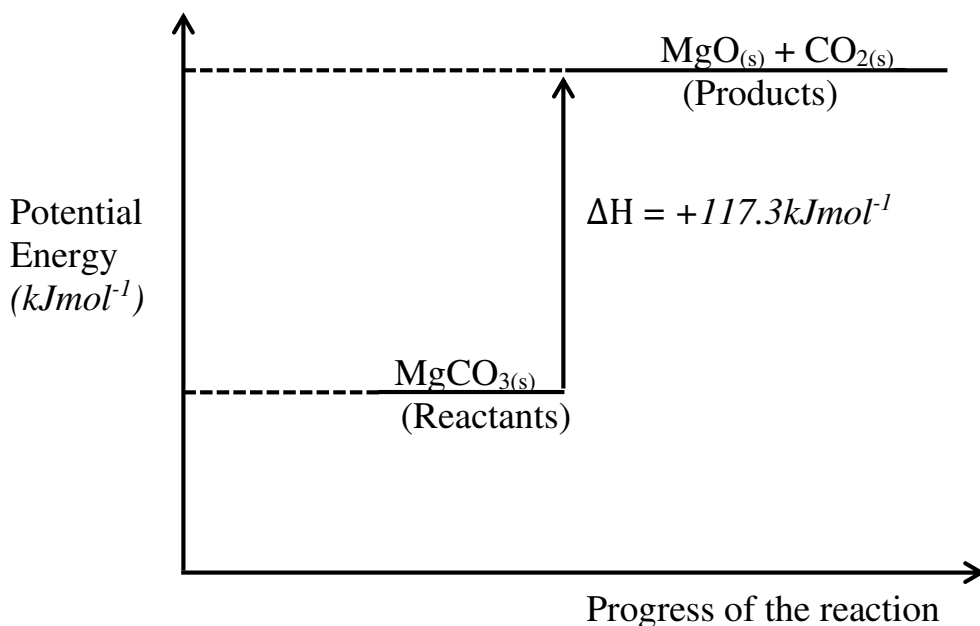
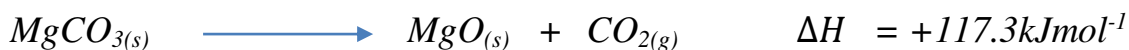
Note: The difference in the energy of products and that of the reactants is called *enthalpy change* or *heat change* and it is donated by ΔH

$$\text{Heat change } (\Delta H) = \text{Energy of products} - \text{Energy of reactants}$$

Energy level diagrams are simplified illustrations that only show the initial and final energies of substances during the reaction.

ENERGY LEVEL DIAGRAM FOR THE DECOMPOSITION OF MAGNESIUM CARBONATE

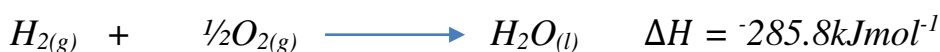
The thermal decomposition of magnesium carbonate is an endothermic reaction, requiring 117.3kJmol^{-1} of heat.

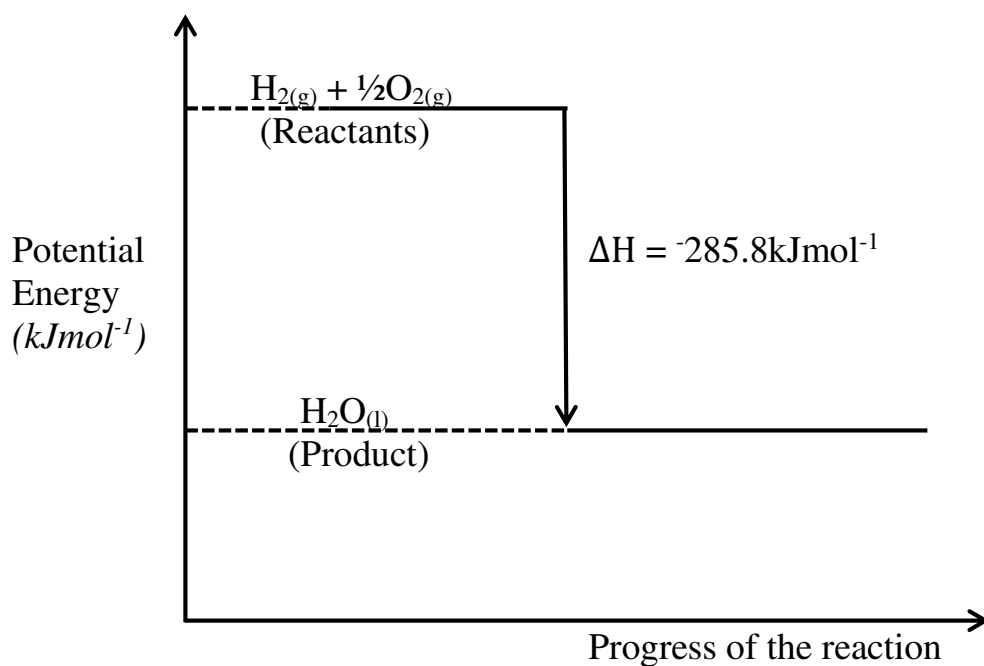
Equation:

From the equation and energy level diagram, we can conclude that during the reaction, energy is transferred from the surrounding to the system, so the heat energy of the products is greater than the heat energy of the reactants, thus, the enthalpy change (ΔH) is positive.

ENERGY LEVEL DIAGRAM FOR THE FORMATION OF A WATER MOLECULE FROM HYDROGEN AND OXYGEN

The reaction of hydrogen gas with oxygen gas to produce water is an exothermic reaction, producing 285.8kJmol^{-1}

Equation



Form the equation and energy level (profile) diagram, we can conclude that during the reaction, energy is transferred from the system to the surroundings, so the enthalpy of the products is lower than the enthalpy of the reactants, thus, the enthalpy change (ΔH) is **negative** implying that is an **exothermic reaction**.

CHEMICALS FOR CONSUMERS

INTRODUCTION

Food is one of the chemicals we use in our daily lives. But there are other chemicals, such as:

- Medicines
- Preservatives
- Washing agents
- Cleansing agents
- Stimulants



Chemicals play various important roles in our everyday lives and almost everything we use is made of chemical substances.

In Uganda, the Uganda National Bureau of Standards (UNBS) is in charge with responsibility of testing and establishing

compliance with specifications and regulation of chemical limits in products such as perfumes and flavors, cosmetics, pharmaceutical products, soaps, detergents and household chemicals.

This effective regulation is an important part of safeguarding public health. Modern laws and regulations place a special responsibility on those trading to ensure the safety of their products, and make sure that they are for a purpose.

Therefore, you will appreciate that the products used in everyday life exist as chemicals and some of them can be prepared at home or in the laboratory.

SOAPS AND DETERGENTS



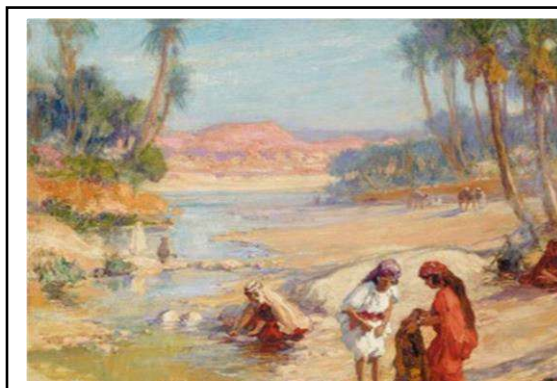
Soap and detergents play an essential role in our daily life. They remove dirt and stains from our clothes, dried on food from our dishes and even germs from our hands. The process of removing dirt, grease and stains found on various substances is called “cleansing action of soap or detergent”.

When we clean, we leave things better than we found them. Cleansing helps to remove dirt and kills microbes such as bacteria and viruses.

SOAP

Soaps are cleansing agents or natural detergent, made from fats and oils of animals and vegetables. Soaps are sodium or potassium salts of fatty acids formed by the reaction of fats and oils with an

The development of soap as cleansing agent and soap making has improved the hygiene and possibly helped to reduce the spread of diseases.



HISTORY OF SOAP MANUFACTURING

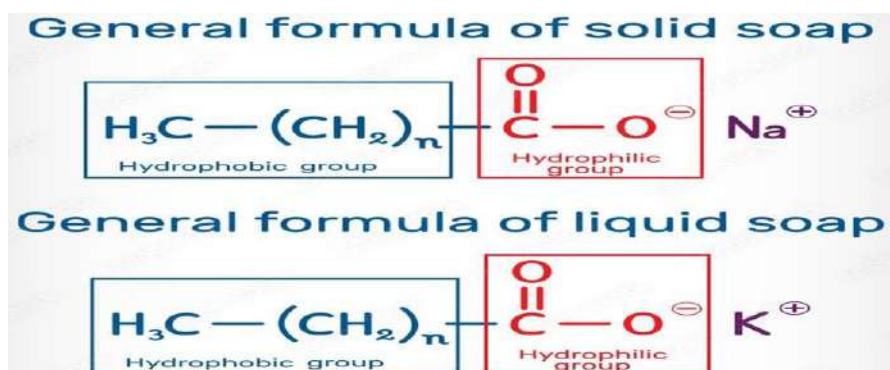
Soap got its name from an ancient Roman legend about Mount Sapo. Legend says that soap was first discovered on soap Hill in Rome when group of Roman women were washing their clothes in the River Tiber at the base of a hill, below which animal fats from the sacrifices ran down into the river and created soapy clay

The earliest recorded evidence of the production of soap-like materials dates back to around 2800 BC in ancient Babylon. A formula for soap consisting of water, alkalis and cassia oil was written on a Babylonian clay tablet around 2200 BC. By the 7th century, soap-making was an established art in Italy, Spain and France.

Chemical nature of soap

Depending on the metal cation, soaps are water-soluble potassium or sodium salts of long-chain carboxylic acid salts.

The exact chemical formula of soap is $C_{17}H_{35}COO^-$ plus a metal cation, either Na^+ or K^+ . the final molecule is called sodium stearate and is a type of salt. Depending on the metal cation, soaps potassium salts or sodium salts of long-chain carboxylic acids.



Physical properties of soap

The physical properties include: odour, colour, texture, hardness, solubility.

Manufacture of soap

During soap preparation, concentrated sodium chloride solution is added to a boiled mixture of concentrated sodium hydroxide solution or potassium hydroxide solution and oil or fats in order to precipitate out soap. The precipitated soap is filtered off and mixed with dyes to make it coloured.

Concentrated sodium hydroxide produces hard soaps whereas concentrated potassium hydroxide produces soft soaps. Soft soaps dissolve more easily water than hard soaps.

Chemical Properties of Soap

These include pH and lathering ability.

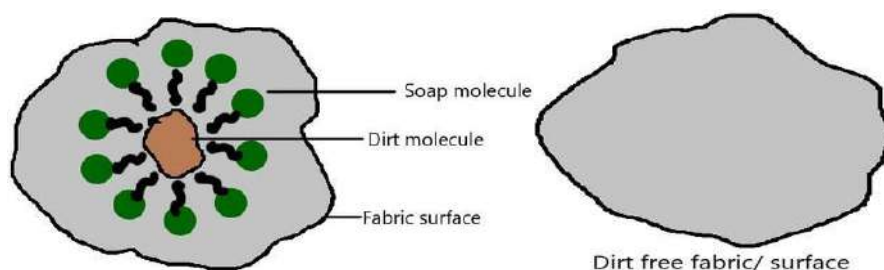
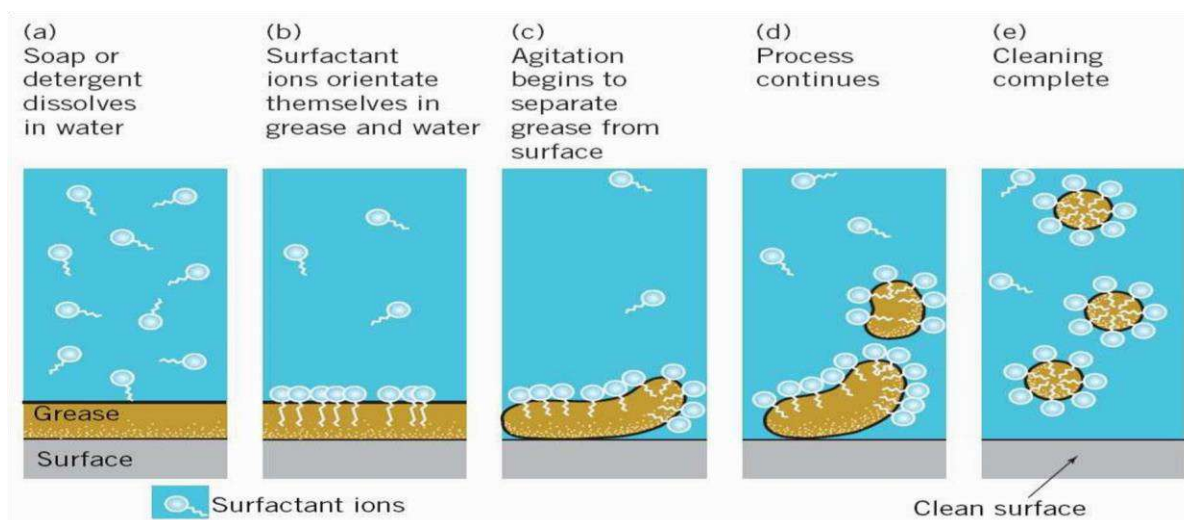
In order to perform as surface-active agents, soaps must have certain chemical structures. The structure of soap consists of a hydrophobic (water-insoluble) part, such as hydrocarbon chain and a hydrophilic (water-soluble) group such as -COONa . See the structure of a soap molecule below.



Toilet soap forms lather with water more easily than hard soap. When a blue litmus paper is dipped into the soap solution, its Colour persists whereas a red litmus paper. This shows that a soap solution is alkaline.

CLEANSING ACTION OF SOAP

Clothes become dirty due to deposition of dust and oily or greasy substances. Plain water is not capable of dissolving oily or greasy substances. However, the hydrophobic end of soap can do so. Therefore, addition of soap makes the removal of dirt and stains easier, this is called cleansing. During cleaning, the hydrophobic tails dissolve in a droplet of oil or grease, whilst the hydrophilic heads face out into the surrounding water, resulting in a ball structure. With agitation or scrubbing, the grease becomes dislodged from the surface. The nonpolar substances such as oil and grease are held inside the ball and suspended in water.



DETERGENTS



Detergents are amphipathic molecules that contain both polar and hydrophobic groups. Detergents are structurally similar to soaps, but differ in the water-soluble portion.

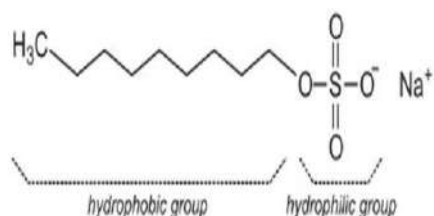
Chemical nature of detergents

Molecules of detergents contain both polar or charged hydrophilic groups (heads) and non-polar or hydrophobic parts (tails). In the same molecule of the detergent, the different parts have different affinities for the solvent and dirt.

One part possesses a high affinity for polar solvents such as water and another part has a strong affinity for nonpolar dirt particles. They are also known as surfactants, because they decrease the surface tension of water.

A detergent may also refer to a surfactant or group of surfactants with cleansing properties in dilute solutions. These substances are usually alkyl benzene sulphonates, a family of compounds that are similar to soap but are more soluble in hard water.

Structure of a detergent molecule



Additives in Detergents

Detergents do not only differ from soap in terms of the main component chemical structure, but rather the other additives they contain. Detergents are made to suit different purposes, such as household detergents, the additives keep metal components clean and free of deposits. Biological enzymes and whitening agents are among the many additives in detergents. Further, detergents neutralize acids that form in the oil. This is key for areas where thorough cleanliness is essential.

Uses of Some Additives in Detergents.

Additive	Use
Optical brighter (eg Fluorescent dye)	Convert stain to colourless substances
Fragrances	Add fragrances to fabrics and detergents
Bleach	Make fabric appear whiter
Biological Enzymes (eg amylase, protease, lipase)	Breakdown fat and protein molecules in food stains
Fabric softening agents	Soften hard water
Drying agents (eg sodium sulphate)	Make the solid detergent to dry and enable the liquid detergent to be poured out easily.

Colourants

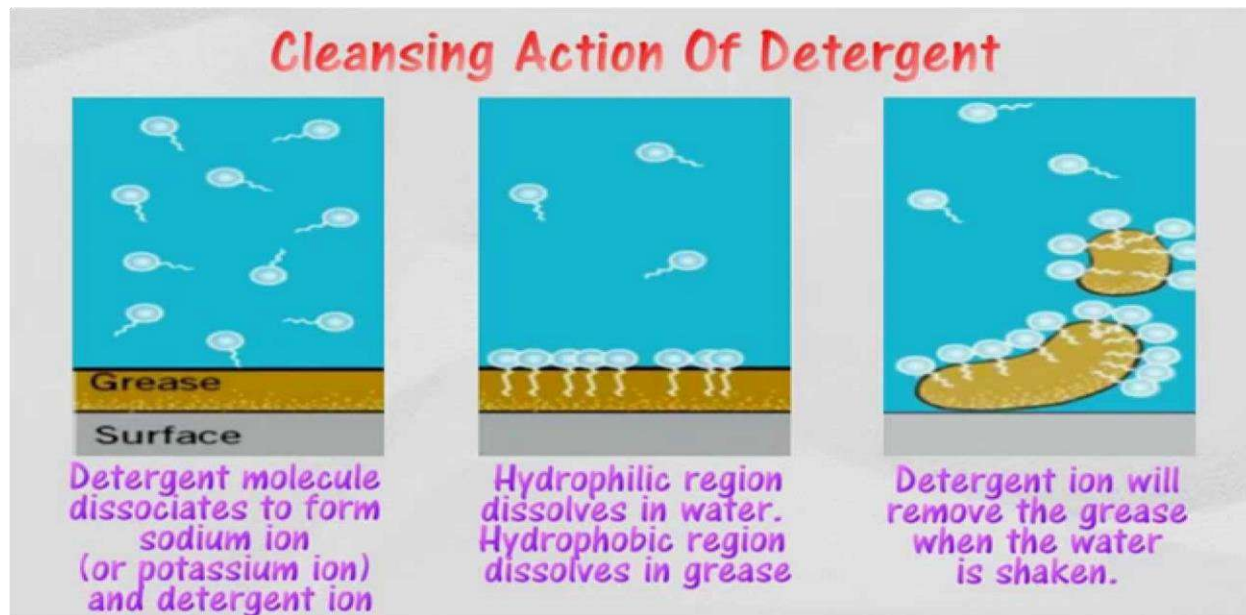
Change and improve on the final Colour of the detergent

Cleansing Action of Detergents

Washing with plain water may not remove all stains in clothes. However, sometimes soap may fail to remove stains in clothes as well. The application of detergents may not spare stains trapped in clothes.

The cleansing action of soap is as follows:

A detergent emulsifies grease into tiny droplets (micelles) in water. The polar end of the detergent dissolves in water and its ions surround fat droplets. Non-polar end of the detergent dissolves in grease that holds dirt on fabric and emulsifies it into a suspension. When washing is done, the oil is pulled off the fabric and floats free and dirt rinsed away by water.



Comparing the Cleansing Action of Soap and Detergents

Soap is potassium or sodium salts of a carboxylic acid attached to a long aliphatic chain whilst detergent is a potassium or sodium salt of a long alkyl chain ending with a sulfonate group. Soap is a metal salt of fatty acid that we use for cleaning and lubrication.

Detergents can work with hard water too while soaps cannot. Detergents can work even in acidic medium while soaps cannot. Synthetic detergents are stronger cleansing agents than soaps. Their solubility is higher than that of soaps.

DANGERS OR SIDE EFFECT OF SOAP AND DETERGENTS.

(a) Soap contains chemicals that can cause:

- Skin burns / blisters / irritation and hence pain or cancer.
- Eye redness and pain; hence loss of vision.

- Mitigation can be done by thoroughly washing the affected areas (or irrigation of the affected areas) like skin or eyes.

(b) Soapless detergents contain phosphates which cause algae bloom/algal bloom and hence water pollution.

N.B. Algae/algal bloom already means accumulation.

(a) Similarities:

- Both soapy detergents and soapless detergents are salts of Organic acids of long carbon chain.
- Both soapy detergents and soapless detergents are effective cleansing agents in soft water / rain water.

(b) Differences;

Soapy detergents	Soapless detergents
Forms scum with hard water	does not form scum with any form of water.
Gentle on skin during cleansing.	not gentle on skin during washing
Sodium salts of carboxylic acid of long chains and cannot be used in strongly acidic solutions.	Sodium salts of long chain benzene sulphonic acids and can be used in strongly acidic solutions.
Biodegradable	Non-biodegradable

Task:

1. Differentiate between soap and detergents
2. What are the advantages of detergents over soap?
3. How is the use of detergents a problem to the environment?
4. Compare the effectiveness of soaps and detergents in soft and hard water

FOOD ADDITIVES

Food, especially that which is processed, comes with attractive colour, appearance and smell. Substances that are added to food to maintain or improve its taste, freshness, texture or appearance are called food additives.

Food additives are any substances the intended use of which results may reasonably be expected to result directly or indirectly in its being a component or otherwise affecting the characteristics of any food. Food additives are applied to food during:

- Production ➤ Treatment ➤ Transportation
- Processing ➤ Packaging ➤ Storage

Food additives need to be checked for potential harmful effects on human health before they can be used. Information regarding food additives is written on packaging label stickers, see figure below.



NUTRITION INFORMATION		
Servings per package: 8		
Serving size: 33g		
	Per serving	Per 100g
Energy	550kJ	1680kJ
Protein	2.4g	7.2g
Fat		
Total	3.6g	11.0g
Saturated	1.3g	4.1g
Carbohydrate		
Total	21.5g	55.1g
Sugars	10.1g	30.7g
Dietary Fibre	1.7g	5.1g
Sodium	20mg	70mg

This tells you how many serves there are in the whole packet.

This tells you the size of one serving of this food.

This tells you the nutrients in 100g of this food. This is the best way to compare similar products.

This tells you the nutrients in a single serve of this food. In this case, it's per 33g.

Direct food additives are those added to food for a specific purpose in that food. For example, xanthan gum used in salad dressings.

Indirect food additives become part of the food in trace amount due to its packaging, storage or other handling. For instance, minute amount of packaging substances like microplastics may find their way into food during packaging.

Types of food additives and their functions

Some food additives have been in use for centuries for preservation. For example: salt (in meats such as bacon or dried fish), sugar (in marmalade) and Sulphur dioxide (in wine).

Many food additives have been developed over time to meet the needs of food production, as making food on a large scale is very different from making them on a small scale at home. Additives are needed to ensure processed food remains safe and in good condition throughout its journey from factories, during transportation to warehouses and shops, and finally to consumers.

The use of food additives is only justified when their use has a technological use, does not mislead consumers and serves as well defined technological function, such as to preserve the nutritional quality of the food or enhance the stability of the food.

Food additives can be derived from plants, animals or minerals or they can be synthetic. They are added intentionally to food to perform certain technological purposes which consumers often take for granted. There are several food additives used, all of which are designed to do a specific job in making food safer.

or more appealing. WHO together with FAO groups food additives into 3 broad categories based on their function.

a) Flavouring agents

These are added to food to improve aroma or taste. These are applied to foods like soft drinks, cake, yoghurt and cereals. Natural flavouring agents include; nut, fruit and spice blends, as well as those derived from vegetables and wine. In addition, there are flavourings that imitate natural flavours.

b) Enzyme preparations

Enzyme preparations are a type of additive that may or may not end up in the final food product. Enzymes are naturally occurring proteins that boost biochemical reactions by breaking down larger molecules into their small building blocks. They can be extracted from plants or animal products or from micro-organisms such as bacteria and are used as alternatives to chemical-based technology. They are mainly used in baking (to improve dough), for manufacturing fruit juices (to increase yields), in wine making and brewing (to improve fermentation)

c) Other additives

Other food additives are used for a variety of reasons, such as preservation, colouring and sweetening. They are added when food is prepared, packaged, transported or stored and they eventually become a component of the food.

Summary

Food additive	Function
Sweeteners	Add sweetness without extra calories
Flavours	Add specific flavours to food
Nutrients	Replace vitamins and minerals
Emulsifiers	Allow smooth mixing of ingredients
Stabilizers and thickeners	Produce uniform texture
Preservatives	Prevent food spoilage Prevents changes in colour, flavour and texture
Food colour	Is added to improve its visual appeal
Anti-caking agents	Used to keep the product from clumping together
Antioxidants	Added to products to prevent rancidity

Examples of Food Additives.

Food additive	Example
Preservatives	Sodium benzoate, potassium propionate, benzoate, calcium ethylene oxide.
Antioxidants	Sulphur dioxide, ascorbic acid
Sequestrants	Iron or copper
Flavouring agents	Amyl acetate for banana flavour, Benz aldehyde for cherry flavour, Ethyl butylate for pineapple flavour.
Anti-caking agents	Magnesium carbonate, Calcium phosphate
Stabilizers and thickeners	Pectin, Amylose.
Food colour	Turmeric, caramel, carotene
Bleaching and maturing	Hydrogen peroxide, Benzoyl peroxide

BANNED FOOD ADDITIVES.

Some food additives can trigger allergic reactions and other mild health problems. However, some food additives pose serious health risks and have been banned for consumption. If food contains any of such chemicals, warning is always given on the label.



Monosodium glutamate, or MSG is a common food additive used to intensify and enhance the flavour of savory dishes. It's found in a variety of processed foods like frozen dinners, salty snacks and canned soups. It's also often added to foods at restaurants and fastfood places. MSG has been a subject of heated controversy since a 1969 study of mice found that large amounts found harmful neurological effects and impaired growth and development.

Cancers have also been raised about the potential cancer-causing effects of certain food dyes. Red 3, also known as erythrosine, has been shown to increase the risk of thyroid tumours in some animal studies, causing it to be replaced by Red 40 in most foods.

Frequently found in processed meats is sodium nitrite which acts as a preservative to prevent the growth of bacteria while also adding a salty flavour and reddish-pink colour. When exposed to high heat and in the presence of amino acids, nitrites can turn into nitrosamine, a compound that can have many negative effects on health.

Summary

Food additive	Reasons for banning
Bisphenol A	Acts like estrogen in the body and change the timing of puberty, decreases fertility and increase body fat.
Phthalates	Affect male genital development and increase childhood obesity
Nitrates/nitrites	Cause tumours in the digestive and nervous system
Synthetic artificial food colours	They have effects on child behavior and attention

IMPACT OF FOOD ADDITIVES TO HEALTH AND ENVIRONMENT

- Some food additives like artificial colours can cause allergies, headache, stomach upsets when consumed by some people.
- Some could be carcinogenic.
- The non biodegradable nature of food additives.
- Some food additives cause asthma and respiratory issues eg sulphites which are commonly used as preservatives.
- Some additives such emulsifiers and thickeners may cause gastrointestinal discomfort to people.
- Once poorly disposed can cause environmental pollution.
- Containers having food additives can cause environmental pollution especially if they are not recycled

Mitigations

- Regulatory agencies like Uganda National Bureau of Standards (UNBS) should assess the safety of food additives before use.
- Consumers make informed choices by reading ingredient labels before consuming substances

CHEMICALS IN MEDICINE

One of the major contributions of Chemistry is the healthcare sector. Medicines are made from chemicals. New drugs are developed from chemical analysis and synthesis of new compounds. This field is so vast that a new field in chemistry known as medical chemistry has emerged.

Medical chemistry is a discipline that surrounds the design, analysis, development and synthesis of pharmaceutical drugs. This discipline needs expertise from synthetic organic chemistry, pharmacology including biological sciences. There are numerous amount of chemicals in medicine.

When one feels unwell, the doctor finds necessary to prescribe some drugs to help him/her get better. Someone's healthcare greatly depends on the work of chemists.

Medicine and drugs.

Medicine or drugs can be classified in a number of ways. For example, they can be grouped according to the;

- Body system they treat
- Chemical compounds in them
- Disease or illness they treat

Classification based on chemical compounds would classify medicine as;

- Antibiotics • Haematics • Sedatives
- Antiseptics • Antifungals • Antipyretics
- Analgesics • Antipsychotics

Each category of drugs has specific chemical components which are useful in treating a particular illness.

TRADITIONAL MEDICINE

Traditional medicine	Function
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Garlic extract	Controls cholesterol level Regulates blood pressure Reduces cancer risks. Treats fungal and bacterial infections, heart burn and nausea
Aloe vera	Treats burns, malaria, ulcers, allergic reactions, fever, diabetes, indigestion, skin diseases.
Lemon juice	Treats hypertension, obesity, fever and bronchitis
Guava leaf and flower extract	Treats malaria, ulcers, coughs and diarrhoea
Pawpaw leaf and root extract	A natural contraceptive. Improves liver and kidney function
Bombo (Momordica foetida)	Used for fever and malaria treatment
Cofta tablets	Treats flu and cough

The herbal medicines are administered through squeezing or crushing and mixing with water. Then, the resulting solution is decanted and drunk. Herbal medicine was discovered from the Holy books, for example, herbs were mentioned in Genesis.

To improve on traditional medicine, herbal medicines should be prescribed by qualified and registered practitioners. The methods of extraction and preparation should be improved in order to extract a reasonable quantity of the active ingredient. Methods of packaging and conservation should also be improved.

Side effects of traditional/herbal medicine

Being “natural” does not necessarily mean they are safe for people to take. Just like conventional medicine, herbal medicine will have an effect on the body, and can be potentially harmful if not used correctly. Herbal medicine contains an active ingredient or chemicals got from plant parts such as leaves, roots or flowers. The chemical can be toxic and result into body reactions.

Task:

How can the side effects of traditional medicine be minimized?

Modern (conventional) Medicine

Modern medicine are chemical compounds used to cure diseases or relieve pain. They are artificially synthesized in the laboratory. Modern medicine has undesirable side effects. It should also be taken in a therapeutically prescribed manner to avoid any side effects. They can be classified as:

a) Analgesics medicine

Paracetamol such as Panadol, hedex and others are drugs we take to relieve pain. The term analgesic refers to a medication that provides relief from pain without outing someone to sleep or making one lose consciousness. They are also known as pain killers. They work by interfering with the transmission of pain signals to the brain.

They are divided into two classes:

i. Non-narcotic analgesic

Antipyretics are medicine that reduce fever by lowering body temperature. The most famous example of this class of medicine are aspirin and paracetamol. Aspirin is chemically known as acetylsalicyclic acid. It is used to reduce fever, pain or inflammation. Paracetamol is also used to treat moderate pain, body aches and fever. They are nonnarcotic analgesics ie. They are non-addictive.

ii. Narcotic analgesics

Produce desired pharmacological effects like they reduce fever and induce sleep. Morphine narcotics (obtained from the opium poppy) are mostly used to get relief from pain after medical operation, cardiac pain and labour pain during child birth, several stage of cancer and many more.

Chemical component in analgesics

Analgesics are also known as anti-inflammatory drugs, due to its action to reduce local inflammatory responses. Most anti-inflammatory analgesics are derived from two compounds: salicyclic acid and phenacetin. Acetylsalicyclic acid or aspirin, which is derived from salicyclic acid, is most widely used mild analgesic.

b) Antibiotics medicine

Antibiotics being less toxic for humans and animals are used as medicine to treat bacterial infections. Antibiotics refer to a substance produced wholly or partially by chemical synthesis which inhibits the growth or destroys the pathogens.

The growth of antibiotics was started by Paul Ehrlich, a German bacteriologist. The real evolution in the field of antibiotics was brought by the discovery of penicillin from fungus by Alexander Fleming in 1929. Both traditional and modern medicine are used to treat common illness like cough and diarrhoea mostly caused by bacteria.

Task:

1. Which medicine do people take to get relief from bacterial infections?
2. What chemical component do antibiotics medicine contain to suppress bacterial activity?

Chemical components antibiotic medicine

With advances in medicinal chemistry, most modern antibacterials are semisynthetic modifications of various natural compounds. For example, the beta lactam antibiotics, which include the penicillin, cephalosporins and carbapenems.

Antibiotics can be classified in several ways. The most common method classifies them according to their chemical structure as antibiotics sharing the same or similar chemical structure will generally show similar patterns of antibacterial activity, effectiveness, toxicity and allergic potential.

Class antibiotics (chemical component)	of	Description or Function		Example
B-lactam		Inhibit bacterial cell synthesis	wall	Penicillin: - Amoxicillin, flucloxacillin, penicillin G
				Cephalosporins: - cefoxitin, cefotaxime ceftriaxone
				Carbapenem: - Imipenem
Macrolides		Inhibit bacterial synthesis	protein	Erythromycin, Azithromycin, Clarithromycin
Tetracyclines		Inhibit bacterial synthesis	protein	Tetracycline, Minocycline, Doxycycline, Lymecycline
Fluoroquinolones		Inhibit bacterial synthesis	DNA	Ciprofloxacin, Ofloxacin, Enoxacin
Sulphonamides		Block bacterial metabolism	cell	Trimethoprim + sulphamethoxazole
Peptides		Inhibit bacterial cell wall synthesis		Bacitracin

Anti-fungal Medicine

Fungal infections such as athletes feet, ringworms and candidiasis are some of the most common.

Task:

1. How do we contract fungal infections?
2. Which medicines do you know that can treat fungal infections?
3. Which chemical component in anti-fungal medicine combat infectious fungi?

Chemical components in anti-fungal medicine

- Ketoconazole ➤ Clotrimazole
- Miconazole ➤ Fluconazole

Herbal medicine used to treat bacterial infections

- Garlic • Honey • Ginger
- Neem • Clove • Echinacea

Herbal medicine used to treat fungal infections

- ✓ Garlic ✓ Grape fruit seed extract ✓ Black walnut ✓ Echinacea

c) Psychotherapeutical Medicine

Mood and behavior can be a sign of mental illness. Emotional and mental disorders for example depression, schizophrenia, manic-depressive disorders are treated with Psychotherapeutical medicine.

Psychotherapeutical medicine can be grouped into; stimulants, antidepressants and antipsychotics.

i. Stimulant medicine

Stimulants, sometimes called “uppers”, temporarily increase alertness and energy. The most common used street drugs that fall into this category are cocaine and amphetamines.

Energy drinks marketed as providing mental and physical stimulation contain stimulant chemical compounds like caffeine.

Chemical components in stimulant medicine

Fungicides are used both in agriculture and to fight fungal infections in animals. Chemicals used to control oomycetes, which are not fungi, are also referred to as fungicides, as oomycetes use the same mechanisms as fungi to infect plants. Fungicides can either be contact, translaminar or systemic.

Theazole anti-fungal agents have five-membered organic rings that contain either 2 or 3 nitrogen molecules (the imidazole and the triazoles respectively). The clinically useful imidazole are clotrimazole, Miconazole and ketoconazole. Two important triazoles are itraconazole and fluconazole.

ii. Depressant medicine

Fear, anxiety and stress are some of the common conditions that cause serious mental conditions. The conditions can lead to sleep disorders and eventually mental breakdown. Central Nervous System (CNS) depressants are medicine that include sedatives, tranquilizers and hypnotics. Drugs that slow brain

activity and therefore useful in treating anxiety, panic, acute stress reactions and sleep disorders are called depressant medicine.

Depressant medicine can be categorized as:

- a. Tranquilizers or anti-psychotics eg Zyprexa, Seroquel and Haldol. They reduce symptoms of mental illness.
- b. Benzos such as Xanax, Halcion and Librium.
- c. Barbiturates are used as sedatives and sleeping pills such are Seconal, Numbutal and Amytal.

Chemical components in depressant medicine

Benzodiazepines enhance the effect of the neurotransmitter gammaaminobutyric acid (GABA) at the GABAA receptor, resulting in sedative, hypnotic and muscle relaxant properties. These chemicals include serotonin, dopamine.

Antiseptic and Disinfectants

Antiseptics are chemicals that prevent the growth or kill the microorganisms without harming the living tissues. They can be applied to wounds, cuts and infected skin surface. They are also used in deodorants, mouthwashes and mouth fresheners to reduce unpleasant odours caused by bacterial decomposition in the body.

Difference between antiseptics and disinfectants

The basic difference between antiseptics and disinfectants is that an antiseptic is applied to the living cells, while disinfectants are applied to nonliving surfaces to kill germs. A commonly used antiseptic is Dettol; it is a mixture of chloroxylenol and terpineol. Chlorine and phenol solutions are known as disinfectants. **Chemical components in antiseptics and disinfectants**

Antiseptics:

Alcohols, chlorhexidine, hypochlorites, antibacterial dyes, inorganic iodine compounds, halogenated phenol derivatives, quinoline derivatives.

Disinfectants:

Alcohol, chlorine and chlorine compounds, formaldehyde, hydrogen peroxide, peracetic acid.

IMPORTANCE AND CONTRIBUTION OF CHEMICAL INDUSTRY TO PEOPLE'S LIVES

Chemical industries are the prime factors to convert the raw materials into desired products that we use in our day-to-day life. This has brought a great change in the way the things operate. The chemicals from the chemical industry serve as the raw materials for other industries. The chemical industry supports industries like food, pharmaceuticals, textile, agriculture, environment, hygiene, decoration and transportation. It has also significantly been used in recycling industries to control the use of new products.

Sector	Importance to people's lives
Transportation	Source of fuel which provides energy for moving vehicles
Decoration	Chemical dyes have pigments of different colours for colouring objects
Food	Additives in making food stuffs to improve appearance, taste and quality.
Water treatment	Provides safe water for domestic use
Hygiene	Sanitizers improve health of people by killing disease causing organisms on the body
Agriculture	Manufacture of fertilizers and pesticides which improve crop yield

Date . Tues 2nd April /2024

CHEMISTRY DEPARTMENT

Senior four Topic: Chemicals for consumers.

Subtopic: Chemicals in medicine.



Lesson objective. By the end of this lesson, you should be able to understand the importance of chemicals in medicine.

One of the chemistry's most significant contribution is in the field of medicine. Chemicals are used to produce medicines. The effective use of chemicals in medicine has revolutionized our ability to prevent and cure diseases, and Will continue to be a pivotal component of medical advancement.

Categories of medicine.

Medicine is any substance used in treating a disease or relief from pain **OR.** Medicine are chemicals or compounds used to halt, prevent disease, ease symptoms, or help in the diagnosis of illnesses . Medicines are classified as,

- (a) **Traditional medicine** , Is the one that utilizes culture and spiritual treatments , mostly obtained locally as extracts from mainly plants, applied singularly or in combination to treat , diagnose and prevent illnesses or cure symptoms.
- (b) **Modern medicine**, is the one that utilizes scientific knowledge, technology and evidence based practices to diagnose, treat and prevent illnesses and diseases.

Students group activity.

In your respective groups, use Internet or any other source of information such as textbooks, and make research about the different traditional medicines you use in every day life, their uses(functions) and the different modern medicines, the classifications of different modern medicine, their examples and uses (functions) of each example of medicine. Share your group answers to the rest of other groups in class.

(a) Traditional medicine.

	Traditional medicine	Use (function)
1.	Aloe vera	Treats malaria, allergic reactions, treats diabetes, skin disease. Treats burns.
2.	Guava leaf extract.	Treats malaria, ulcers, cough, diarrhea
3.	Paw paw leaf extract	Improves liver and kidney function.
4.	Garlic extract	Controls cholesterol levels, regulates blood sugar, reduces cancer risks, treats fungal and bacterial infection.
5	Momordica foetida (Bombo)	Used for fever, malaria infection, treating body odour.
6	Vernonia amygdalina. / Bitter leaf (omululuza)	Treats malaria, relieves fever, regulates blood sugars (treats diabetes) , lowers cholesterol, improves digestion.

Traditional medicines contain **active ingredients that provide relief from pain** (especially those for that work on pain) and others for bacterial infections work by **killing the bacteria thus preventing them from multiplying.**

Side effects associated with uses of traditional medicines.

The use of traditional medicines though often perceived as natural and safe, can have a dark side with potential side effects that can have serious and even life threatening. These side effects include,

- Allergic reactions and rashes
- Headache.
- Stomachache.
- Diarrhea.
- Kidney and liver damage

Mitigations to above side effects.

- Use of small amounts of traditional medicines other than excess.
- Use of modern medicine to counteract the effect of traditional medicine.

(b) Modern medicines. These are classified into three types,

- (I) Analgesics.
- (II) Antibiotics.
- (III) Psychotherapeutic medicine

What are Analgesics?

Analgesics are medicines that are used to relieve pain. They are also known as painkillers or pain relievers. Technically, the term analgesic refers to a medication that provides relief from pain without putting you to sleep or making you lose consciousness.

Antibiotics

Antibiotics are used to treat or prevent some types of bacterial infection. They kill bacteria or prevent them from reproducing and spreading.

What are Psychotherapeutic agents?

Psychotherapeutic agents are used to treat psychosis, which refers to a group of mental disorders for example depression, schizophrenia, manic-depressive disorders and so on. They affect mood and behavior.

1) Examples of analgesics, their uses(functions) , their side effects and mitigations.

Medicine .	Use of medicine	Side effects	Mitigations
1) Aspirin	Used to relieve pain from conditions of muscle aches and fever .	Abdominal pain , chest pain	-Taking right prescription. - Seek immediate Medical attention. Switch to different medicine.
2) Paracetamol	Used for pain relief and reducing high temperatures	Loss of appetite. Excessive sweating. Nausea Allergic reactions.	Follow recommended dose. Switch to different medicine. Seek immediate Medical attention.
3) Codeine	Used for pain relief and cough suppression.	Nausea. Vomiting. Fatigue. Headache.	Taking enough rest. Seek immediate Medical attention. Switch to different medicine.

How do analgesics work (function).

Analgesics work by inhibiting the sensation of pain, thus providing relief from pain.

Analgesics work by inhibiting the sensation of pain.

2) Examples of antibiotics.

Example of Antibiotics	Use (function)	Side effects	Mitigations.
1) Penicillin	Treat infections bacterial infections	Fatigue, dizziness, Nausea, vomiting, allergic reactions.	Seeking immediate Medical attention. Taking plenty of fluids. Taking enough rest.
2) Streptomycin	Treating bacterial infections for example tuberculosis	Chest pain, dizziness, Nausea, vomiting,	Seeking immediate Medical attention. Taking plenty of fluids. Taking enough rest.
3) Amoxicillin	Treating chest infections for example Phenomenia.	Nausea, allergic reactions, fatigue, vomiting, dizziness.	Seeking immediate Medical attention. Taking plenty of fluids. Taking enough rest .

How do antibiotics work (function)

Antibiotics work by **killing** the bacteria, thus preventing them from multiplying.

3) Examples of psychotherapeutic medicine, their uses, side effects and mitigations.

Example of psychotherapeutic medicine .	Use (function)	Side effects.	Mitigations
1) Stimulants.	-Awakens alertness, confidence and energy. -Treats attention-deficit hyperactivity disorder (ADHD)	Loss of appetite. Headache. Nausea. Vomiting. Increased blood pressure.	Taking enough rest Seeking immediate Medical attention. Taking medicine in right amount as prescribed.
2) Antidepressants.	Treats generalized body anxiety disorder. Treats stress and depression.	Dizziness Loss of appetite. Feeling headache. Loss of libido. Erectile dysfunction.	Seeking immediate Medical attention. Taking medicine in right amount as prescribed.
3) Antipsychotics.	Treating bipolar disorders such as extreme moods (stabilizing mood) and	Dizziness. Weight gain. Headache. Sexual problems.	Taking medicine at different times of the day. Seeking immediate

	reducing anxiety.		Medical attention.
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Evaluate (similarities and differences between traditional medicines and modern medicines)

(a) Similarities.

- (I) Both aim to prevent and treat illnesses.
- (II) Both use natural ingredients from plants and animals.

(b) Differences.

	<u>Traditional medicine</u>	<u>Modern medicine.</u>
1)	Less effective in healing illnesses.	More effective in healing illnesses.
2)	Increased risks of side effects.	Often less side effects.
3)	Has no dosage instructions.	Has required dosage instructions.
4)	Often lacks standardization in preparation.	Strong standards in production.

SPECIAL RESEARCH.

Contributions of chemical industry to people's lives.

In your respective groups, make research and discuss the different contributions of chemical industry to people's lives. (10 contributions)

END.

NUCLEAR PROCESSES.

Nuclear chemistry is the study of the chemical and physical properties of elements which are influenced by changes in the structure of the atomic nucleus.

The nucleus of an atom is composed of protons and neutrons. The number of protons in the nucleus of an atom is called **Atomic number** while the sum of protons and neutrons in the nucleus of an atom is called **atomic mass**.

Protons and neutrons are collectively called **Nucleons** which are packed together tightly in a very small nucleus.

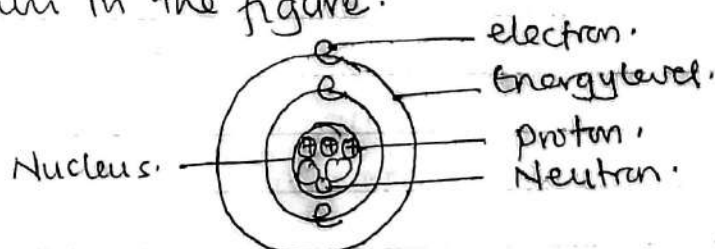
To hold the protons together in the very small volume of a nucleus, a strong nuclear force is required. This is because the tively charged protons repel each other so strongly at those short distances that the repulsion can lead to spontaneous decay of the nucleus, which would result into release of energy.

Atomic structure and Nuclear Reactions.

Structure of an atom.

An atom has 3 subatomic particles which are - Protons - Electrons and - Neutrons.

They are also known as fundamental particles of an atom. Protons and neutrons are found in the nucleus of an atom and are known as **nucleons** while electrons are found in the energy levels or shells surrounding the nucleus as shown in the figure.



- Atoms contain the same number of positively charged particles (protons) and negatively charged particles (the electrons). Thus the 2 opposite total charges cancel out to zero, leaving the atom electrically neutral.

Qn. Natural Boron occurs as $^{10}_5\text{B}$ and $^{11}_5\text{B}$.

- (a). Identify the term used to describe the two forms.
→ Isotopes.
- (b). Describe the difference between them in their sub-atomic particles.
→ They have different numbers of neutrons.
- (c). State the meaning of the term nuclide.
→ Species of atom characterised by the constitution of its nucleus and hence by the number of protons, the number of neutrons and the energy content.

Nuclear Reactions.

This is a process in which two nuclei or a nucleus and an external sub-atomic particle collide to produce one or more new nuclides. The reaction products may contain a different element or a different isotope of the same element.

Radioactive decay is the process by which unstable atomic nuclei emit sub-atomic particles and energy by radiation and disintegrate into one or more smaller nuclei.

Radioactive decay is a nuclear reaction rather than chemical reaction, coz it involves only the nuclei of atoms.

Radiation.

The emission or transmission of energy in the form of waves or particles through space is called radiation. Sunlight is the commonest form of radiation. There are radiations which are invisible like radio waves and television waves.

Exploring the Different forms of Radiations.

There are 3 major forms of radiations i.e

- Alpha radiation (α)
- Beta radiation (β) and
- Gamma radiation (γ).

Properties of Forms of Radiations.

(a) Alpha radiation (α).

- It has least penetrating power.
- It has high ionising power.
- It is deflected by ~~the~~ both electric and magnetic fields.

(b) Beta radiation (β).

- It has moderate penetrating power.
- It has moderate ionising power.
- It is deflected by both electric and magnetic fields but the direction of deflection is opposite to that of alpha radiations/particles.

(c) Gamma rays (γ).

- It has the highest penetrating power.
- It has the least ionising power.
- It is not deflected by both electric and magnetic fields.

Qn. state the changes that occur to an atomic nucleus that emits each of the following

radiations.

- (a). ~~The mass num~~ α -particle
→ The mass number decreases by four (4) and the atomic number decreases by 2 (two).
- (b). β -particle
→ The atomic number increases by one and the mass number remains unchanged.
- (c). γ -particle
→ The atomic number and mass number remain unchanged.

Nuclear fission.

This is a reaction in which the nucleus of an atom splits into two or more smaller nuclei. It is initiated by bombarding a heavy unstable nucleus by a slow-moving neutron which results in a large amount of energy being released.

The elements commonly used in nuclear fission reactions are, majorly two;

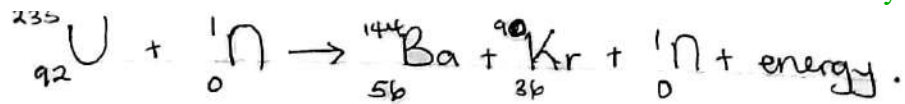
- Uranium
- Plutonium.

How nuclear fission occurs.

The fission process is caused by bombarding a heavy unstable nucleus by a slow-moving neutron. The process releases more neutrons which escape and move at a great speed.

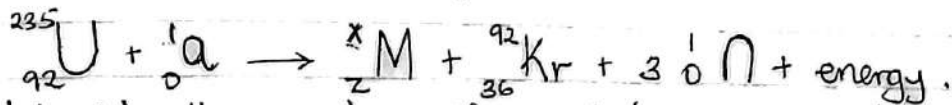
Once they collide with another radioactive heavy nucleus, they cause it to undergo fission with production of more neutrons, leading to a series of repeated steps.

Example: Nuclear fission of Uranium is as follows;



Sample Question.

Uranium atom decomposed as shown below.



- (a) Identify the radioactive which caused the nuclear fission.

→ Neutron.

- (b) Find the value of x and Z .

$$\Rightarrow 235 + 1 = x + 92 + 3$$

$$\Rightarrow 236 = x + 95$$

$$\Rightarrow x = 236 - 95 \\ = 141.$$

$$\Rightarrow 92 + 0 = Z + 36 + 0.$$

$$\Rightarrow 92 = Z + 36$$

$$Z = 92 - 36$$

$$Z = 56.$$

- (c) Use a periodic table to identify element M shown in the decay equation.

Hint: Here, we base on the atomic number coz it is always unique for every element.

So the element is Barium.

- (d) How can the above equation be Nuclear fusion?

Nuclear fusion is a reaction in which two or more atomic nuclei are combined to form one or more atomic nuclei and sub-atomic particles such as neutrons and protons.

The results of nuclear fusion are; large amounts of energy, heavier nuclei and sub-atomic particles mostly a neutron.

In presence of the sun, high temperature and pressure causes nuclear fusion.

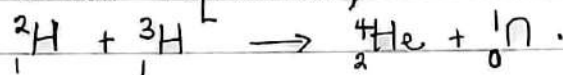
How nuclear fusion occurs.

The ~~an~~ neutron increases the mass of the atom by raising the nucleon number and since the atom must neutrally contain a nucleus with equal number of neutrons and protons, it becomes slightly unstable and thus splits up to ~~dis~~ dis-integrate releasing energy.

There are 2 major elements that undergo nuclear fusion and these are;

- Deuterium (${}^2_1\text{H}$).
- Tritium (${}^3_1\text{H}$).

Reaction equation;



Advantages of nuclear reactions.

- Provide relatively low-cost energy.

They are ~~reliable~~ reliable.

low pollution production.

Source of fuel.

- Nuclear reactions release more energy per reaction than chemical reactions.

→ **Reliable power Generation;** Nuclear power can provide a stable and continuous supply of electricity, as nuclear reactor can operate independently of weather conditions and fuel availability.

→ **Low Greenhouse Gas Emissions.** Nuclear power plants produce minimal greenhouse gases compared to fossil fuel-based power plants, which helps in reducing climate change impacts.

→ **Reduced Air pollution;** Nuclear energy generation does not emit air pollutants like sulphur dioxide or nitrogen oxides, which contribute to smog

→ Long-term fuel supply.

Uranium, used in many nuclear reactors, is abundant and can supply energy for a long time.

Disadvantages of Nuclear reactions.

→ Nuclear reactions produce radioactive waste that remains hazardous for thousands of years.

→ Nuclear accidents; Accidents at nuclear power plants such as those at Chernobyl, can lead to catastrophic releases of radiation, posing severe health and environmental risks.

→ High initial costs. Building and decommissioning nuclear power plants require substantial capital investment. The construction process can also take many years.

→ Nuclear proliferation risks; The technology and materials used in nuclear reactors can potentially be diverted for the development of nuclear weapons, raising concerns about global security.

→ Limited fuel supply. Although uranium is relatively abundant, its supply is finite. Long-term sustainability depends on alternative nuclear fuels.

→

Precaution to be taken while carrying out nuclear reactions.

- Use gloves always when working with radioactive materials.
- In radioactive laboratory, do not eat, drink, smoke or do not use cosmetics.
- Do not pipette orally.
- Wear disposable clothes if there is substantial contamination risk.
- Keep your lab coat separate from your everyday attire.
- Stringent hygiene is required when handling radioactive materials.

Radioactivity.

This is the spontaneous disintegration of unstable atomic nuclei, giving out radiations and large amounts of energy.

The unstable atomic nucleus gives up some energy in order to attain stability. The process is spontaneous (occurs on its own, without any energy input from the outside) and so is not influenced by external conditions like temperature and pressure.

Radioactivity is also known as Radioactive decay. It occurs at random; it is impossible to predict when it will occur or decay.

NB. Radioactive decay can either be natural or artificial.

Types of Radioactive Decay.

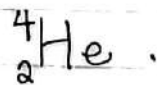
There are 3 types of Radioactive decay 'ie',

- Alpha decay;
- Beta decay.
- Gamma decay.

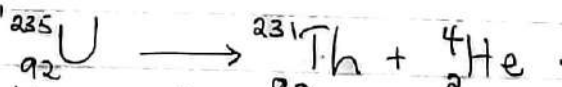
Alpha Decay.

This is a type of radioactive decay in which atomic nucleus emits an alpha particle and transforms into a different atomic nucleus with a mass ^{number} that is reduced by four and atomic number reduced by two.

An alpha particle is a helium nucleus. It has a mass of 4 atomic mass unit and atomic number of 2. It is represented as



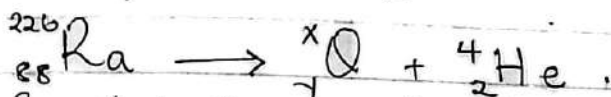
Example:



Uranium disintegrated to form Thorium and Helium.

Qn. Radium 226 undergoes radioactive decay as

shown in the equation below.



- (a) Complete the equation by finding the value of X and Z .

$$\Rightarrow 226 = X + 4$$

$$\Rightarrow X = 222$$

The value of X is 222.

$$\Rightarrow 88 = Z + 2$$

$$\Rightarrow Z = 86$$

The value of Z is 86.

- (b) Identify the element Q, which radium decays to (Use the periodic table)

$\rightarrow$ Radon is the element with the atomic number of 86.

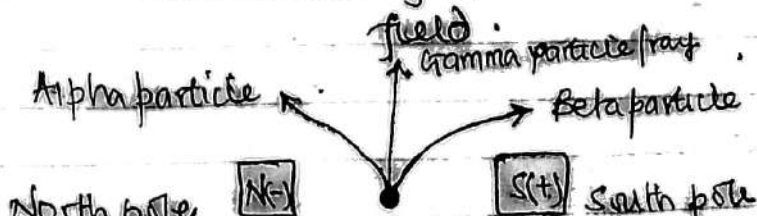
- (c) Explain why we consider the atomic number, not mass number in determining the new element formed.

$\rightarrow$ This is coz the atomic number remains the same for all isotopes of an element but the mass number changes with each isotope.

Properties of alpha particles.

- They are positively charged. Each alpha particle has a charge of +2 and a mass of 4 units.
- Alpha particles are deflected towards the negative electric plate and repelled by the positive plate.
- Alpha particles are deflected by a magnetic field.

Illustration of Radiations in a magnetic field.



- The penetrating power of alpha particles is very poor since they are large particles. They can be stopped by a sheet of paper.
- Because of high charge, alpha particles have a great ionizing power (the ability to change atoms into ions due to radiation effect).

Beta particle Decay.

Beta decay is a type of radioactive decay in which an atomic nucleus emits a beta particle, transforming into an isobar (isotopes of different elements but with the same atomic mass (mass number) of that nuclide).

A beta particle is a fast-moving electron. It has negligible mass and a charge of -1 . When an atomic nucleus emits a beta particle, the atomic mass remains the same while the atomic number increases by 1. eg

Equation: e

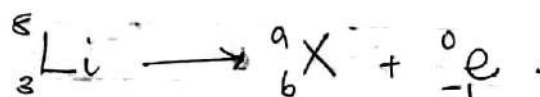


Properties of beta particles.

- They are negatively charged. Each beta particle has a charge of -1 and negligible mass.
- Beta particles are deflected towards the positive electric plate and ~~repe~~ repelled by the negative plate.
- Beta particles are deflected by a magnetic field.
- The penetrating power of beta particles is moderate. They can be stopped by a block of wood.
- Because of the low charge, beta particles have an average ionizing power.

Sample Question.

Lithium-8 undergoes radioactive decay as shown below.



- (a) Copy and complete the equation. (Calculate the values of a and b and re-write the equation with known values).
- (b) Identify element X, which lithium decays to.

Gamma Decay.

This is a type of radioactivity in which some unstable atomic nuclei dissipate excess energy by a spontaneous electromagnetic process.

Gamma rays are not particles but electromagnetic waves with very short wave length.

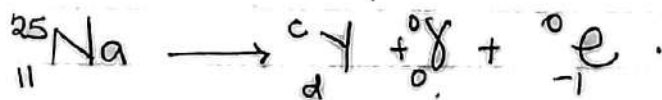
When an atomic nucleus emits gamma rays, there is no change in the atomic number or mass number but the energy associated with gamma emission leads to an extra stability of the atomic nucleus.

Properties of Gamma rays.

- They are not charged, hence are not deflected by magnetic or electric field.
- Their penetrating power is very great.
- Their ionizing power is very low.

Question.

Sodium -25 undergoes gamma decay as shown in the equation.



- (a) Copy and complete the equation.
- (b) Identify element X and give a reason for your answer.

HALF - LIFE.

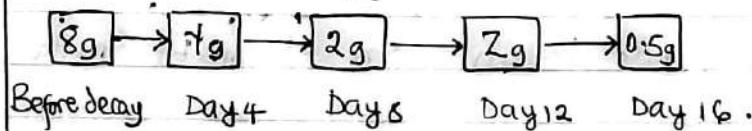
This is the time taken for a given mass of a radioactive

Different atomic nuclei undergo radioactivity at different rates. Some disintegrate very fast while others decay slowly.

eg 2g of Carbon-14 isotope take about 5730 years to reduce to half of the initial quantity, i.e. 1g while the same mass of Sodium-25 takes only 1.0 minute to reduce to 1g.

The concept of half-life:-

Study and Interpret the figure below



(a) What was the half-life of the radioactive material?

→ Four decay.

(b) Work out the masses x and z obtained at day 4 and day 12, respectively.

→ $x = 4g$.

→ $z = 1g$.

(c) What percentage of the radioactive material had been lost at day 8?

Lost mass = $(8 - 2)g$.

= $6g$.

Total mass = $8g$.

Percentage loss = $\left(\frac{6}{8} \times 100\right)$

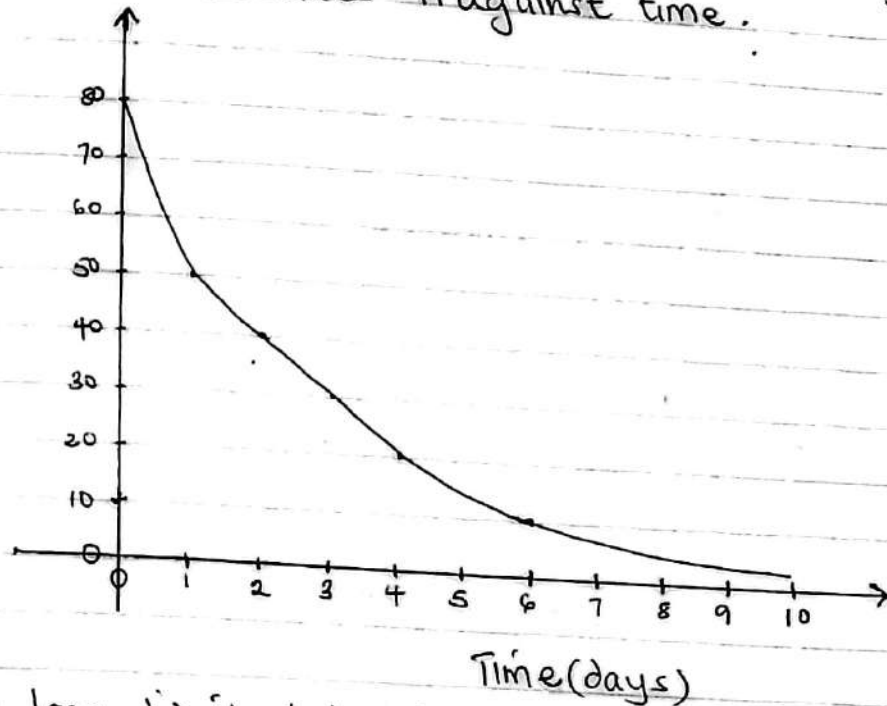
= 75%.

(d) Predict the mass of the radioactive substance that will remain at day 20.

→ Since the half-life is 4 after every four days, we consider our initial mass at day 16 which is $0.5g$ and will reduce to $0.25g$.

measuring the time taken for an element to reduce to half of its original amount.

Study the graph below showing count rate of a radioactive substance H against time.



How long did it take for the count rate to reduce to reduce from;

- (a) 80 to 40
- (b) 40 to 20
- (c) 20 to 10
- (d) 10 to 5

Qn. Eight grams (8g) of substance D were initially present in a sample. The mass of D in the sample reduced to half in 8 days. Find the mass remaining after 24 days.

Soln:

After 8 days, 4 grams ~~reduced~~ remained.

After 16 days, 2 grams remained.

After 24 days, 1 gram remained.

There 1 gram remained after 24 days.

Calculating Half life.

After one half-life, one half-life of the original amount of a radioactive substance remains.

After 2 half-lives, one half of the previous half remains or one quarter and so on.

After n half-lives, $\frac{1}{2^n}$ of the original amount of a radioactive substance remains.

Example.

A radioactive isotope has activity measured on a Geiger-Muller tube at 10,000 becquerels (Bq). What is the activity of the isotope after 3 half-lives?

Soln:

The activity after 3 half-lives = $\frac{1}{2^3}$ x original activity.

$$\Rightarrow \frac{1}{2^3} \times 10,000.$$

$$= 1,250 \text{ Bq.}$$

Qn. The table below shows for a substance to decay and the remaining percentages.

Time (yrs)	%age remaining	Time (yrs)	%age remaining
0	100	1.02	87.5
5.27	50	7.46	37.5
10.54	25	15.81	12.5
21.08	6.25	26.35	3.13

- plot an appropriate graph of percentage remaining against time.
- from the graph they have drawn, let them determine the half-life of the substance.

Qn. The half-life of a certain radioactive isotope is 24 days. If the initial mass is 0.64g. Find;

- the mass remaining after 120 days.
- how long it will take to have only 0.04g remaining undecayed.

Soln:

$$(a) \text{ Mass of the isotope remaining after 24 days} = \frac{0.64}{2}$$

$$= 0.32 \text{ g.}$$

$$\text{Mass of the isotope remaining after 48 days} = \frac{0.32}{2}$$

$$= 0.16 \text{ g.}$$

Mass of the Isotope remaining after 72 days

$$= \frac{0.16}{2}$$

$$= 0.08 \text{ g.}$$

Mass of the Isotope remaining after 96 days

$$= \frac{0.08}{2}$$

$$= 0.04 \text{ g.}$$

Mass of the Isotope remaining after 120 days

$$= \frac{0.04}{2}$$

$$= 0.02 \text{ g.}$$

$$\text{Number of half-lives in summary} = \frac{120}{24} = 5.$$

(b) 96 days.

Applications of Radioactivity.

1. Radiotherapy; Gamma rays are used in the treatment of cancer and detecting breakages in bones.
2. Sterilization; Gamma rays are used to sterilise medical instruments.
3. Archeology; living plants and animals take in radioactive carbon, when a plant or animal dies, the carbon starts to decay. By measuring the radioactive activity, the age of the remains can be obtained.
4. Environment; Radioisotopes can be used to trace and analyse pollutants.
5. Biochemistry and genetics; radioisotopes may be used as tracers to label molecules so that chemical processes like DNA replication (action of reproducing) can be traced.

Note: Radioactive Isotopes are used for diagnosis of some organs as shown in the table below.

Dangers of Radioactivity.

- Causes cancer
- Affect eyesight.
- Causes radiation skin burns.
- Damage the body cells and tissues.
-

Safe Safety precautions.

- Exposure time should be short.
- Radioisotopes can be handled with tongs with remote control.
- Personnel working in Radioactive plants should wear lead jackets.
- Radiation should be directed away from the people around.

Sample paper

SECTION A

(Answer all items in this section)

ITEM 1

In one of the societies in Kampala, there is an outbreak of bacterial diseases such as typhoid. The society members rely on each other for medical advice. Recently, a resident complained about persistent stomach ache followed by diarrhoea, but there was no one to advise on proper medication the society members should use.

Task: As a learner of chemistry;

- a) Advise society members on the possible types of products to use?

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- b) Carefully educate them on what the products exactly do.

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- c) Advise them on the challenges associated with the product use and consequently how to select the products to use

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ITEM 2

Sharon was using a mobile phone while studying at home, nearby, there was a glass containing a solution prepared by dissolving salt in water. While reaching for the phone, Sharon accidentally knocked over the glass, and the salt solution spilled onto the phone. Within a short time, the phone screen went off and device stopped working completely. Curiously about what happened, Sharon reported the incident to a friend but couldn't convince her about the incidence.

Task; As a Chemistry student;

- a) Suggest two precautions that should be taken to prevent damage to electronic devices from such substances.

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- b) Explain in detail what happened when the salt solution came into contact with the mobile phone.

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- c) State one other example of a substance that behaves like table salt in solution and explain why.

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SECTION B

PART I

(Respond to only one item from this part)

ITEM 3

A parent of a senior four candidate in one of the villages in Masaka District belongs to a “women’s development group”. The members of the group have come up with a project idea of making soap which they think they can sell to the nearby shops and obtain revenue. However, none of the members has knowledge about soap making. But the parent suggested a name of an s.4 candidate to the executive committee of the group as a rescuer and the committee has approached him/her for guidance.

TASK; As a chemistry learner, Write a brief message that candidate will deliver to the group members.

ITEM 4

Due to increased demand for the construction of strong and modern houses, iron bars are used but are few in supply. To ensure the availability of iron bars, the government is considering setting up an iron production plant in a certain area. However, the residents of the area need to be sensitized about the industrial process, social benefits, side effects and how they can be overcome.

Your head teacher has been identified to sensitize the residents but has no knowledge about the matter.

TASK: As a learner of chemistry, prepare a write up that your head teacher will use to sensitize the residents.

PART II

(Respond to only one item from this part)

ITEM 5

In many parts of the country, there has been rapid urbanization, modernization and infrastructure development such as construction of roads, and expansion of towns.

While these developments have improved living standards, they have also led to degradation of Natural resources, including poor waste management.

This situation has attracted the attention of officials from the National Environment Management Authority (NEMA), who are planning to conduct sensitization workshops in different urban and peri-urban communities,

Task; As a chemistry student, you have been invited to participate in the workshop, prepare a well-organised presentation that you will deliver to the community. (18 scores)

ITEM 6

Natural resources especially rocks, hill and mountains have been destroyed as a result of increasing population and human activities. This has attracted the attention of the officials from the National Environment Management Authority (NEMA).

The officials are planning to create awareness to the people of the country through sensitization workshops organized in different communities.

TASK: As a learner of chemistry, make a write up of the presentation the NEMA officials would deliver during the sensitization workshop. **(18 Scores)**

END